CHAPTER 01

Nomenclature

Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: 42d83a6177ccbbbd04c60c595e30172dd4d45b28a69e518658557693be440796

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

Coverage checklist

ch01:exercise:gp1a | gp1a:stage-1, gp1a:stage-2, gp1a:stage-3

ch01:exercise:gp1b | gp1b:stage-1, gp1b:stage-2, gp1b:stage-3

ch01:exercise:gp2a | gp2a:stage-1, gp2a:stage-2, gp2a:stage-3

ch01:exercise:gp2b | gp2b:stage-1, gp2b:stage-2, gp2b:stage-3

ch01:exercise:gp3a | gp3a:stage-1, gp3a:stage-2, gp3a:stage-3

ch01:exercise:gp3b | gp3b:stage-1, gp3b:stage-2, gp3b:stage-3

ch01:exercise:gp4a | gp4a:stage-1, gp4a:stage-2

ch01:exercise:gp4b | gp4b:stage-1, gp4b:stage-2

ch01:exercise:gp5a | gp5a:stage-1, gp5a:stage-2, gp5a:stage-3

ch01:exercise:gp5b | gp5b:stage-1, gp5b:stage-2, gp5b:stage-3

ch01:exercise:gp6a | gp6a:stage-1, gp6a:stage-2, gp6a:stage-3

ch01:exercise:gp6b | gp6b:stage-1, gp6b:stage-2, gp6b:stage-3

ch01:lesson:naming | naming:rule, naming:cue, naming:trap, naming:app, naming:step-1, naming:step-2, naming:step-3, naming:step-4

ch01:lesson:alcohols | alcohols:rule, alcohols:cue, alcohols:trap, alcohols:app, alcohols:step-1, alcohols:step-2, alcohols:step-3, alcohols:step-4

ch01:lesson:rings | rings:rule, rings:cue, rings:trap, rings:app, rings:step-1, rings:step-2, rings:step-3

ch01:lesson:diols | diols:rule, diols:cue, diols:trap, diols:app, diols:step-1, diols:step-2

ch01:lesson:carbonyls | carbonyls:rule, carbonyls:cue, carbonyls:trap, carbonyls:app, carbonyls:step-1, carbonyls:step-2, carbonyls:step-3, carbonyls:step-4

ch01:lesson:derivatives | derivatives:rule, derivatives:cue, derivatives:trap, derivatives:app, derivatives:step-1, derivatives:step-2, derivatives:step-3, derivatives:step-4

ch01:molecule:fig15 | fig15

ch01:molecule:enol | enol

ch01:molecule:cyhex | cyhex

ch01:molecule:vicinal | vicinal

ch01:molecule:geminal | geminal

ch01:molecule:ketone | ketone

ch01:molecule:hyald | hyald

ch01:molecule:hyoxo | hyoxo

ch01:molecule:ester | ester

ch01:molecule:namide | namide

ch01:molecule:anhyd | anhyd

ch01:molecule:branch | branch

ch01:molecule:fake_pent | fake_pent

ch01:molecule:ynol5 | ynol5

ch01:molecule:cyol | cyol

ch01:curriculum | scope-1, scope-2, scope-3, scope-4, scope-5

Curriculum scope

scope-1: 1.1 IUPAC Naming Conventions

scope-2: 1.2 Hydrocarbons and Alcohols

scope-3: 1.3 Aldehydes and Ketones

scope-4: 1.4 Carboxylic Acids and Derivatives

scope-5: 1.5 Summary of Functional Groups

Objective: Choose parents and locants

Objective: Distinguish common functional groups

Objective: Name multifunctional compounds

Supplied Kaplan curriculum: Chapter 1, sections 1.1–1.5 | organic_chemistry_kaplan/OEBPS/chapter01b.html

IUPAC Blue Book: nomenclature conventions | https://iupac.qmul.ac.uk/BlueBook/contents.html

IUPAC Blue Book P-14.5.1–2: simple and compound substituent ordering (inspected 2026-10-19) | https://iupac.qmul.ac.uk/BlueBook/P1.html#1405

OpenStax Organic Chemistry 3.4: Naming Alkanes (inspected 2026-10-19) | https://openstax.org/books/organic-chemistry/pages/3-4-naming-alkanes

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

naming | 1.1 Parent Chains, Locant Tiebreakers & Alphabetization

Principal Group → Longest Chain → Lowest Locants

naming:rule | Core Rules

For the simple saturated acyclic structures here, choose the longest continuous carbon chain; tied lengths favor more substituents. For functionalized structures, first include the principal suffix group and required unsaturation. Compare locant sets at their first difference, not by their sum.

Schematic summary: Principal Group → Longest Chain → Lowest Locants

Principal suffix — Include its group and required unsaturation

Parent chain — Compare valid continuous chains

Locants — First point of difference

naming:cue | Memory Cues

Alphabetize substituent names: Ignore locants and multipliers that count repeated simple substituents: dimethyl is ordered under m. In traditional alkyl names, ignore n-, sec-, and tert-; include iso-, neo-, and cyclo- when part of the substituent name (isopropyl under i; cyclopropyl under c). The parent name is not part of this substituent list. For a compound substituent in parentheses, use its complete name: (1,1-dimethylethyl) is ordered under d. A multiplier within that substituent is retained for alphabetization.

Schematic summary: Choose the case that applies

Simple repeated substituent — Ignore di-, tri- and similar counting multipliers

Traditional alkyl prefix — Ignore n-, sec-, tert-; retain iso-, neo-, cyclo-

Compound substituent — Alphabetize its complete name

naming:trap | Common Mistakes / MCAT Traps

The "Fake Parent Chain" Trap ("2,3-diethylpentane"): An ethyl group at C2 of a pentane chain extends the longest chain to 6 carbons (3-ethyl-4-methylhexane). Always draw out suspect names!

Schematic summary: Shortcut versus a better check

Avoid — Trusting a written parent chain without tracing it

Check — A branch may extend the longest continuous chain

naming:app | MCAT Applications

Branched Amino Acids (Val, Leu, Ile): Recognizing isopropyl (Valine), isobutyl (Leucine), and sec-butyl (Isoleucine) lets you classify hydrophobic side chains on sight.

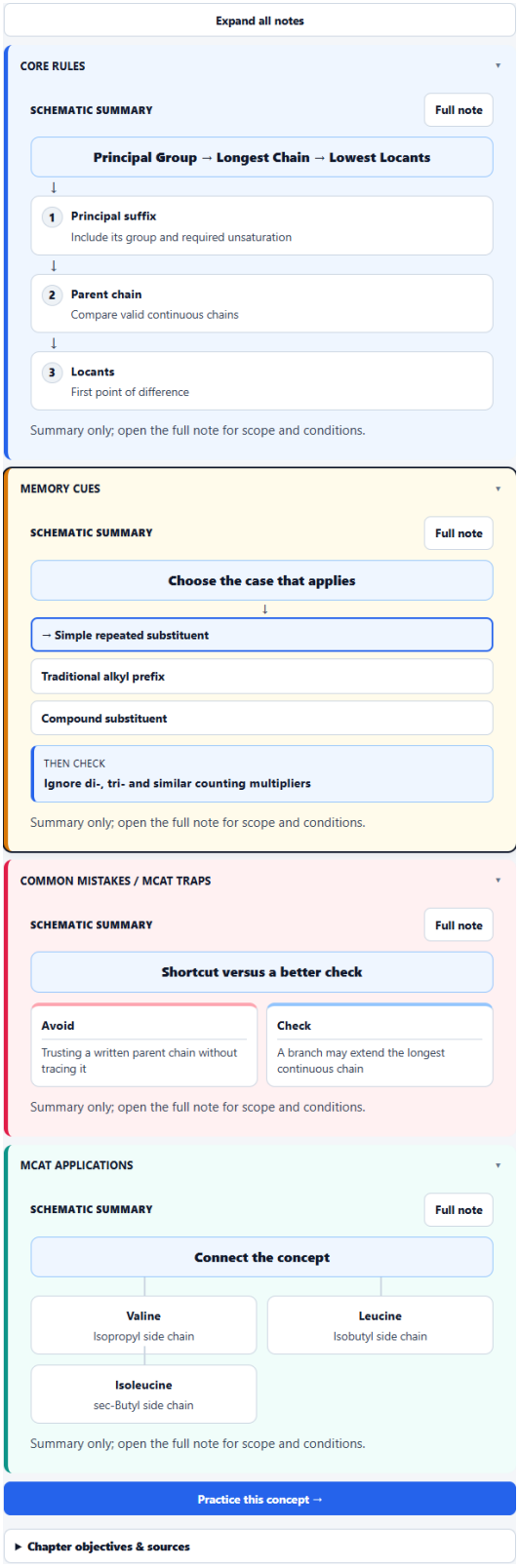
Schematic summary: Connect the concept

Valine — Isopropyl side chain

Leucine — Isobutyl side chain

Isoleucine — sec-Butyl side chain

Screenshot: naming-visuals



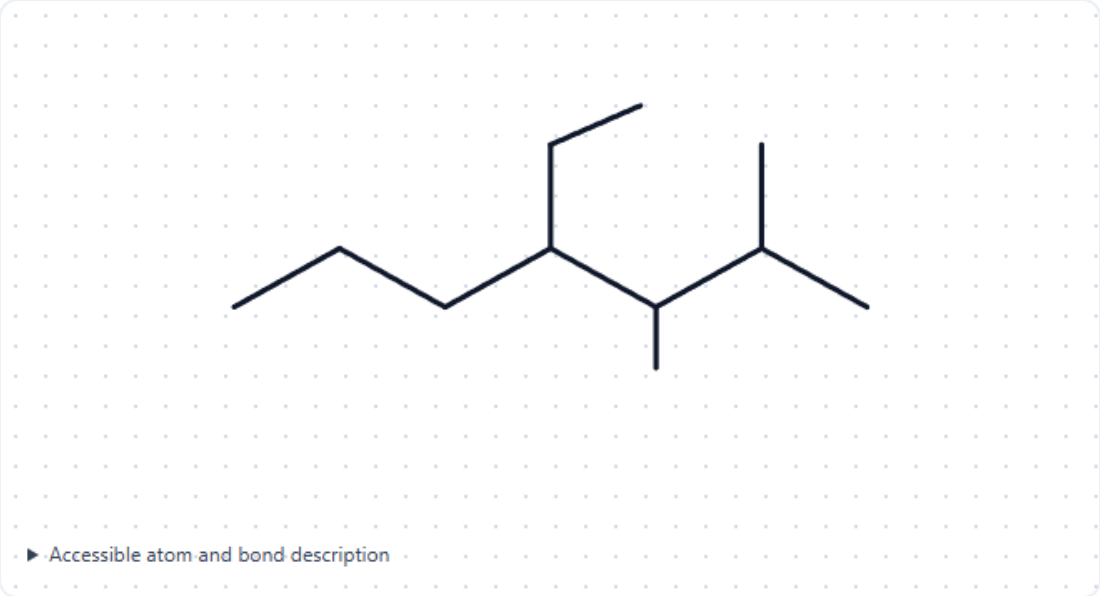
naming:step-1 | 1. Inspect

Step 1 (Inspect): Unnumbered hydrocarbon with 11 total carbons. Trace every continuous path before choosing the parent chain.

Displayed molecule ID: fig15

{"highlights": {}, "revealName": false}

Screenshot: naming-step-1

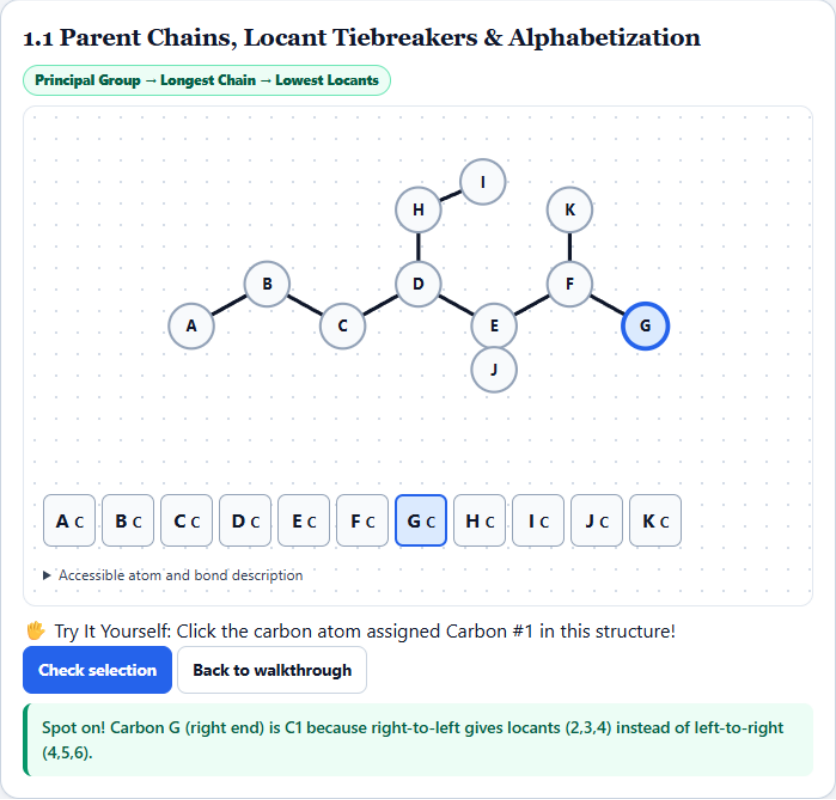


{"nameAssembly": [["np-sub", "4-ethyl-", "C4 branch (E 1st)"], ["np-sub", "2,3-dimethyl", "C2 & C3 (ignore di-)"], ["np-root", "hept", "7C parent"], ["np-suf", "ane", "alkane"]]}

Embedded challenge: Hands-on: Try It Yourself: Click the carbon atom assigned Carbon #1 in this structure!

{"prompt": "Hands-on: Try It Yourself: Click the carbon atom assigned Carbon #1 in this structure!", "answer": [6], "explanation": "Spot on! Carbon G (right end) is C1 because right-to-left gives locants (2,3,4) instead of left-to-right (4,5,6)."}

Screenshot: naming-step-1-challenge



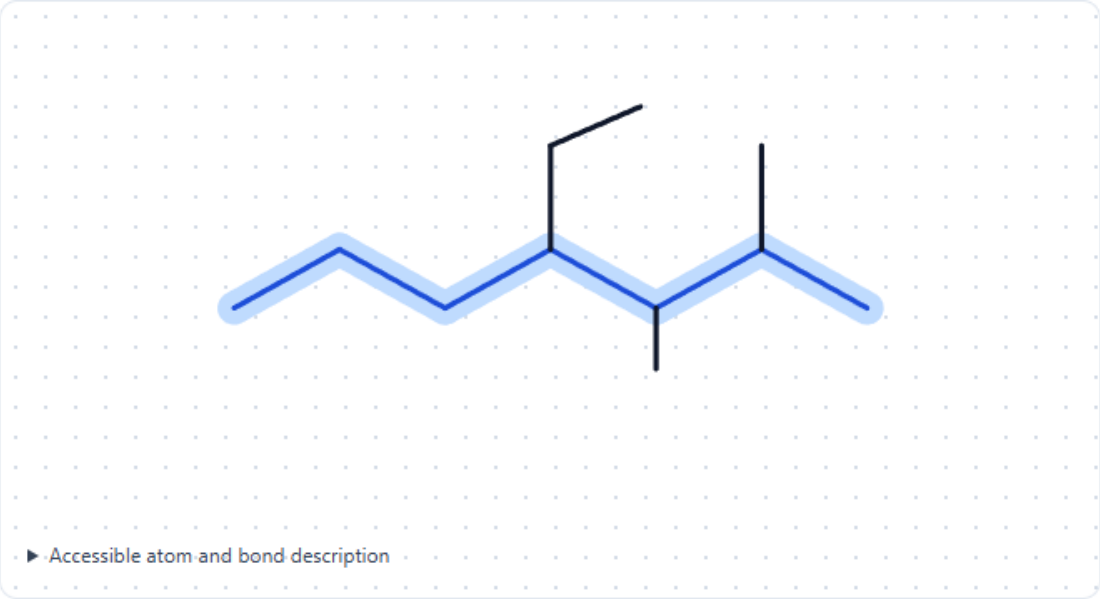
naming:step-2 | 2. Parent Chain

Step 2 (Longest Chain + Tiebreaker): The longest continuous path has 7 carbons (heptane) . Going through the ethyl branch only reaches 6 carbons, so the 7-carbon path wins. (If two chains did tie, pick the one with more substituents .)

Displayed molecule ID: fig15

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Screenshot: naming-step-2

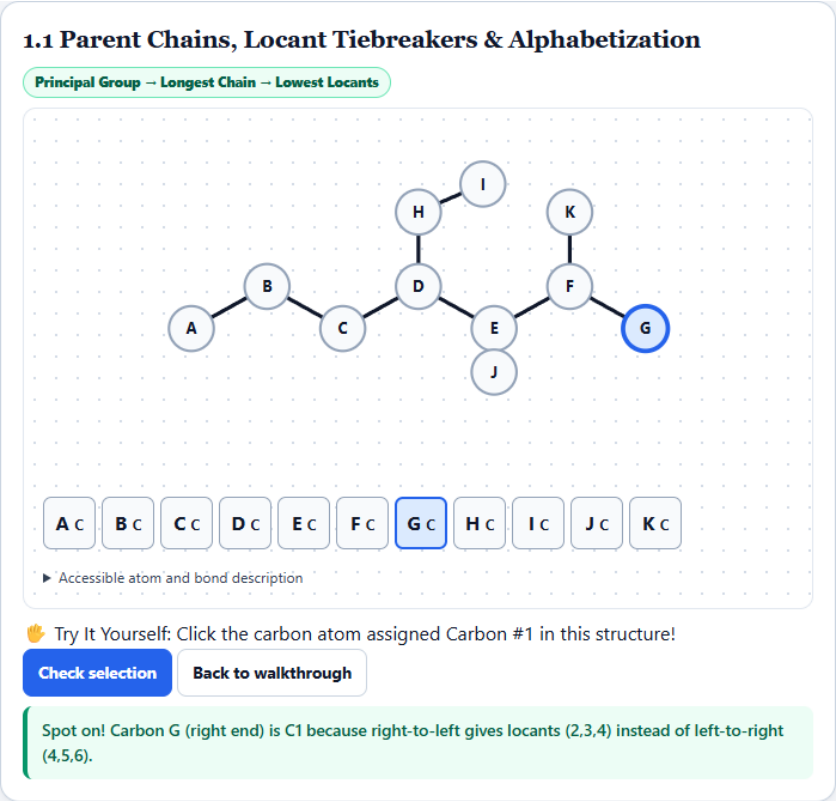


{"nameAssembly": [["np-sub", "4-ethyl-", "C4 branch (E 1st)"], ["np-sub", "2,3-dimethyl", "C2 & C3 (ignore di-)"], ["np-root", "hept", "7C parent"], ["np-suf", "ane", "alkane"]]}

Embedded challenge: Hands-on: Try It Yourself: Click the carbon atom assigned Carbon #1 in this structure!

{"prompt": "Hands-on: Try It Yourself: Click the carbon atom assigned Carbon #1 in this structure!", "answer": [6], "explanation": "Spot on! Carbon G (right end) is C1 because right-to-left gives locants (2,3,4) instead of left-to-right (4,5,6)."}

Screenshot: naming-step-2-challenge



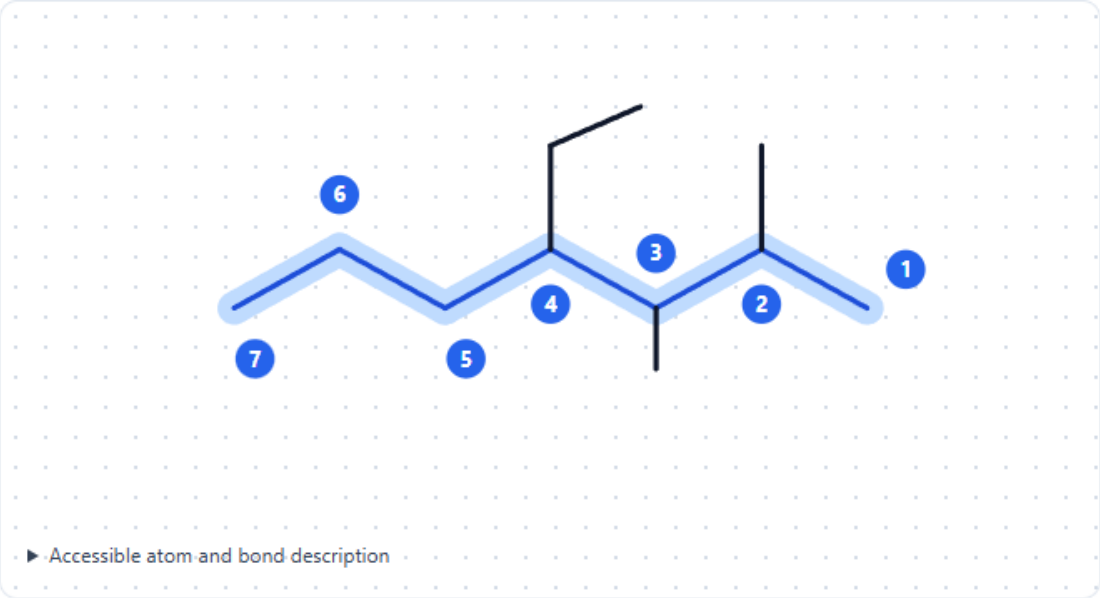
naming:step-3 | 3. Number Carbons

Step 3 (Animated Carbon Numbering): Numbering right-to-left gives (2,3,4) ; left-to-right gives (4,5,6). Compare at the first point of difference: 2 < 4 .

Displayed molecule ID: fig15

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Screenshot: naming-step-3

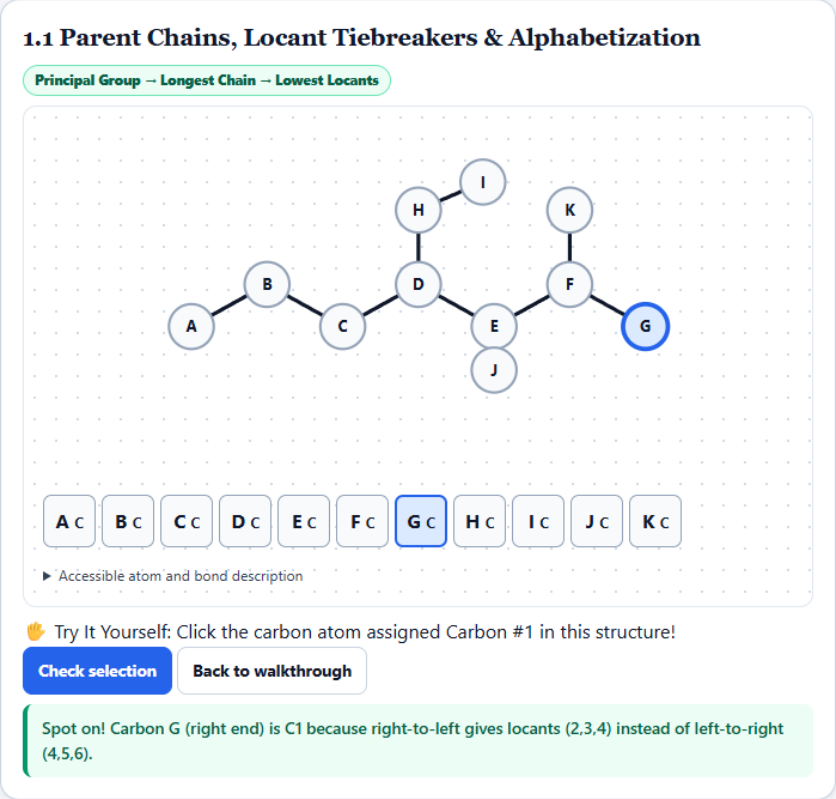


{"nameAssembly": [["np-sub", "4-ethyl-", "C4 branch (E 1st)"], ["np-sub", "2,3-dimethyl", "C2 & C3 (ignore di-)"], ["np-root", "hept", "7C parent"], ["np-suf", "ane", "alkane"]]}

Embedded challenge: Hands-on: Try It Yourself: Click the carbon atom assigned Carbon #1 in this structure!

{"prompt": "Hands-on: Try It Yourself: Click the carbon atom assigned Carbon #1 in this structure!", "answer": [6], "explanation": "Spot on! Carbon G (right end) is C1 because right-to-left gives locants (2,3,4) instead of left-to-right (4,5,6)."}

Screenshot: naming-step-3-challenge



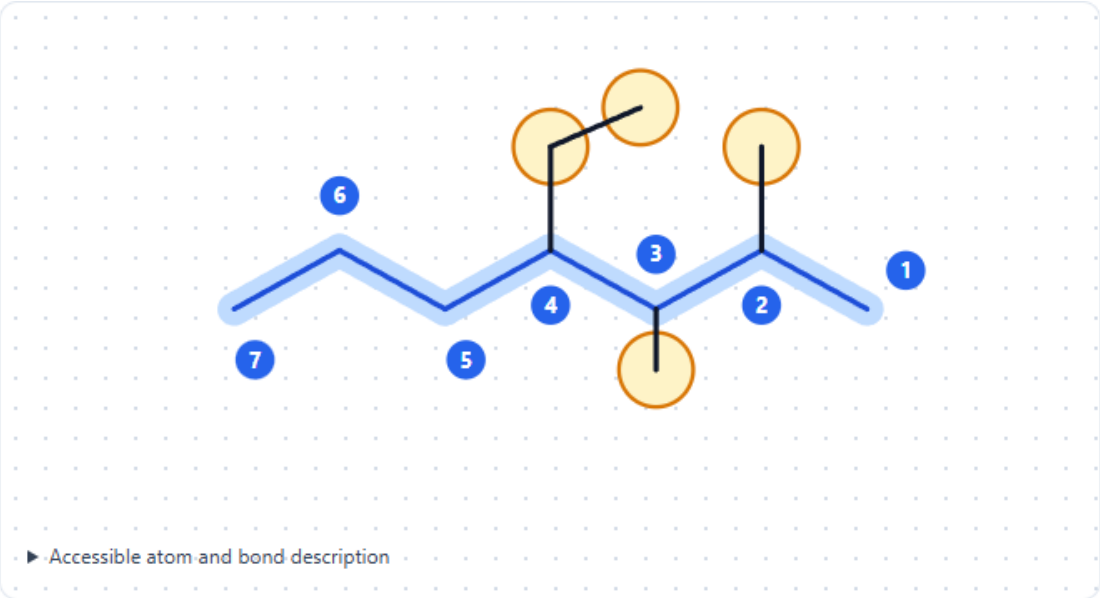
naming:step-4 | 4. Assemble Name

Step 4 (IUPAC Name Assembly): Alphabetize 4-ethyl before 2,3-dimethyl (ignore di-) → 4-ethyl-2,3-dimethylheptane .

Displayed molecule ID: fig15

{"highlights": {"path": [0, 1, 2, 3, 4, 5, 6], "num": [6, 5, 4, 3, 2, 1, 0], "sub": [7, 8, 9, 10]}, "revealName": true}

Screenshot: naming-step-4

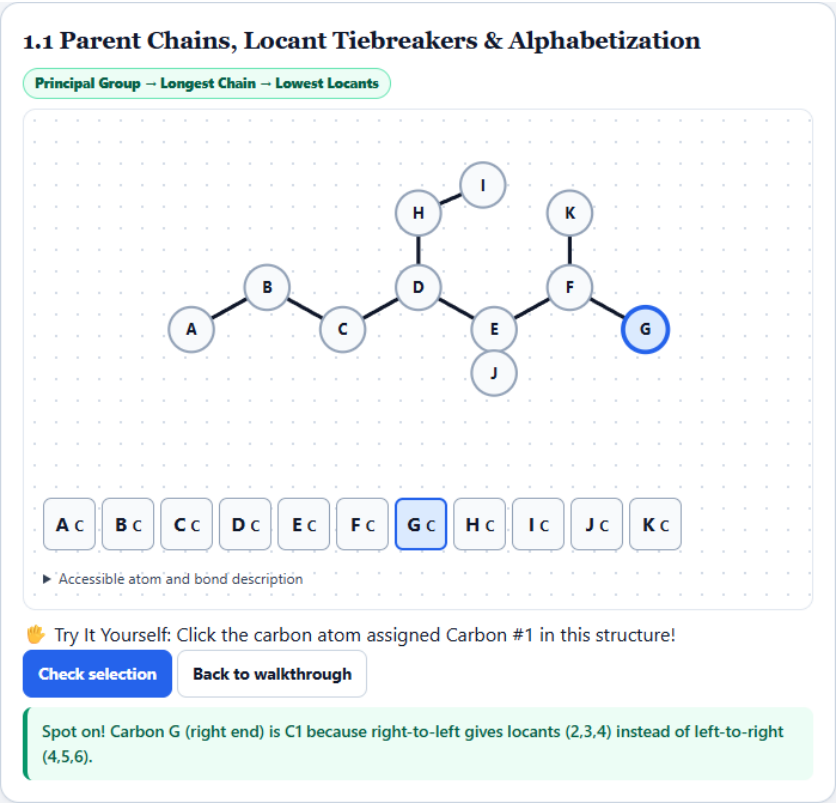


{"nameAssembly": [["np-sub", "4-ethyl-", "C4 branch (E 1st)"], ["np-sub", "2,3-dimethyl", "C2 & C3 (ignore di-)"], ["np-root", "hept", "7C parent"], ["np-suf", "ane", "alkane"]]}

Embedded challenge: Hands-on: Try It Yourself: Click the carbon atom assigned Carbon #1 in this structure!

{"prompt": "Hands-on: Try It Yourself: Click the carbon atom assigned Carbon #1 in this structure!", "answer": [6], "explanation": "Spot on! Carbon G (right end) is C1 because right-to-left gives locants (2,3,4) instead of left-to-right (4,5,6)."}

Screenshot: naming-step-4-challenge



alcohols | 1.2 Hydrocarbons, Unsaturation & Alcohol Suffix Priority

Alcohol (–ol) Outranks Alkene (–en–) & Alkyne (–yn–)

alcohols:rule | Core Rules

An alcohol (–OH) has higher IUPAC naming seniority than carbon–carbon double or triple bonds. Replace the final –e with –ol and number the chain to give the –OH carbon the lowest possible locant.

Schematic summary: Alcohol (–ol) Outranks Alkene (–en–) & Alkyne (–yn–)

Principal OH — Outranks C=C and C≡C here

Number the chain — Give the principal OH the lower locant

Build the name — Retain en/yn in the stem; add ol

alcohols:cue | Memory Cues

Infix + Suffix Pattern (Root + en/yn + ol): Unsaturation inside the parent chain stays in the stem as –en– or –yn– (e.g., but-3-en-2-ol).

Schematic summary: Choose the case that applies

Principal group — Use the suffix -ol

Subordinate OH — Use the prefix hydroxy-

alcohols:trap | Common Mistakes / MCAT Traps

Give the principal alcohol the lowest locant before the C=C or C≡C. If multiple-bond locant sets tie in an enyne, the double bond receives the lower locant.

Schematic summary: Shortcut versus a better check

Avoid — Giving unsaturation priority over the principal OH

Check — If enyne locant sets tie, the double bond gets the lower locant

alcohols:app | MCAT Applications

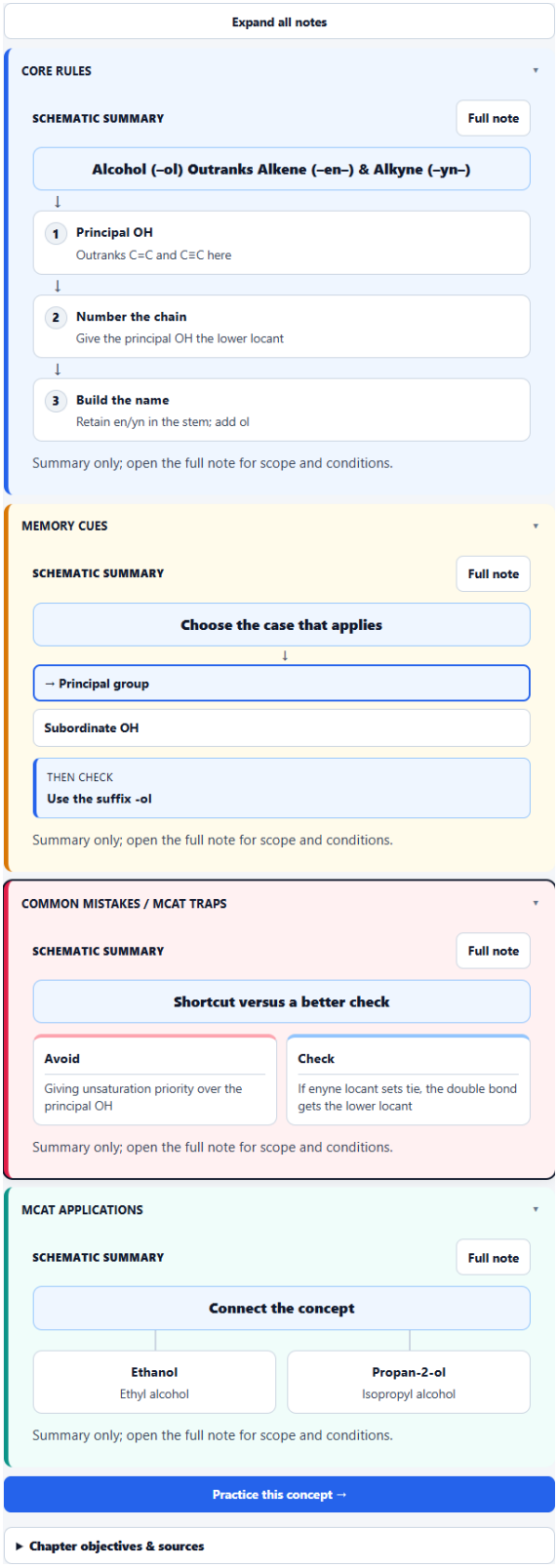
Common Alcohol Names: Ethanol = ethyl alcohol ; propan-2-ol (2-propanol) = isopropyl alcohol . Subordinate –OH is named hydroxy– .

Schematic summary: Connect the concept

Ethanol — Ethyl alcohol

Propan-2-ol — Isopropyl alcohol

Screenshot: alcohols-visuals



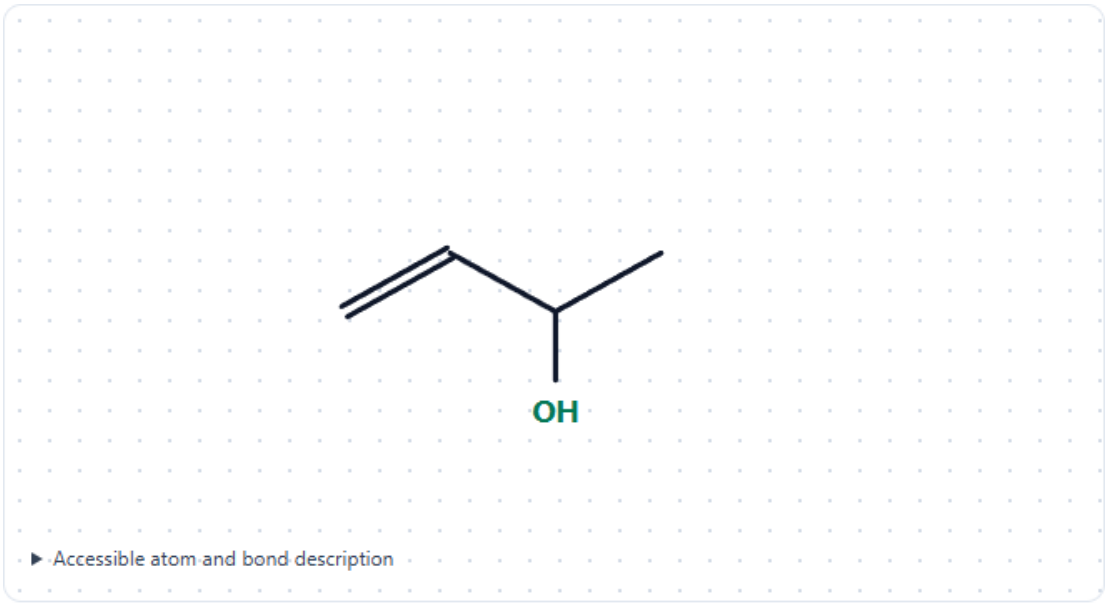
alcohols:step-1 | 1. Inspect

Step 1 (Inspect): Unnumbered skeletal structure. Identify the alcohol (–OH) and the double bond (C=C).

Displayed molecule ID: enol

{"highlights": {}, "revealName": false}

Screenshot: alcohols-step-1

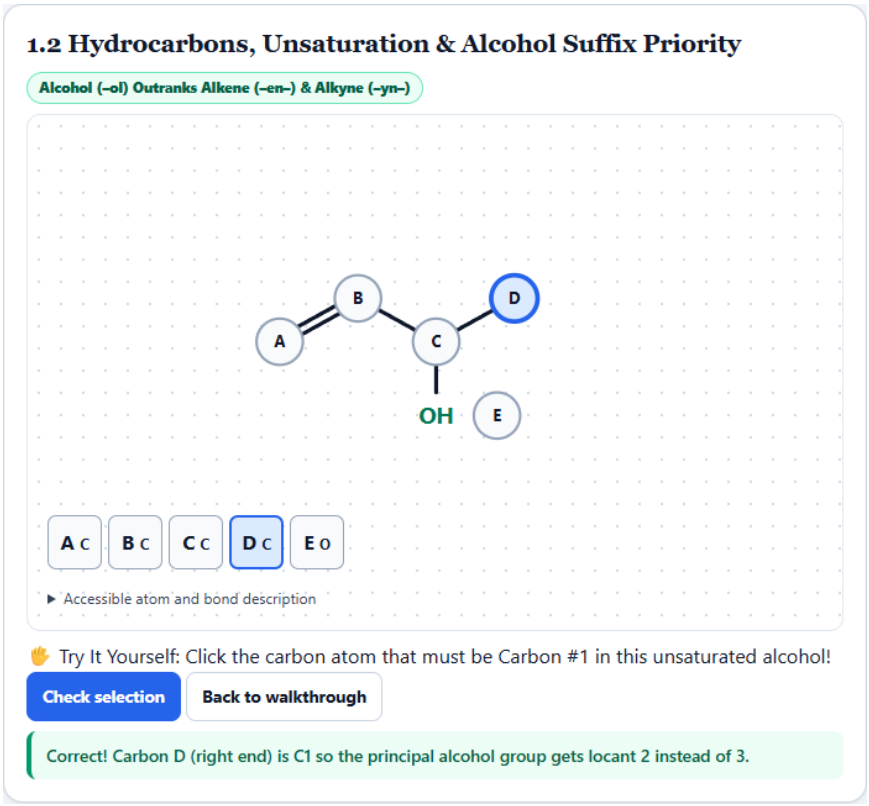


{"nameAssembly": [["np-root", "but", "4C parent"], ["np-inf", "-3-en-", "C3 alkene infix"], ["np-suf", "-2-ol", "C2 alcohol"]]}

Embedded challenge: Hands-on: Try It Yourself: Click the carbon atom that must be Carbon #1 in this unsaturated alcohol!

{"prompt": "Hands-on: Try It Yourself: Click the carbon atom that must be Carbon #1 in this unsaturated alcohol!", "answer": [3], "explanation": " Correct! Carbon D (right end) is C1 so the principal alcohol group gets locant 2 instead of 3."}

Screenshot: alcohols-step-1-challenge



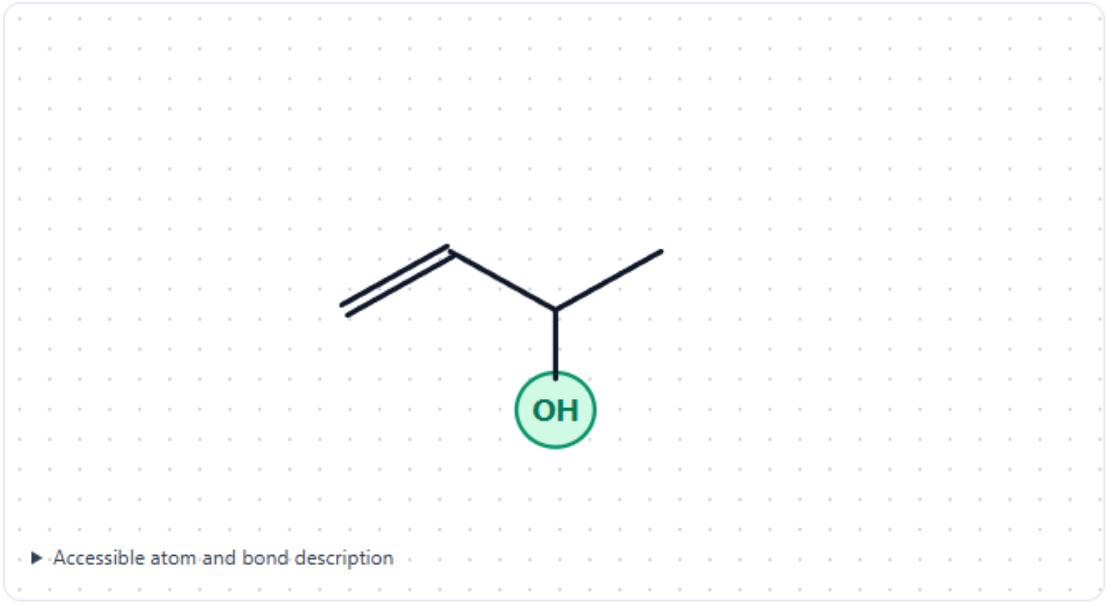
alcohols:step-2 | 2. Principal Group

Step 2 (Identify Principal Group): By IUPAC seniority, alcohols (–OH) outrank alkenes and alkynes and supply the suffix –ol .

Displayed molecule ID: enol

{"highlights": {"prin": [4]}, "revealName": false}

Screenshot: alcohols-step-2

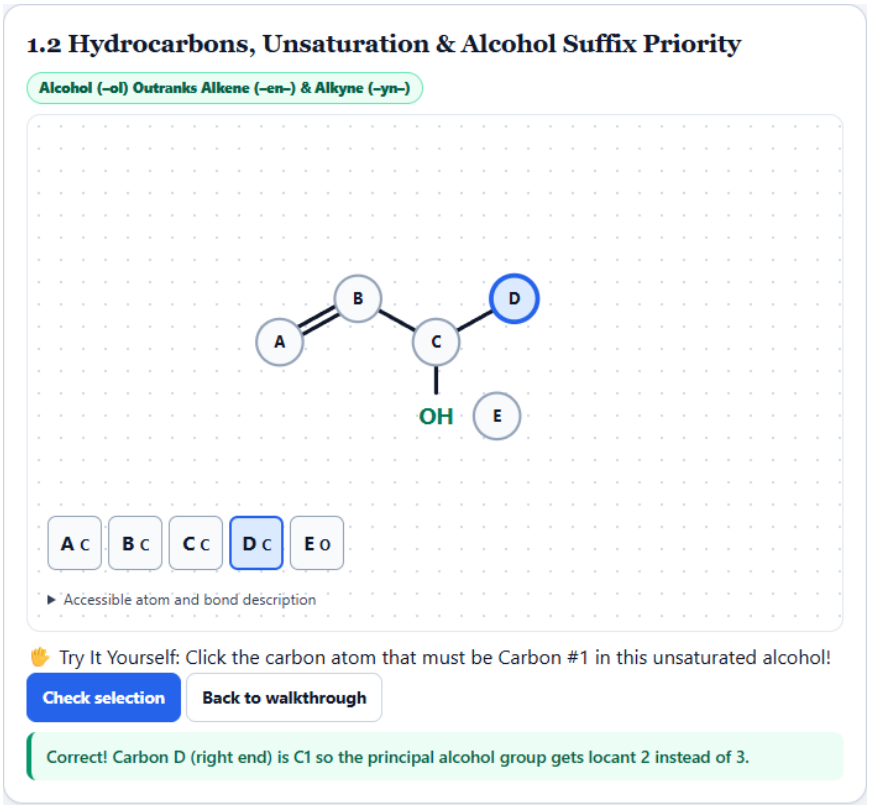


{"nameAssembly": [["np-root", "but", "4C parent"], ["np-inf", "-3-en-", "C3 alkene infix"], ["np-suf", "-2-ol", "C2 alcohol"]]}

Embedded challenge: Hands-on: Try It Yourself: Click the carbon atom that must be Carbon #1 in this unsaturated alcohol!

{"prompt": "Hands-on: Try It Yourself: Click the carbon atom that must be Carbon #1 in this unsaturated alcohol!", "answer": [3], "explanation": " Correct! Carbon D (right end) is C1 so the principal alcohol group gets locant 2 instead of 3."}

Screenshot: alcohols-step-2-challenge



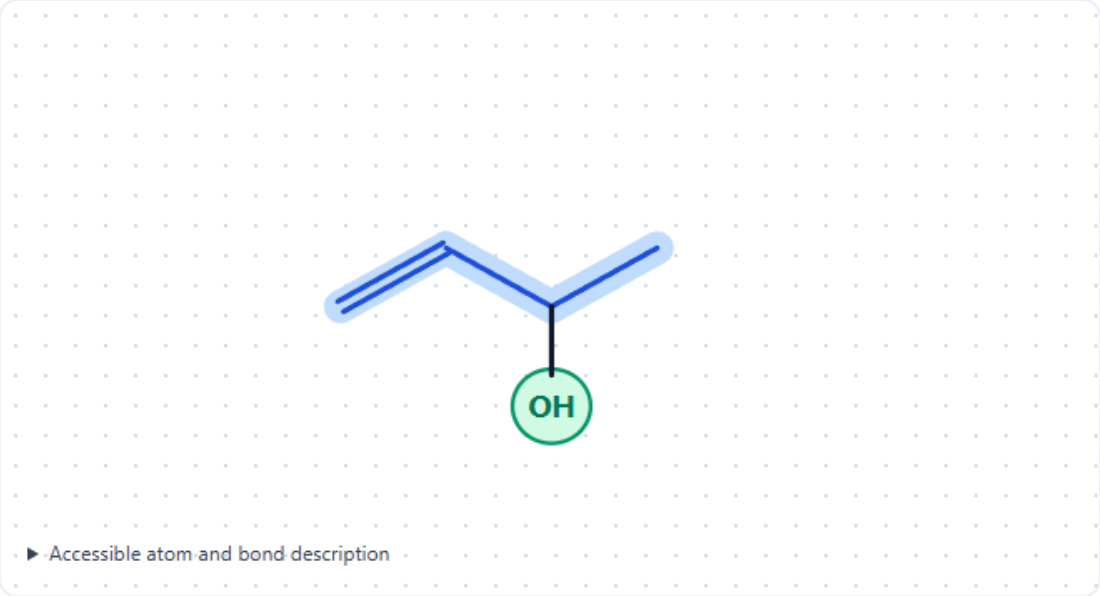
alcohols:step-3 | 3. Parent Chain

Step 3 (Parent Chain & Infix): Both the –OH carbon and C=C bond lie in the 4-carbon chain, so C=C becomes the infix –en– .

Displayed molecule ID: enol

{"highlights": {"path": [0, 1, 2, 3], "prin": [4]}, "revealName": false}

Screenshot: alcohols-step-3

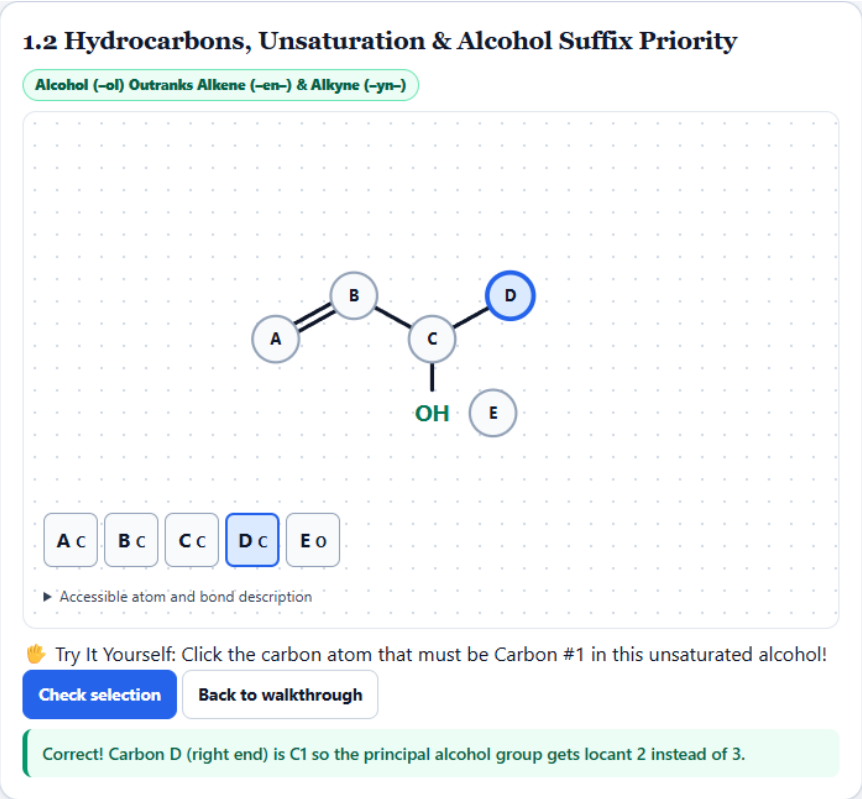


{"nameAssembly": [["np-root", "but", "4C parent"], ["np-inf", "-3-en-", "C3 alkene infix"], ["np-suf", "-2-ol", "C2 alcohol"]]}

Embedded challenge: Hands-on: Try It Yourself: Click the carbon atom that must be Carbon #1 in this unsaturated alcohol!

{"prompt": "Hands-on: Try It Yourself: Click the carbon atom that must be Carbon #1 in this unsaturated alcohol!", "answer": [3], "explanation": " Correct! Carbon D (right end) is C1 so the principal alcohol group gets locant 2 instead of 3."}

Screenshot: alcohols-step-3-challenge



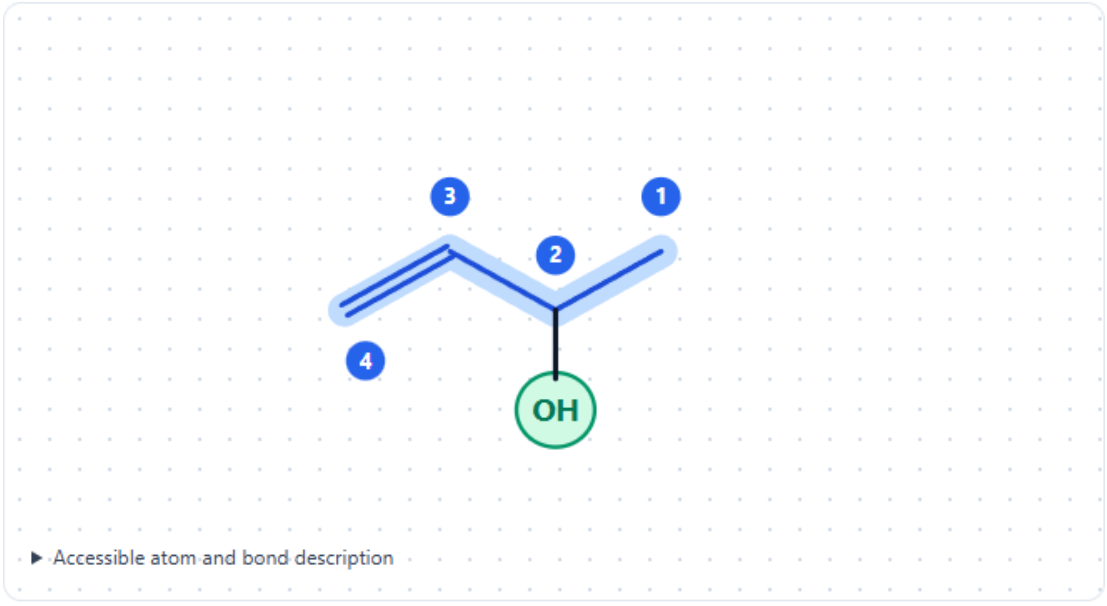
alcohols:step-4 | 4. Number & Name

Step 4 (Number & Assemble): Right-to-left gives –OH locant 2 (rather than 3). The double bond starts at C3 → but-3-en-2-ol .

Displayed molecule ID: enol

{"highlights": {"path": [0, 1, 2, 3], "num": [3, 2, 1, 0], "prin": [4]}, "revealName": true}

Screenshot: alcohols-step-4

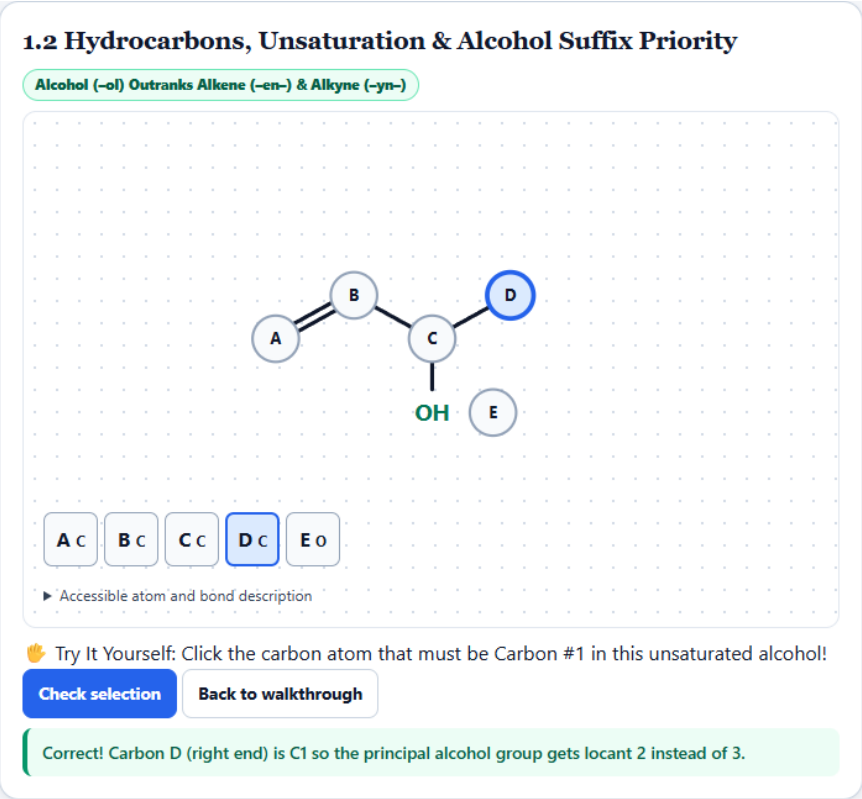


{"nameAssembly": [["np-root", "but", "4C parent"], ["np-inf", "-3-en-", "C3 alkene infix"], ["np-suf", "-2-ol", "C2 alcohol"]]}

Embedded challenge: Hands-on: Try It Yourself: Click the carbon atom that must be Carbon #1 in this unsaturated alcohol!

{"prompt": "Hands-on: Try It Yourself: Click the carbon atom that must be Carbon #1 in this unsaturated alcohol!", "answer": [3], "explanation": " Correct! Carbon D (right end) is C1 so the principal alcohol group gets locant 2 instead of 3."}

Screenshot: alcohols-step-4-challenge



rings | 1.3 Cyclic Hydrocarbons & Ring Numbering Hierarchy

Principal suffix → Multiple bonds → Substituents

rings:rule | Core Rules

A senior suffix such as –ol determines the starting locant first. In a simple cycloalkene with no senior suffix, the double bond occupies C1 and C2. Then minimize substituent locants at the first difference; use alphabetization only for a remaining appropriate tie.

Schematic summary: Principal suffix → Multiple bonds → Substituents

Senior suffix — First numbering priority

Simple cycloalkene — Without a senior suffix: double bond at C1–C2

Substituents — Minimize locants, then resolve genuine ties

rings:cue | Memory Cues

"Never Jump Across a Ring Double Bond!" If you assign one alkene carbon C1, C2 must be the other carbon of that double bond.

Schematic summary: Choose the case that applies

C1 is an alkene carbon — C2 must be the other alkene carbon

Senior suffix present — Apply its locant priority first

rings:trap | Common Mistakes / MCAT Traps

Why "Most Substituted Carbon First" Fails: In 4-propylcyclohexene , the alkene takes C1–C2 priority over the propyl substituent. Only use substitution when priority tiers tie (1,1,2-trimethylcyclobutane beats 1,2,2-).

Schematic summary: Shortcut versus a better check

Avoid — Starting at the most substituted carbon regardless of priority

Check — Suffix, multiple bonds, then substituent locants

rings:app | MCAT Applications

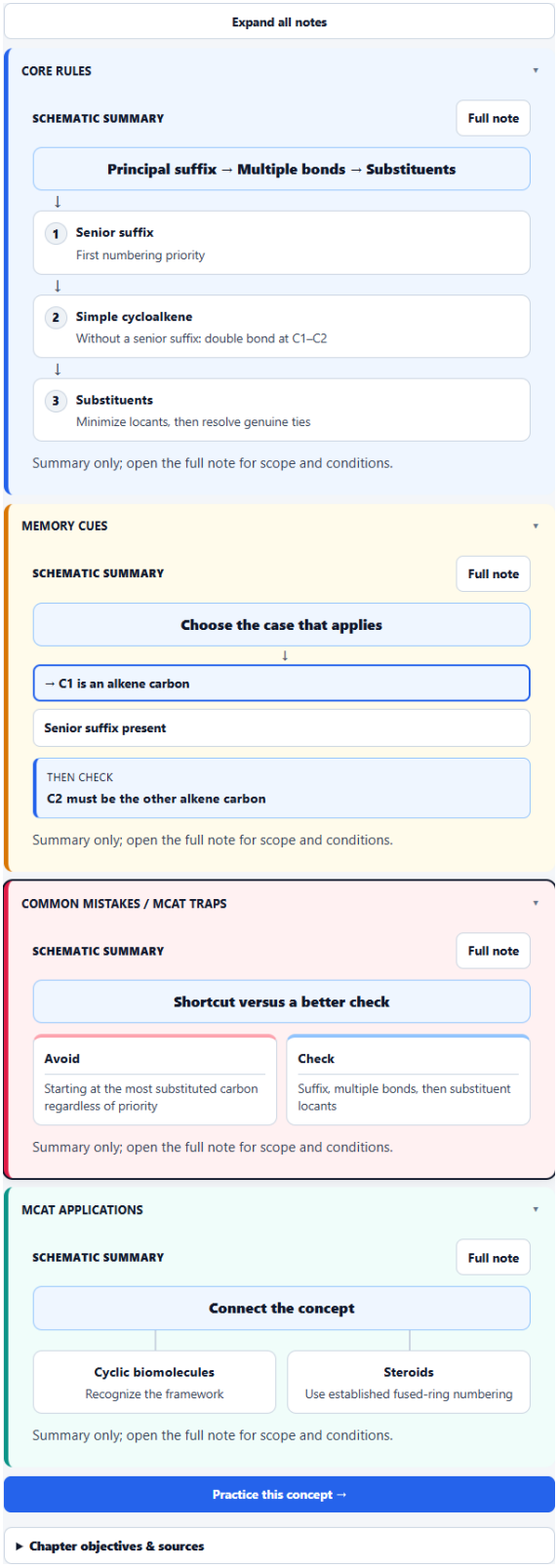
Ring recognition helps interpret cyclic biomolecules. Retinal is an aldehyde related to vitamin A. Steroids use established fused-ring parent numbering; do not extend simple cyclohexene numbering mechanically to steroid skeletons.

Schematic summary: Connect the concept

Cyclic biomolecules — Recognize the framework

Steroids — Use established fused-ring numbering

Screenshot: rings-visuals



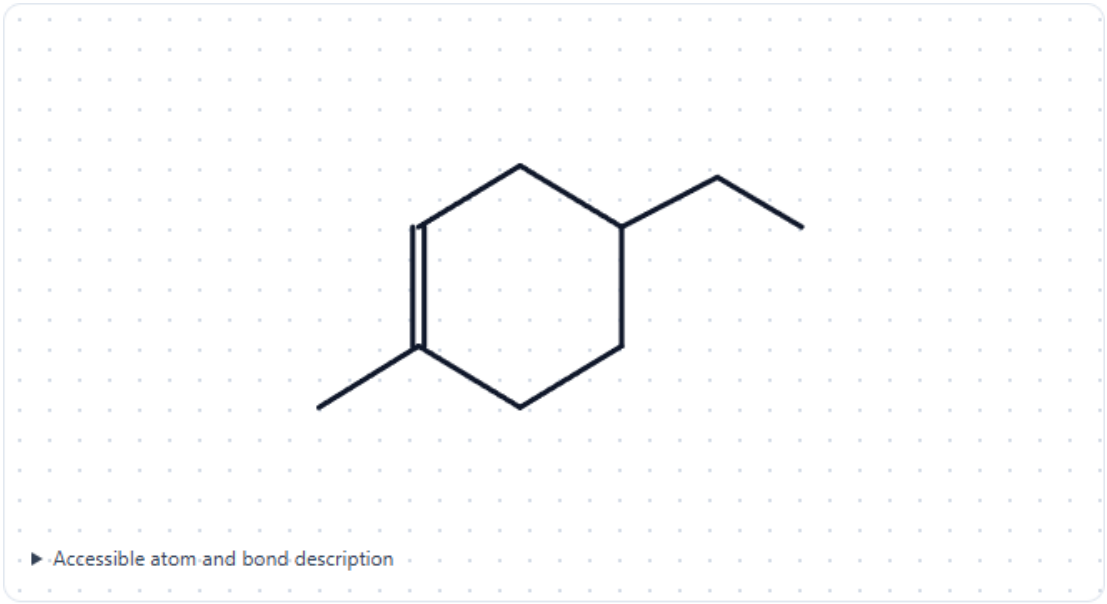
rings:step-1 | 1. Inspect Ring

Step 1 (Inspect): A 6-membered ring with a double bond, a methyl group on one alkene carbon, and an ethyl group across the ring.

Displayed molecule ID: cyhex

{"highlights": {}, "revealName": false}

Screenshot: rings-step-1

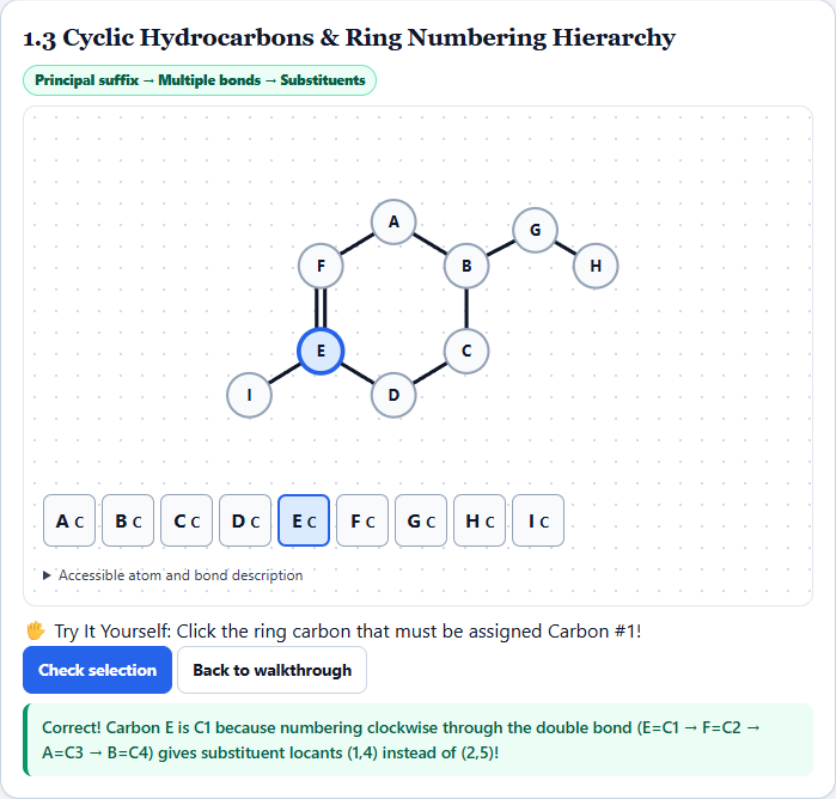


{"nameAssembly": [["np-sub", "4-ethyl-1-methyl", "alphabetized prefixes (1,4)"], ["np-root", "cyclohex", "6C ring"], ["np-suf", "-1-ene", "C1=C2 ring alkene"]]}

Embedded challenge: Hands-on: Try It Yourself: Click the ring carbon that must be assigned Carbon #1!

{"prompt": "Hands-on: Try It Yourself: Click the ring carbon that must be assigned Carbon #1!", "answer": [4], "explanation": "Correct! Carbon E is C1 because numbering clockwise through the double bond (E=C1 → F=C2 → A=C3 → B=C4) gives substituent locants (1,4) instead of (2,5)!"}

Screenshot: rings-step-1-challenge



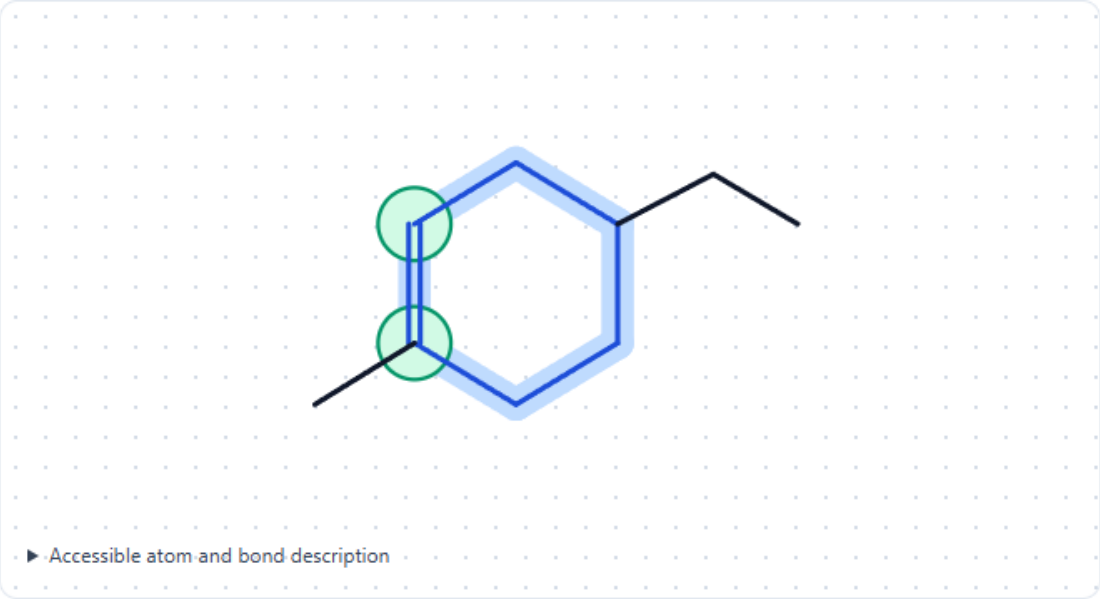
rings:step-2 | 2. Lock C1–C2 on C=C

Step 2 (Cycloalkene Rule): Unsaturation outranks alkyl substituents, so the two carbons of the double bond must be C1 and C2 .

Displayed molecule ID: cyhex

{"highlights": {"path": [0, 1, 2, 3, 4, 5], "prin": [4, 5]}, "revealName": false}

Screenshot: rings-step-2

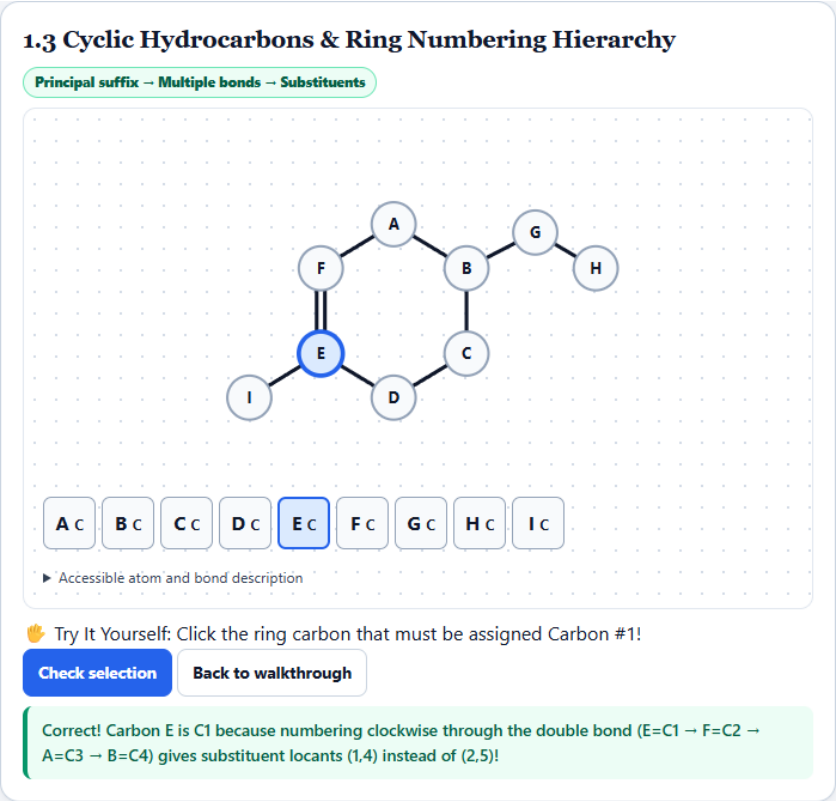


{"nameAssembly": [["np-sub", "4-ethyl-1-methyl", "alphabetized prefixes (1,4)"], ["np-root", "cyclohex", "6C ring"], ["np-suf", "-1-ene", "C1=C2 ring alkene"]]}

Embedded challenge: Hands-on: Try It Yourself: Click the ring carbon that must be assigned Carbon #1!

{"prompt": "Hands-on: Try It Yourself: Click the ring carbon that must be assigned Carbon #1!", "answer": [4], "explanation": "Correct! Carbon E is C1 because numbering clockwise through the double bond (E=C1 → F=C2 → A=C3 → B=C4) gives substituent locants (1,4) instead of (2,5)!"}

Screenshot: rings-step-2-challenge



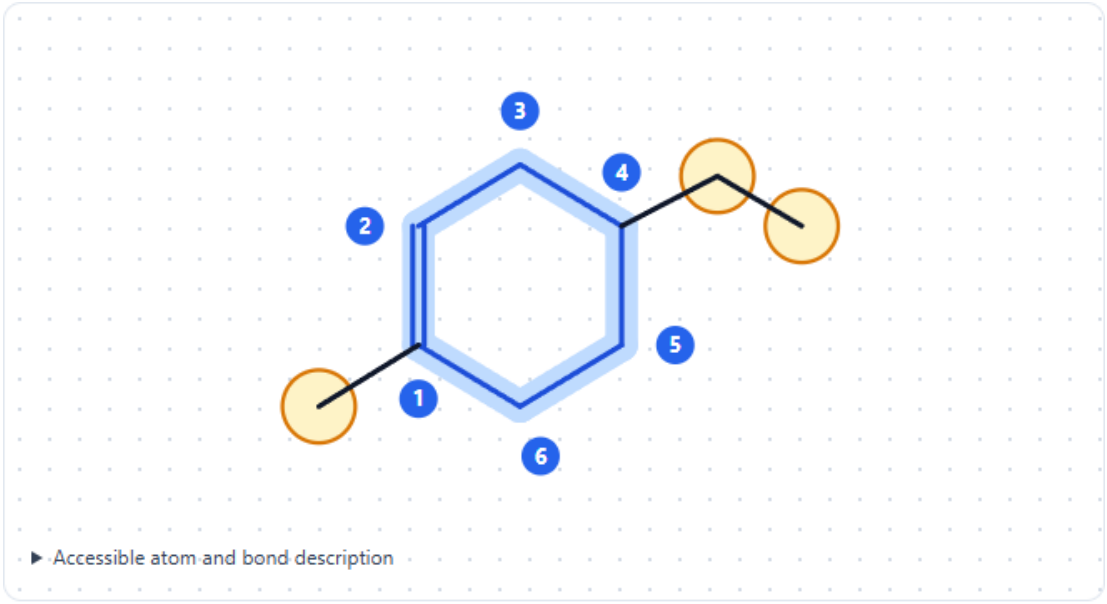
rings:step-3 | 3. Number & Name

Step 3 (Compare Locant Sets): Clockwise from the methylated alkene carbon (E=C1 → F=C2 → A=C3 → B=C4) gives substituent locants (1,4) , beating counter-clockwise (2,5) → 4-ethyl-1-methylcyclohex-1-ene .

Displayed molecule ID: cyhex

{"highlights": {"path": [0, 1, 2, 3, 4, 5], "num": [4, 5, 0, 1, 2, 3], "sub": [6, 7, 8]}, "revealName": true}

Screenshot: rings-step-3

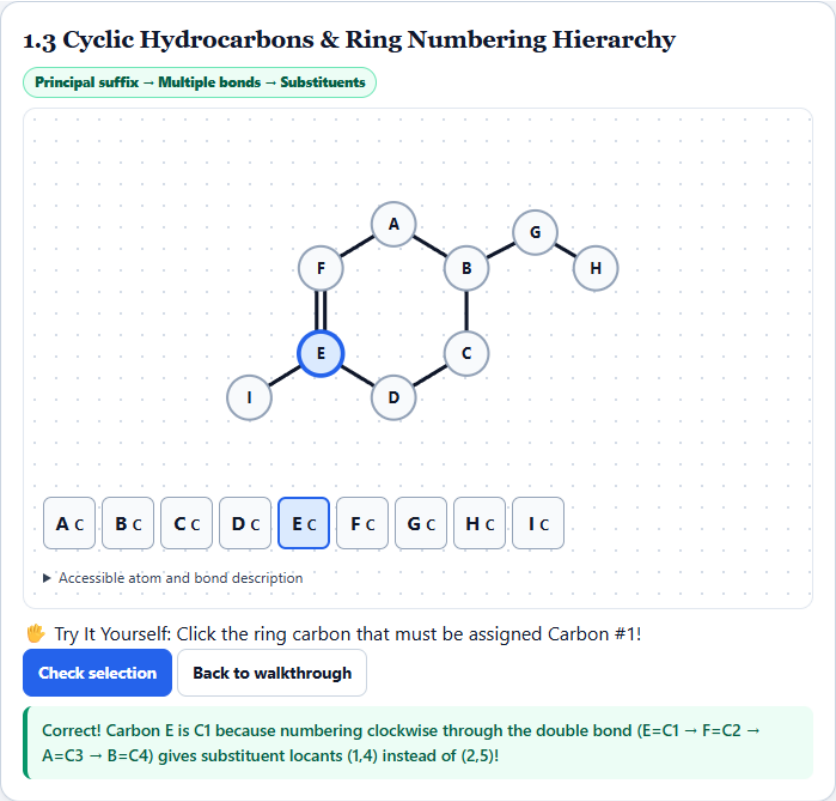


{"nameAssembly": [["np-sub", "4-ethyl-1-methyl", "alphabetized prefixes (1,4)"], ["np-root", "cyclohex", "6C ring"], ["np-suf", "-1-ene", "C1=C2 ring alkene"]]}

Embedded challenge: Hands-on: Try It Yourself: Click the ring carbon that must be assigned Carbon #1!

{"prompt": "Hands-on: Try It Yourself: Click the ring carbon that must be assigned Carbon #1!", "answer": [4], "explanation": "Correct! Carbon E is C1 because numbering clockwise through the double bond (E=C1 → F=C2 → A=C3 → B=C4) gives substituent locants (1,4) instead of (2,5)!"}

Screenshot: rings-step-3-challenge



diols | 1.4 Geminal vs. Vicinal Diols & Reversible Hydration

Vicinal = 1,2-Diol (Adjacent) · Geminal = 1,1-Diol (Same C)

diols:rule | Core Rules

Alcohols with two hydroxyl groups are named alkane-x,y-diols (preserving the –e of the parent alkane before "d"). Every hydroxyl gets a locant number.

Schematic summary: Vicinal = 1,2-Diol (Adjacent) · Geminal = 1,1-Diol (Same C)

Two OH groups — Use -diol; retain the alkane e

Locants — Number every OH

Arrangement — Same carbon versus adjacent carbons

diols:cue | Memory Cues

Memory cue: • Vicin al = in the vicin ity (adjacent carbons, 1,2). • Gemin al = Gemin i twins (same carbon, 1,1).

Schematic summary: Choose the case that applies

Adjacent carbons — Vicinal: 1,2-diol

Same carbon — Geminal: 1,1-diol

diols:trap | Common Mistakes / MCAT Traps

Carbonyl hydration is reversible. Electron-withdrawing substituents and less crowding generally favor the hydrate. Formaldehyde is strongly hydrated in water; acetone predominantly remains a carbonyl. The equilibrium depends on conditions.

Schematic summary: Shortcut versus a better check

Avoid — Treating all carbonyl hydrates as equally favored

Check — Substrate and conditions control the hydration equilibrium

diols:app | MCAT Applications

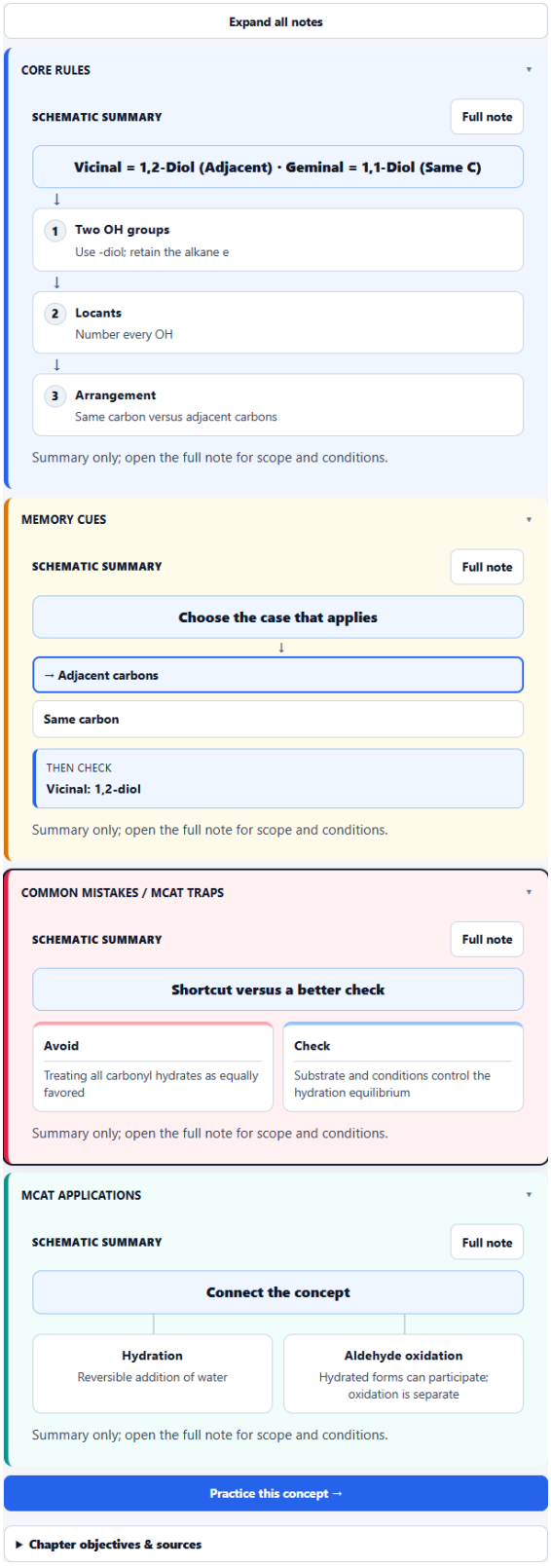
In aqueous aldehyde oxidation, the hydrated form can participate in oxidation to the carboxylic acid. Hydration and oxidation are distinct steps.

Schematic summary: Connect the concept

Hydration — Reversible addition of water

Aldehyde oxidation — Hydrated forms can participate; oxidation is separate

Screenshot: diols-visuals



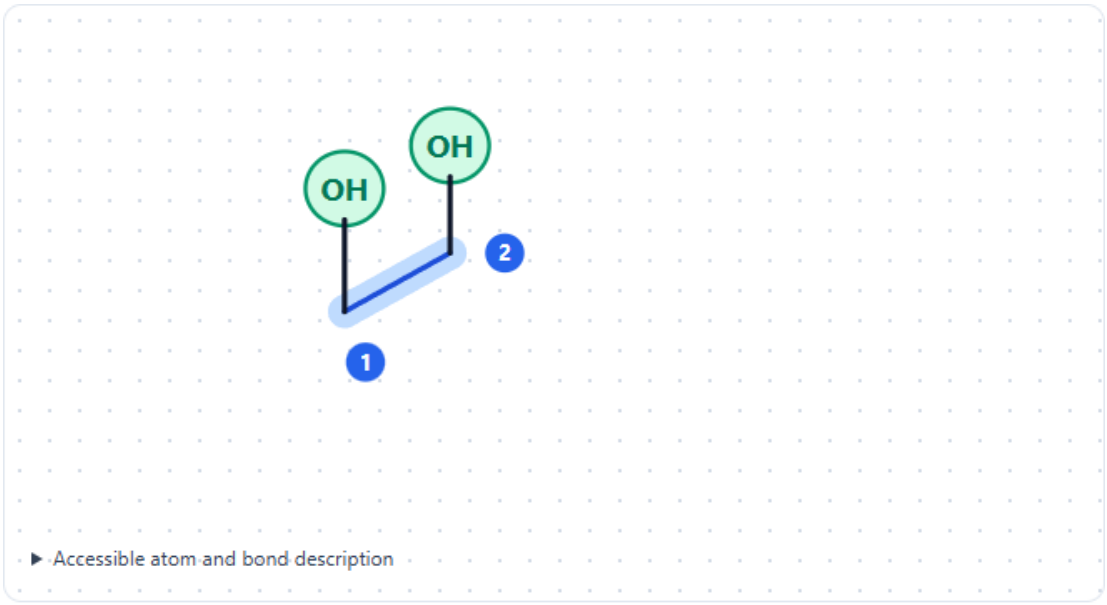
diols:step-1 | 1. Vicinal Diol (1,2)

Vicinal Diol (Glycol): Two –OH groups on adjacent carbons : ethane-1,2-diol (ethylene glycol).

Displayed molecule ID: vicinal

{"highlights": {"path": [0, 1], "prin": [2, 3], "num": [0, 1]}, "revealName": true}

Screenshot: diols-step-1

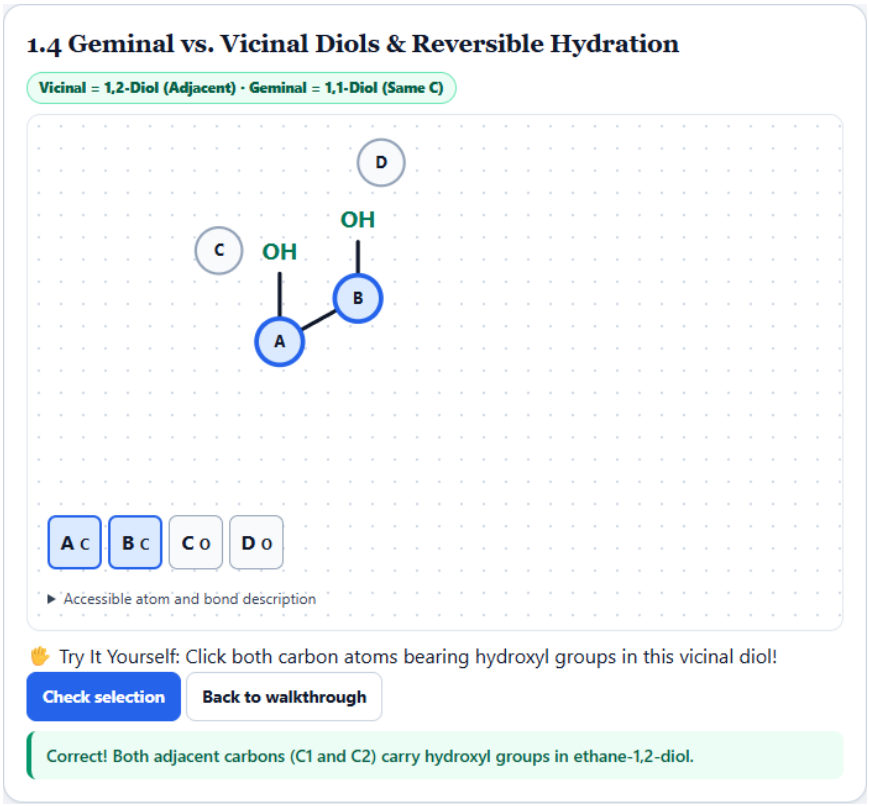


{"nameAssembly": [["np-root", "ethane", "retain final -e!"], ["np-inf", "-1,2-di", "adjacent C1 & C2"], ["np-suf", "ol", "diol suffix"]]}

Embedded challenge: Hands-on: Try It Yourself: Click both carbon atoms bearing hydroxyl groups in this vicinal diol!

{"prompt": "Hands-on: Try It Yourself: Click both carbon atoms bearing hydroxyl groups in this vicinal diol!", "answer": [0, 1], "explanation": " Correct! Both adjacent carbons (C1 and C2) carry hydroxyl groups in ethane-1,2-diol."}

Screenshot: diols-step-1-challenge



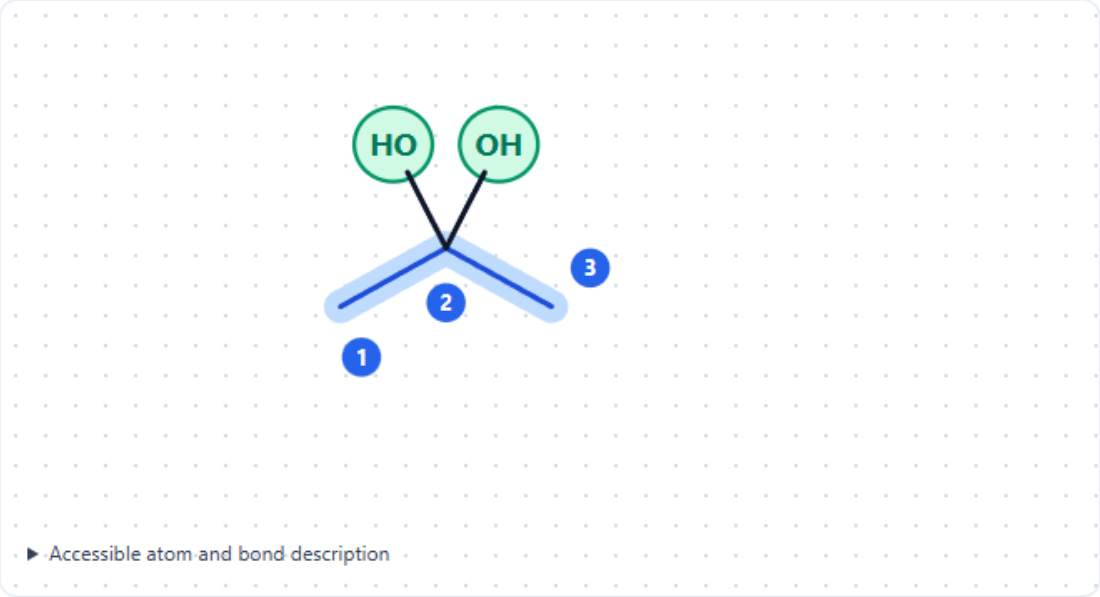
diols:step-2 | 2. Geminal Diol (1,1)

Geminal Diol (Hydrate): Two –OH groups on the same carbon (propane-2,2-diol), in reversible equilibrium with a carbonyl + H_2O.

Displayed molecule ID: geminal

{"highlights": {"path": [0, 1, 2], "prin": [3, 4], "num": [0, 1, 2]}, "revealName": true}

Screenshot: diols-step-2

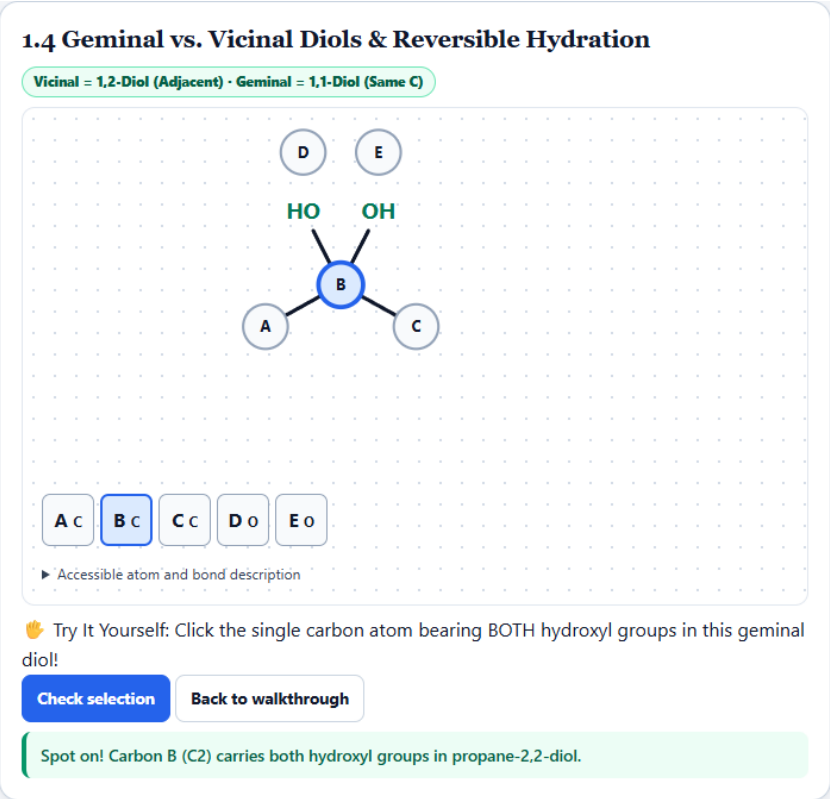


{"nameAssembly": [["np-root", "propane", "3C parent (keep -e)"], ["np-inf", "-2,2-di", "paired on SAME carbon"], ["np-suf", "ol", "geminal diol (hydrate)"]]}

Embedded challenge: Hands-on: Try It Yourself: Click the single carbon atom bearing BOTH hydroxyl groups in this geminal diol!

{"prompt": "Hands-on: Try It Yourself: Click the single carbon atom bearing BOTH hydroxyl groups in this geminal diol!", "answer": [1], " explanation": "Spot on! Carbon B (C2) carries both hydroxyl groups in propane-2,2-diol."}

Screenshot: diols-step-2-challenge



carbonyls | 1.5 Carbonyls: Aldehydes (–al), Ketones (–one) & α-Carbons

Aldehyde (Terminal C1) Outranks Ketone (Internal)

carbonyls:rule | Core Rules

Aldehydes have a carbonyl carbon bonded to at least one H; ketones have two carbon substituents. The simple acyclic examples here use –al and –one. A formyl group attached to a ring may instead require a –carbaldehyde name. Subordinate carbonyl naming depends on whether its carbon belongs to the selected parent; oxo– and formyl– are not universally interchangeable.

Schematic summary: Aldehyde (Terminal C1) Outranks Ketone (Internal)

Aldehyde — Carbonyl carbon has at least one H

Ketone — Carbonyl carbon has two carbon substituents

Naming context — Simple acyclic -al/-one; ring attachment may use -carbaldehyde

carbonyls:cue | Memory Cues

Common Names (1C–3C): • 1C: Formaldehyde (methanal) • 2C: Acetaldehyde (ethanal) • 3C: Propionaldehyde (propanal) & Acetone (propanone)

Schematic summary: Choose the case that applies

Formaldehyde — Methanal: one carbon

Acetaldehyde — Ethanal: two carbons

Acetone — Propanone: three carbons

carbonyls:trap | Common Mistakes / MCAT Traps

Acetone Carbon-Count Trap: Although acet– usually means 2 carbons, acetone (propanone) has 3 carbons because a ketone requires a carbon on both sides of C=O!

Schematic summary: Shortcut versus a better check

Avoid — Assuming acetone has two carbons from acet-

Check — Count its complete three-carbon skeleton

carbonyls:app | MCAT Applications

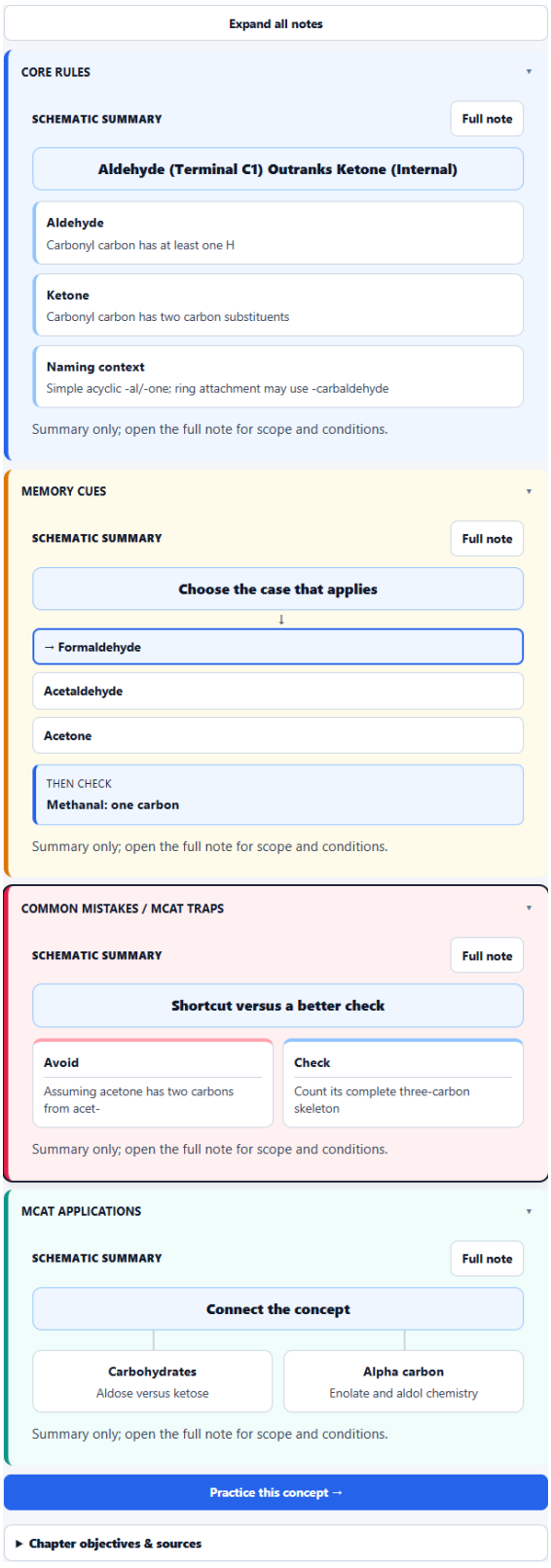
Aldoses, Ketoses & Enolates: Carbohydrates are aldoses or ketoses; acidic α-hydrogens on α-carbons form nucleophilic enolates in aldol reactions.

Schematic summary: Connect the concept

Carbohydrates — Aldose versus ketose

Alpha carbon — Enolate and aldol chemistry

Screenshot: carbonyls-visuals



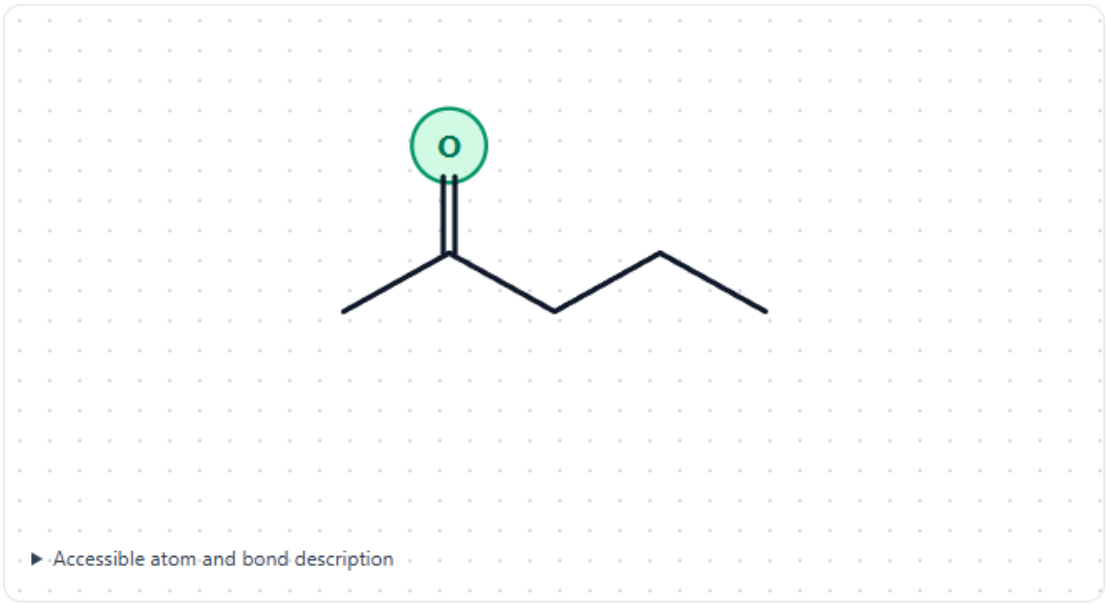
carbonyls:step-1 | 1. Inspect Carbonyl

Step 1 (Classify): C=O bonded to two carbons is an internal ketone (–one) ; terminal C=O bonded to H is an aldehyde (–al) .

Displayed molecule ID: ketone

{"highlights": {"prin": [5]}, "revealName": false}

Screenshot: carbonyls-step-1

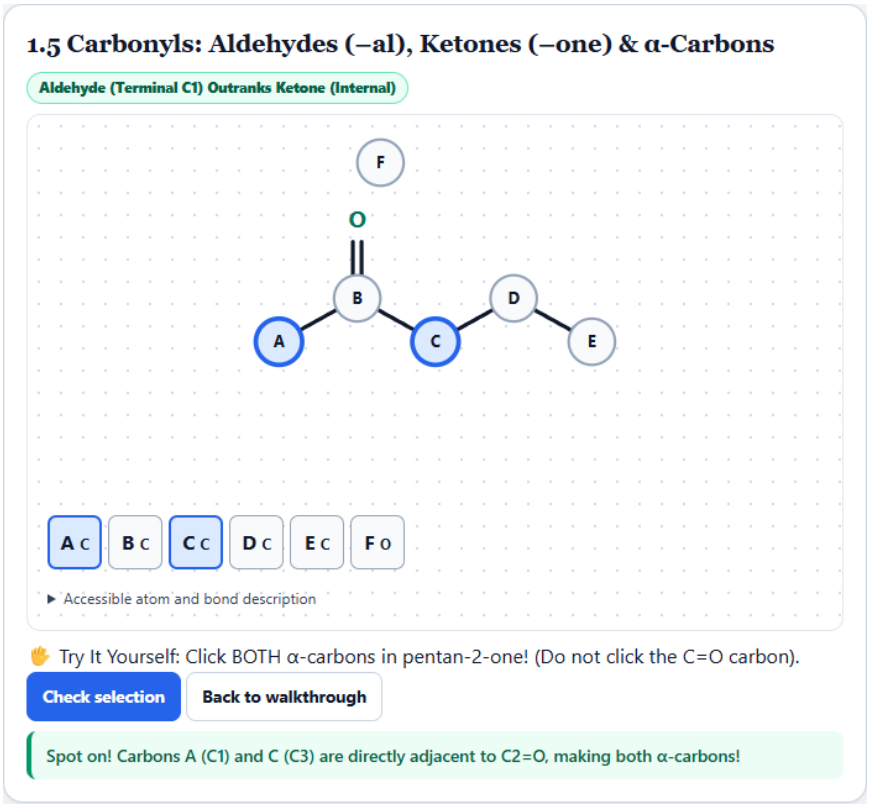


{"nameAssembly": [["np-root", "pent", "5C parent"], ["np-inf", "an", "saturated chain"], ["np-suf", "-2-one", "C2 ketone"]]}

Embedded challenge: Hands-on: Try It Yourself: Click BOTH α-carbons in pentan-2-one! (Do not click the C=O carbon).

{"prompt": "Hands-on: Try It Yourself: Click BOTH α-carbons in pentan-2-one! (Do not click the C=O carbon).", "answer": [0, 2], "explanation": " Spot on! Carbons A (C1) and C (C3) are directly adjacent to C2=O, making both α-carbons!"}

Screenshot: carbonyls-step-1-challenge



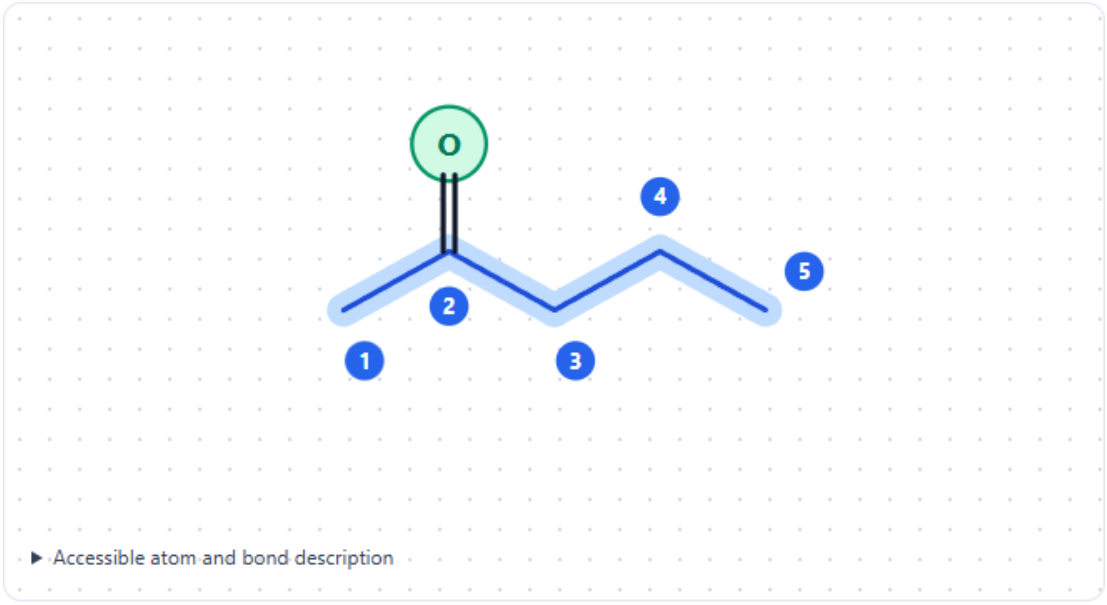
carbonyls:step-2 | 2. Number Ketone

Step 2 (Numbering): Left-to-right gives C=O C2 (rather than C4).

Displayed molecule ID: ketone

{"highlights": {"path": [0, 1, 2, 3, 4], "num": [0, 1, 2, 3, 4], "prin": [5]}, "revealName": false}

Screenshot: carbonyls-step-2

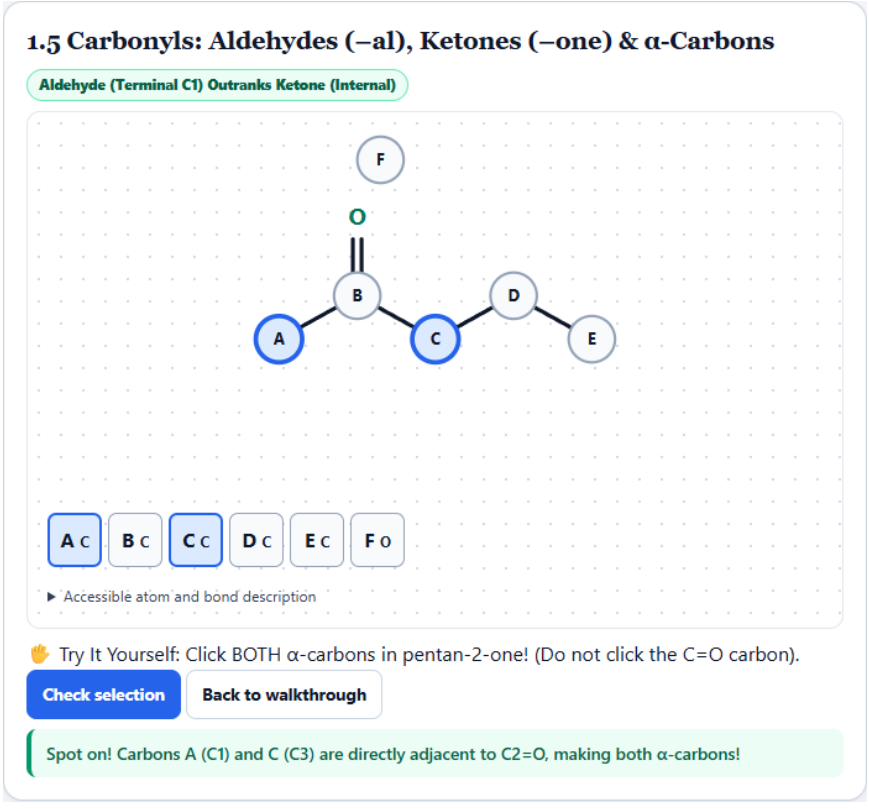


{"nameAssembly": [["np-root", "pent", "5C parent"], ["np-inf", "an", "saturated chain"], ["np-suf", "-2-one", "C2 ketone"]]}

Embedded challenge: Hands-on: Try It Yourself: Click BOTH α-carbons in pentan-2-one! (Do not click the C=O carbon).

{"prompt": "Hands-on: Try It Yourself: Click BOTH α-carbons in pentan-2-one! (Do not click the C=O carbon).", "answer": [0, 2], "explanation": "Spot on! Carbons A (C1) and C (C3) are directly adjacent to C2=O, making both α-carbons!"}

Screenshot: carbonyls-step-2-challenge



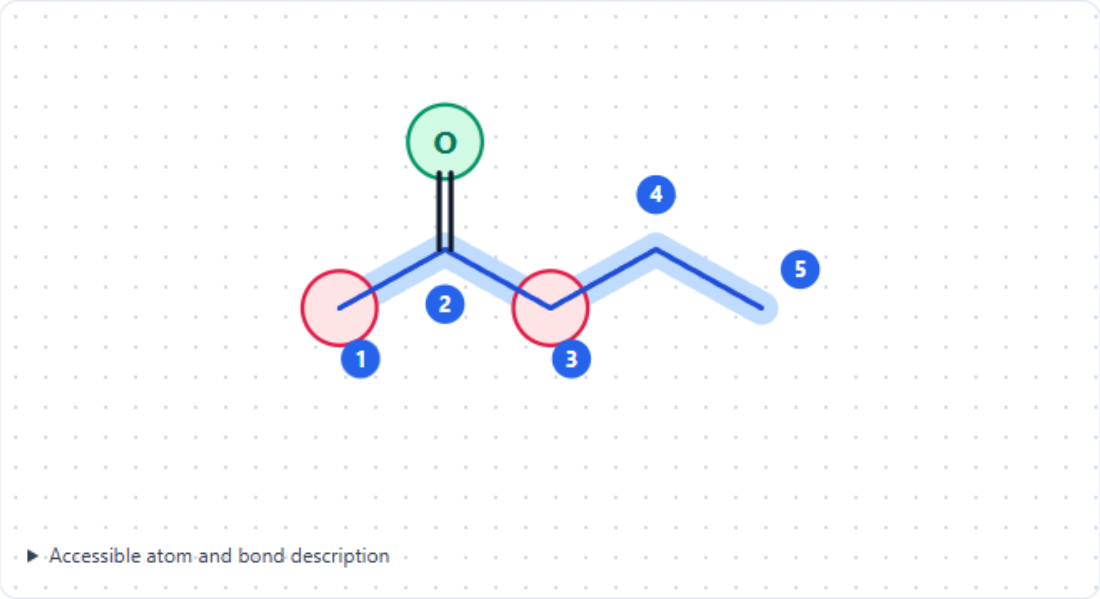
carbonyls:step-3 | 3. α-Carbons & Name

Step 3 (Greek Letters & Name): Carbons directly adjacent to C=O are α-carbons (C1 and C3), followed by β (C4) and γ (C5) → pentan-2-one .

Displayed molecule ID: ketone

{"highlights": {"path": [0, 1, 2, 3, 4], "num": [0, 1, 2, 3, 4], "alpha": [0, 2], "prin": [5]}, "revealName": true}

Screenshot: carbonyls-step-3

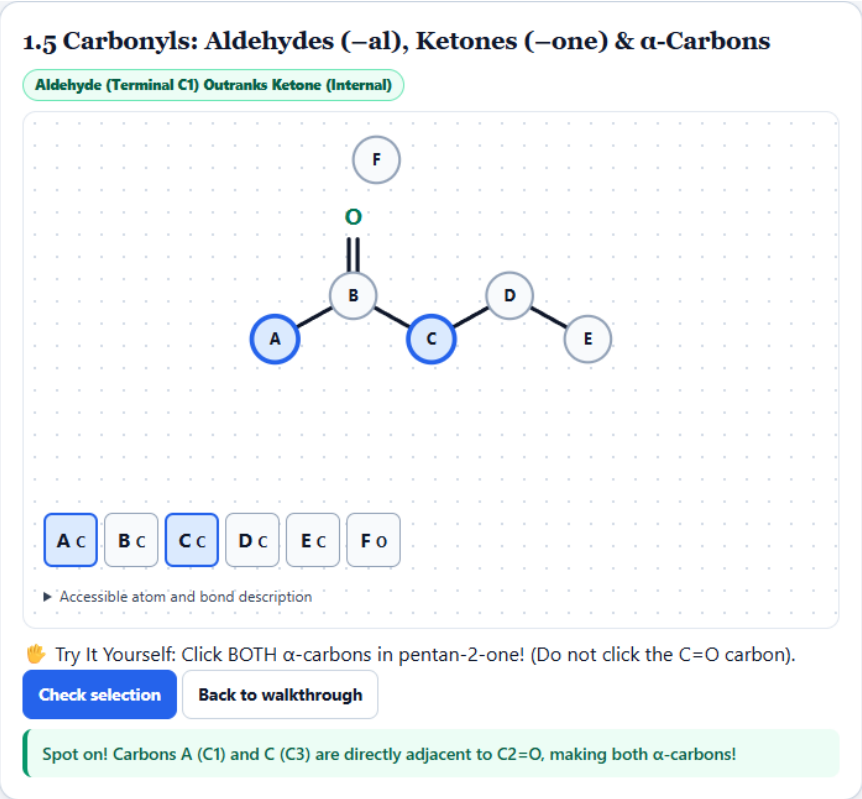


{"nameAssembly": [["np-root", "pent", "5C parent"], ["np-inf", "an", "saturated chain"], ["np-suf", "-2-one", "C2 ketone"]]}

Embedded challenge: Hands-on: Try It Yourself: Click BOTH α-carbons in pentan-2-one! (Do not click the C=O carbon).

{"prompt": "Hands-on: Try It Yourself: Click BOTH α-carbons in pentan-2-one! (Do not click the C=O carbon).", "answer": [0, 2], "explanation": " Spot on! Carbons A (C1) and C (C3) are directly adjacent to C2=O, making both α-carbons!"}

Screenshot: carbonyls-step-3-challenge



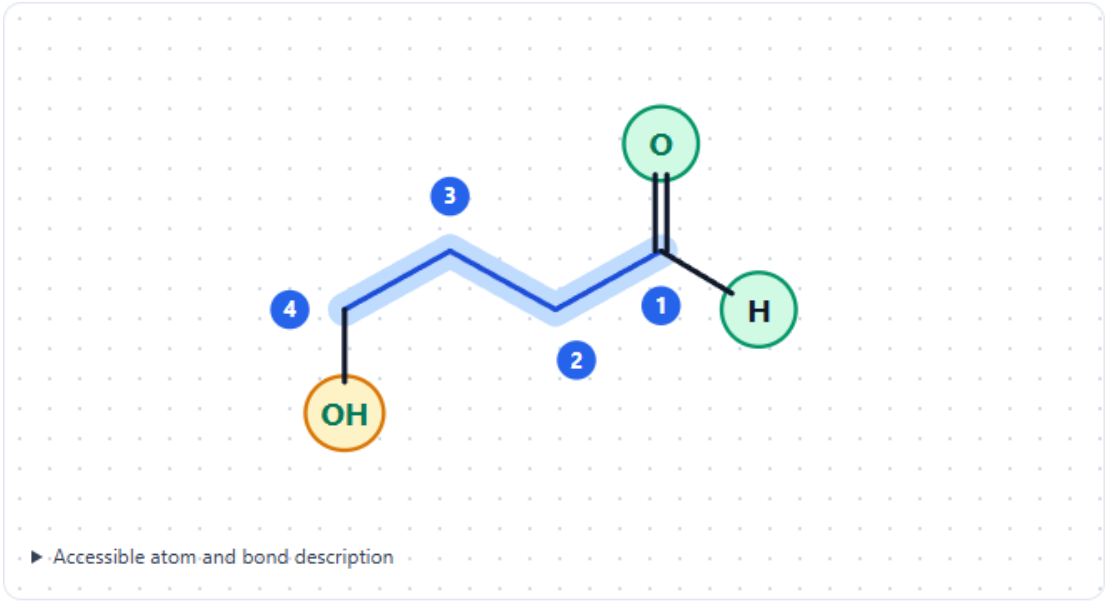
carbonyls:step-4 | 4. Aldehyde Priority

Step 4 (4-hydroxybutanal): Terminal aldehyde outranks alcohol, taking C1 (–al) while the C4 alcohol becomes 4-hydroxy– .

Displayed molecule ID: hyald

{"highlights": {"path": [0, 1, 2, 3], "num": [3, 2, 1, 0], "prin": [4, 5], "sub": [6]}, "revealName": true}

Screenshot: carbonyls-step-4

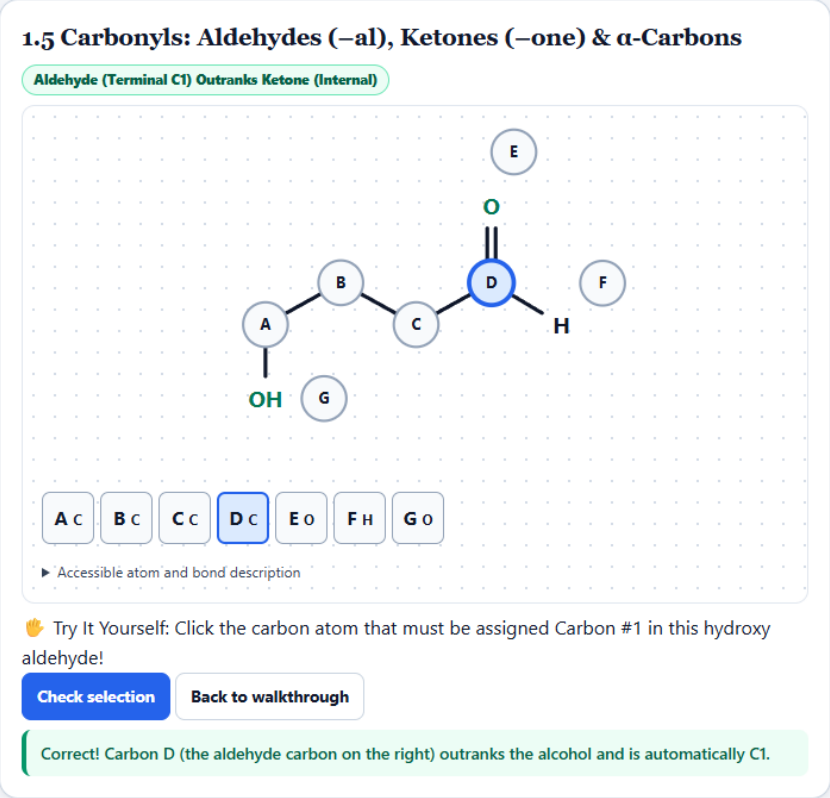


{"nameAssembly": [["np-sub", "4-hydroxy", "subordinate C4 alcohol"], ["np-root", "butan", "4C parent chain"], ["np-suf", "al", "terminal C1 aldehyde"]]}

Embedded challenge: Hands-on: Try It Yourself: Click the carbon atom that must be assigned Carbon #1 in this hydroxy aldehyde!

{"prompt": "Hands-on: Try It Yourself: Click the carbon atom that must be assigned Carbon #1 in this hydroxy aldehyde!", "answer": [3], "
explanation": "Correct! Carbon D (the aldehyde carbon on the right) outranks the alcohol and is automatically C1."}

Screenshot: carbonyls-step-4-challenge



derivatives | 1.6 Carboxylic Acids, Derivatives & Master Priority Hierarchy

Acid > Anhydride > Ester > Amide > Aldehyde > Ketone > Alcohol

derivatives:rule | Core Rules

For these examples, naming seniority is acid > anhydride > ester > amide > aldehyde > ketone > alcohol. Naming seniority is a convention, not an oxidation-state calculation. Unsaturation is retained in the parent name. This is a chapter subset, not the complete IUPAC hierarchy.

Schematic summary: Acid > Anhydride > Ester > Amide > Aldehyde > Ketone > Alcohol

Senior group — Select the suffix using this chapter hierarchy

Other groups — Retain prefixes and unsaturation

Separate questions — Naming seniority is not oxidation-state ranking

derivatives:cue | Memory Cues

Common Acid Roots: • 1C: Formic acid (methanoic) → methyl formate • 2C: Acetic acid (ethanoic) → acetic anhydride • 3C: Propionic acid (propanoic)

Schematic summary: Choose the case that applies

Formic acid — Methanoic: one carbon

Acetic acid — Ethanoic: two carbons

Propionic acid — Propanoic: three carbons

derivatives:trap | Common Mistakes / MCAT Traps

High-Yield Distinctions: • Ester vs. Ether: Esters have C=O next to O; ethers (R–O–R′) do not. • NADH Trap: NADH contains an amide (nicotinamide) and phosphoanhydride (P–O–P), NOT a carboxylic anhydride!

Schematic summary: Shortcut versus a better check

Avoid — Equating an ester with an ether

Check — An ester has an adjacent carbonyl; distinguish P–O–P from carboxylic anhydrides

derivatives:app | MCAT Applications

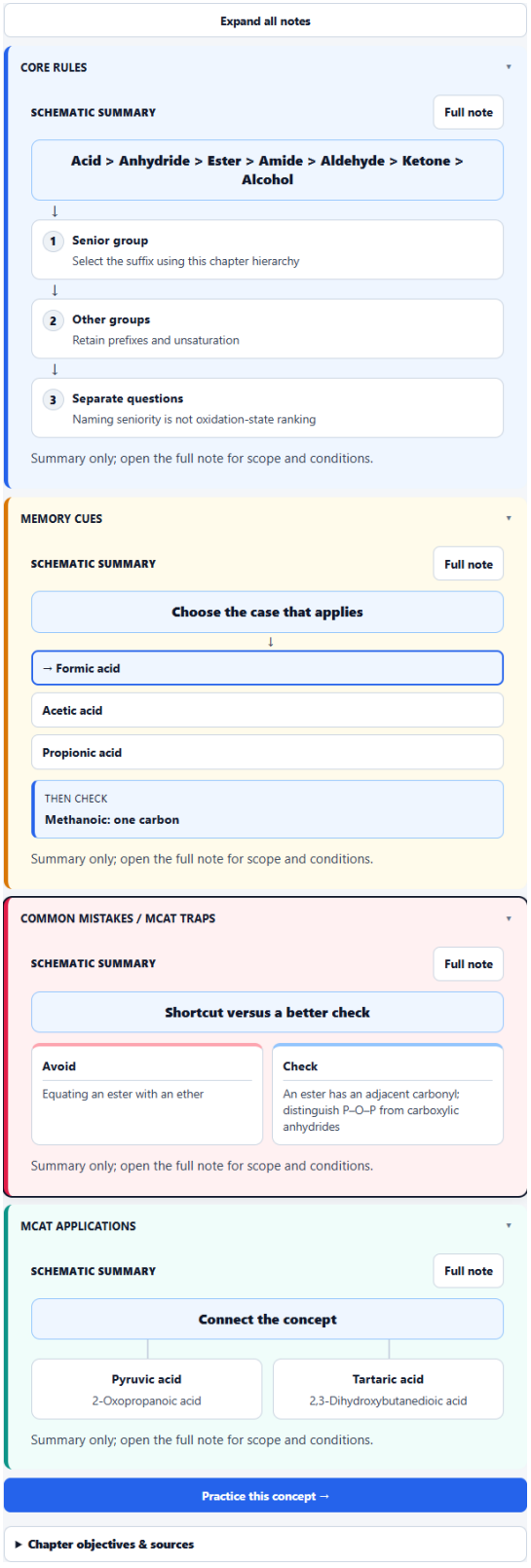
Pyruvic & Tartaric Acid: • Pyruvic acid (acetylformic acid) = 2-oxopropanoic acid . • Tartaric acid = 2,3-dihydroxybutanedioic acid .

Schematic summary: Connect the concept

Pyruvic acid — 2-Oxopropanoic acid

Tartaric acid — 2,3-Dihydroxybutanedioic acid

Screenshot: derivatives-visuals



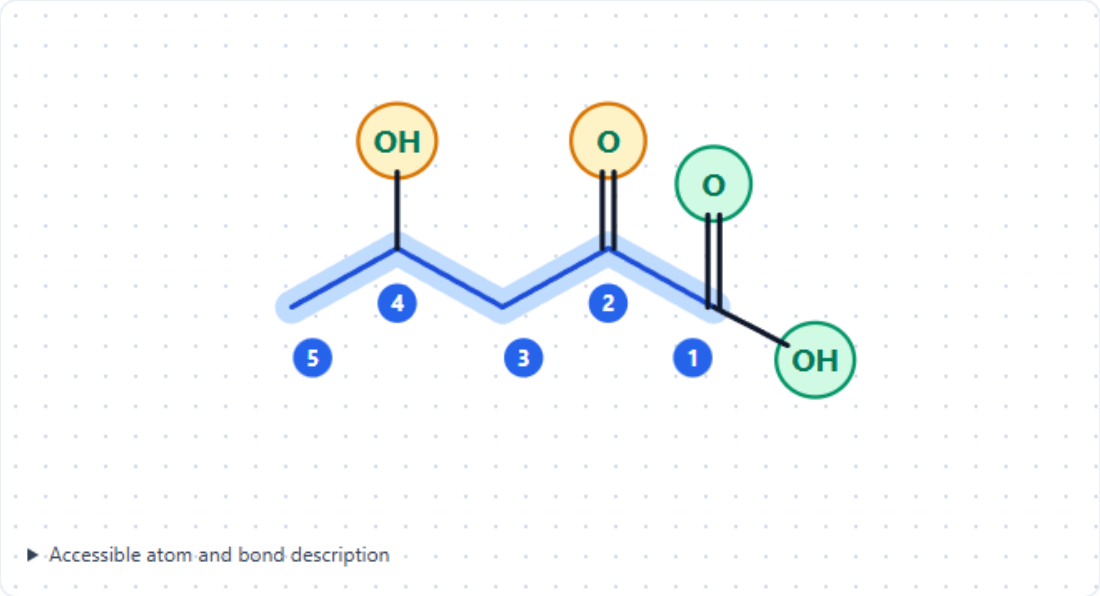
derivatives:step-1 | 1. Multifunctional Acid

Priority Winner: Carboxylic acid takes C1 and –oic acid . Alphabetize 4-hydroxy– before 2-oxo– → 4-hydroxy-2-oxopentanoic acid .

Displayed molecule ID: hyoxo

{"highlights": {"path": [0, 1, 2, 3, 4], "num": [4, 3, 2, 1, 0], "prin": [7, 8], "sub": [5, 6]}, "revealName": true}

Screenshot: derivatives-step-1

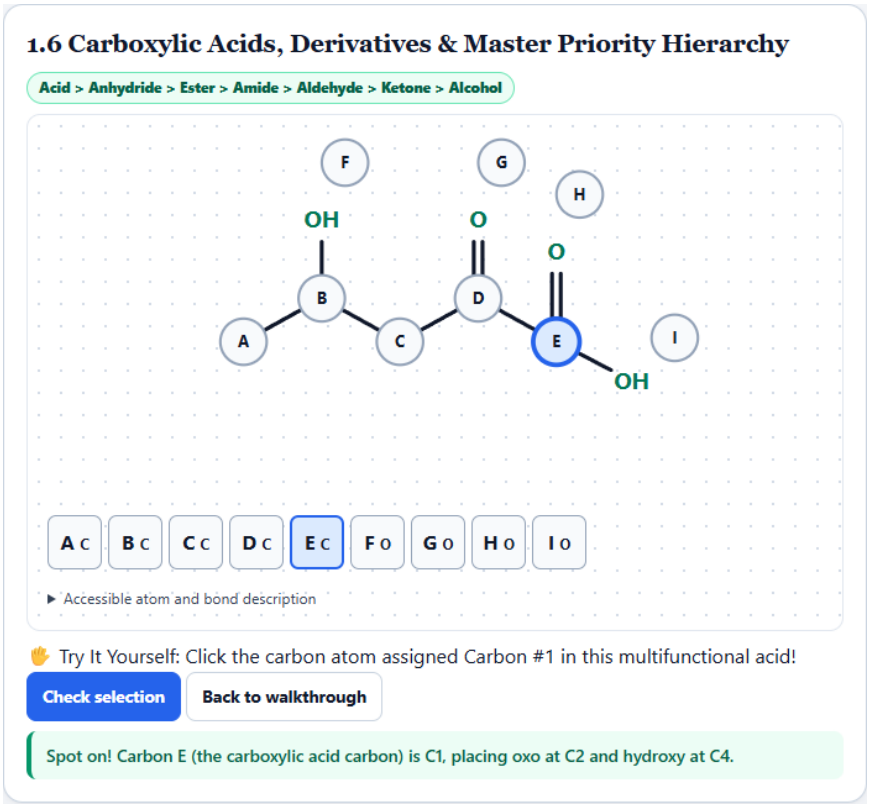


{"nameAssembly": [["np-sub", "4-hydroxy-", "C4 alcohol (H 1st)"], ["np-sub", "2-oxo", "C2 ketone (O 2nd)"], ["np-root", "pentan", "5C parent"], ["np-suf", "oic acid", "C1 acid suffix"]]}

Embedded challenge: Hands-on: Try It Yourself: Click the carbon atom assigned Carbon #1 in this multifunctional acid!

{"prompt": "Hands-on: Try It Yourself: Click the carbon atom assigned Carbon #1 in this multifunctional acid!", "answer": [4], "explanation": "Spot on! Carbon E (the carboxylic acid carbon) is C1, placing oxo at C2 and hydroxy at C4."}

Screenshot: derivatives-step-1-challenge



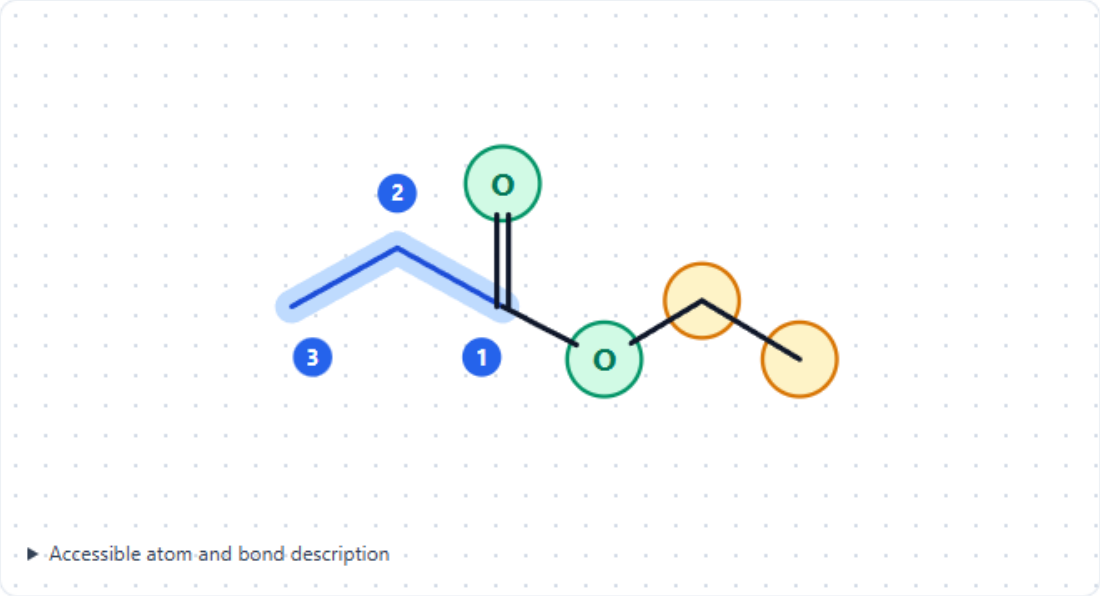
derivatives:step-2 | 2. Ester (alkyl –oate)

Ester Naming: O-attached alkyl group first (ethyl) + 3C acid parent (propanoate) → ethyl propanoate .

Displayed molecule ID: ester

{"highlights": {"path": [2, 1, 0], "num": [2, 1, 0], "prin": [3, 4], "sub": [5, 6]}, "revealName": true}

Screenshot: derivatives-step-2

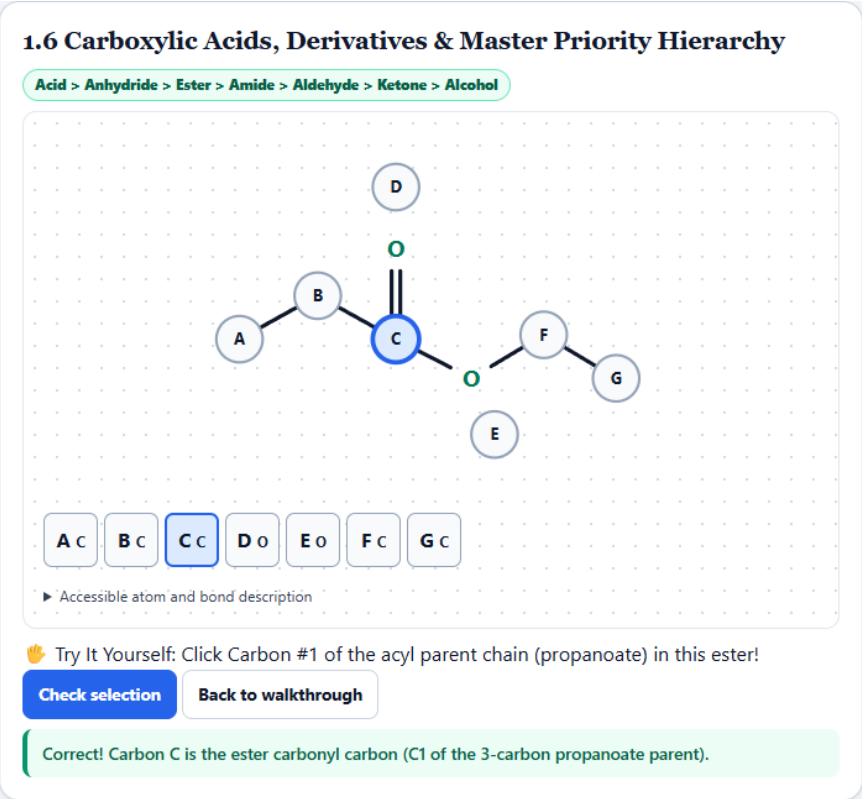


{"nameAssembly": [["np-sub", "ethyl ", "O-attached alkyl (1st word)"], ["np-root", "propan", "3C acyl chain"], ["np-suf", "oate", "ester suffix"]]}

Embedded challenge: Hands-on: Try It Yourself: Click Carbon #1 of the acyl parent chain (propanoate) in this ester!

{"prompt": "Hands-on: Try It Yourself: Click Carbon #1 of the acyl parent chain (propanoate) in this ester!", "answer": [2], "explanation": "Correct! Carbon C is the ester carbonyl carbon (C1 of the 3-carbon propanoate parent)."}

Screenshot: derivatives-step-2-challenge



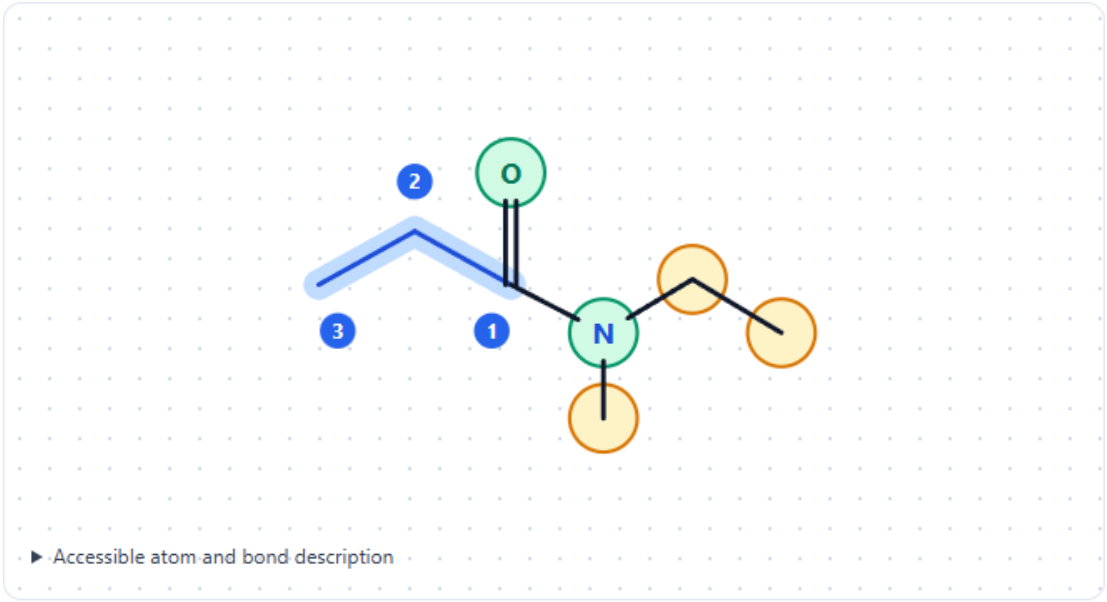
derivatives:step-3 | 3. N-Amide (N– prefix)

Amide Naming: Nitrogen groups use capital N– and are alphabetized → N-ethyl-N-methylpropanamide .

Displayed molecule ID: namide

{"highlights": {"path": [2, 1, 0], "num": [2, 1, 0], "prin": [3, 4], "sub": [5, 6, 7]}, "revealName": true}

Screenshot: derivatives-step-3

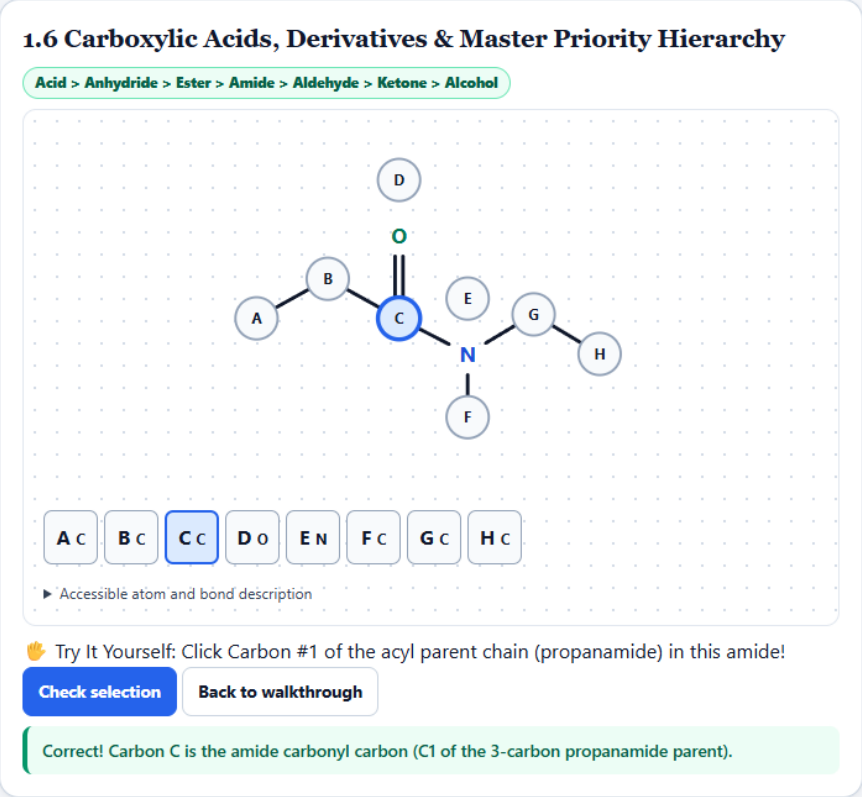


{"nameAssembly": [["np-sub", "N-ethyl-N-methyl", "N-alkyl groups (alphabetical)"], ["np-root", "propan", "3C acyl parent"], ["np-suf", "amide", "amide suffix"]]}

Embedded challenge: Hands-on: Try It Yourself: Click Carbon #1 of the acyl parent chain (propanamide) in this amide!

{"prompt": "Hands-on: Try It Yourself: Click Carbon #1 of the acyl parent chain (propanamide) in this amide!", "answer": [2], "explanation": " Correct! Carbon C is the amide carbonyl carbon (C1 of the 3-carbon propanamide parent)."}

Screenshot: derivatives-step-3-challenge



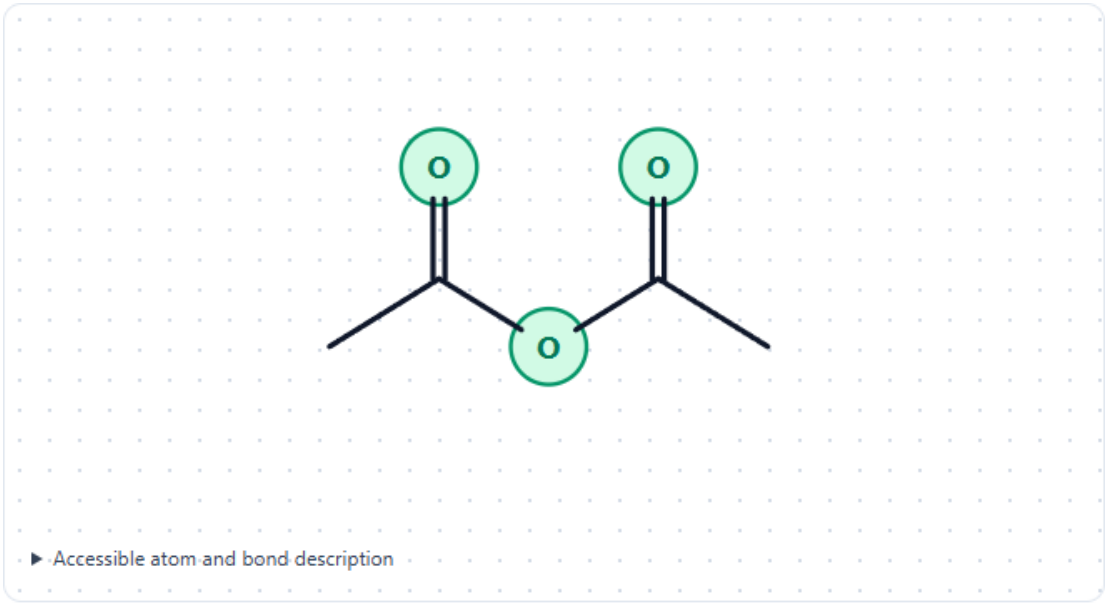
derivatives:step-4 | 4. Anhydrides

Anhydride Naming: Two acyl groups bridged by O (ethanoic anhydride = acetic anhydride).

Displayed molecule ID: anhyd

{"highlights": {"prin": [2, 5, 6]}, "revealName": true}

Screenshot: derivatives-step-4

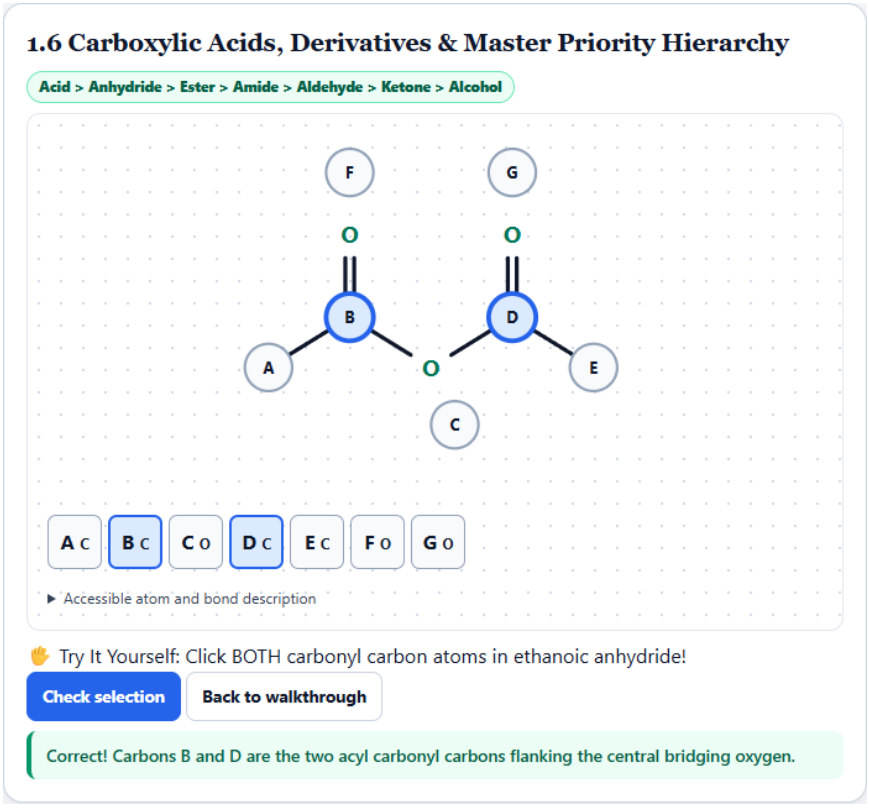


{"nameAssembly": [["np-root", "ethanoic ", "2C acid root (symmetric)"], ["np-suf", "anhydride", "replaces \"acid\""]]}

Embedded challenge: Hands-on: Try It Yourself: Click BOTH carbonyl carbon atoms in ethanoic anhydride!

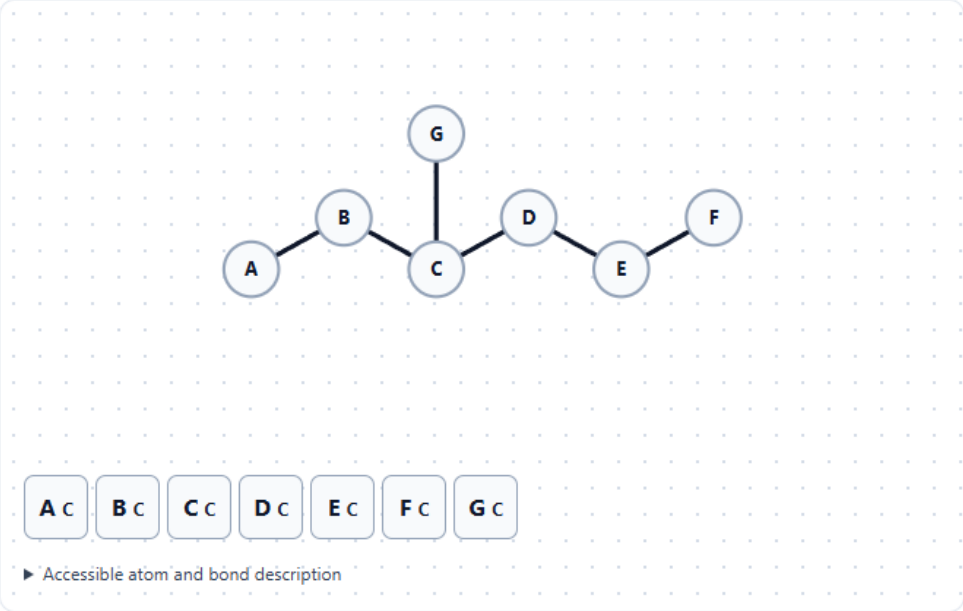
{"prompt": "Hands-on: Try It Yourself: Click BOTH carbonyl carbon atoms in ethanoic anhydride!", "answer": [1, 3], "explanation": "Correct! Carbons B and D are the two acyl carbonyl carbons flanking the central bridging oxygen."}

Screenshot: derivatives-step-4-challenge



gp1a | 1.1A · Parent Chain & Numbering Direction

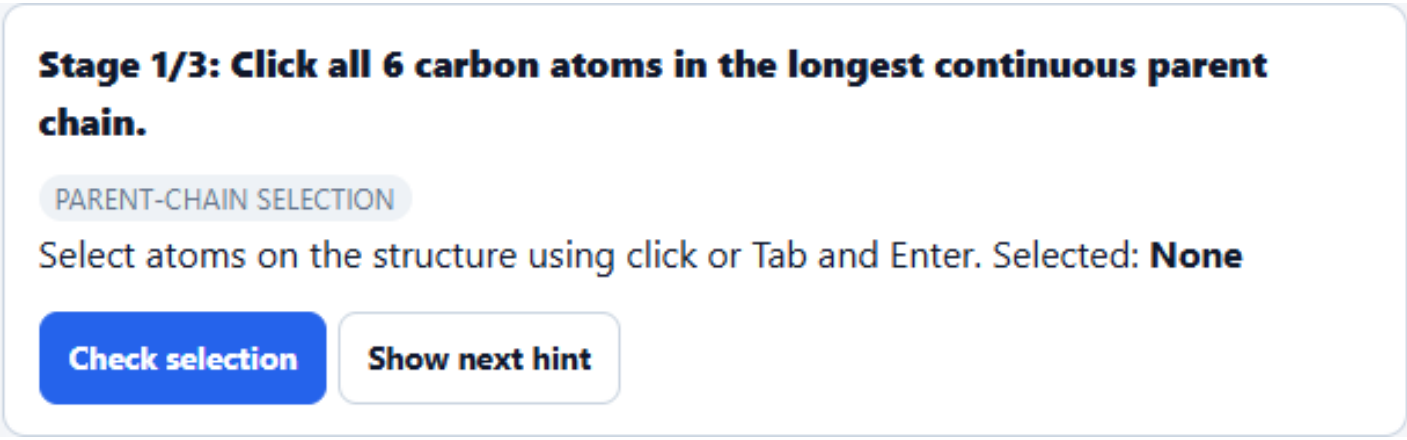
Screenshot: gp1a-stage-1-molecule



gp1a:stage-1 | Stage 1/3: Click all 6 carbon atoms in the longest continuous parent chain.

Answer type: atoms

Screenshot: gp1a-stage-1-question



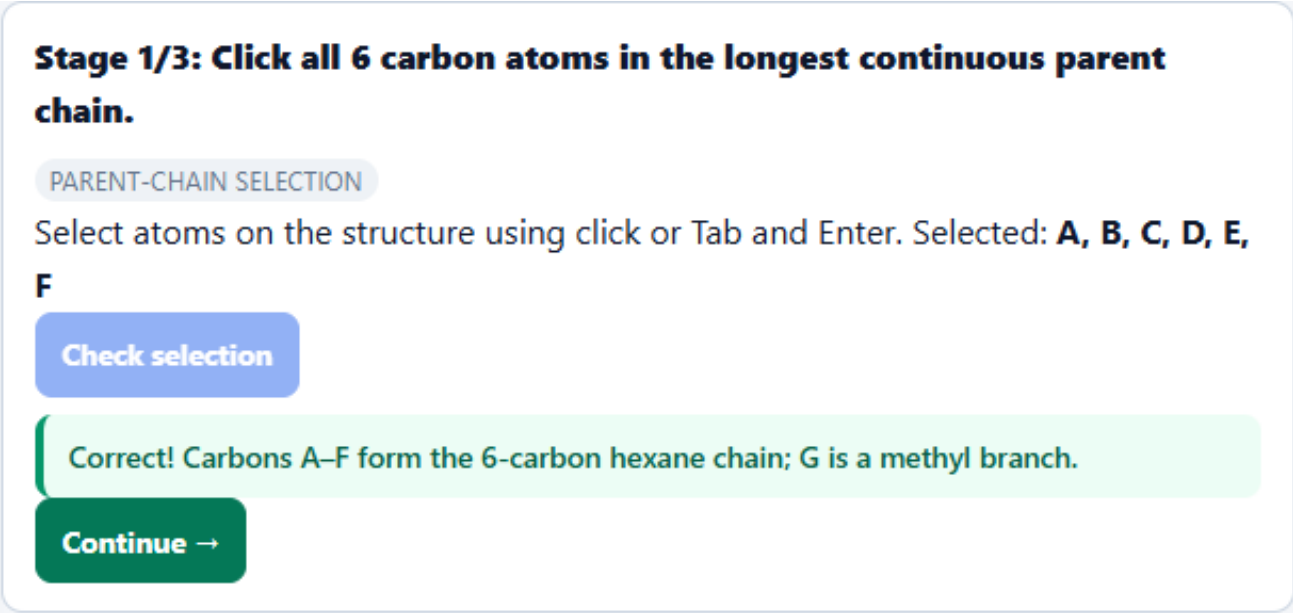
{"authorAnswer": [0, 1, 2, 3, 4, 5], "tolerance": null, "highlightsAfterCorrectAnswer": {"path": [0, 1, 2, 3, 4, 5]}}

Hint 1: Trace a continuous route between two terminal carbons without visiting any carbon twice.

Hint 2: The parent contains six carbons; at the branch point, follow the longer route and leave the short methyl arm outside the parent.

Explanation: Correct! Carbons A–F form the 6-carbon hexane chain; G is a methyl branch.

Screenshot: gp1a-stage-1-feedback



gp1a:stage-2 | Stage 2/3: Which numbering direction gives the methyl branch its lowest locant?

Answer type: choice

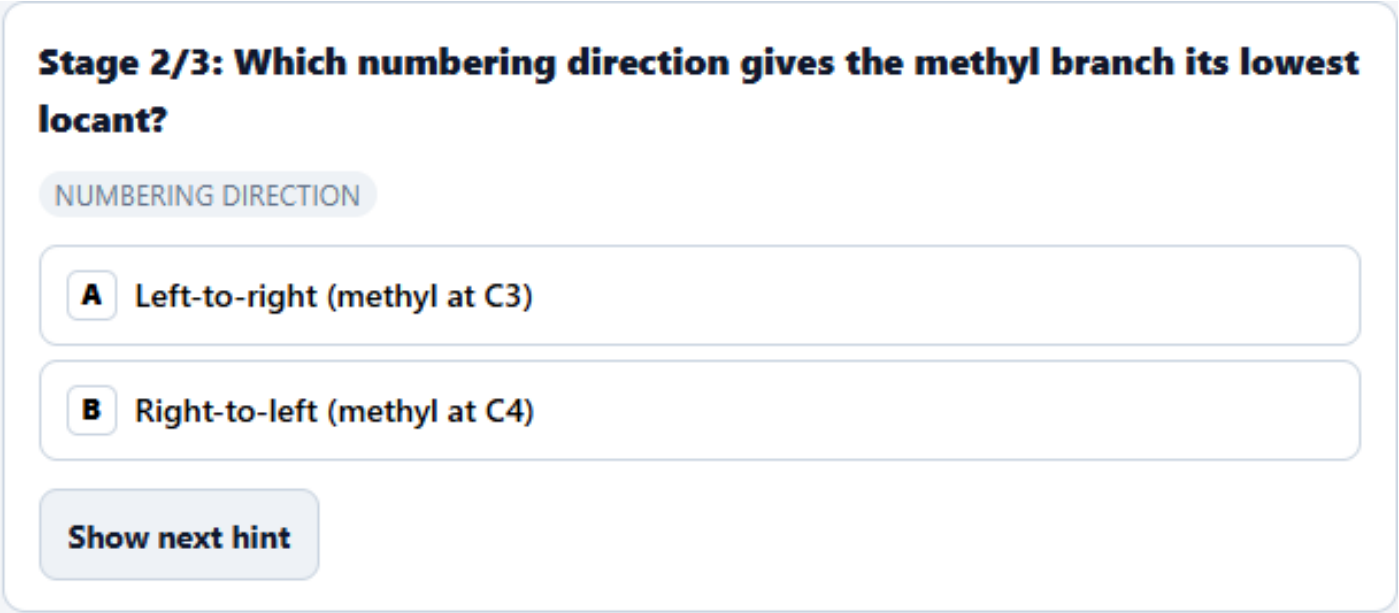
Option 0 (A): Left-to-right (methyl at C3)

Option 1 (B): Right-to-left (methyl at C4)

Screenshot: gp1a-stage-2-molecule



Screenshot: gp1a-stage-2-question



{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {"path": [0, 1, 2, 3, 4, 5], "num": [0, 1, 2, 3, 4, 5], "sub": [6]}}

Hint 1: Keep the same parent chain and count the branch position from each end.

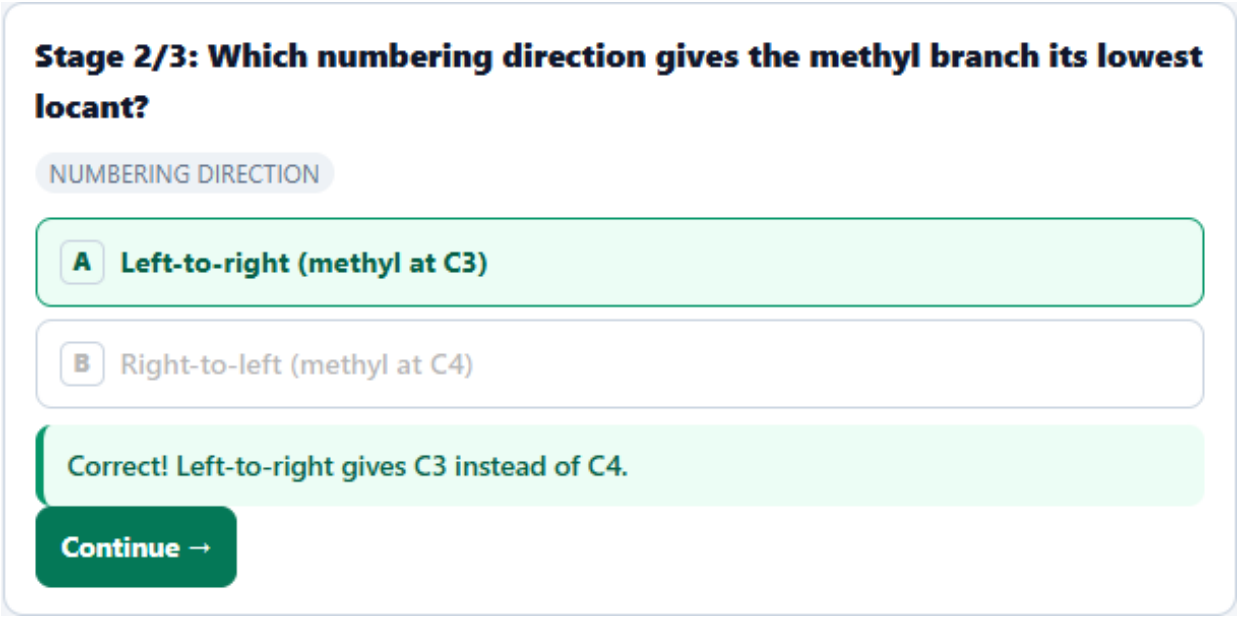
Hint 2: Compare locants 3 and 4; choose the direction with the smaller branch locant.

Explanation: Correct! Left-to-right gives C3 instead of C4.

Option 0 feedback: Correct! Left-to-right gives C3 instead of C4.

Option 1 feedback: Right-to-left puts the methyl group at C4 instead of C3!

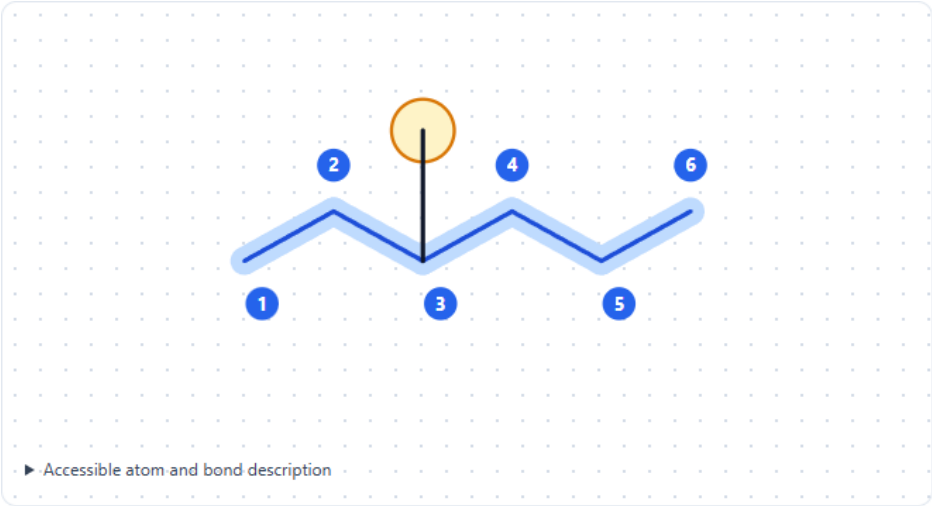
Screenshot: gp1a-stage-2-feedback



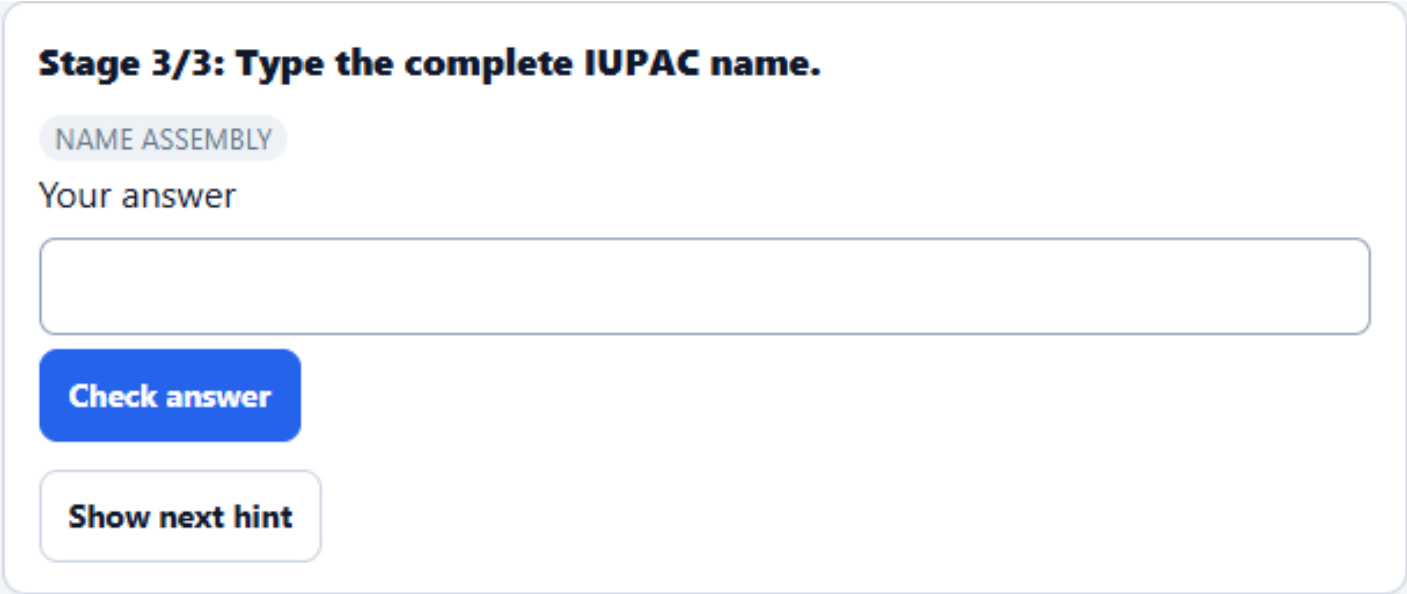
gp1a:stage-3 | Stage 3/3: Type the complete IUPAC name.

Answer type: text

Screenshot: gp1a-stage-3-molecule



Screenshot: gp1a-stage-3-question



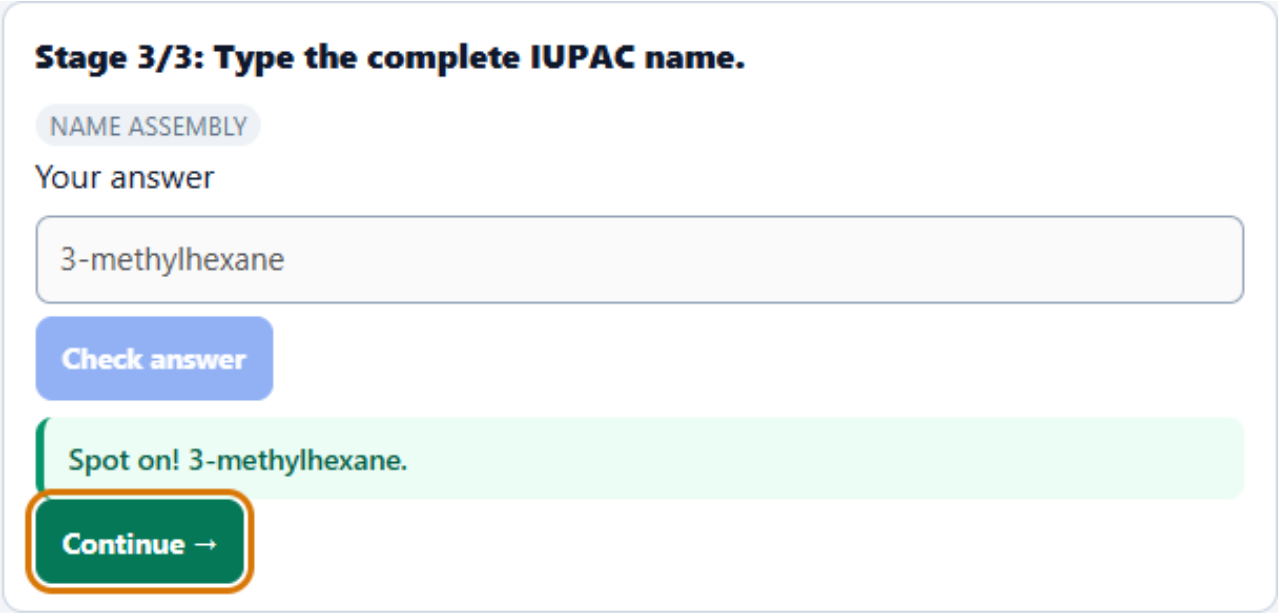
{"authorAnswer": ["3-methylhexane"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Hint 1 (Rule): Format as [locant]-[substituent][parent root][suffix] with a hyphen between the number and letter.

Hint 2: Hint 2 (Structure): There is a methyl group at C3 on a 6-carbon alkane parent (hexane).

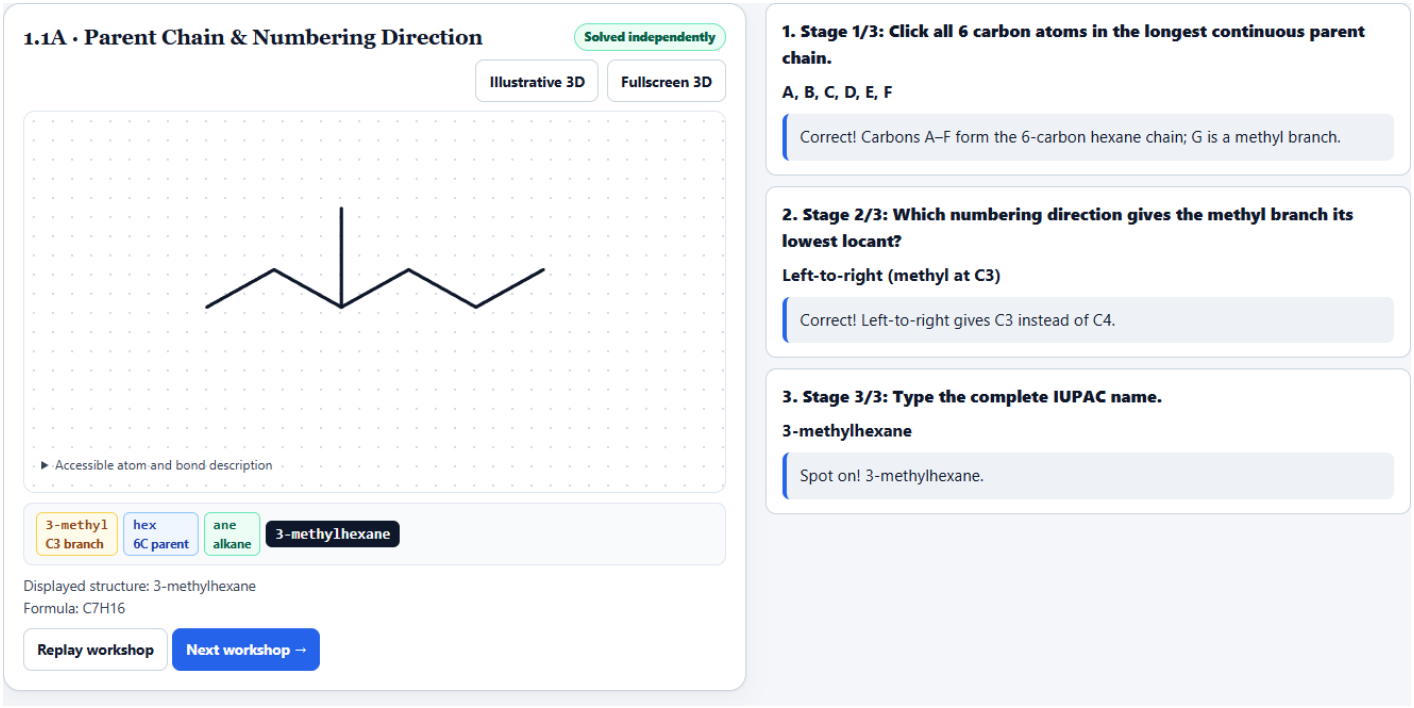
Explanation: Spot on! 3-methylhexane.

Screenshot: gp1a-stage-3-feedback



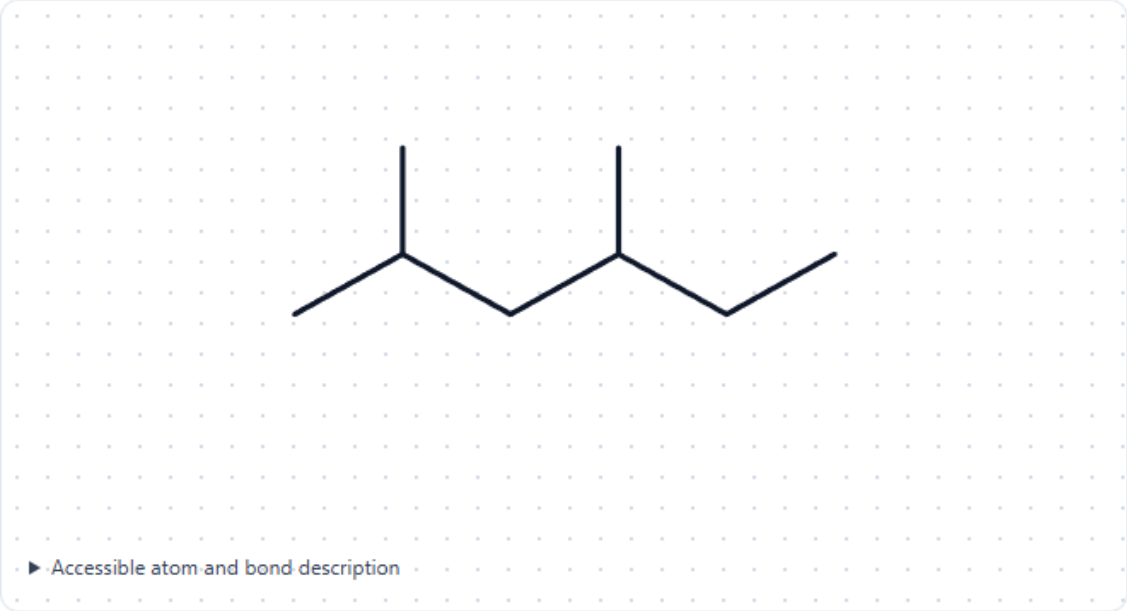
Completion summary: 3-methylhexane

Screenshot: gp1a-complete



gp1b | 1.1B · Diagnosing a Hidden Parent Chain

Screenshot: gp1b-stage-1-molecule



gp1b:stage-1 | Stage 1/3: Why is "2-ethyl-4-methylpentane" an invalid name for this structure?

Answer type: choice

Option 0 (A): The 2-carbon branch extends the longest continuous chain to 6 carbons (hexane)

Option 1 (B): It has 8 carbons in a straight chain

Screenshot: gp1b-stage-1-question



{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {"path": [0, 1, 2, 3, 4, 5]}}

Hint 1: A parent chain may turn at a branch; it does not have to follow the most horizontal line.

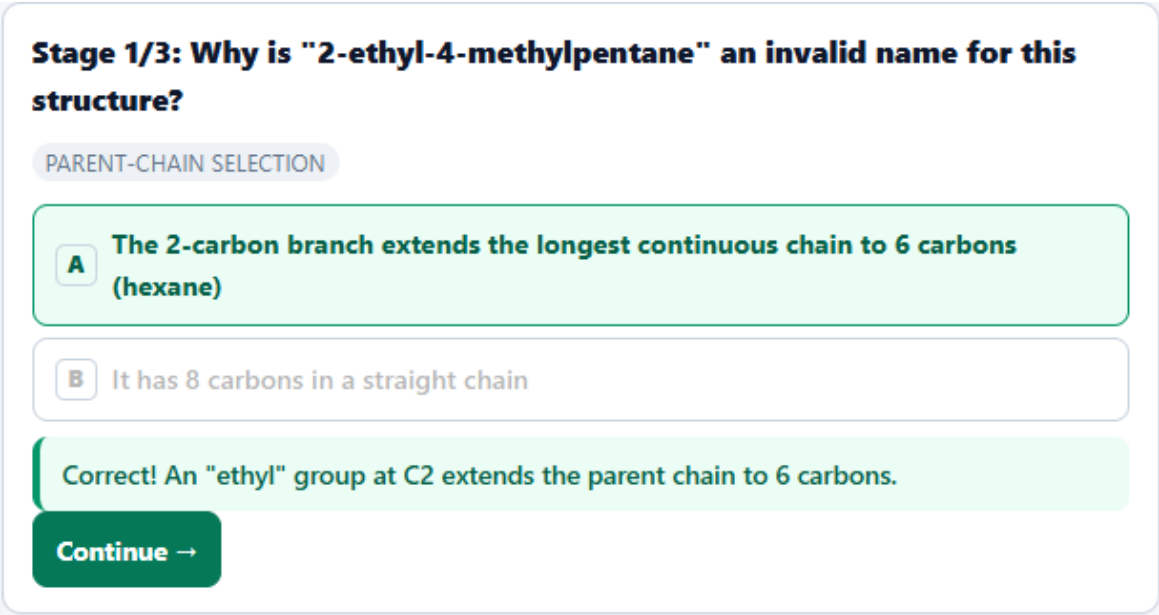
Hint 2: Include the two-carbon arm in a continuous six-carbon route. The proposed pentane parent is too short.

Explanation: Correct! An "ethyl" group at C2 extends the parent chain to 6 carbons.

Option 0 feedback: Correct! An "ethyl" group at C2 extends the parent chain to 6 carbons.

Option 1 feedback: There are 8 total carbons, but a continuous chain cannot fork (longest path = 6 carbons).

Screenshot: gp1b-stage-1-feedback



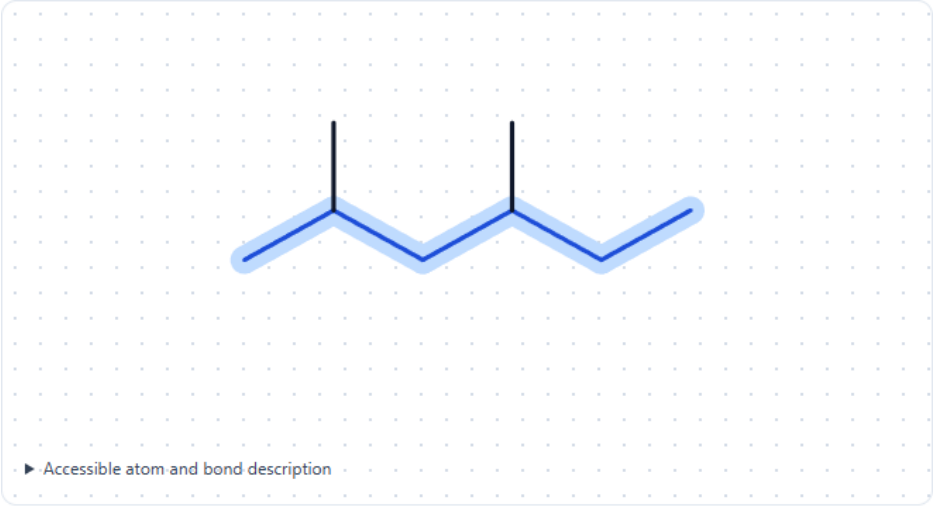
gp1b:stage-2 | Stage 2/3: Which locant set is lower for the two methyl substituents?

Answer type: choice

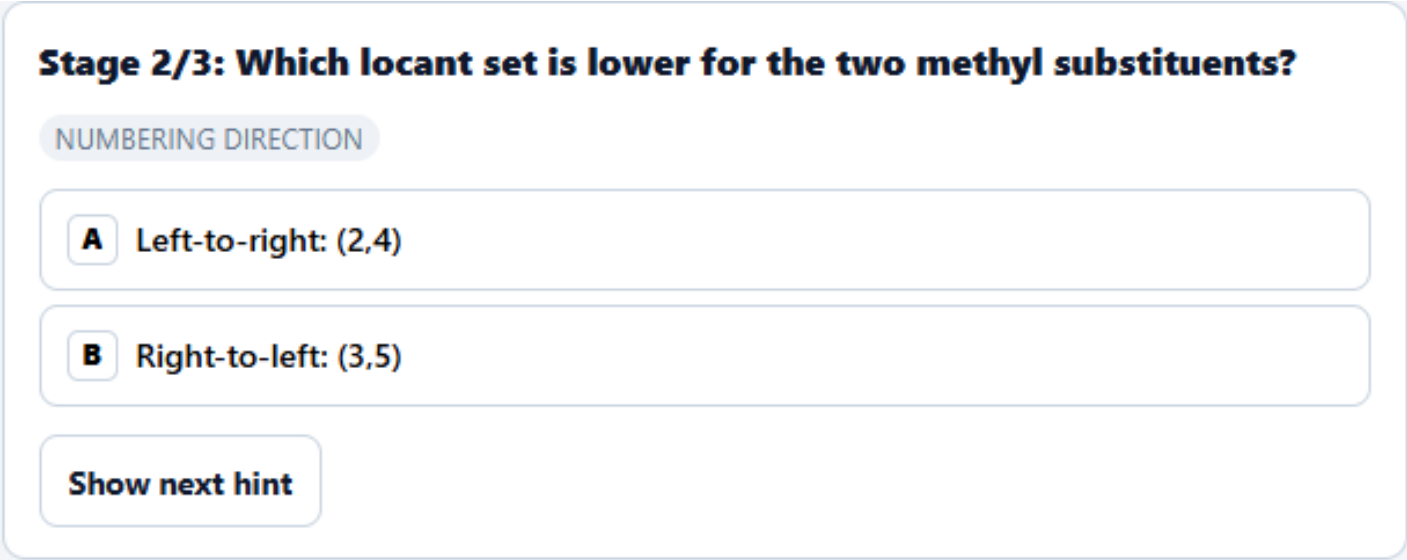
Option 0 (A): Left-to-right: (2,4)

Option 1 (B): Right-to-left: (3,5)

Screenshot: gp1b-stage-2-molecule



Screenshot: gp1b-stage-2-question



{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {"path": [0, 1, 2, 3, 4, 5], "num": [0, 1, 2, 3, 4, 5], "sub": [6, 7]}}

Hint 1: List the methyl locants in increasing order for each numbering direction.

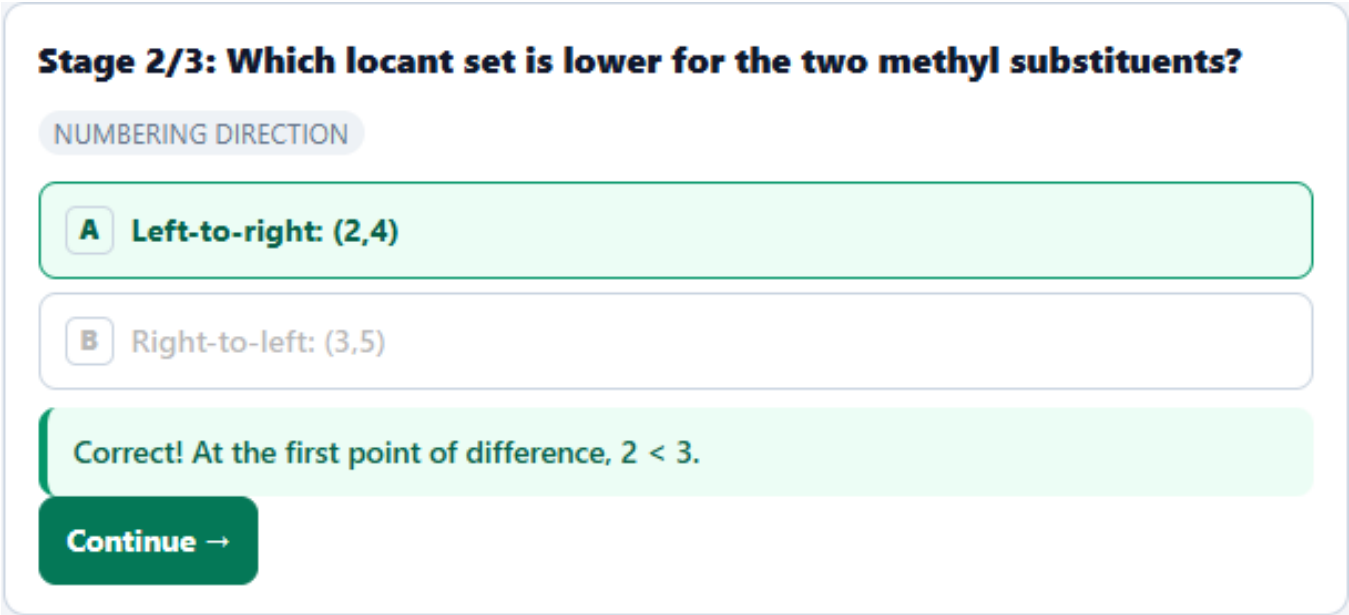
Hint 2: Compare (2,4) with (3,5) at the first differing entry; do not compare their sums.

Explanation: Correct! At the first point of difference, 2 < 3.

Option 0 feedback: Correct! At the first point of difference, 2 < 3.

Option 1 feedback: Right-to-left gives locants (3,5), which loses to (2,4) at the first point of difference!

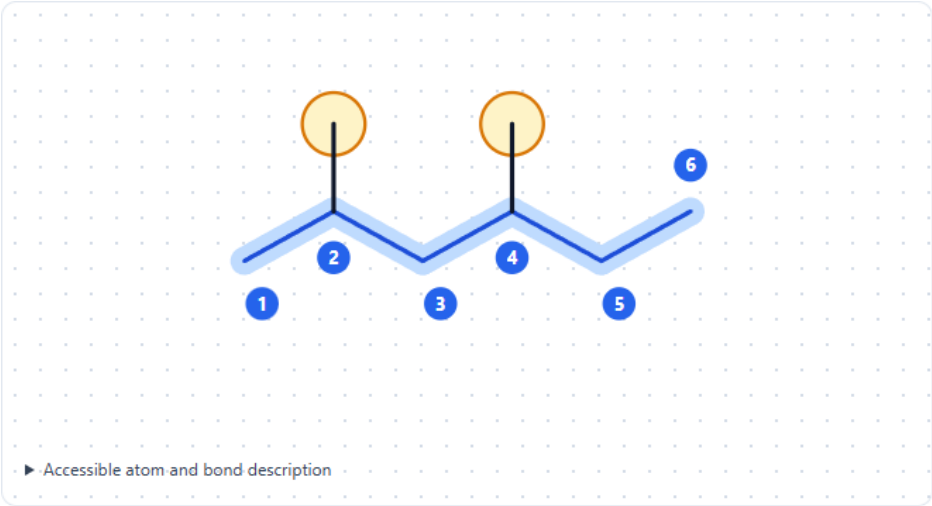
Screenshot: gp1b-stage-2-feedback



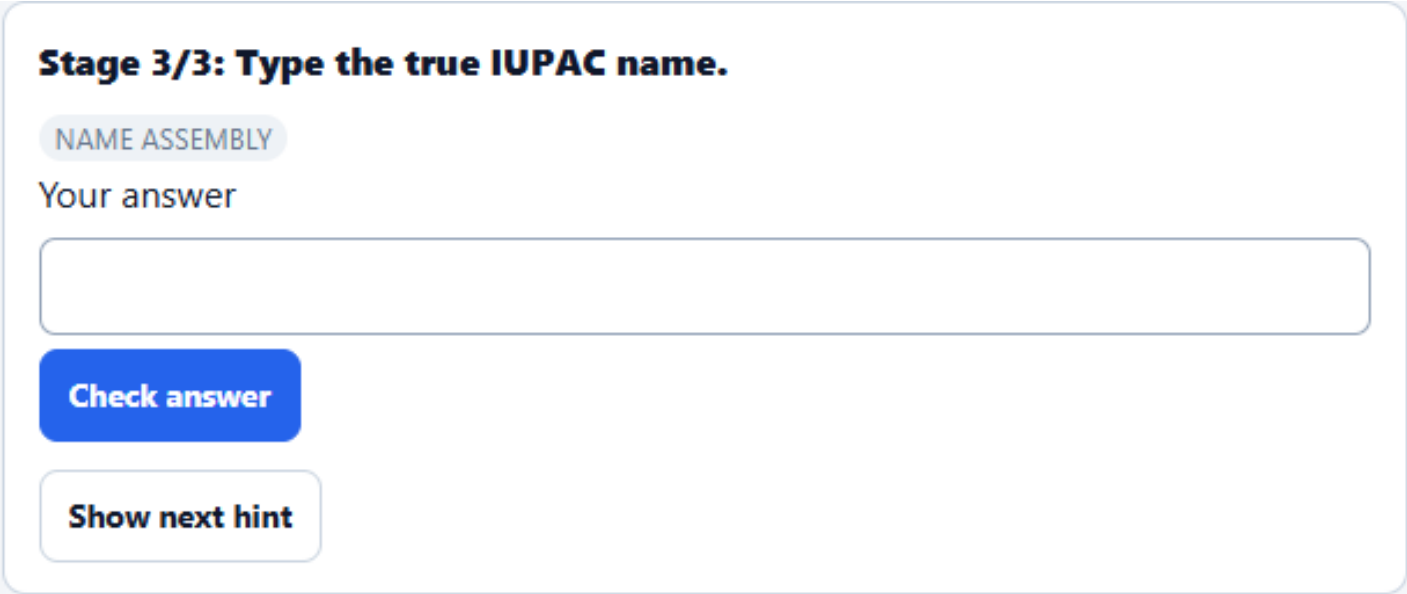
gp1b:stage-3 | Stage 3/3: Type the true IUPAC name.

Answer type: text

Screenshot: gp1b-stage-3-molecule



Screenshot: gp1b-stage-3-question



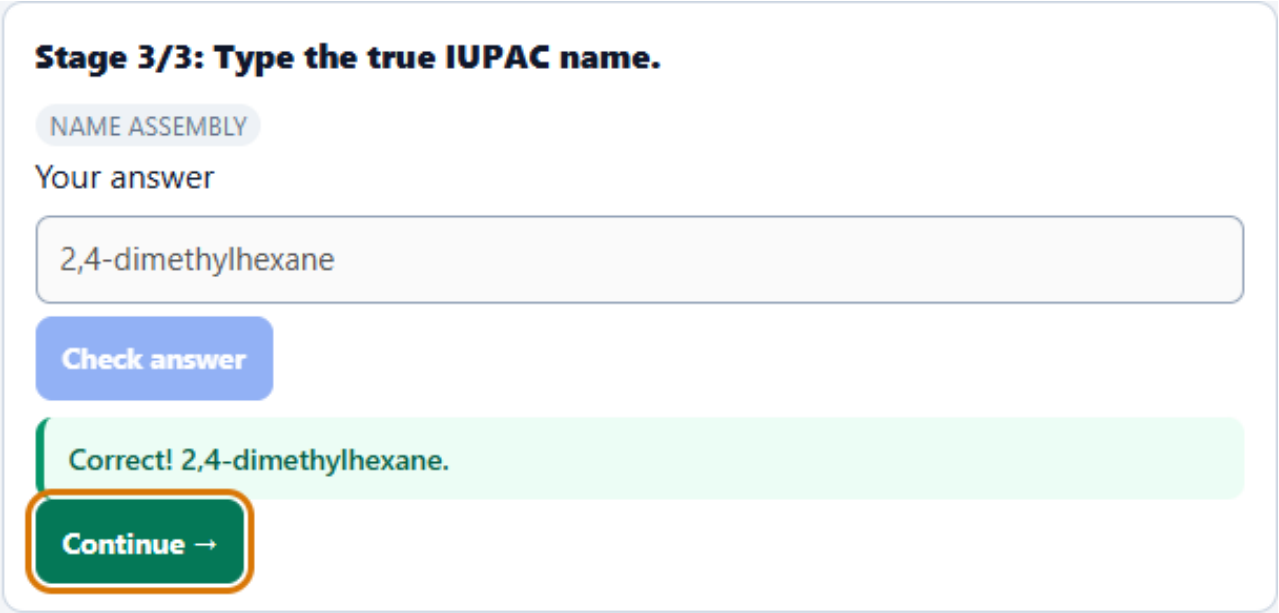
{"authorAnswer": ["2,4-dimethylhexane"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Hint 1 (Rule): Two identical substituents require two locants separated by a comma and the multiplier prefix di–.

Hint 2: Hint 2 (Structure): Two methyl groups sit at C2 and C4 on a 6-carbon hexane chain.

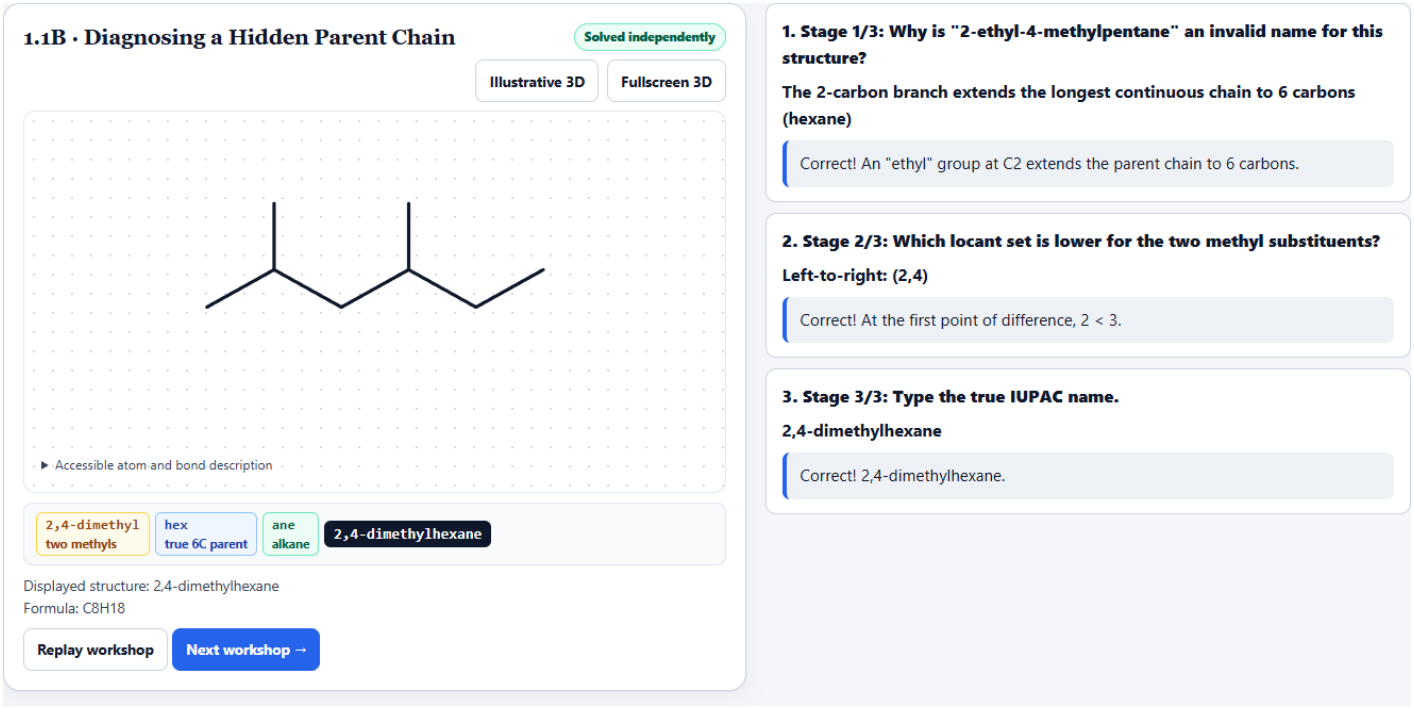
Explanation: Correct! 2,4-dimethylhexane.

Screenshot: gp1b-stage-3-feedback



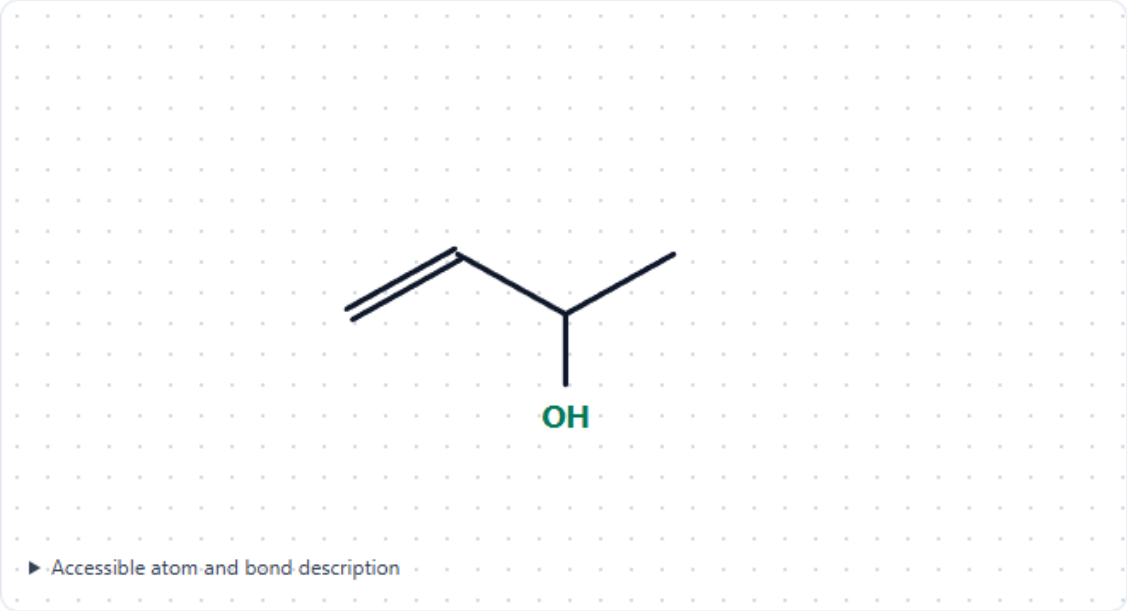
Completion summary: 2,4-dimethylhexane

Screenshot: gp1b-complete



gp2a | 1.2A · Alcohol vs. Alkene Priority

Screenshot: gp2a-stage-1-molecule



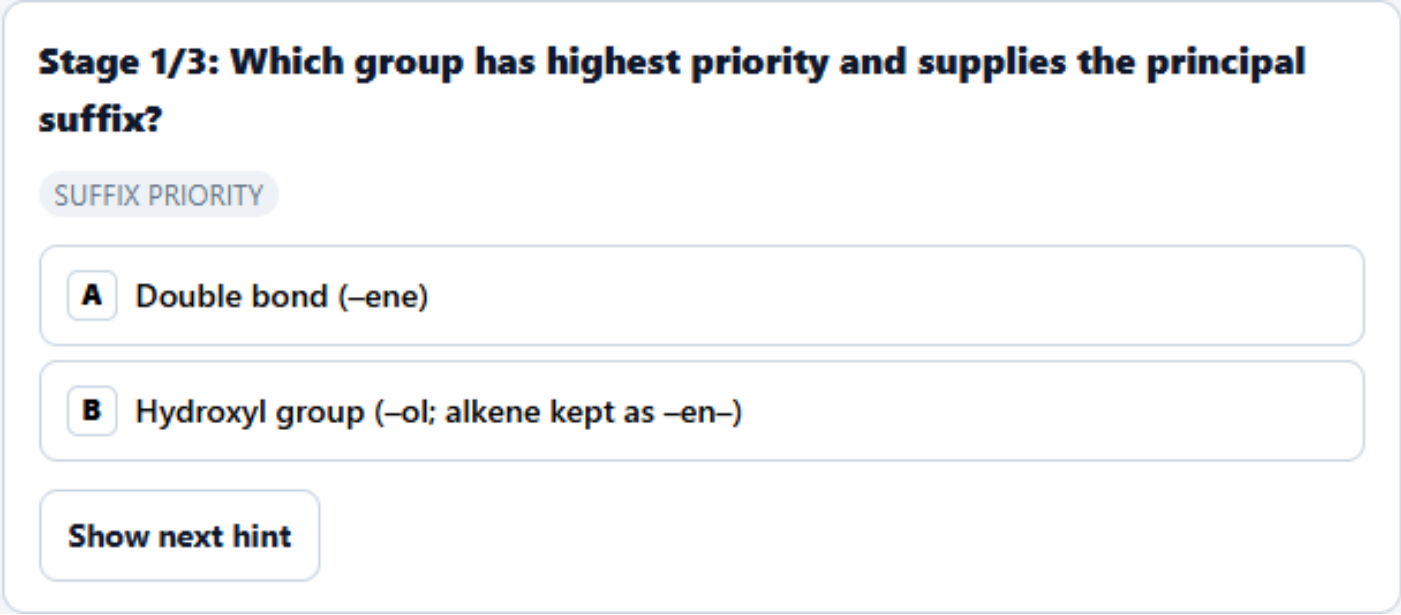
gp2a:stage-1 | Stage 1/3: Which group has highest priority and supplies the principal suffix?

Answer type: choice

Option 0 (A): Double bond (–ene)

Option 1 (B): Hydroxyl group (–ol; alkene kept as –en–)

Screenshot: gp2a-stage-1-question



{"authorAnswer": 1, "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [4]}}

Hint 1: Decide which characteristic group takes the principal suffix when an alcohol and an alkene are both present.

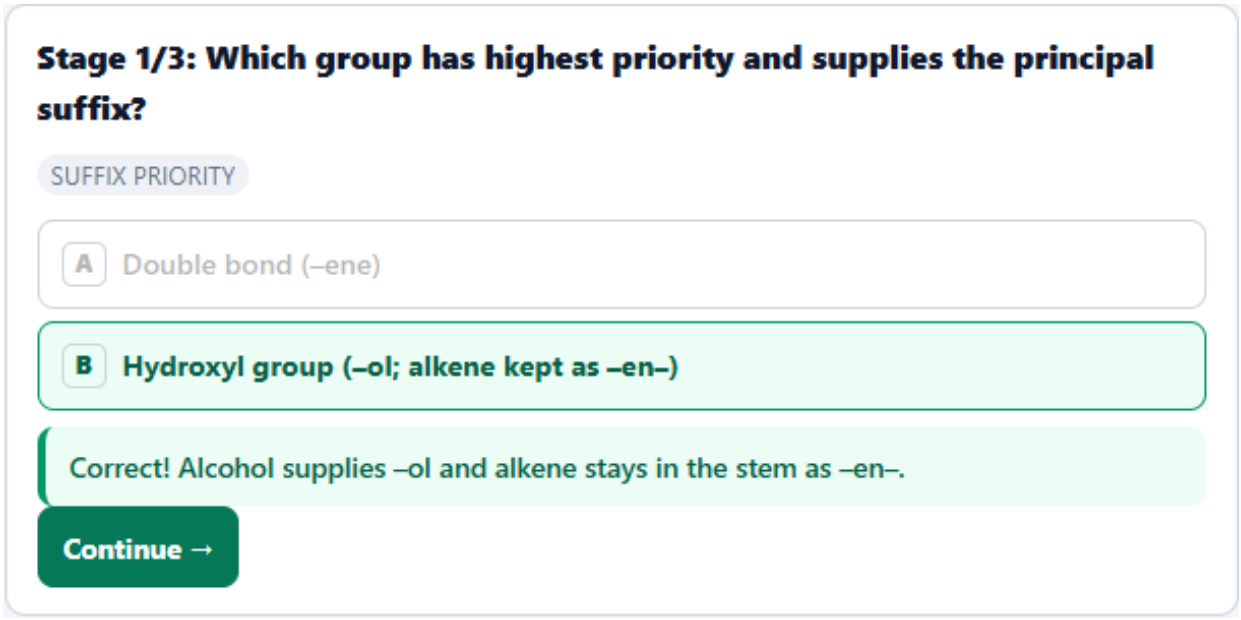
Hint 2: The hydroxyl group supplies -ol; the double bond is still represented by -en-.

Explanation: Correct! Alcohol supplies –ol and alkene stays in the stem as –en–.

Option 0 feedback: Priority Error: Alcohols have higher IUPAC seniority than alkenes.

Option 1 feedback: Correct! Alcohol supplies –ol and alkene stays in the stem as –en–.

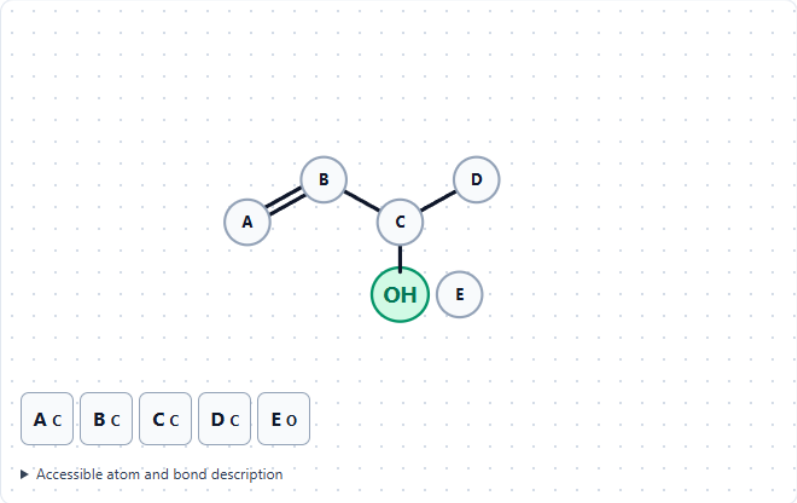
Screenshot: gp2a-stage-1-feedback



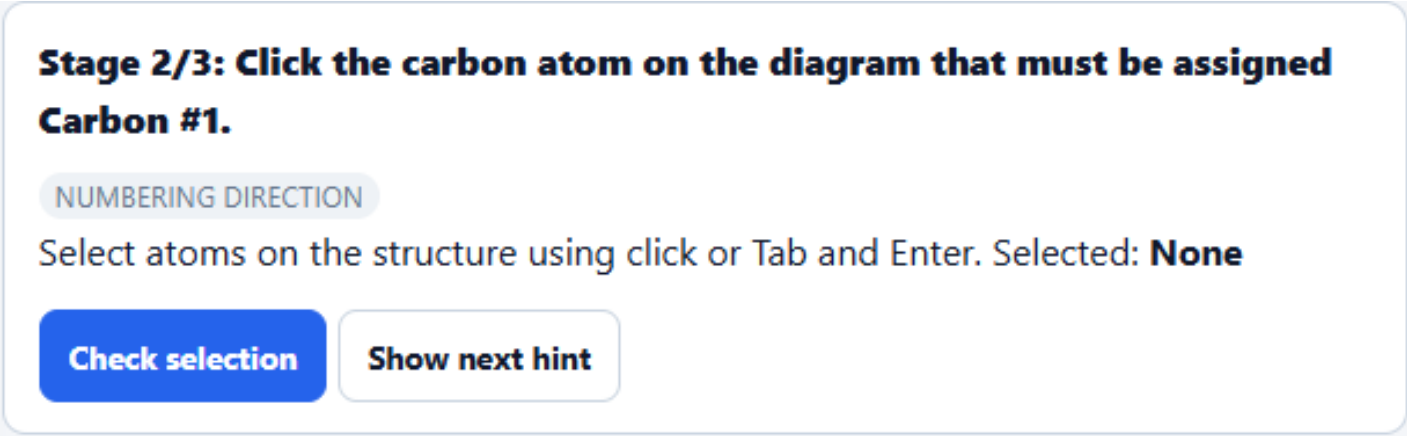
gp2a:stage-2 | Stage 2/3: Click the carbon atom on the diagram that must be assigned Carbon #1.

Answer type: atoms

Screenshot: gp2a-stage-2-molecule



Screenshot: gp2a-stage-2-question



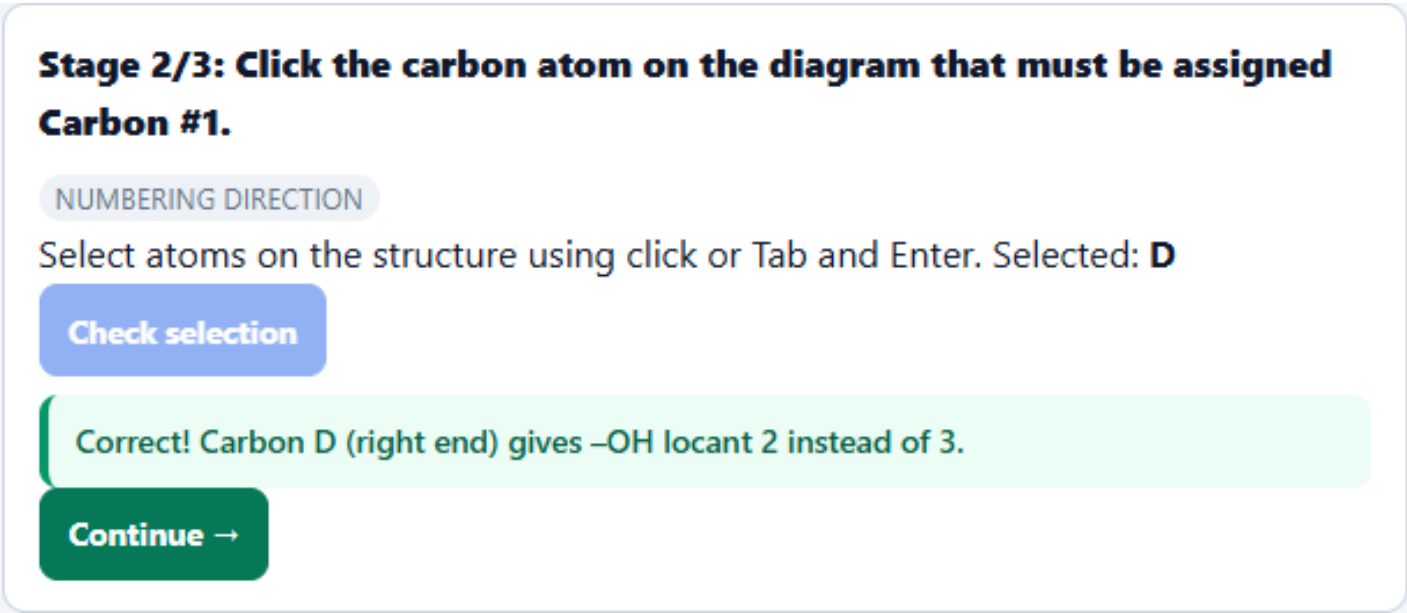
{"authorAnswer": [3], "tolerance": null, "highlightsAfterCorrectAnswer": {"path": [0, 1, 2, 3], "num": [3, 2, 1, 0], "prin": [4]}}

Hint 1: Number the four-carbon parent from both ends and compare the hydroxyl locant first.

Hint 2: Start at the terminal carbon nearest the hydroxyl-bearing carbon, even though that gives the double bond a larger locant.

Explanation: Correct! Carbon D (right end) gives –OH locant 2 instead of 3.

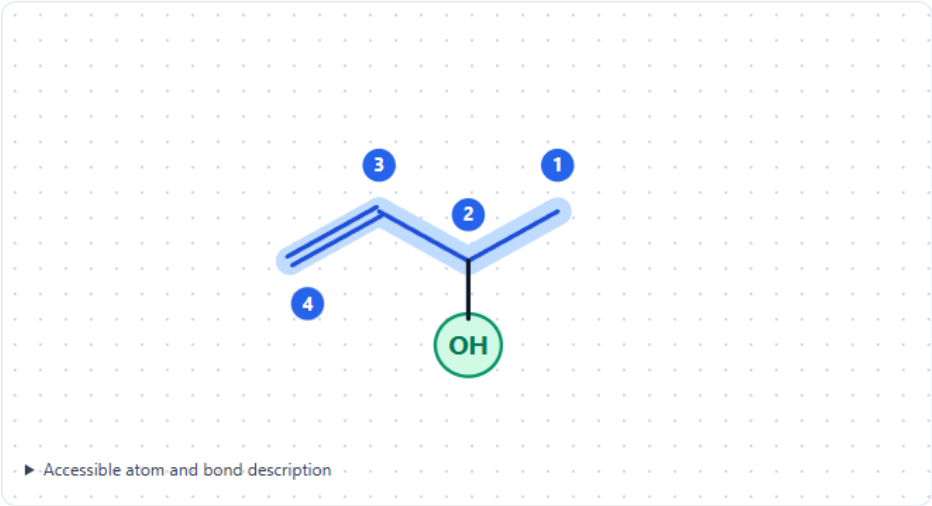
Screenshot: gp2a-stage-2-feedback



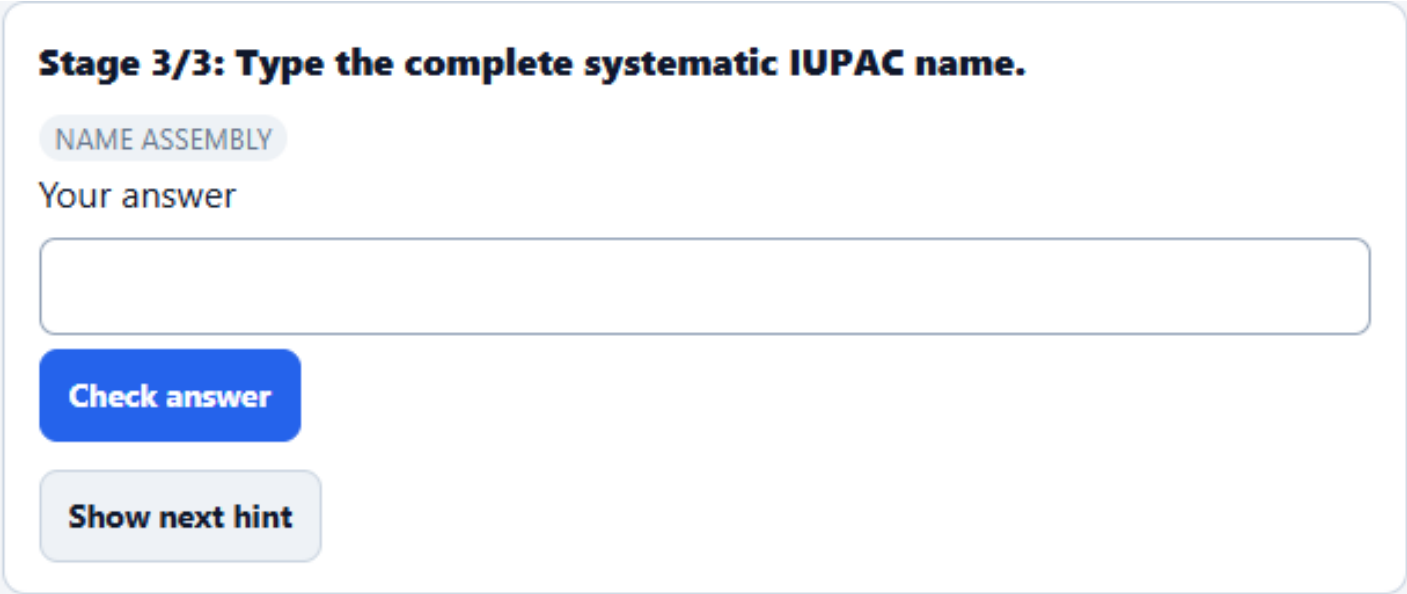
gp2a:stage-3 | Stage 3/3: Type the complete systematic IUPAC name.

Answer type: text

Screenshot: gp2a-stage-3-molecule



Screenshot: gp2a-stage-3-question



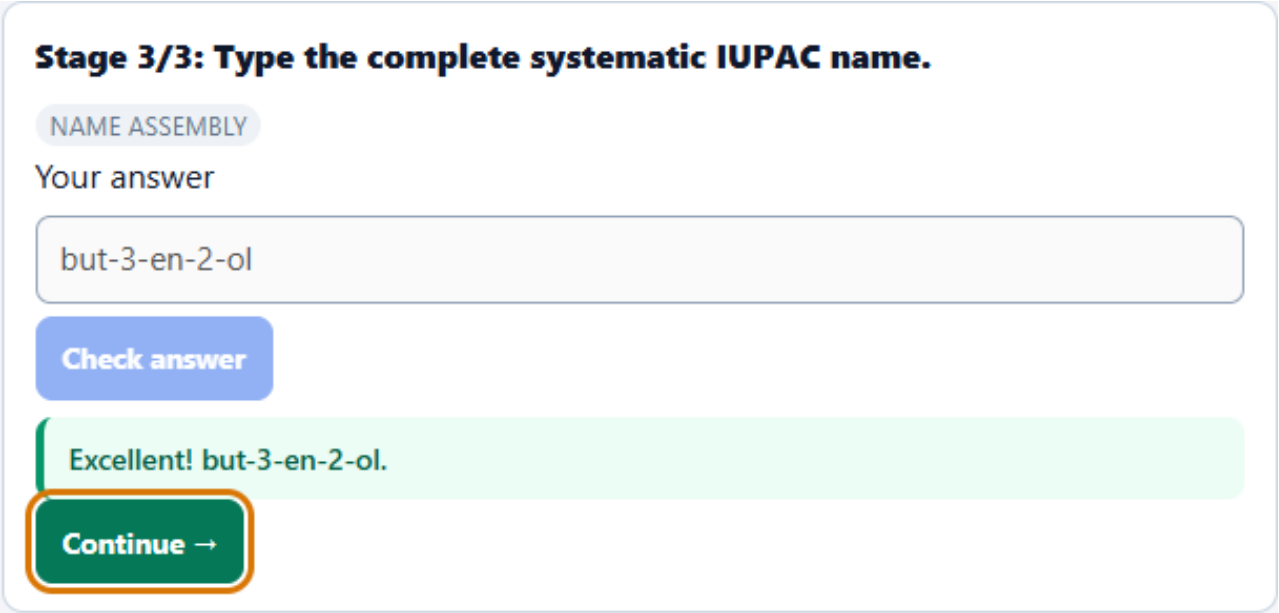
{"authorAnswer": ["but-3-en-2-ol", "3-buten-2-ol"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Hint 1 (Rule): Combine the carbon root, the unsaturation locant + infix (–en–), and the principal alcohol locant + suffix (–ol).

Hint 2: Hint 2 (Structure): 4-carbon root (but), double bond starting at C3, and alcohol at C2.

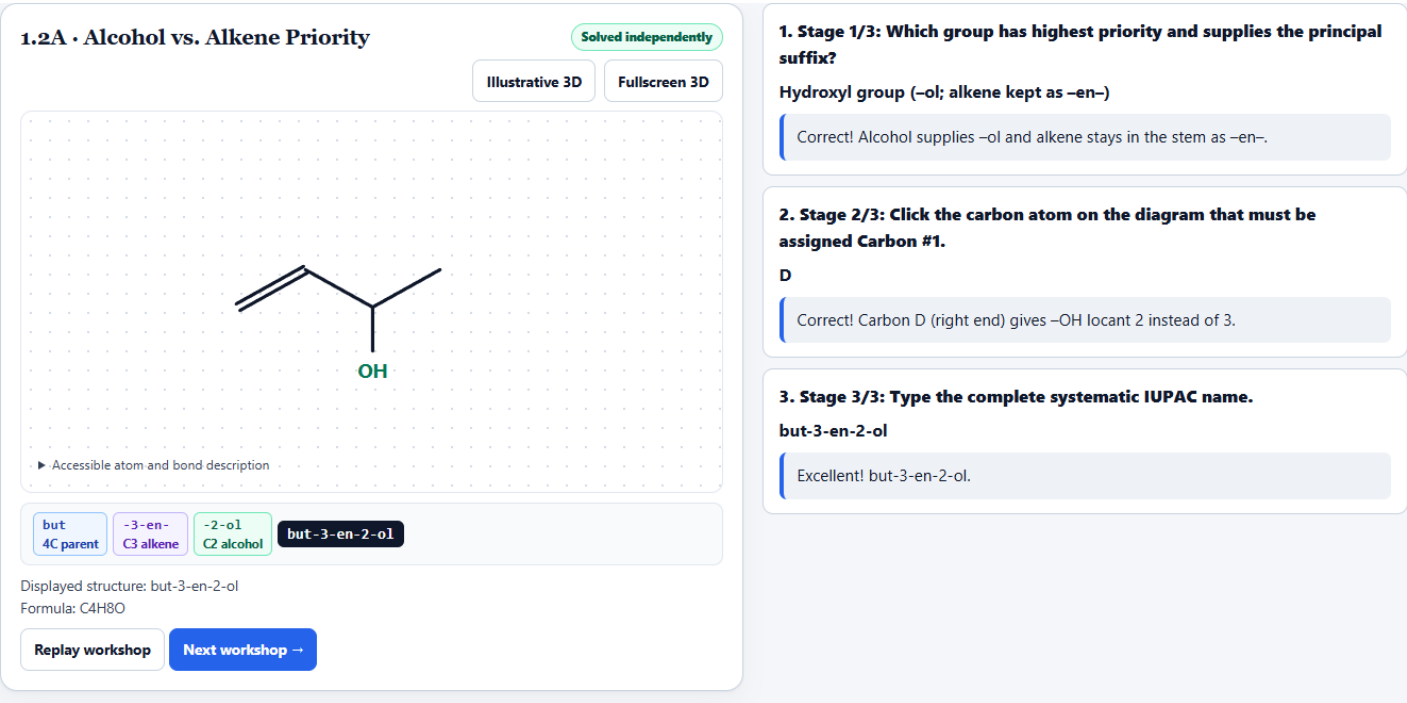
Explanation: Excellent! but-3-en-2-ol.

Screenshot: gp2a-stage-3-feedback



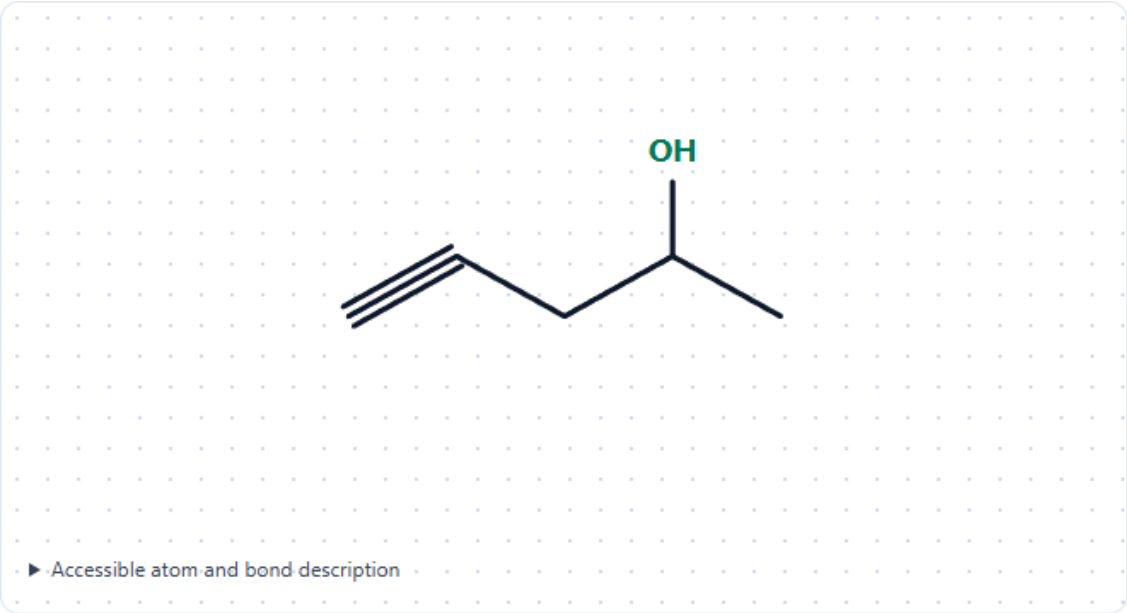
Completion summary: but-3-en-2-ol

Screenshot: gp2a-complete



gp2b | 1.2B · Alcohol vs. Alkyne Priority

Screenshot: gp2b-stage-1-molecule



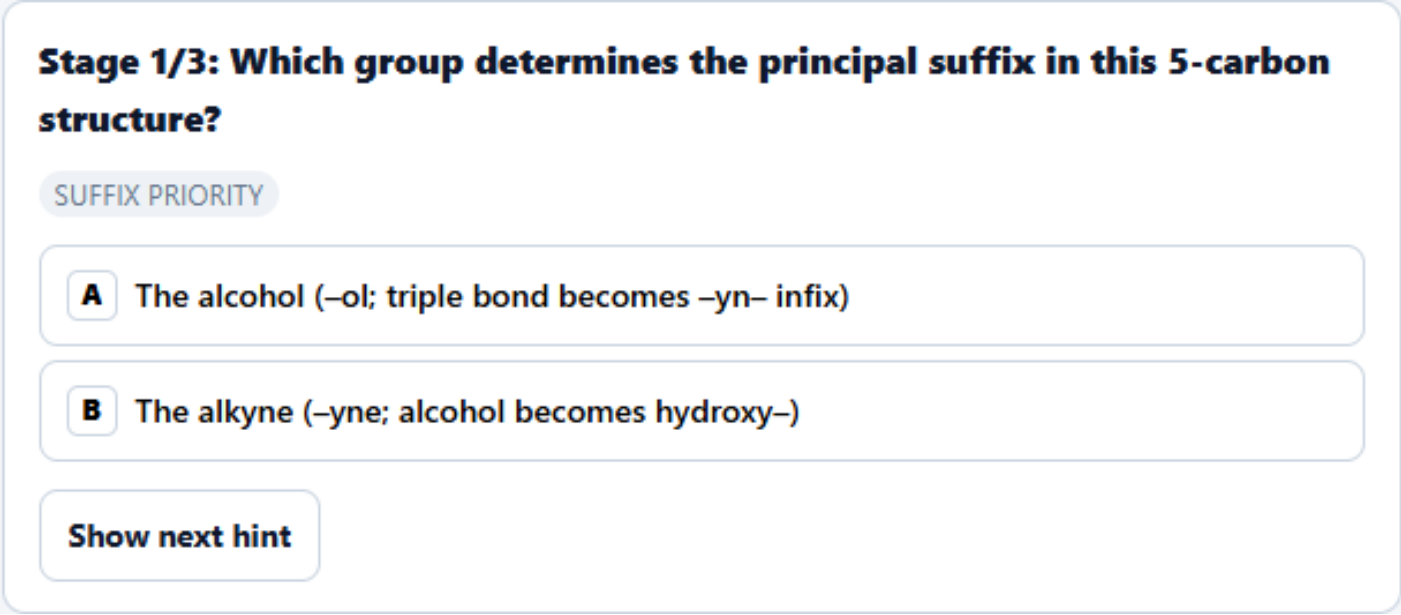
gp2b:stage-1 | Stage 1/3: Which group determines the principal suffix in this 5-carbon structure?

Answer type: choice

Option 0 (A): The alcohol (–ol; triple bond becomes –yn– infix)

Option 1 (B): The alkyne (–yne; alcohol becomes hydroxy–)

Screenshot: gp2b-stage-1-question



{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [5]}}

Hint 1: An alkyne does not outrank an alcohol for the principal suffix.

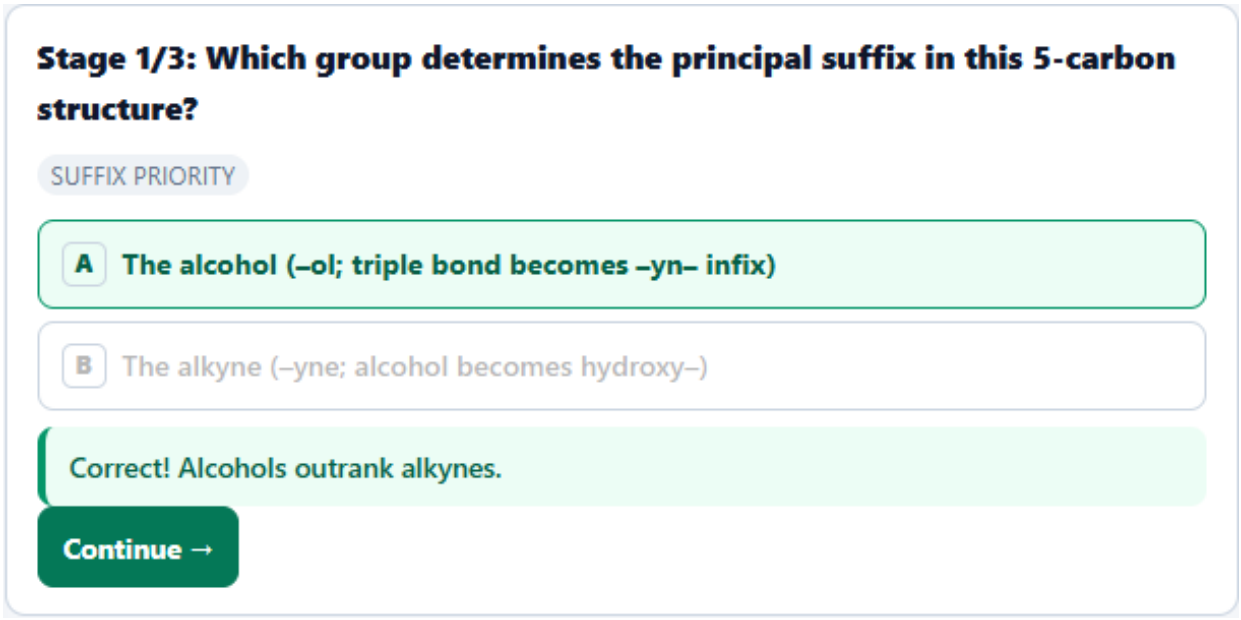
Hint 2: Use -ol for the principal group and retain -yn- for the triple bond.

Explanation: Correct! Alcohols outrank alkynes.

Option 0 feedback: Correct! Alcohols outrank alkynes.

Option 1 feedback: Priority Error: Alcohols outrank both alkenes and alkynes.

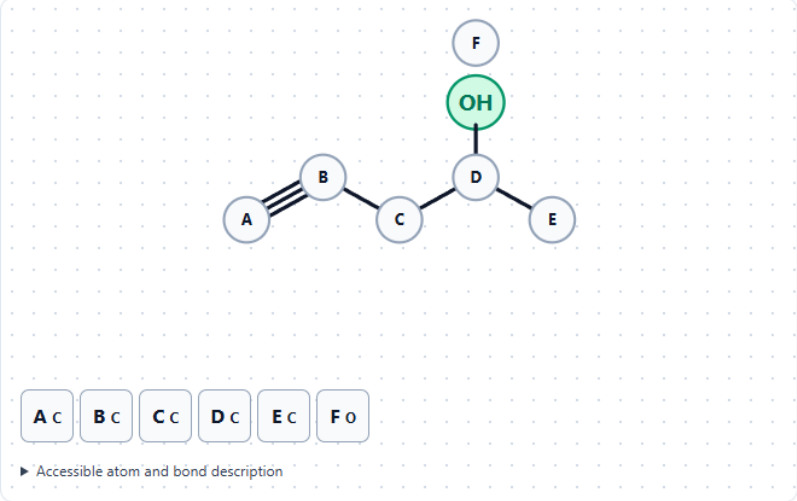
Screenshot: gp2b-stage-1-feedback



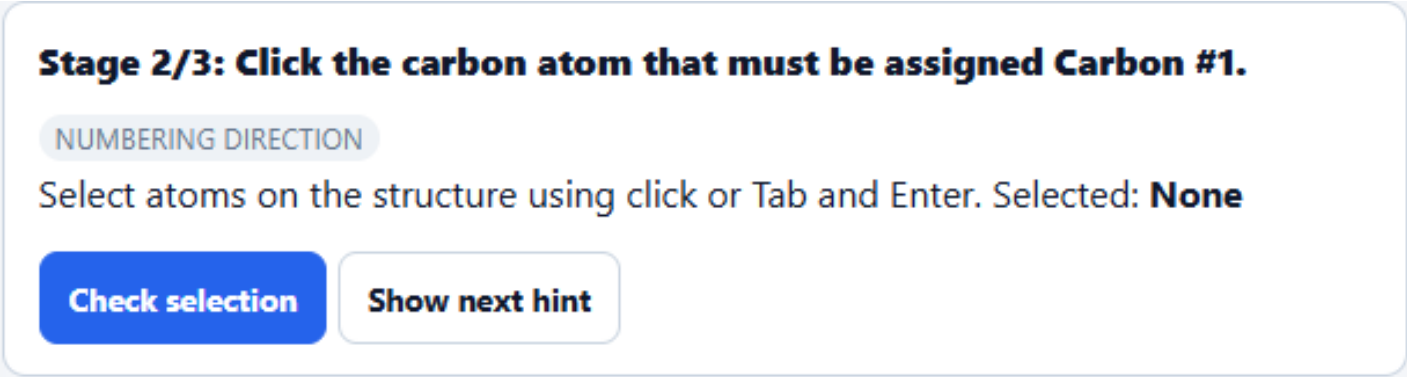
gp2b:stage-2 | Stage 2/3: Click the carbon atom that must be assigned Carbon #1.

Answer type: atoms

Screenshot: gp2b-stage-2-molecule



Screenshot: gp2b-stage-2-question



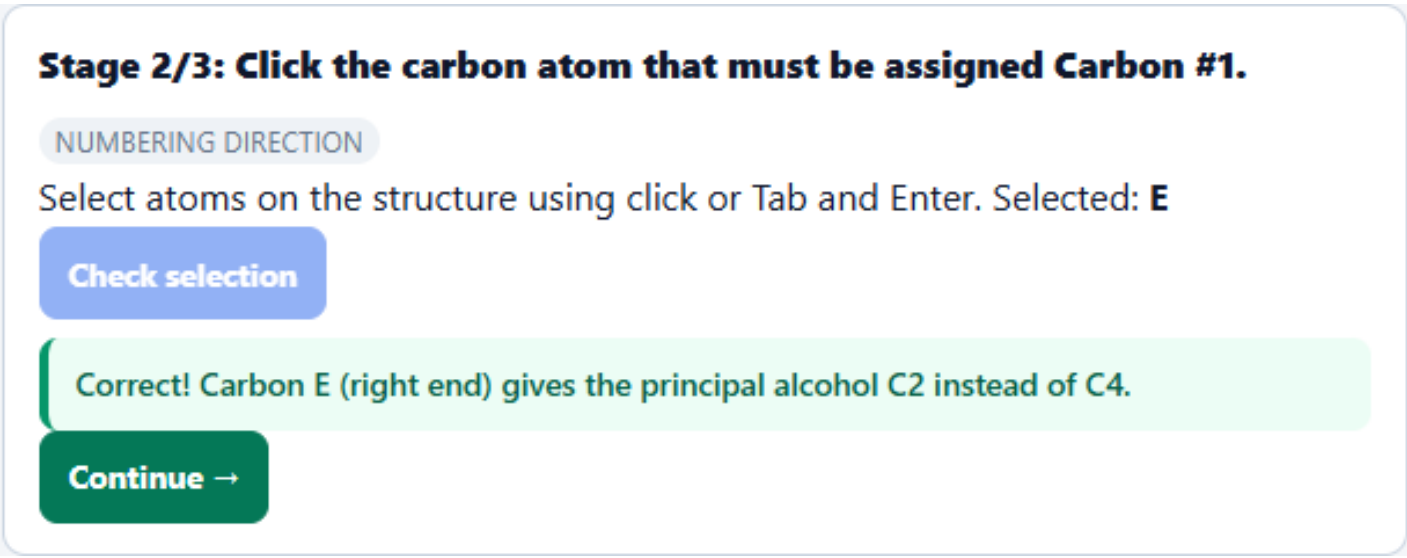
{"authorAnswer": [4], "tolerance": null, "highlightsAfterCorrectAnswer": {"path": [0, 1, 2, 3, 4], "num": [4, 3, 2, 1, 0], "prin": [5]}}

Hint 1: Compare the hydroxyl position when the five-carbon chain is numbered from each end.

Hint 2: The end nearest the hydroxyl gives it locant 2 rather than 4; choose that end as C1.

Explanation: Correct! Carbon E (right end) gives the principal alcohol C2 instead of C4.

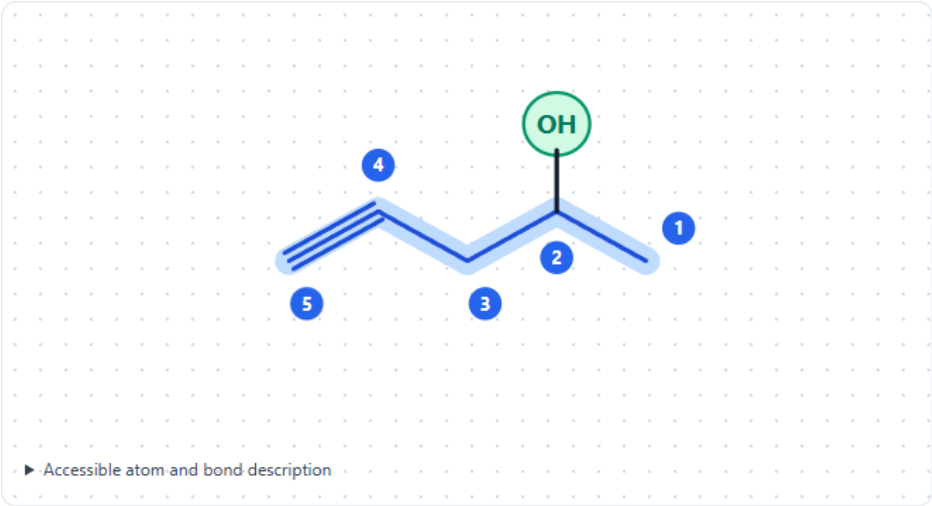
Screenshot: gp2b-stage-2-feedback



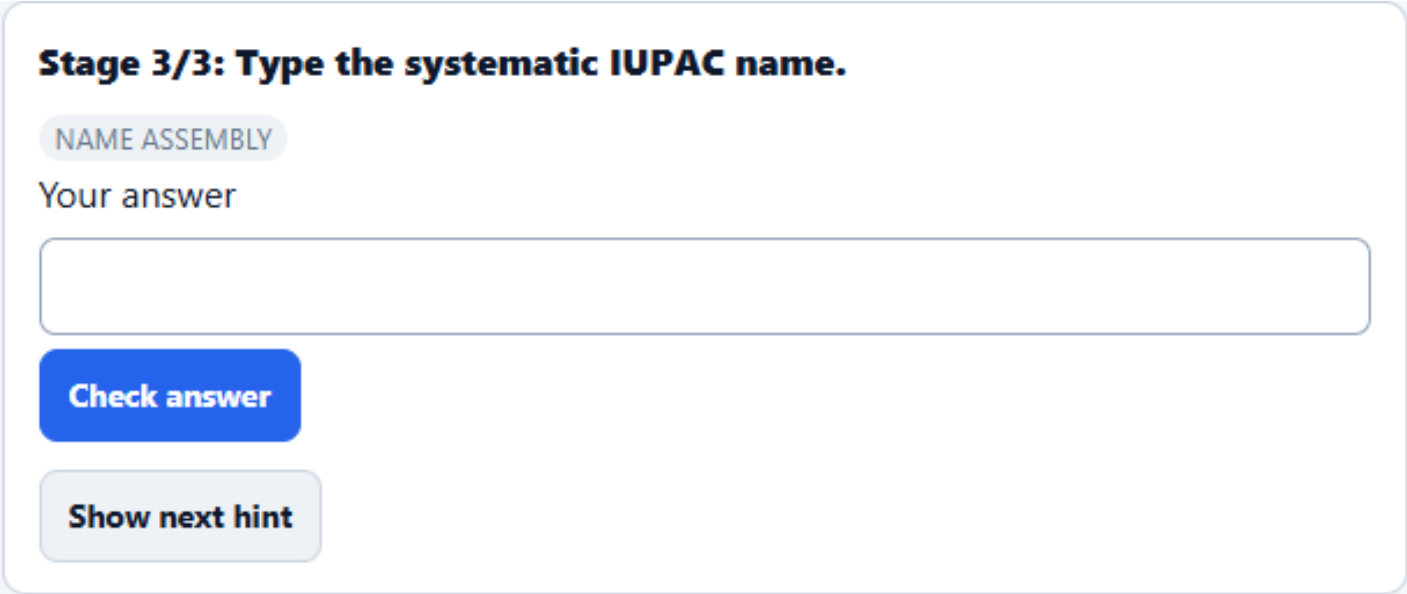
gp2b:stage-3 | Stage 3/3: Type the systematic IUPAC name.

Answer type: text

Screenshot: gp2b-stage-3-molecule



Screenshot: gp2b-stage-3-question



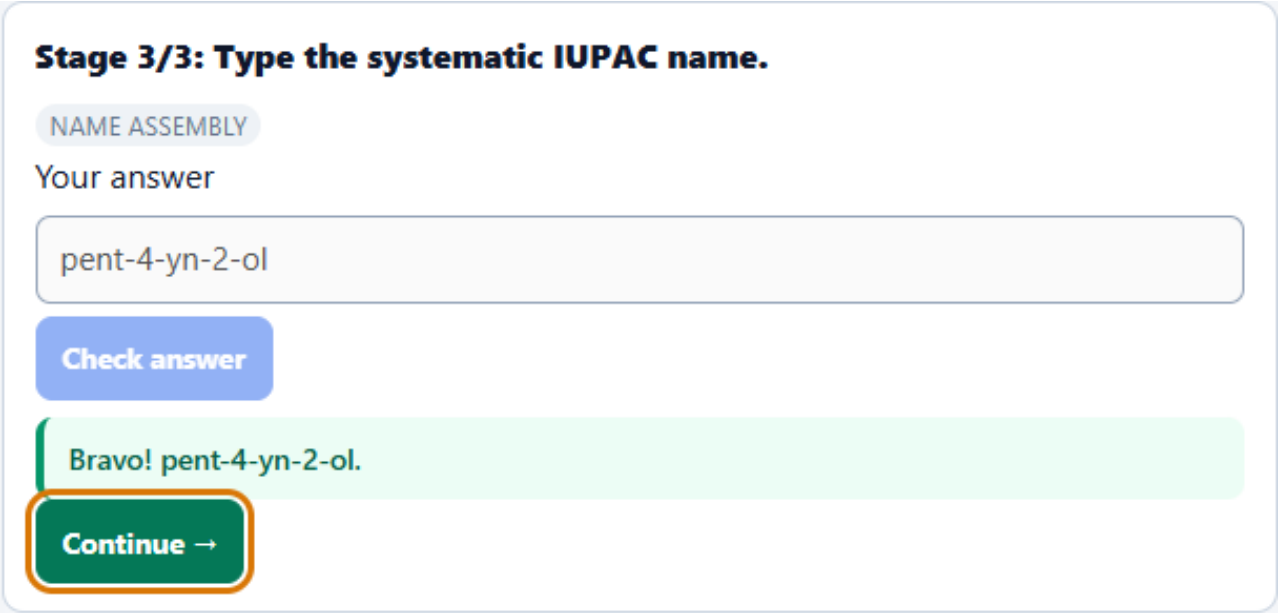
{"authorAnswer": ["pent-4-yn-2-ol", "4-pentyn-2-ol"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Hint 1 (Rule): Use the triple-bond infix –yn– before the principal alcohol suffix –ol.

Hint 2: Hint 2 (Structure): 5-carbon chain (pent), triple bond at C4, and alcohol at C2.

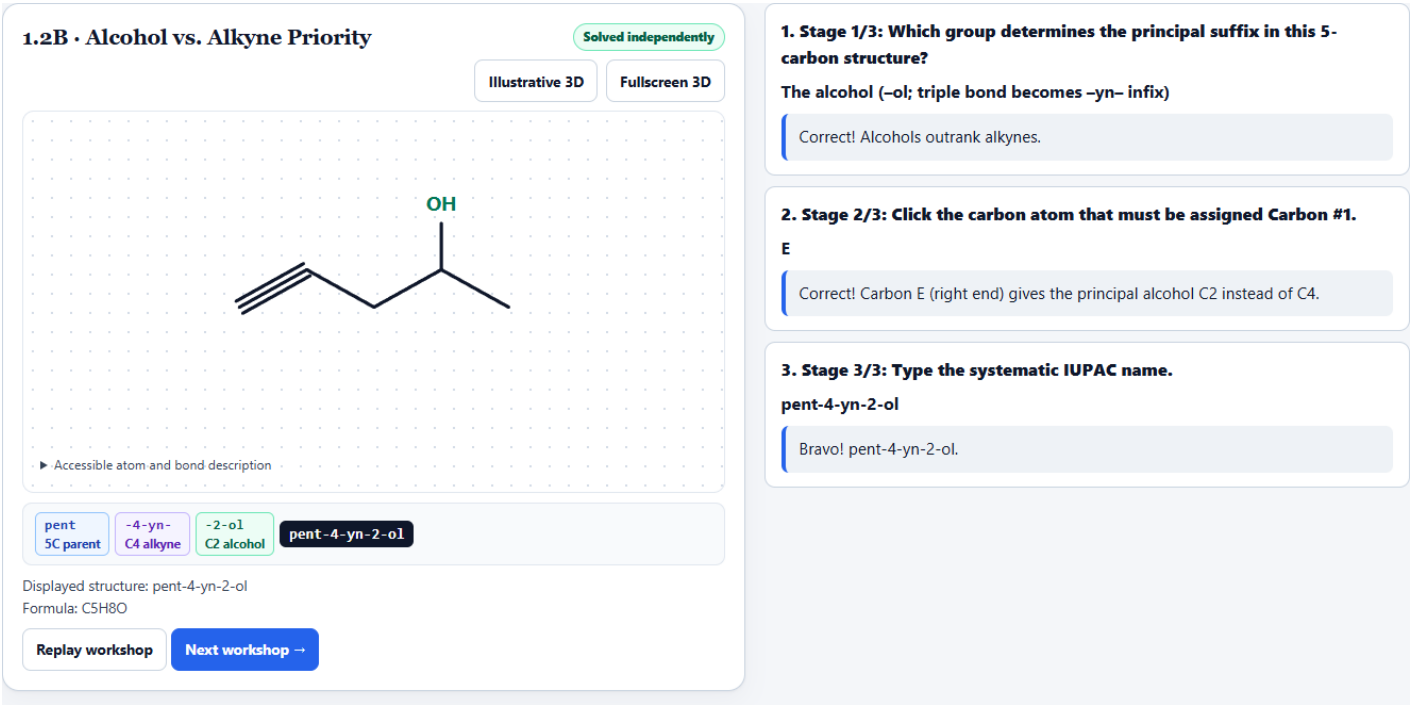
Explanation: Bravo! pent-4-yn-2-ol.

Screenshot: gp2b-stage-3-feedback



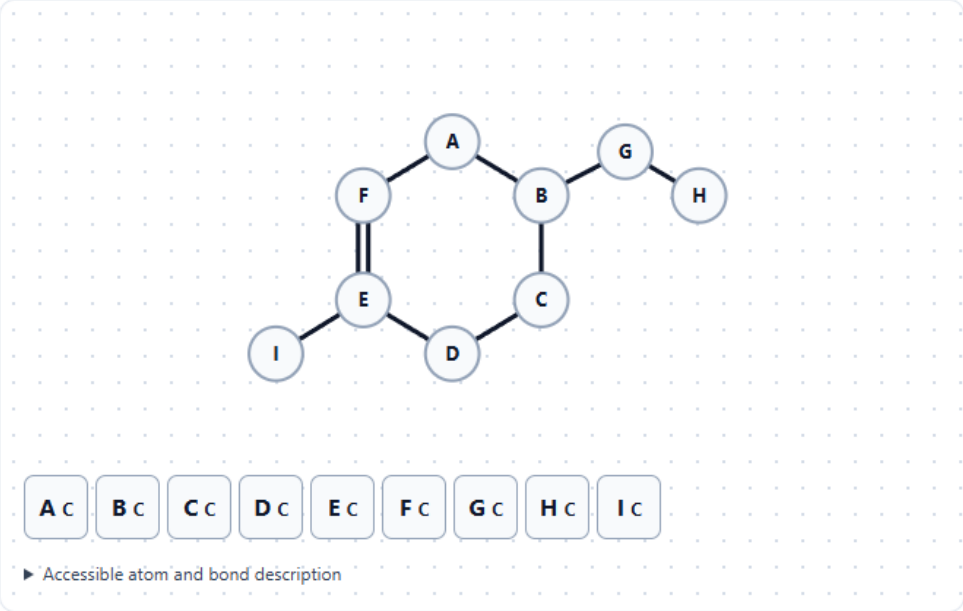
Completion summary: pent-4-yn-2-ol

Screenshot: gp2b-complete



gp3a | 1.3A · Cycloalkene Numbering Rules

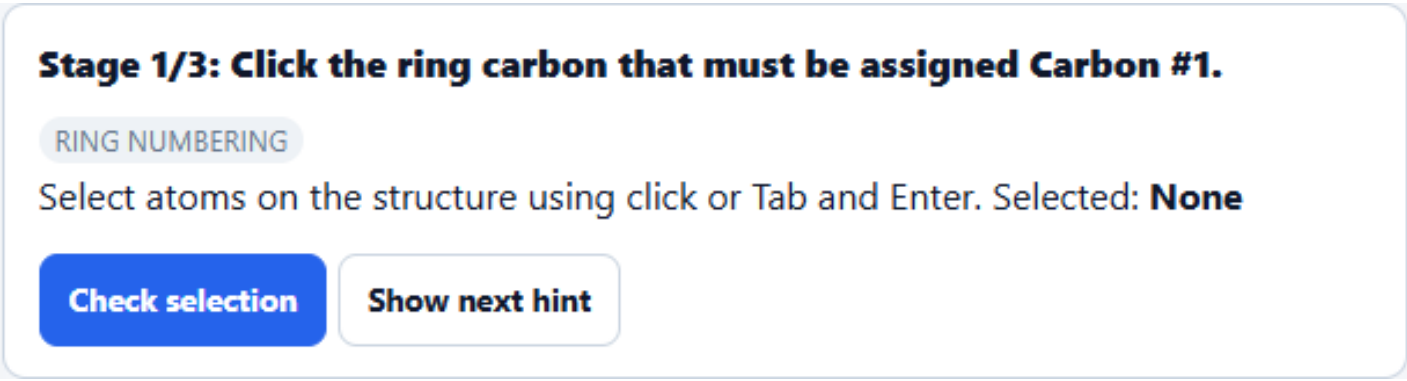
Screenshot: gp3a-stage-1-molecule



gp3a:stage-1 | Stage 1/3: Click the ring carbon that must be assigned Carbon #1.

Answer type: atoms

Screenshot: gp3a-stage-1-question



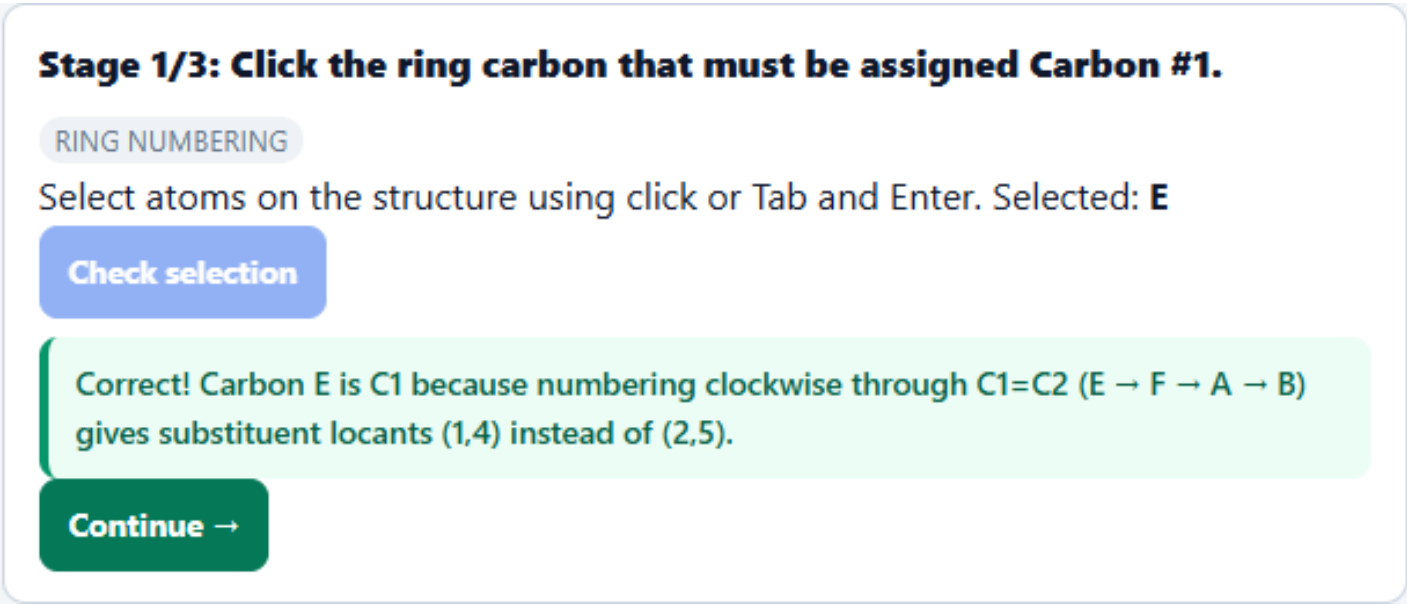
{"authorAnswer": [4], "tolerance": null, "highlightsAfterCorrectAnswer": {"path": [0, 1, 2, 3, 4, 5], "num": [4, 5, 0, 1, 2, 3]}}

Hint 1: In this cycloalkene, the two double-bond carbons receive locants 1 and 2.

Hint 2: Choose the alkene carbon bearing methyl as C1; this also lowers the substituent locants.

Explanation: Correct! Carbon E is C1 because numbering clockwise through C1=C2 (E → F → A → B) gives substituent locants (1,4) instead of (2,5).

Screenshot: gp3a-stage-1-feedback



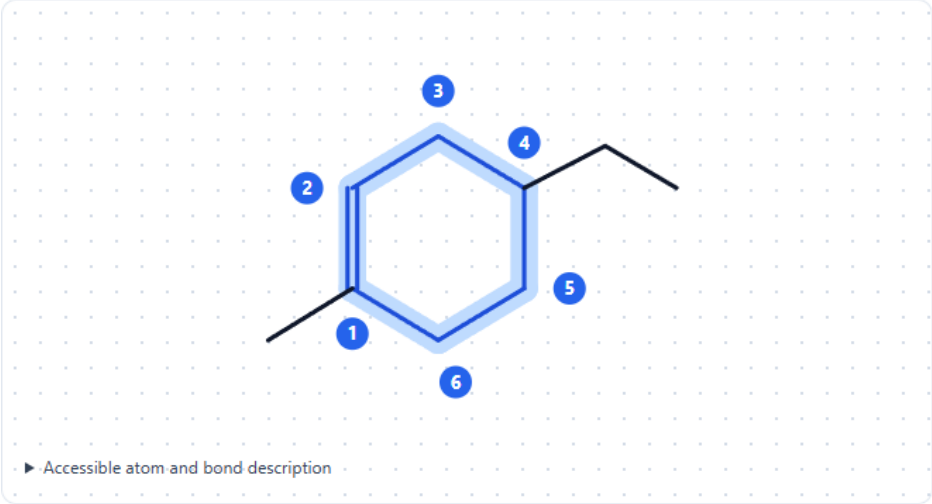
gp3a:stage-2 | Stage 2/3: With C1 at the methylated alkene carbon and C2 at the other alkene carbon, what is the ethyl locant?

Answer type: choice

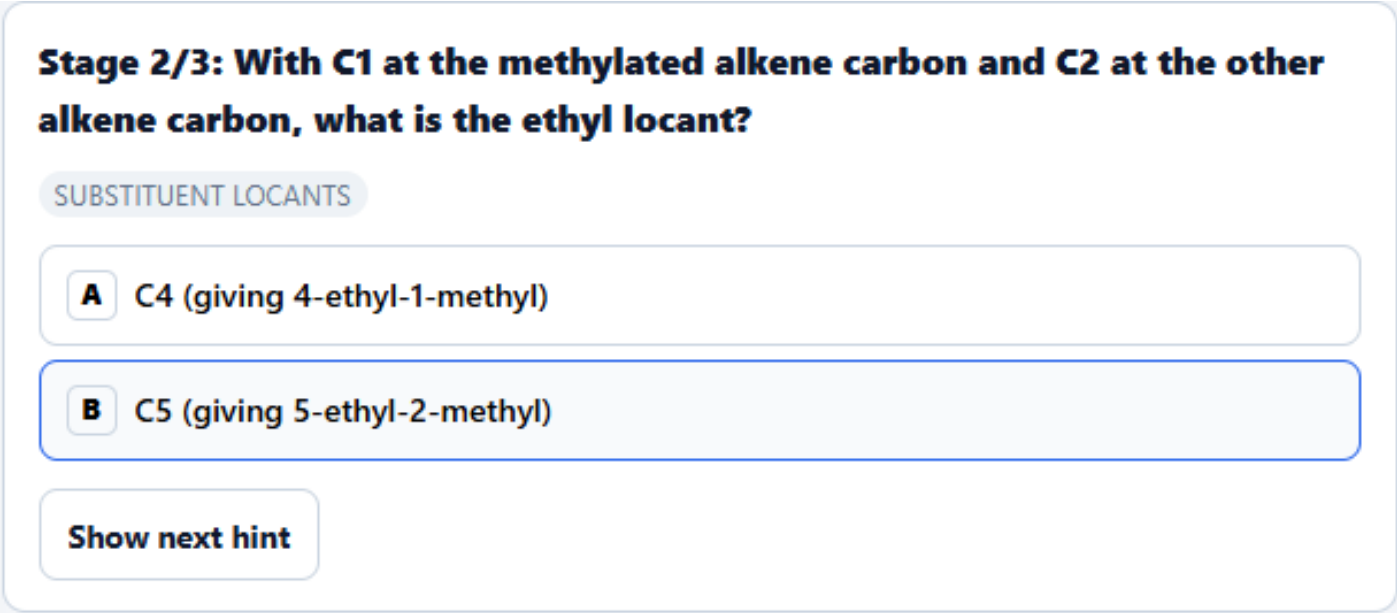
Option 0 (A): C4 (giving 4-ethyl-1-methyl)

Option 1 (B): C5 (giving 5-ethyl-2-methyl)

Screenshot: gp3a-stage-2-molecule



Screenshot: gp3a-stage-2-question



{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {"path": [0, 1, 2, 3, 4, 5], "num": [4, 5, 0, 1, 2, 3], "sub": [6, 7, 8]}}

Hint 1: Follow the fixed C1-to-C2 direction around the ring to the ethyl attachment.

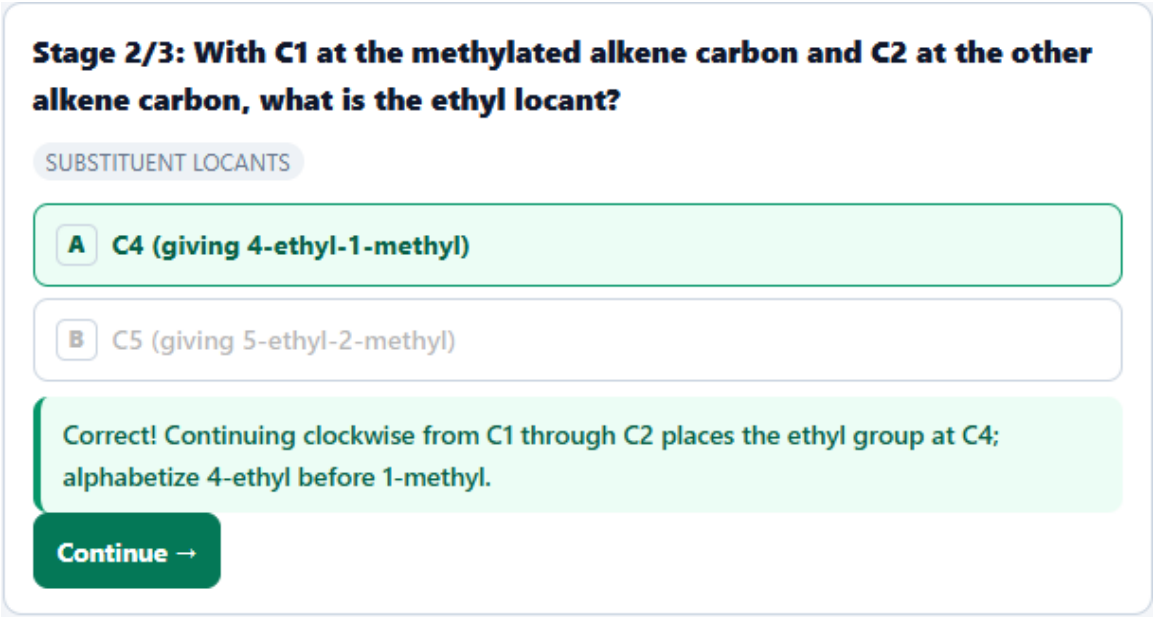
Hint 2: Count ring carbons only for the locant; the carbons inside the ethyl branch do not extend the ring numbering.

Explanation: Correct! Continuing clockwise from C1 through C2 places the ethyl group at C4; alphabetize 4-ethyl before 1-methyl.

Option 0 feedback: Correct! Continuing clockwise from C1 through C2 places the ethyl group at C4; alphabetize 4-ethyl before 1-methyl.

Option 1 feedback: (1,4) beats (2,5) at the first point of difference!

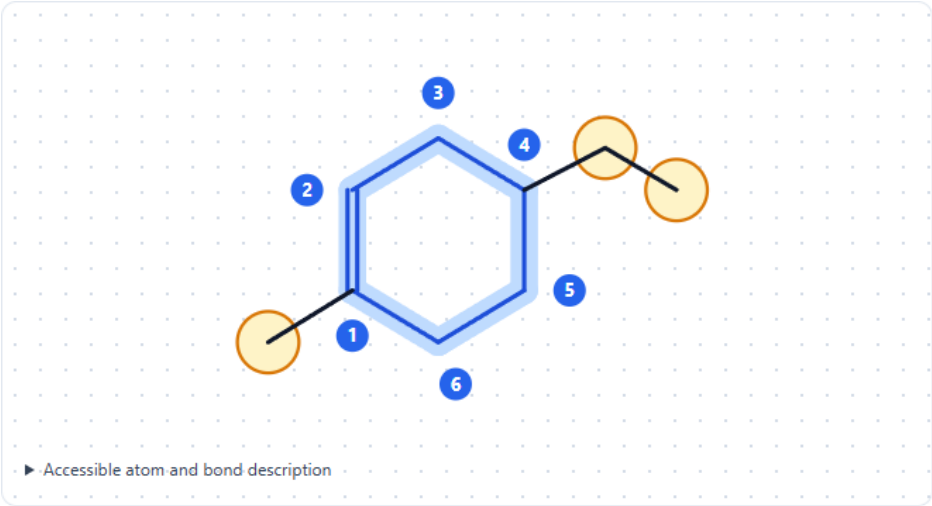
Screenshot: gp3a-stage-2-feedback



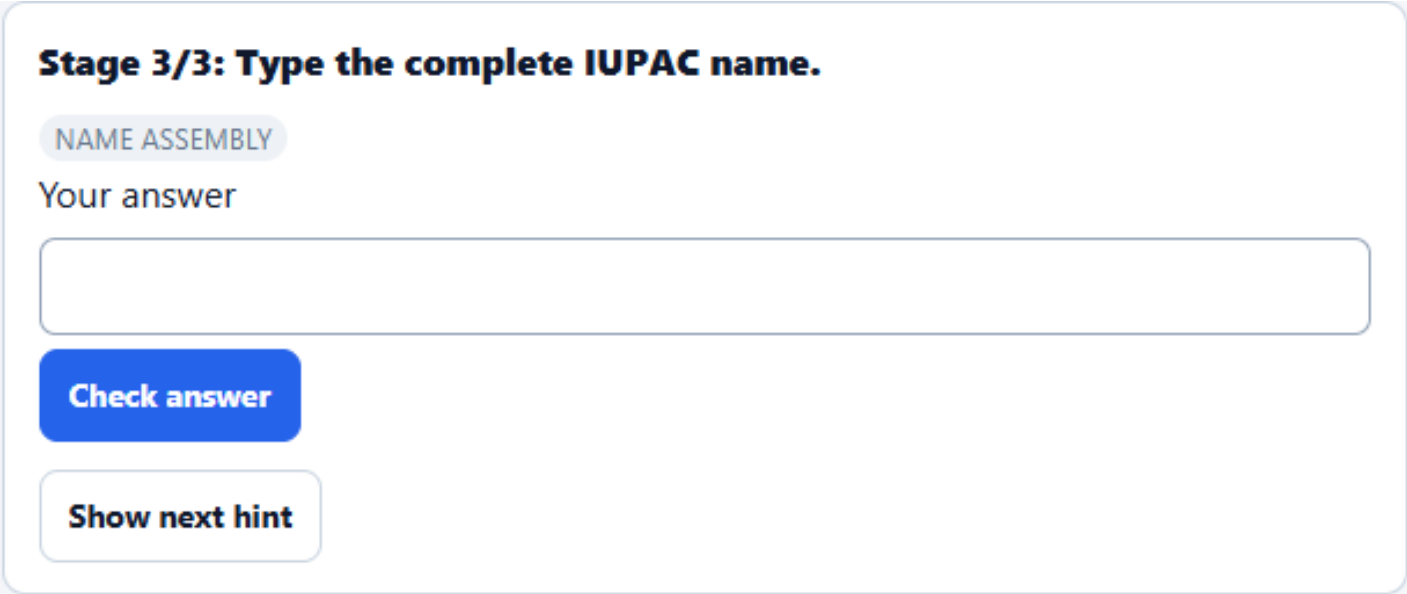
gp3a:stage-3 | Stage 3/3: Type the complete IUPAC name.

Answer type: text

Screenshot: gp3a-stage-3-molecule



Screenshot: gp3a-stage-3-question



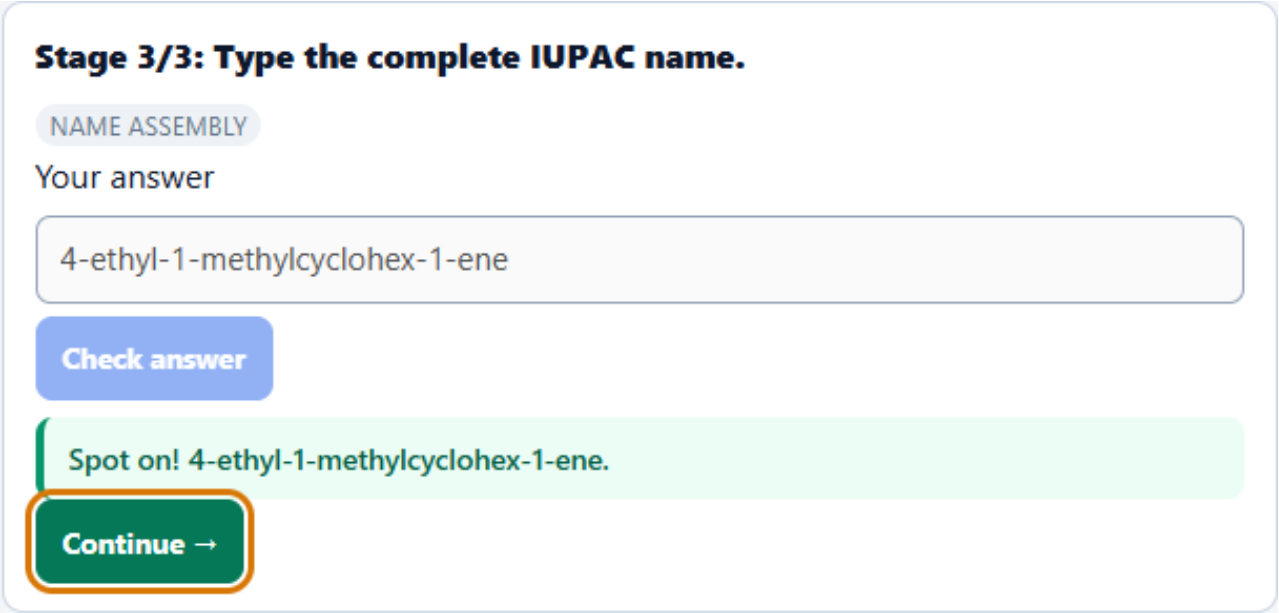
{"authorAnswer": ["4-ethyl-1-methylcyclohex-1-ene", "4-ethyl-1-methylcyclohexene"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Hint 1 (Rule): Alphabetize the substituents (ethyl before methyl), followed by the cycloalkene parent name.

Hint 2: Hint 2 (Structure): Ethyl is at C4 and methyl is at C1 on cyclohex-1-ene (or cyclohexene).

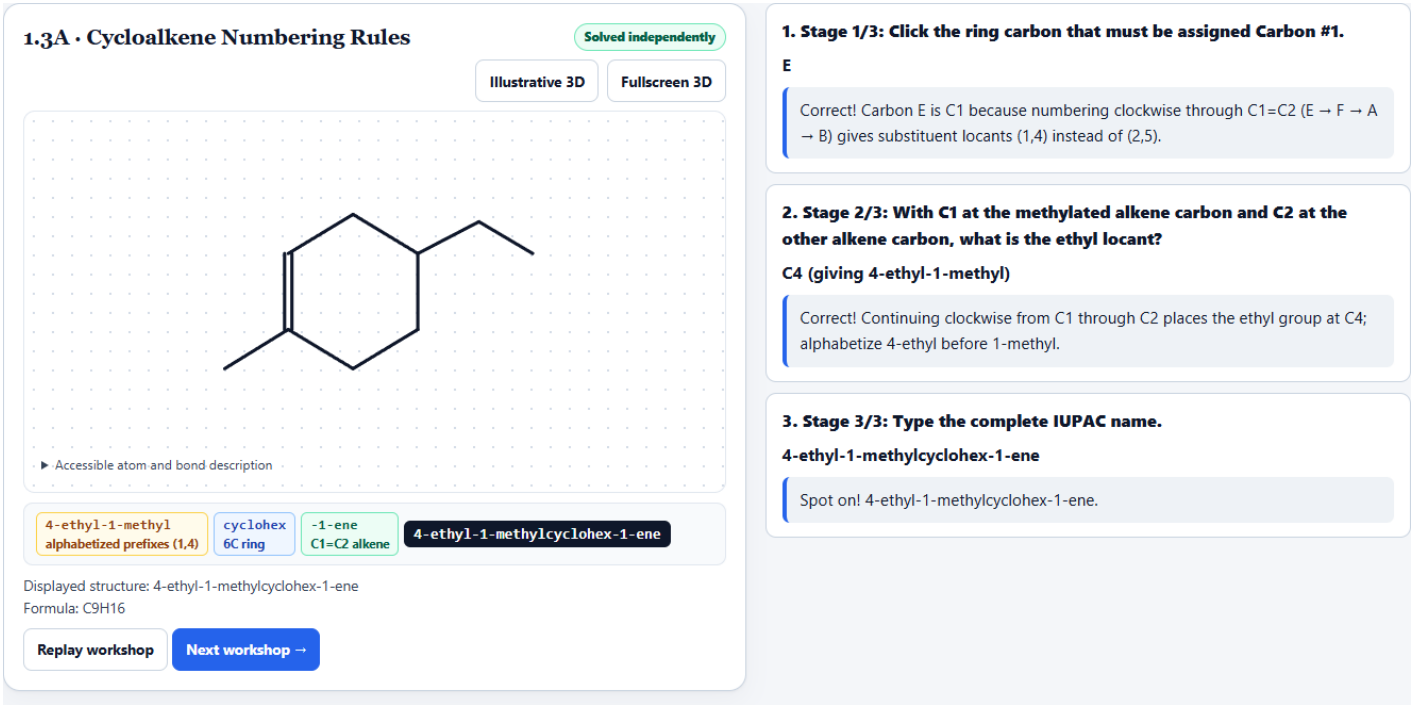
Explanation: Spot on! 4-ethyl-1-methylcyclohex-1-ene.

Screenshot: gp3a-stage-3-feedback



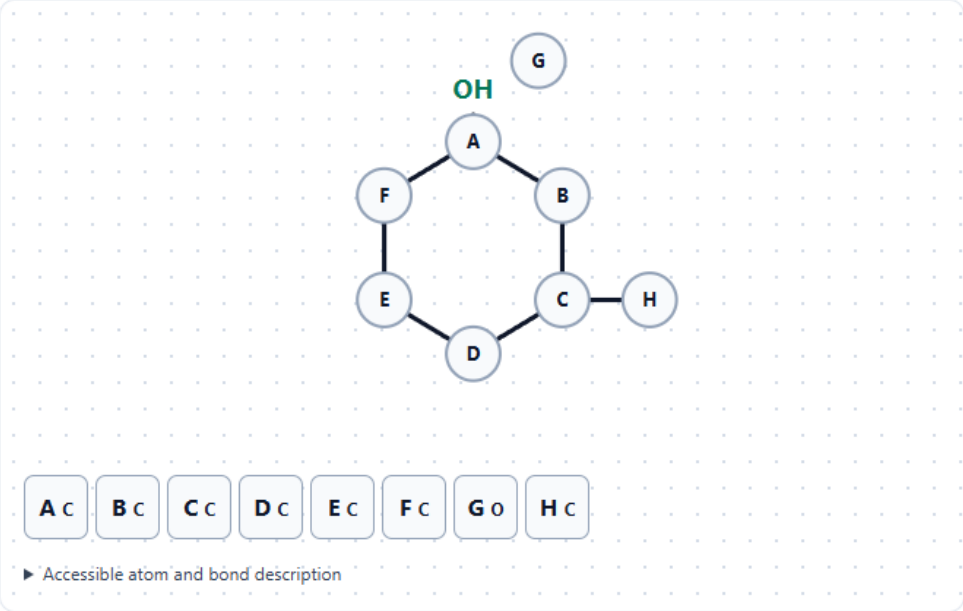
Completion summary: 4-ethyl-1-methylcyclohex-1-ene

Screenshot: gp3a-complete



gp3b | 1.3B · Cyclic Alcohol Priority & Numbering

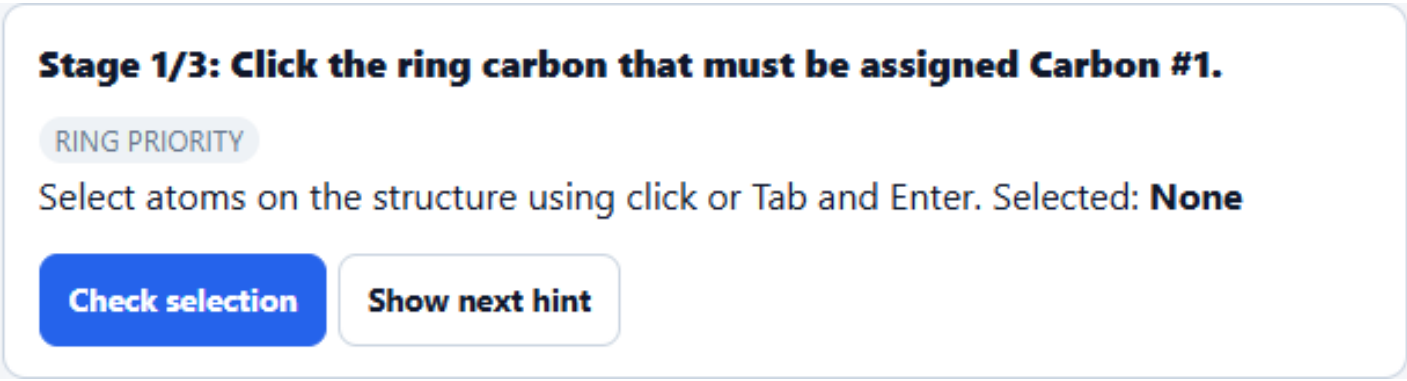
Screenshot: gp3b-stage-1-molecule



gp3b:stage-1 | Stage 1/3: Click the ring carbon that must be assigned Carbon #1.

Answer type: atoms

Screenshot: gp3b-stage-1-question



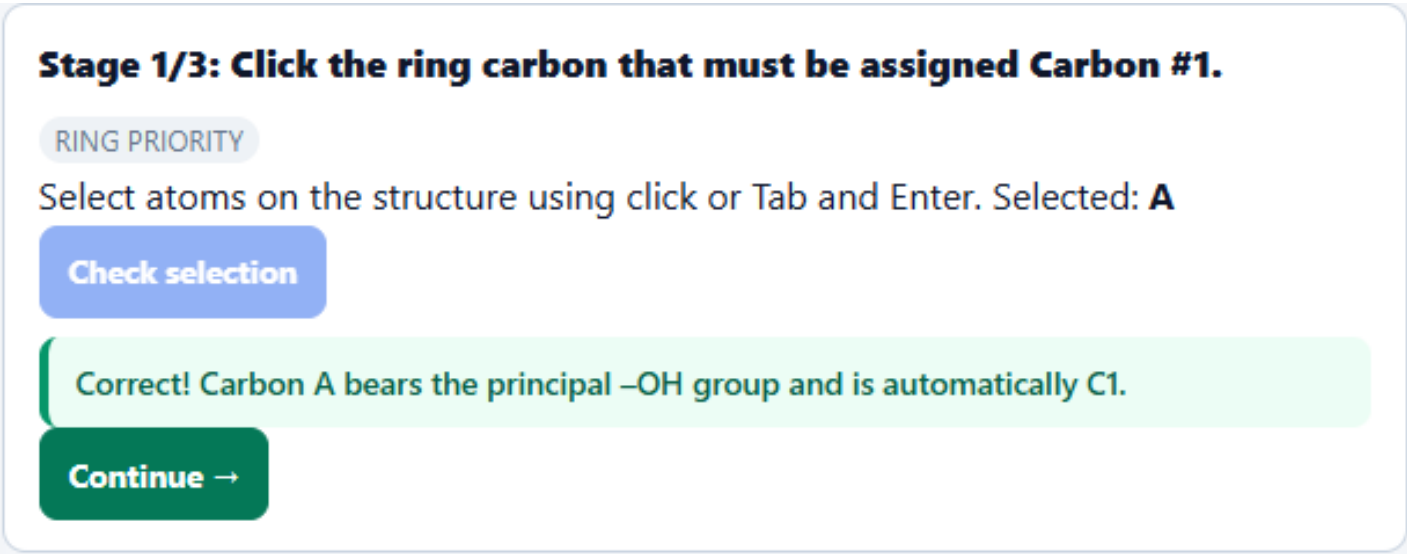
{"authorAnswer": [0], "tolerance": null, "highlightsAfterCorrectAnswer": {"path": [0, 1, 2, 3, 4, 5], "prin": [6]}}

Hint 1: The hydroxyl group supplies the ring alcohol suffix.

Hint 2: The ring carbon directly bonded to oxygen receives locant 1.

Explanation: Correct! Carbon A bears the principal –OH group and is automatically C1.

Screenshot: gp3b-stage-1-feedback



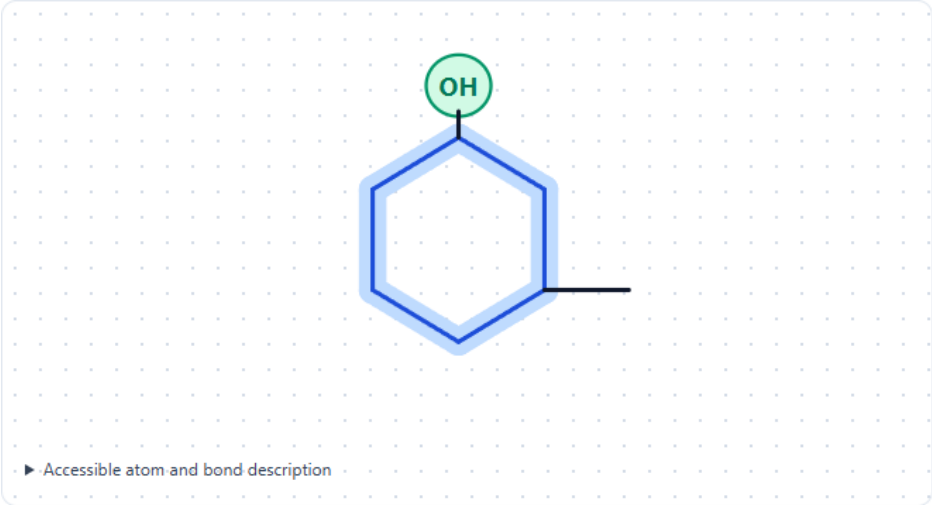
gp3b:stage-2 | Stage 2/3: From C1 (at –OH), which direction gives the methyl group its lowest locant?

Answer type: choice

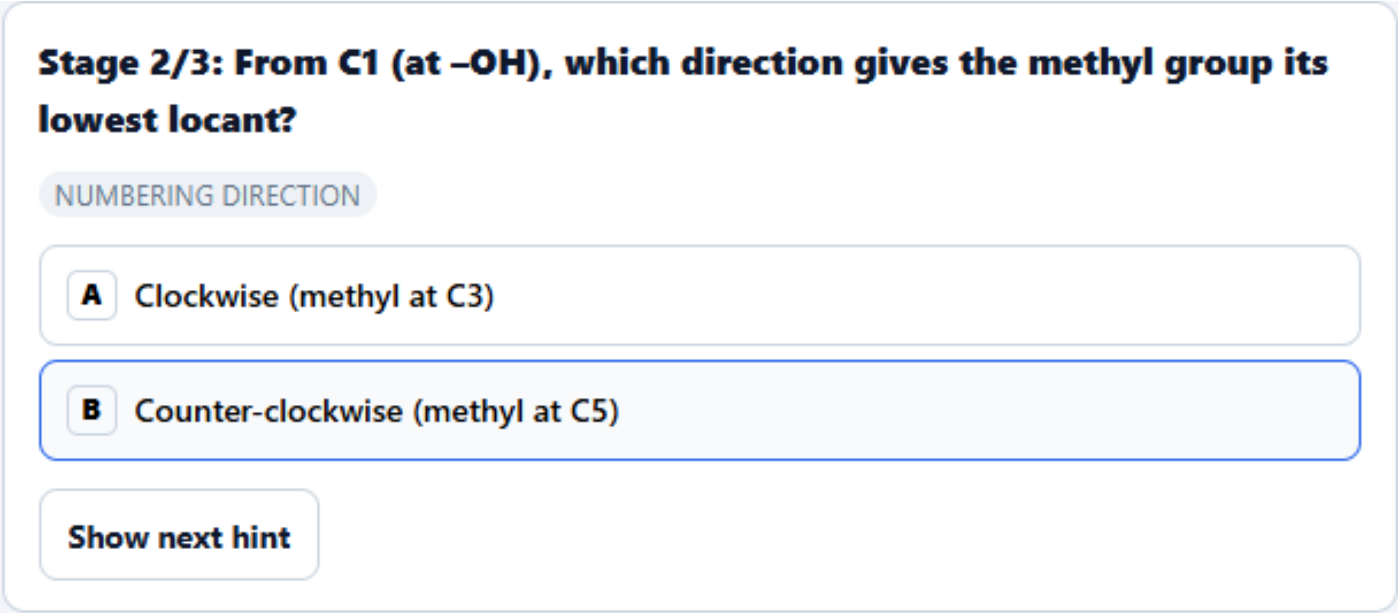
Option 0 (A): Clockwise (methyl at C3)

Option 1 (B): Counter-clockwise (methyl at C5)

Screenshot: gp3b-stage-2-molecule



Screenshot: gp3b-stage-2-question



{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {"path": [0, 1, 2, 3, 4, 5], "num": [0, 1, 2, 3, 4, 5], "prin": [6], "sub": [7]}}

Hint 1: Keep the hydroxyl-bearing carbon at C1 and compare both routes around the ring.

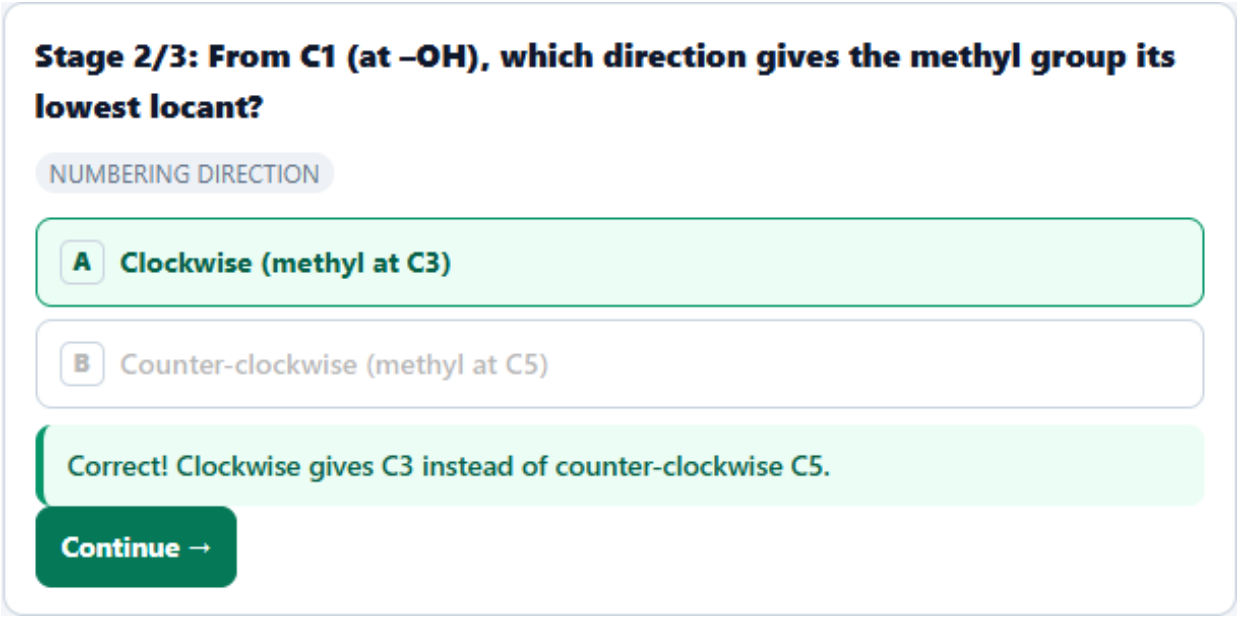
Hint 2: Choose methyl at C3 rather than C5; the ring orientation on the page does not override the lower-locant rule.

Explanation: Correct! Clockwise gives C3 instead of counter-clockwise C5.

Option 0 feedback: Correct! Clockwise gives C3 instead of counter-clockwise C5.

Option 1 feedback: Counter-clockwise puts methyl at C5 instead of C3!

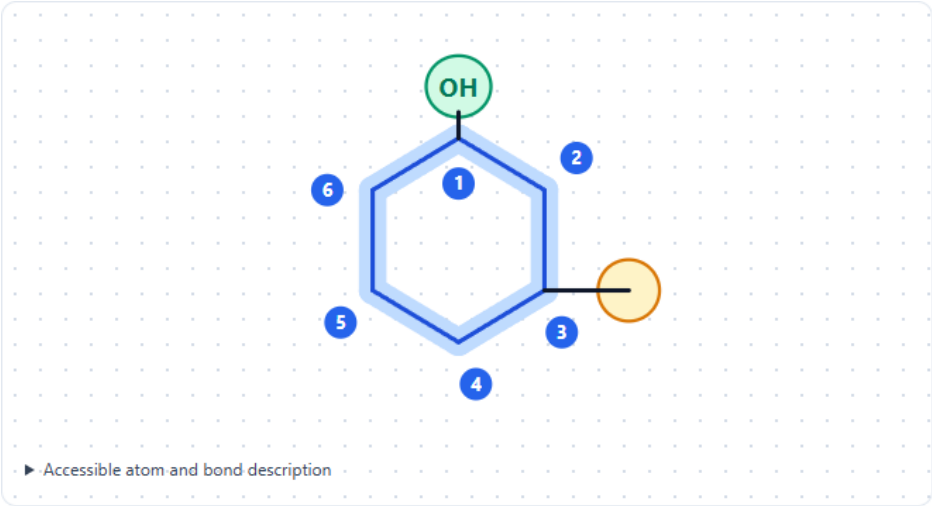
Screenshot: gp3b-stage-2-feedback



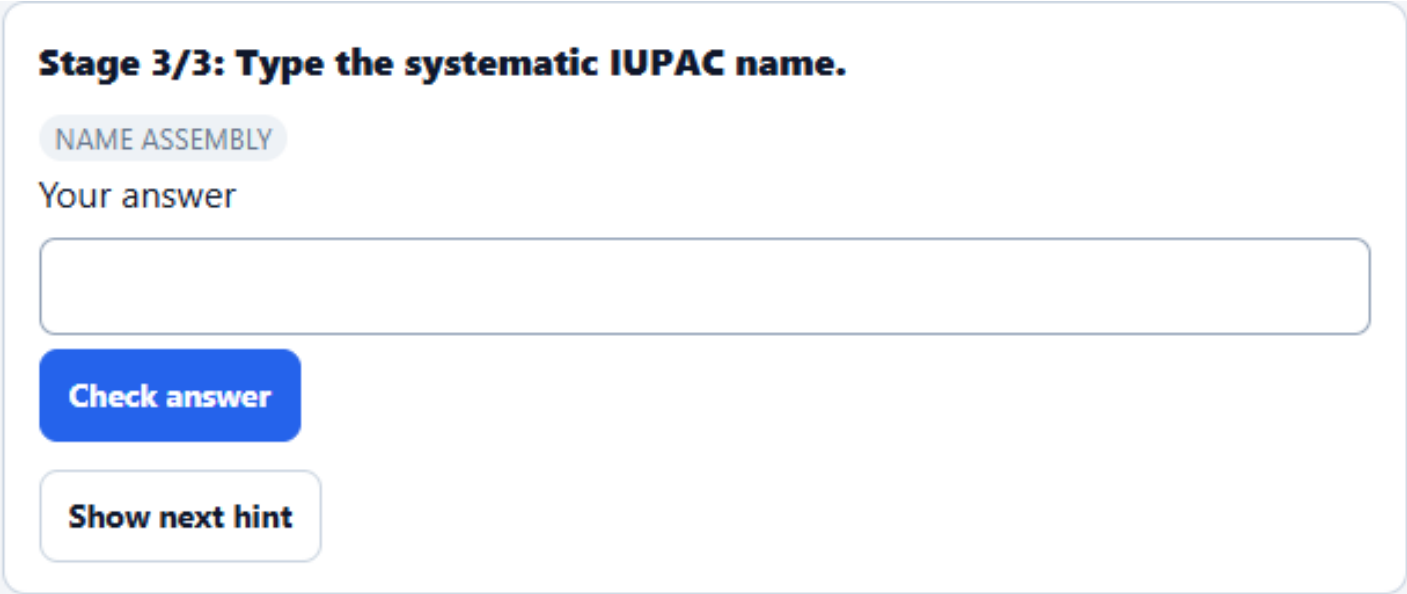
gp3b:stage-3 | Stage 3/3: Type the systematic IUPAC name.

Answer type: text

Screenshot: gp3b-stage-3-molecule



Screenshot: gp3b-stage-3-question



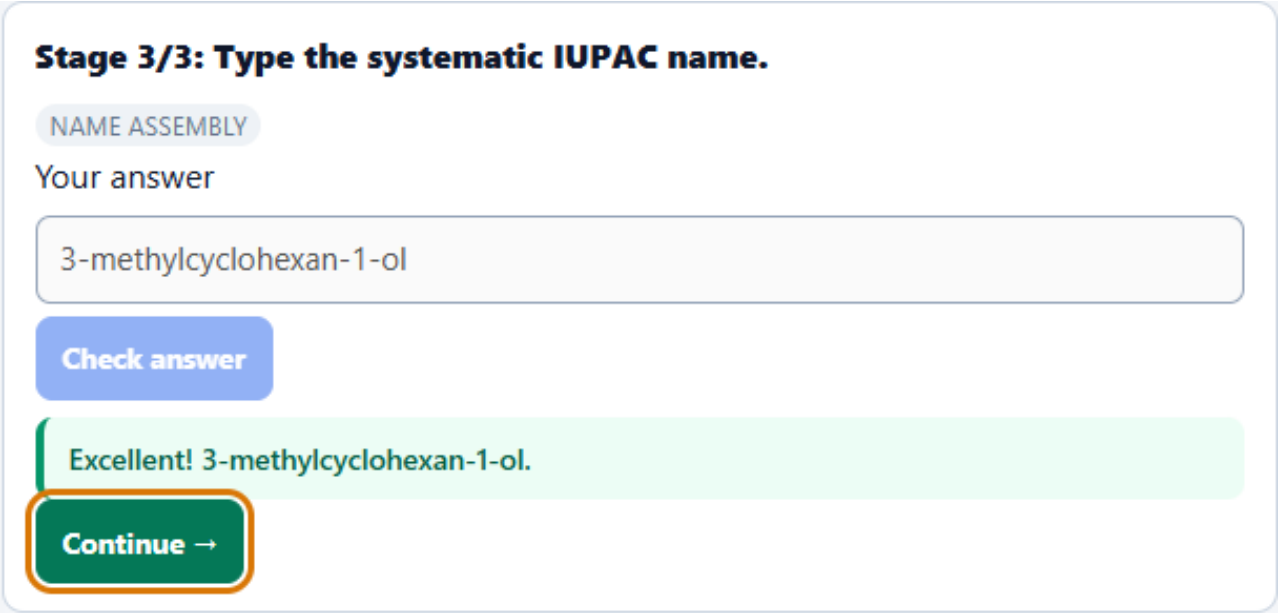
{"authorAnswer": ["3-methylcyclohexan-1-ol", "3-methylcyclohexanol"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Hint 1 (Rule): Prefix the methyl substituent and its locant before the cyclic alcohol parent name.

Hint 2: Hint 2 (Structure): Methyl at C3 on a cyclohexan-1-ol (or cyclohexanol) ring.

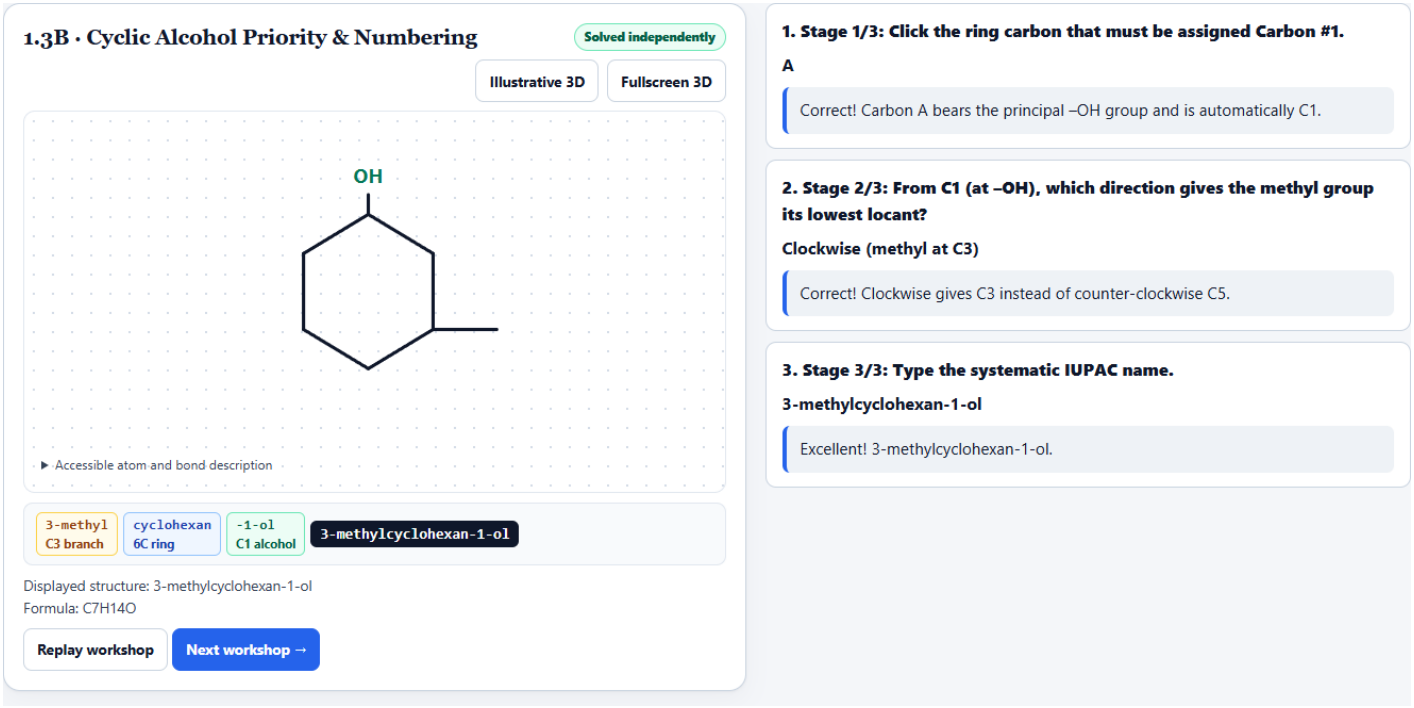
Explanation: Excellent! 3-methylcyclohexan-1-ol.

Screenshot: gp3b-stage-3-feedback



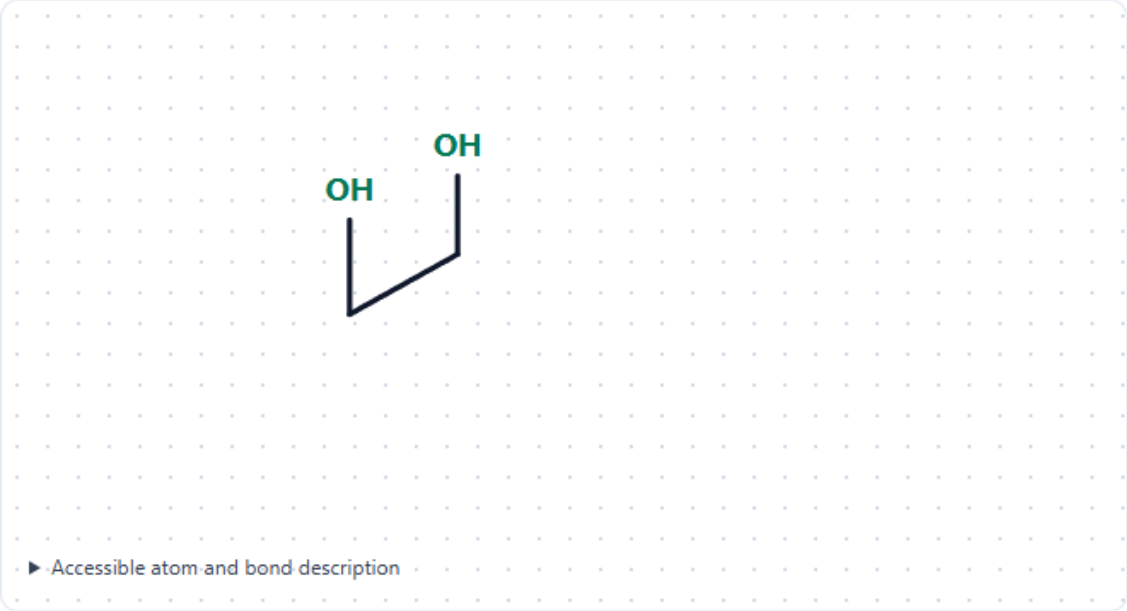
Completion summary: 3-methylcyclohexan-1-ol

Screenshot: gp3b-complete



gp4a | 1.4A · Vicinal Diol (Glycol) Nomenclature

Screenshot: gp4a-stage-1-molecule



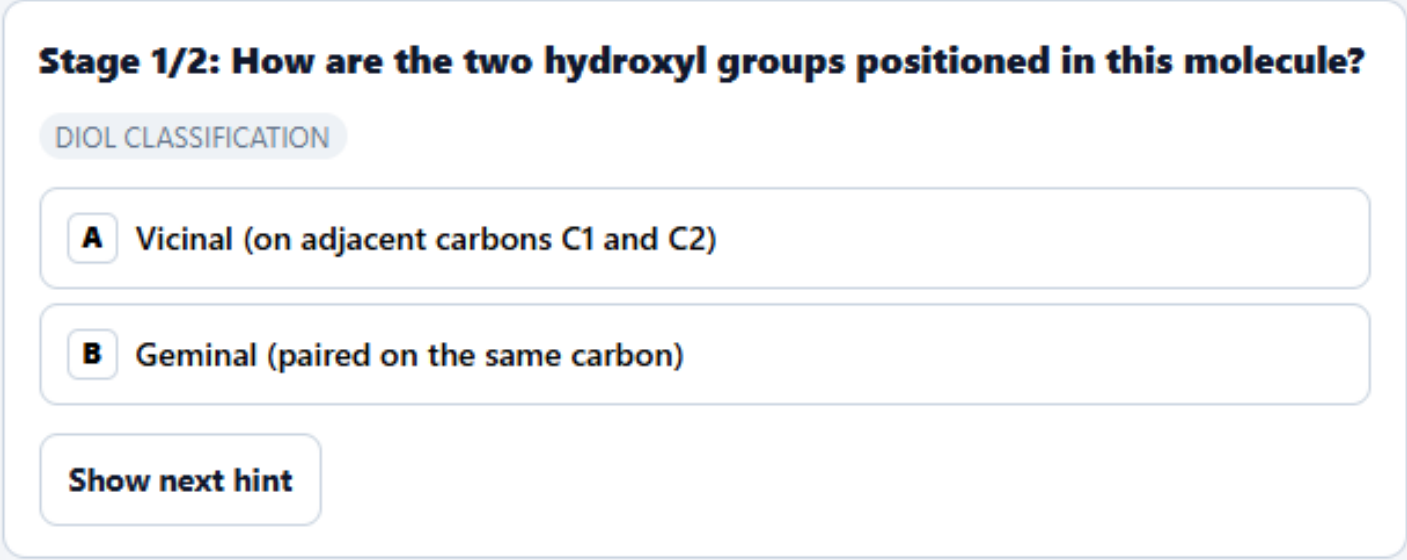
gp4a:stage-1 | Stage 1/2: How are the two hydroxyl groups positioned in this molecule?

Answer type: choice

Option 0 (A): Vicinal (on adjacent carbons C1 and C2)

Option 1 (B): Geminal (paired on the same carbon)

Screenshot: gp4a-stage-1-question



{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {"path": [0, 1], "num": [0, 1], "prin": [2, 3]}}

Hint 1: Find the carbon attached to each hydroxyl oxygen.

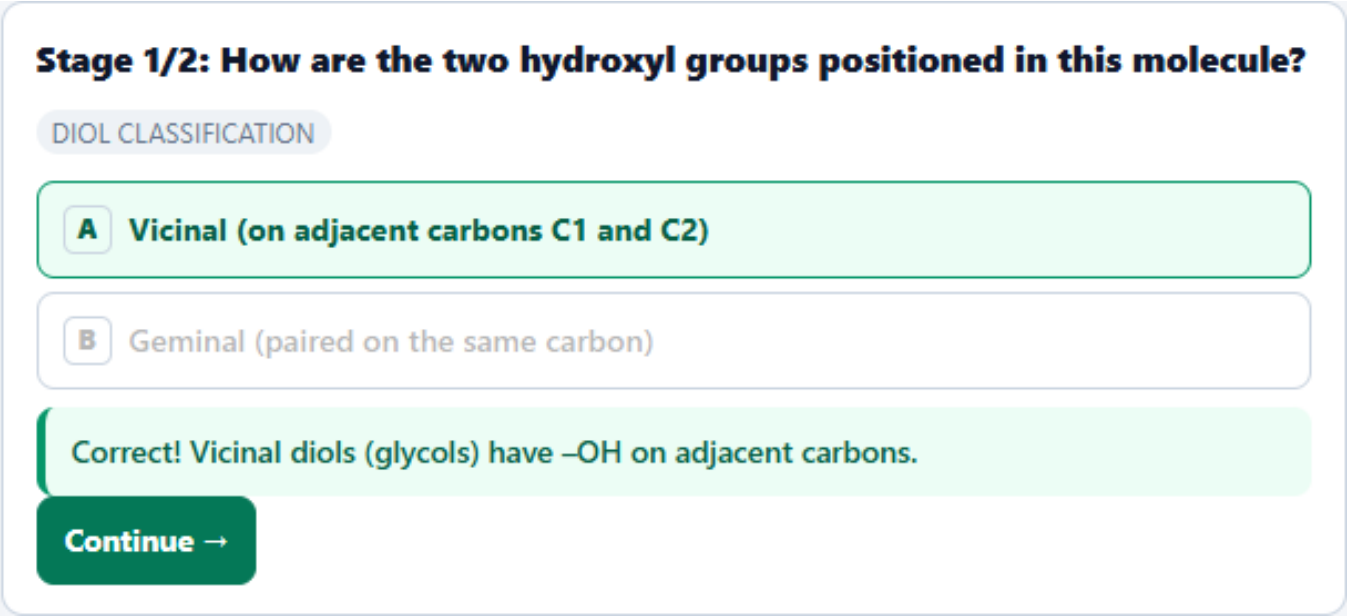
Hint 2: Hydroxyls on neighboring carbons are vicinal; hydroxyls on the same carbon are geminal.

Explanation: Correct! Vicinal diols (glycols) have –OH on adjacent carbons.

Option 0 feedback: Correct! Vicinal diols (glycols) have –OH on adjacent carbons.

Option 1 feedback: Geminal diols have both –OH groups on the same carbon.

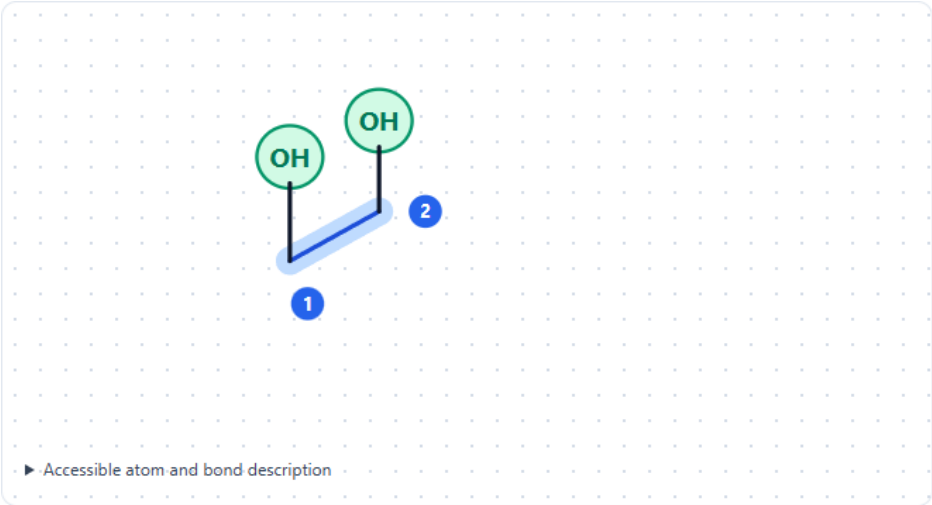
Screenshot: gp4a-stage-1-feedback



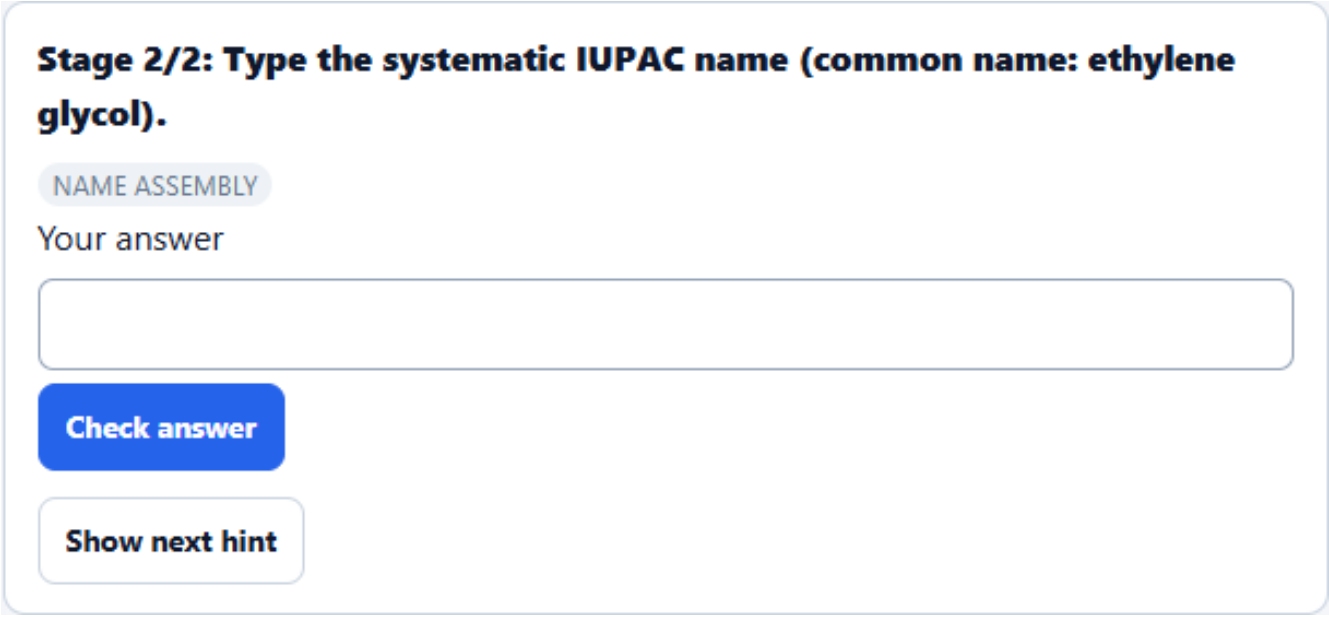
gp4a:stage-2 | Stage 2/2: Type the systematic IUPAC name (common name: ethylene glycol).

Answer type: text

Screenshot: gp4a-stage-2-molecule



Screenshot: gp4a-stage-2-question



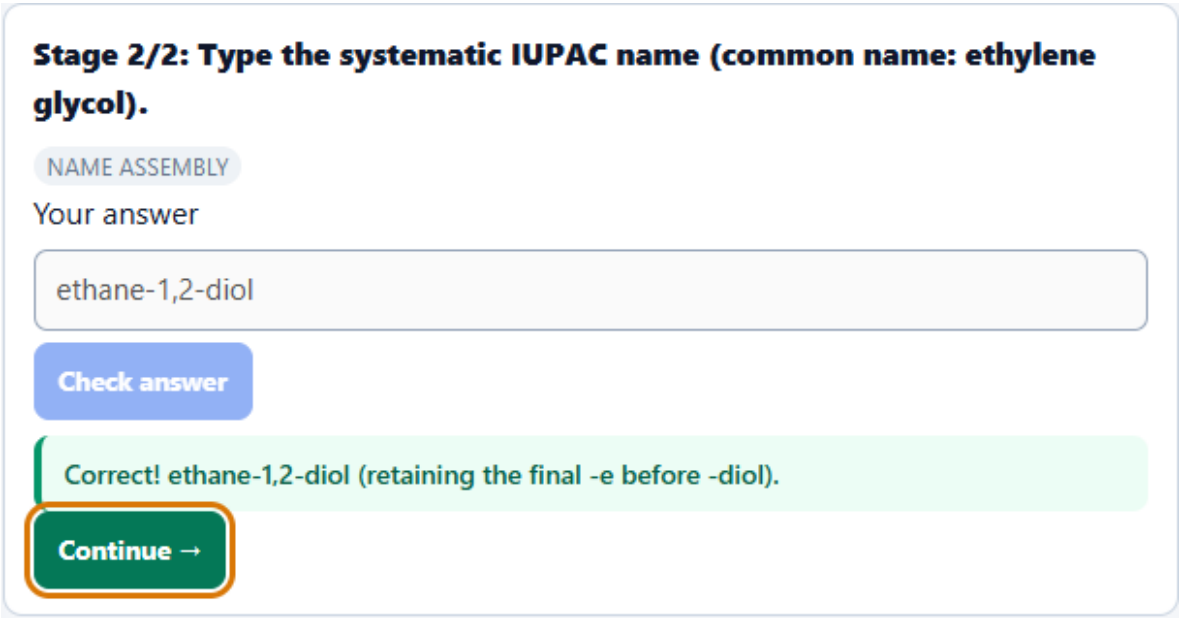
{"authorAnswer": ["ethane-1,2-diol", "1,2-ethanediol"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Hint 1 (Rule): Preserve the full alkane name (including –e) before adding the locants and –diol suffix.

Hint 2: Hint 2 (Structure): 2-carbon alkane (ethane) with hydroxyls at C1 and C2.

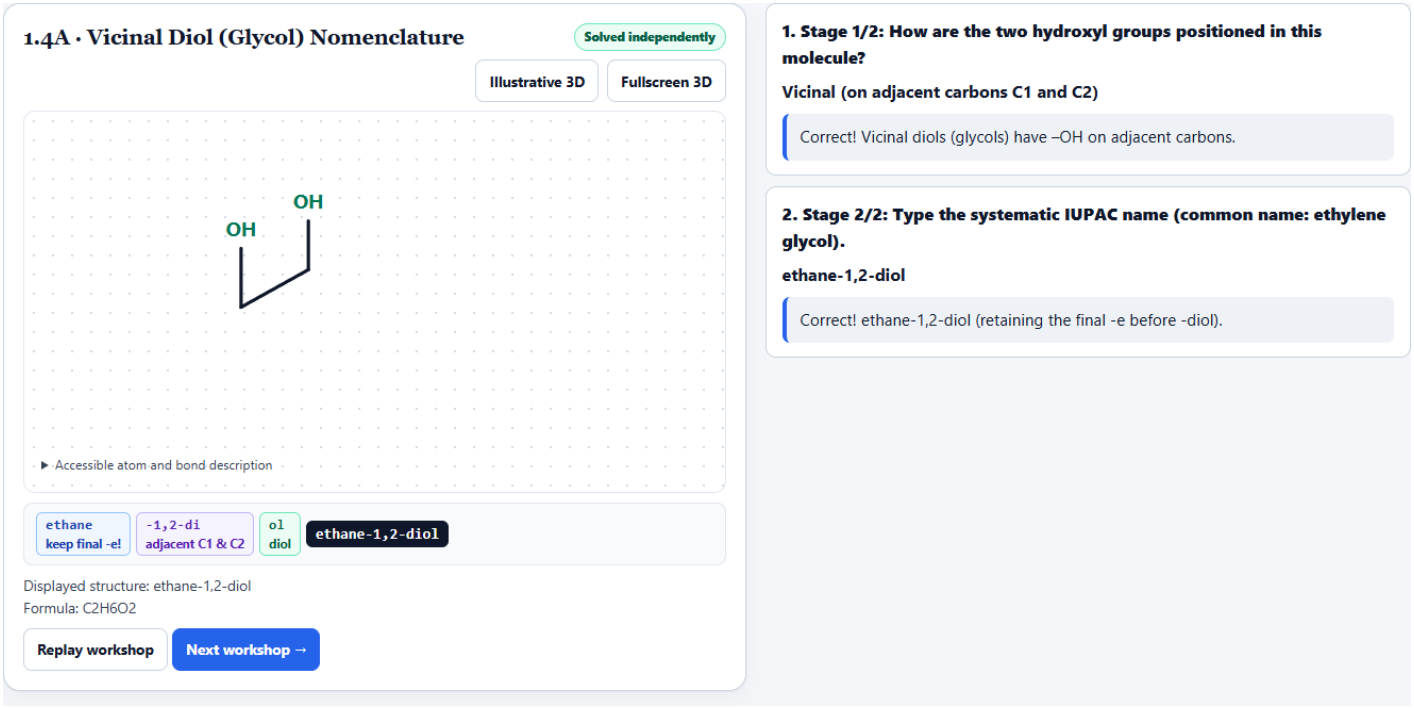
Explanation: Correct! ethane-1,2-diol (retaining the final -e before -diol).

Screenshot: gp4a-stage-2-feedback



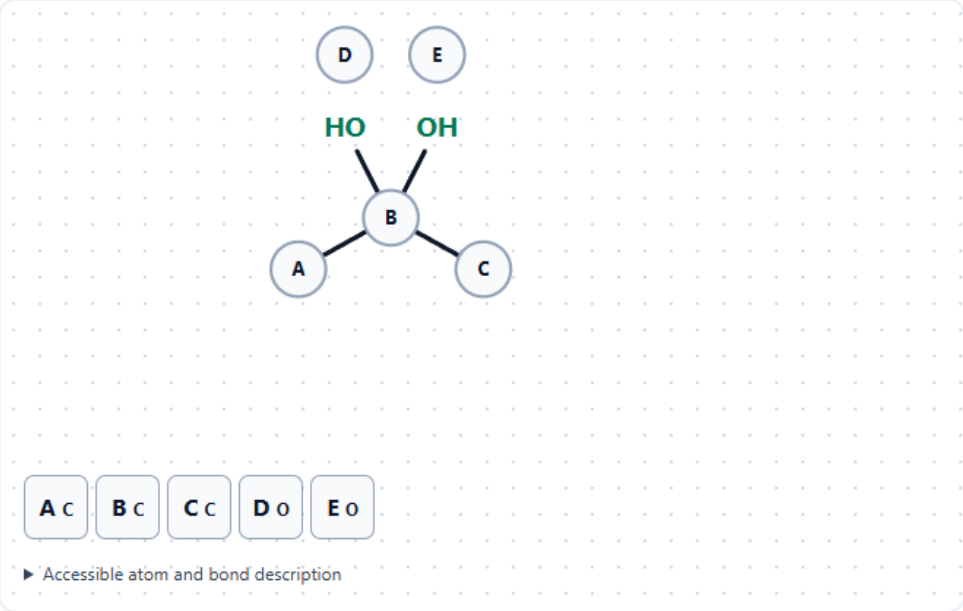
Completion summary: ethane-1,2-diol

Screenshot: gp4a-complete



gp4b | 1.4B · Geminal Diol (Hydrate) Nomenclature

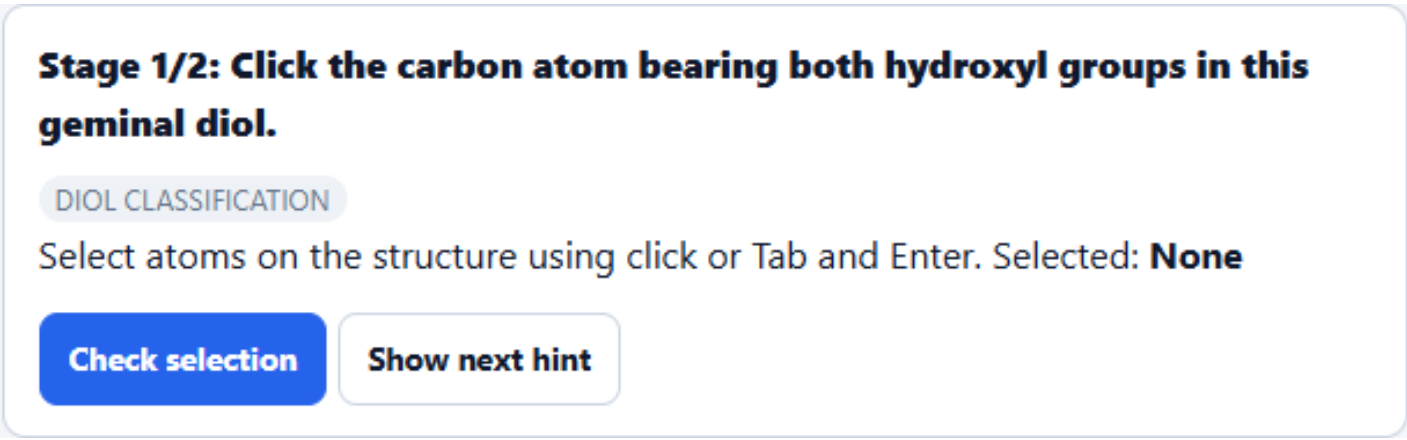
Screenshot: gp4b-stage-1-molecule



gp4b:stage-1 | Stage 1/2: Click the carbon atom bearing both hydroxyl groups in this geminal diol.

Answer type: atoms

Screenshot: gp4b-stage-1-question



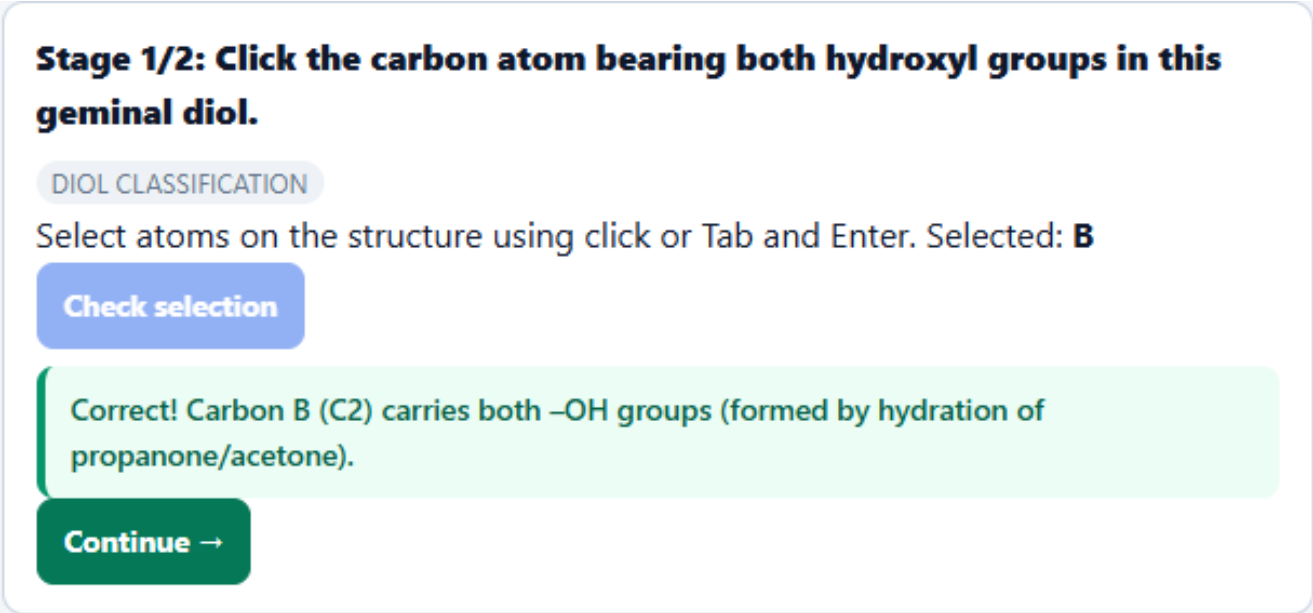
{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"path": [0, 1, 2], "num": [0, 1, 2], "prin": [3, 4]}}

Hint 1: Follow both carbon–oxygen single bonds to their carbon endpoints.

Hint 2: Both hydroxyl oxygens connect to the central carbon of the three-carbon chain.

Explanation: Correct! Carbon B (C2) carries both –OH groups (formed by hydration of propanone/acetone).

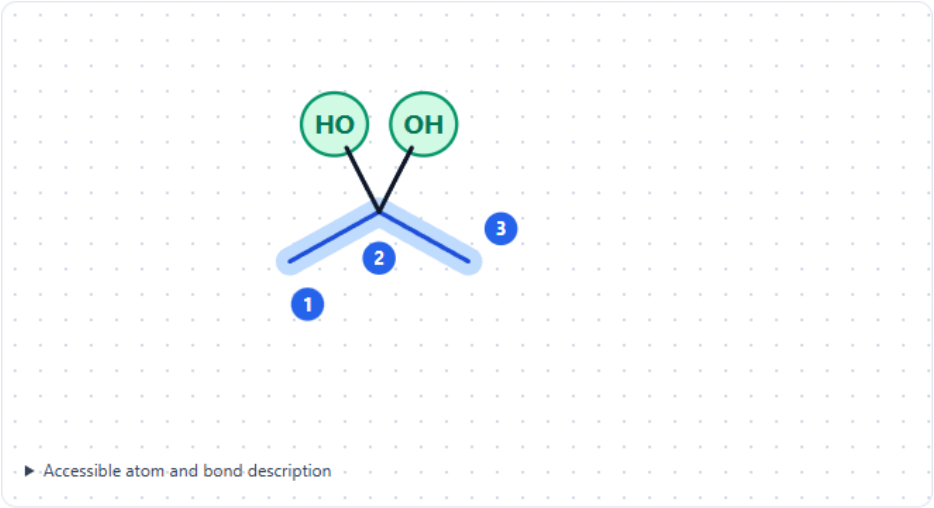
Screenshot: gp4b-stage-1-feedback



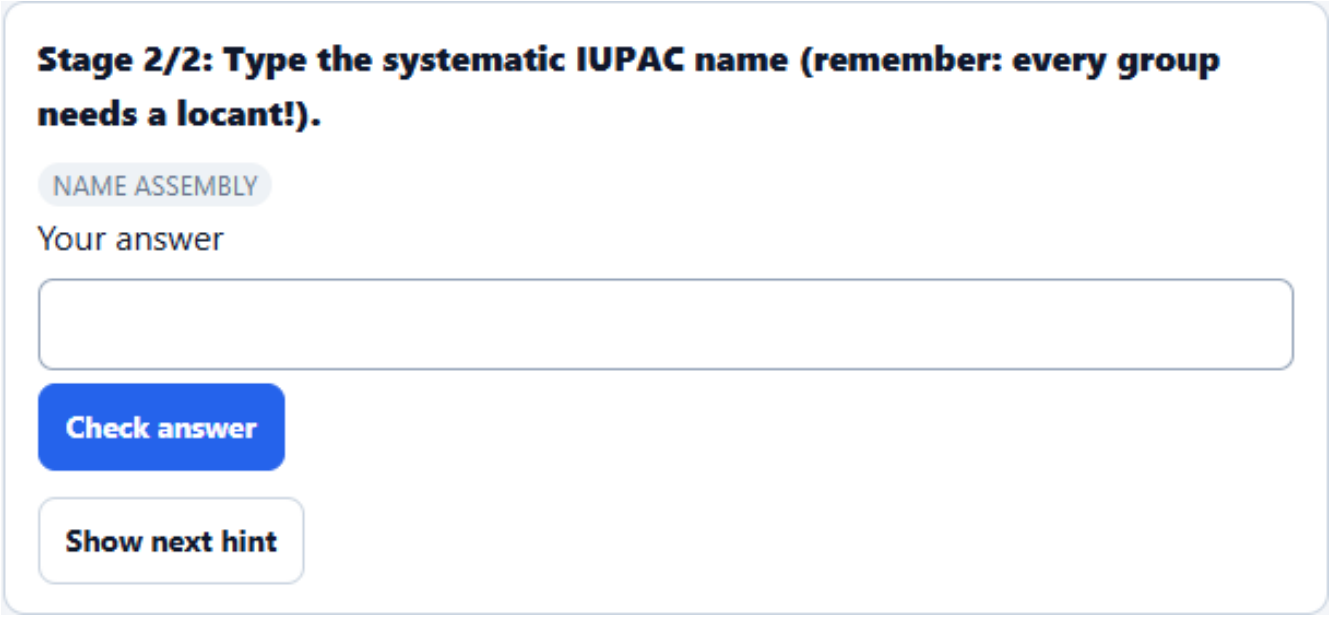
gp4b:stage-2 | Stage 2/2: Type the systematic IUPAC name (remember: every group needs a locant!).

Answer type: text

Screenshot: gp4b-stage-2-molecule



Screenshot: gp4b-stage-2-question



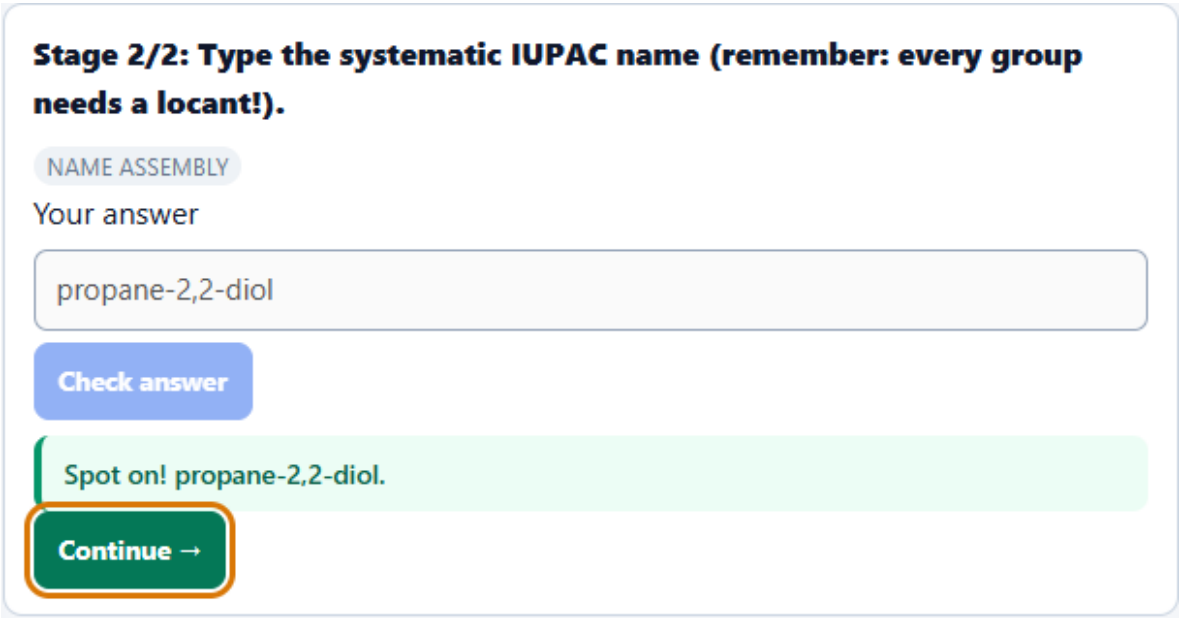
{"authorAnswer": ["propane-2,2-diol", "2,2-propanediol"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Hint 1 (Rule): Even when both –OH groups sit on the same carbon, repeat the locant number before –diol.

Hint 2: Hint 2 (Structure): 3-carbon alkane (propane) with both hydroxyls at C2.

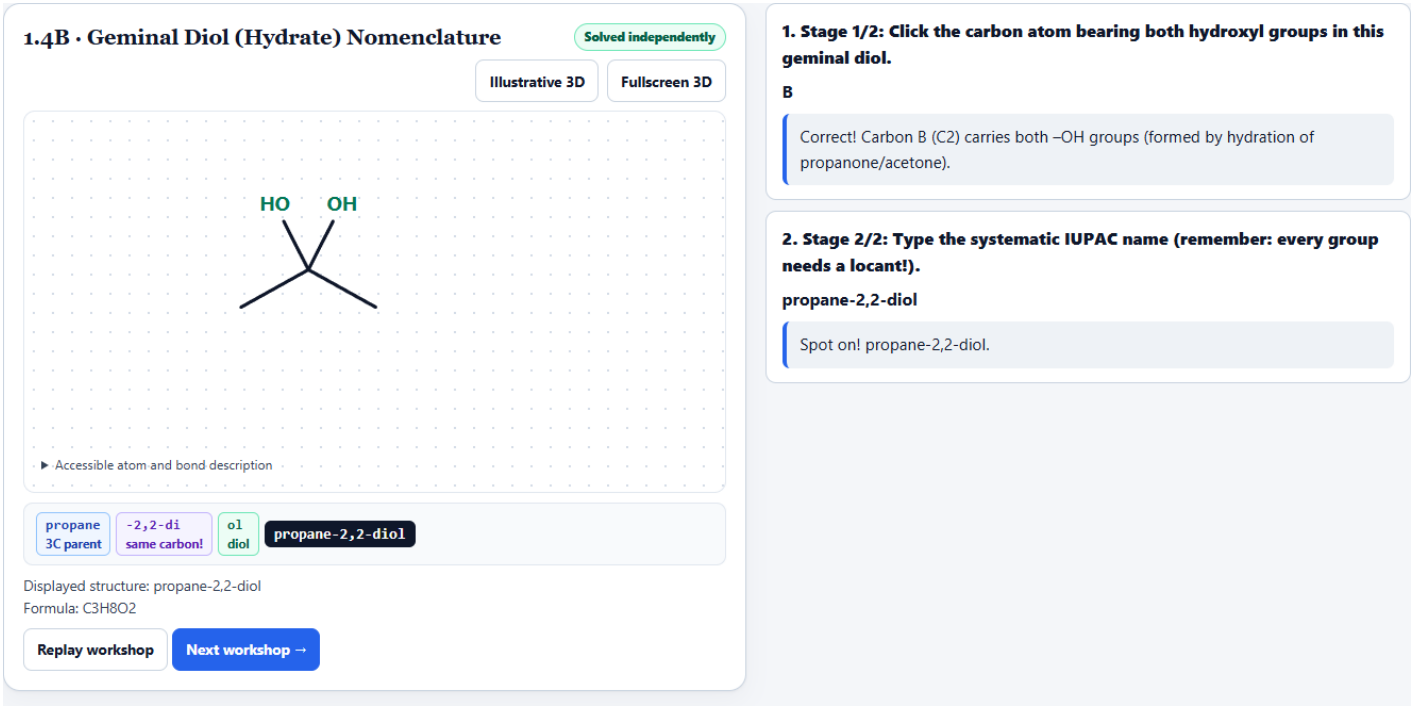
Explanation: Spot on! propane-2,2-diol.

Screenshot: gp4b-stage-2-feedback



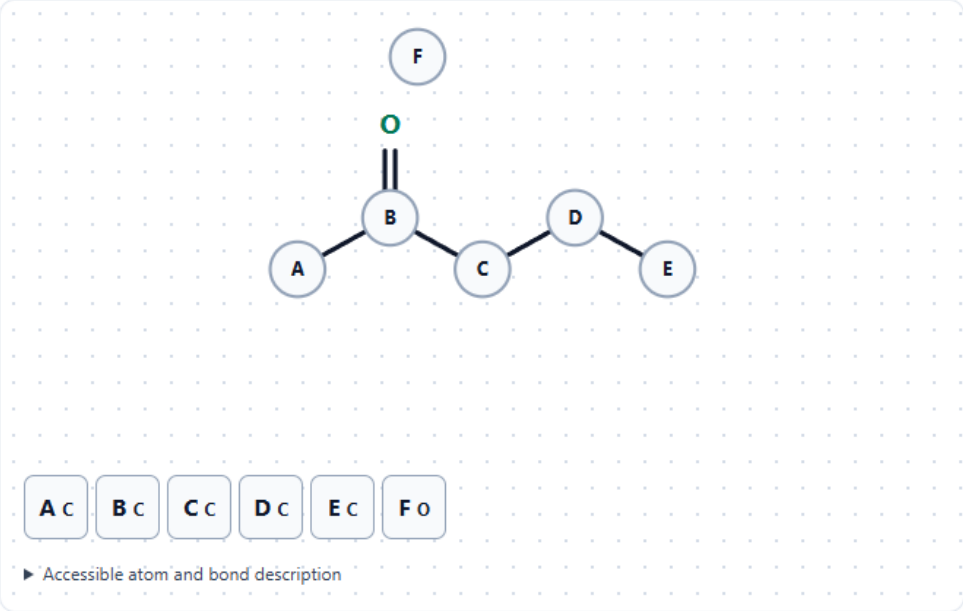
Completion summary: propane-2,2-diol

Screenshot: gp4b-complete



gp5a | 1.5A · Ketone Numbering & Alpha Carbons

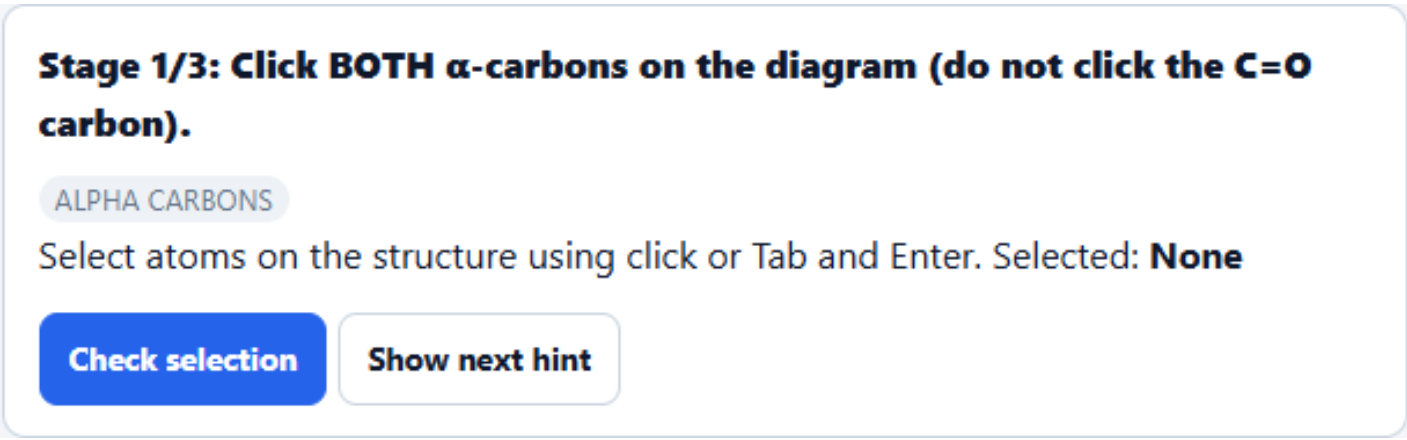
Screenshot: gp5a-stage-1-molecule



gp5a:stage-1 | Stage 1/3: Click BOTH α-carbons on the diagram (do not click the C=O carbon).

Answer type: atoms

Screenshot: gp5a-stage-1-question



{"authorAnswer": [0, 2], "tolerance": null, "highlightsAfterCorrectAnswer": {"alpha": [0, 2], "prin": [5]}}

Hint 1: An alpha carbon is directly bonded to the carbonyl carbon; the carbonyl carbon itself is excluded.

Hint 2: Find the C=O carbon, then select its two carbon neighbors, one on each side.

Explanation: Correct! Carbons A and C flank the C=O carbon and are both α-carbons.

Screenshot: gp5a-stage-1-feedback

Stage 1/3: Click BOTH α-carbons on the diagram (do not click the C=O carbon).

ALPHA CARBONS

Select atoms on the structure using click or Tab and Enter. Selected: **A, C**

Check selection

Correct! Carbons A and C flank the C=O carbon and are both α-carbons.

Continue →

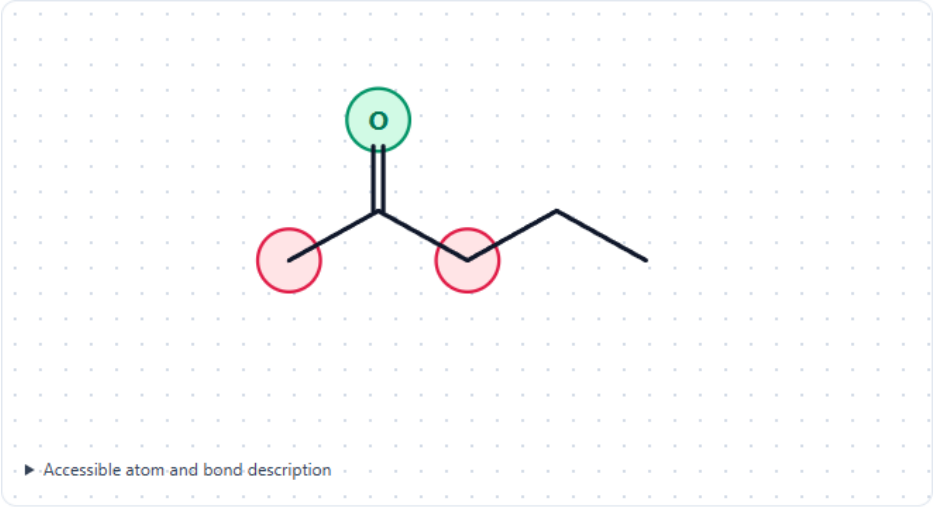
gp5a:stage-2 | Stage 2/3: Which direction gives the ketone carbonyl its lower locant?

Answer type: choice

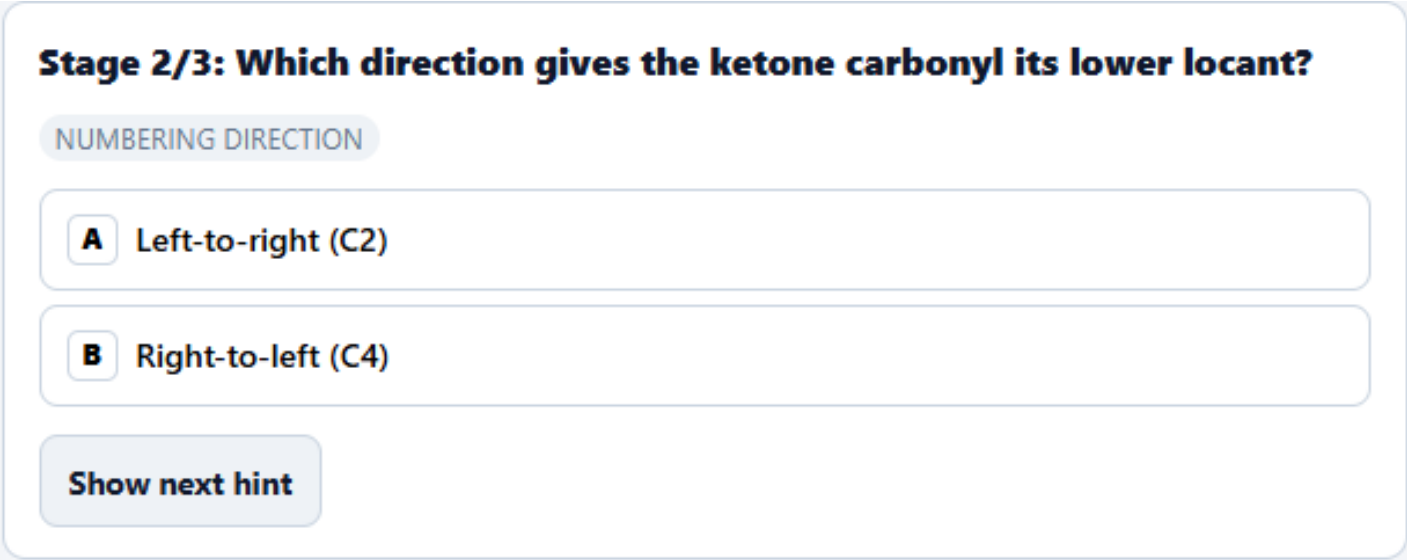
Option 0 (A): Left-to-right (C2)

Option 1 (B): Right-to-left (C4)

Screenshot: gp5a-stage-2-molecule



Screenshot: gp5a-stage-2-question



{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {"path": [0, 1, 2, 3, 4], "num": [0, 1, 2, 3, 4], "alpha": [0, 2], "prin": [5]}}

Hint 1: Compare the carbonyl-carbon locant when counting from either end of the five-carbon chain.

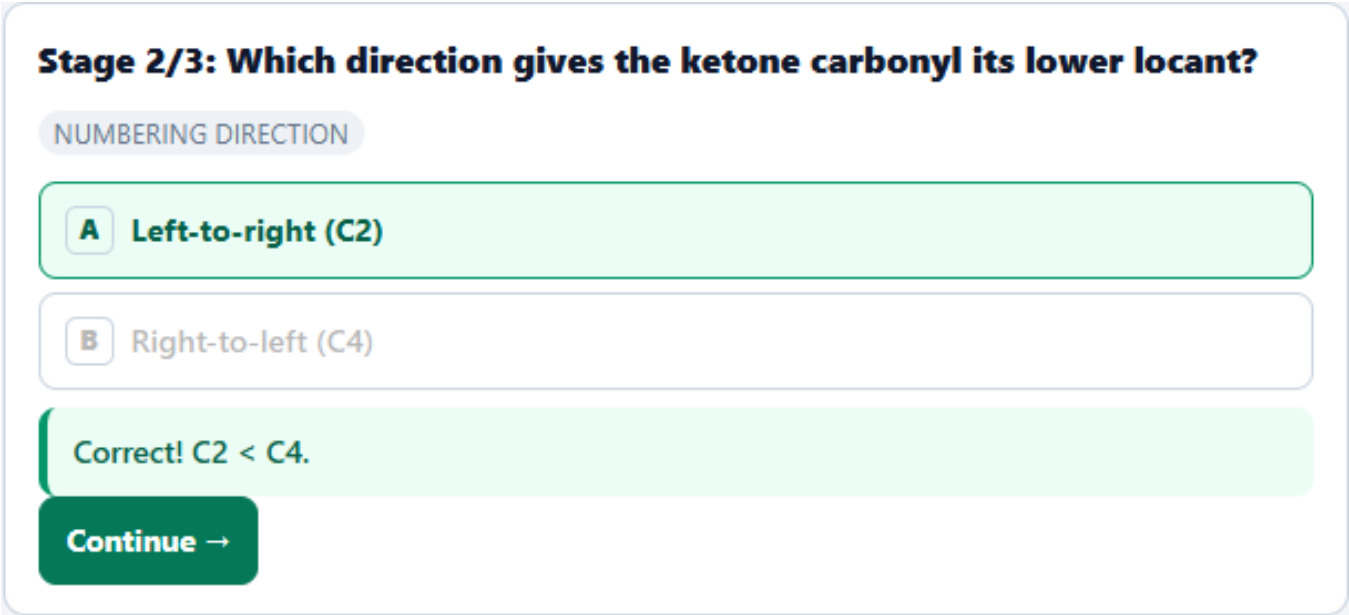
Hint 2: The ketone receives locant 2 from one end and 4 from the other; choose the smaller value.

Explanation: Correct! C2 < C4.

Option 0 feedback: Correct! C2 < C4.

Option 1 feedback: Right-to-left gives the ketone C4 instead of C2!

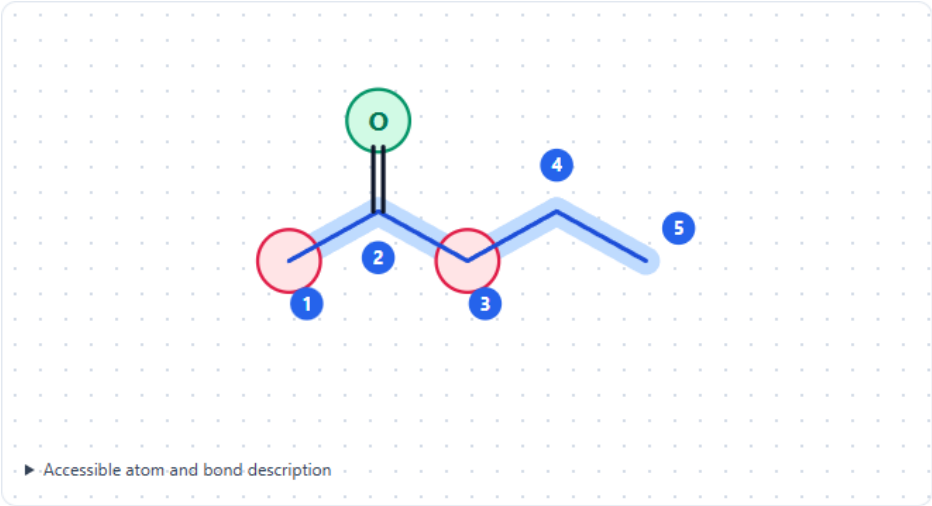
Screenshot: gp5a-stage-2-feedback



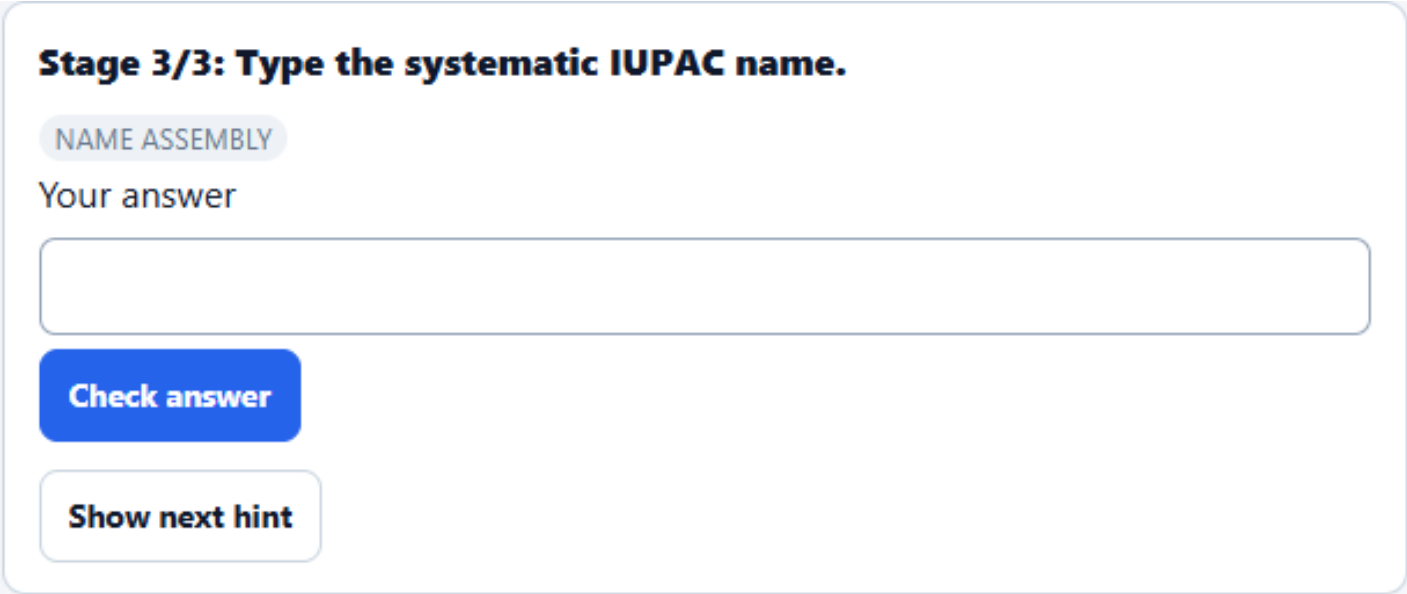
gp5a:stage-3 | Stage 3/3: Type the systematic IUPAC name.

Answer type: text

Screenshot: gp5a-stage-3-molecule



Screenshot: gp5a-stage-3-question



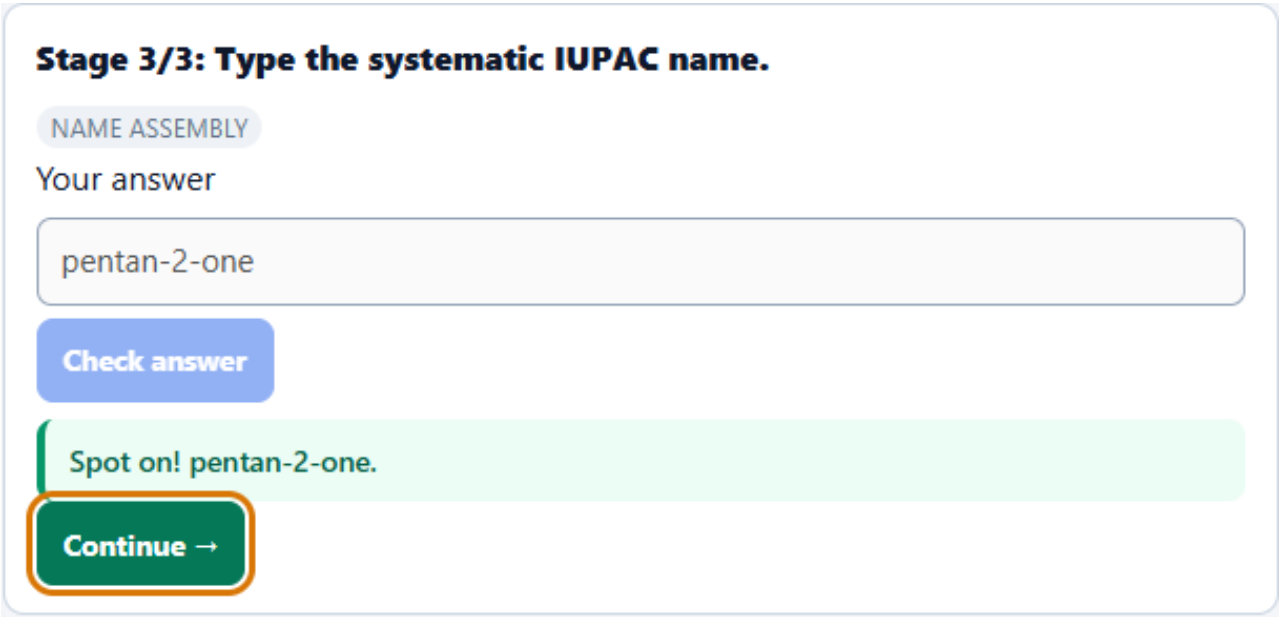
{"authorAnswer": ["pentan-2-one", "2-pentanone"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Hint 1 (Rule): Replace the final –e of the alkane with the ketone locant and suffix –one.

Hint 2: Hint 2 (Structure): 5-carbon alkane (pentan-) with a ketone at C2.

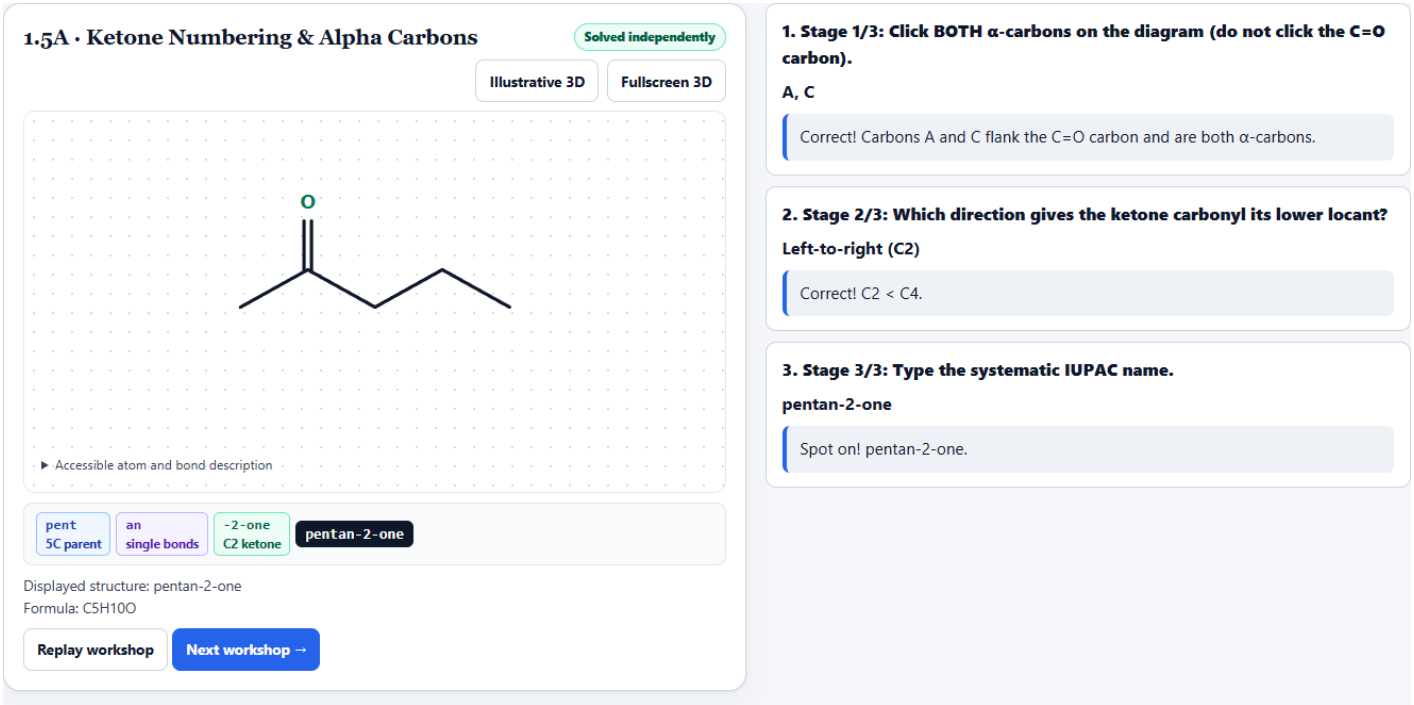
Explanation: Spot on! pentan-2-one.

Screenshot: gp5a-stage-3-feedback



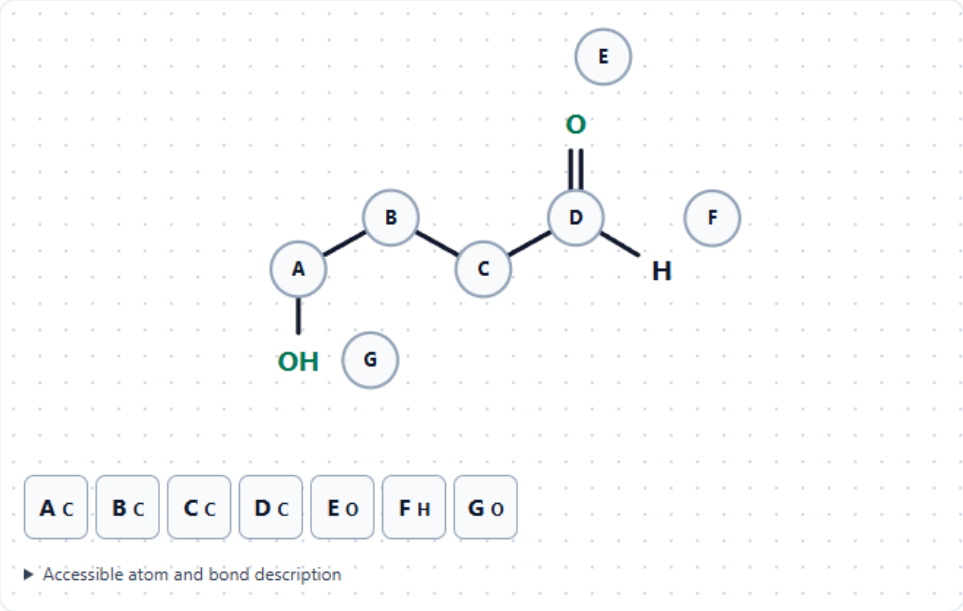
Completion summary: pentan-2-one

Screenshot: gp5a-complete



gp5b | 1.5B · Aldehyde vs. Alcohol Priority

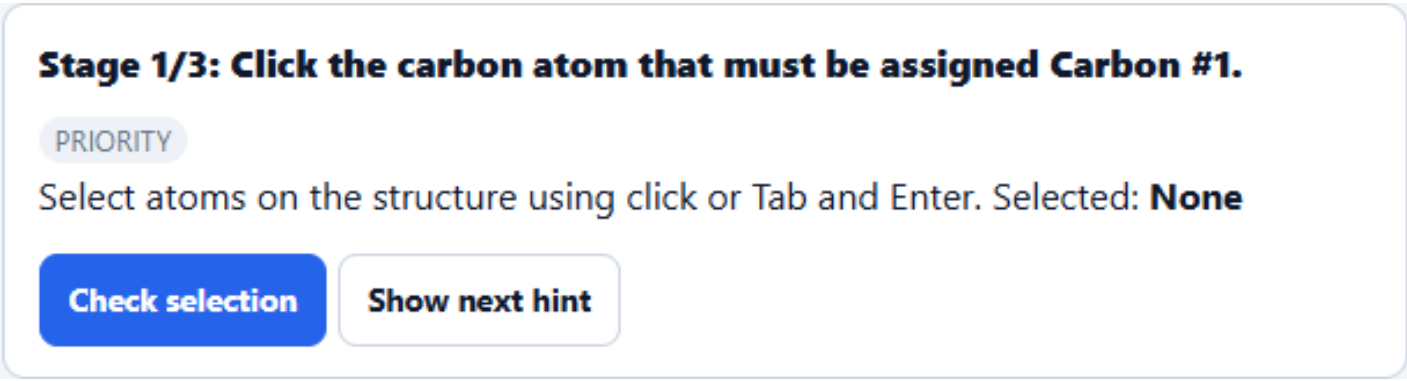
Screenshot: gp5b-stage-1-molecule



gp5b:stage-1 | Stage 1/3: Click the carbon atom that must be assigned Carbon #1.

Answer type: atoms

Screenshot: gp5b-stage-1-question



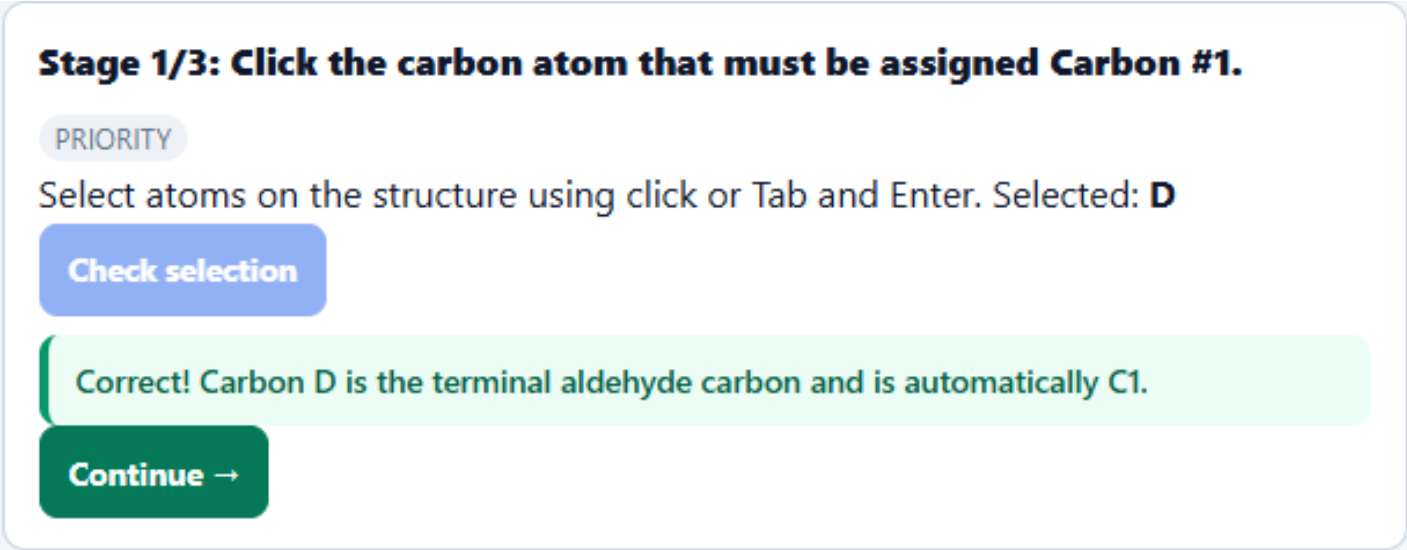
{"authorAnswer": [3], "tolerance": null, "highlightsAfterCorrectAnswer": {"path": [0, 1, 2, 3], "num": [3, 2, 1, 0], "prin": [4, 5]}}

Hint 1: An aldehyde carbon belongs to the parent chain and outranks the alcohol for the suffix.

Hint 2: Select the carbon double-bonded to oxygen and single-bonded to the explicitly drawn H; select the carbon, not that H.

Explanation: Correct! Carbon D is the terminal aldehyde carbon and is automatically C1.

Screenshot: gp5b-stage-1-feedback



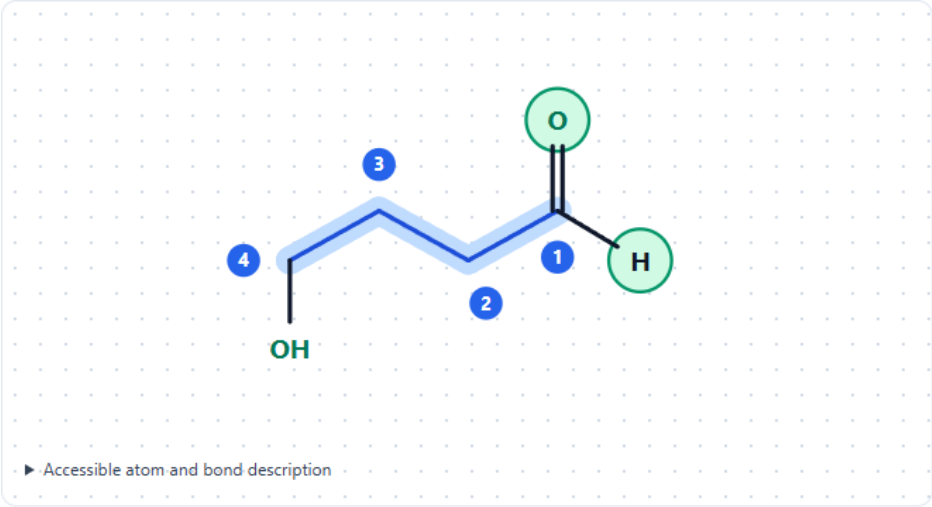
gp5b:stage-2 | Stage 2/3: When an alcohol is subordinate to an aldehyde, how is the C4 –OH named?

Answer type: choice

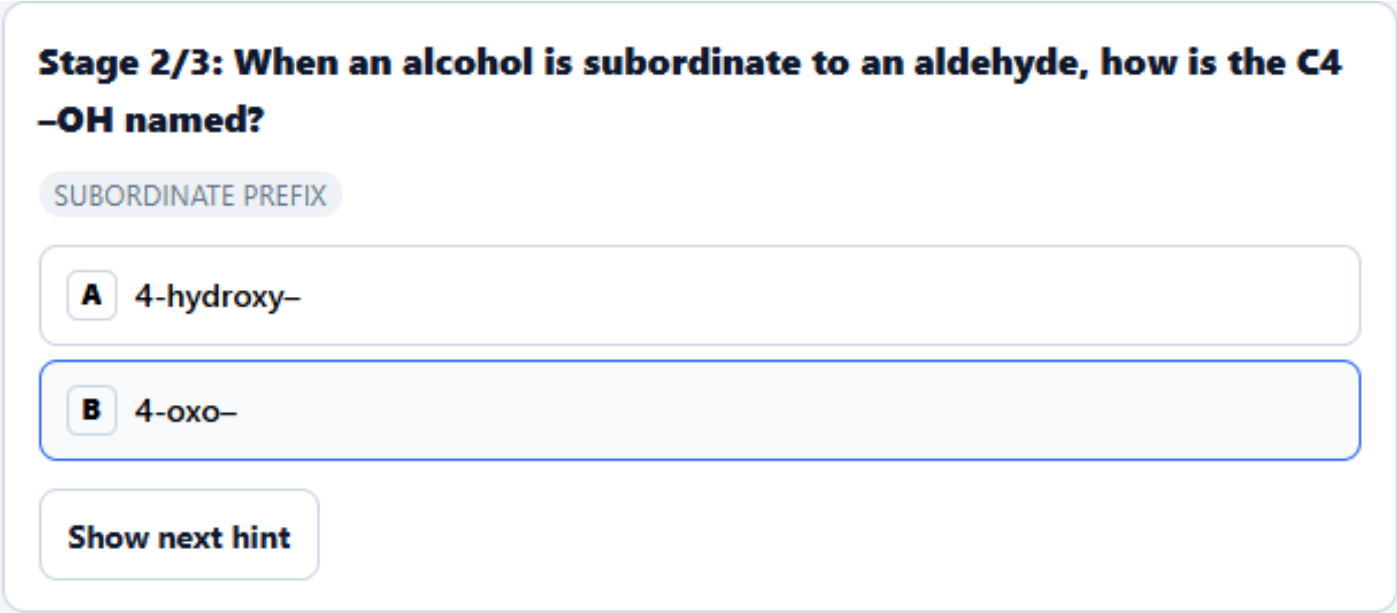
Option 0 (A): 4-hydroxy–

Option 1 (B): 4-oxo–

Screenshot: gp5b-stage-2-molecule



Screenshot: gp5b-stage-2-question



{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {"path": [0, 1, 2, 3], "num": [3, 2, 1, 0], "prin": [4, 5], "sub": [6]}}

Hint 1: The aldehyde already supplies -al, so the alcohol must be expressed as a prefix.

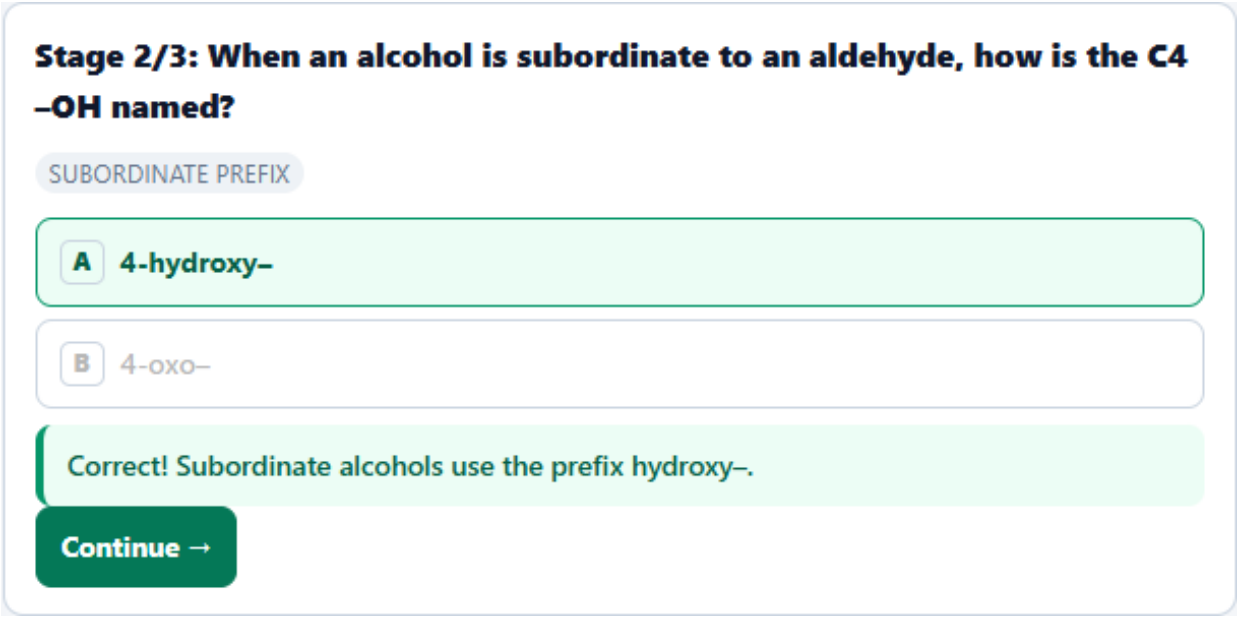
Hint 2: Use hydroxy- with the hydroxyl-bearing carbon locant; do not add a second -ol suffix.

Explanation: Correct! Subordinate alcohols use the prefix hydroxy–.

Option 0 feedback: Correct! Subordinate alcohols use the prefix hydroxy–.

Option 1 feedback: oxo– is for subordinate carbonyls, not alcohols.

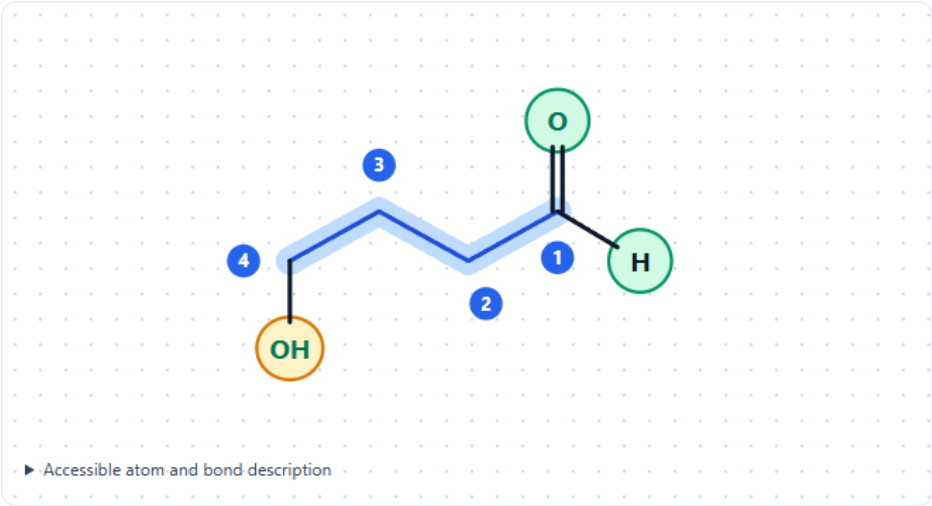
Screenshot: gp5b-stage-2-feedback



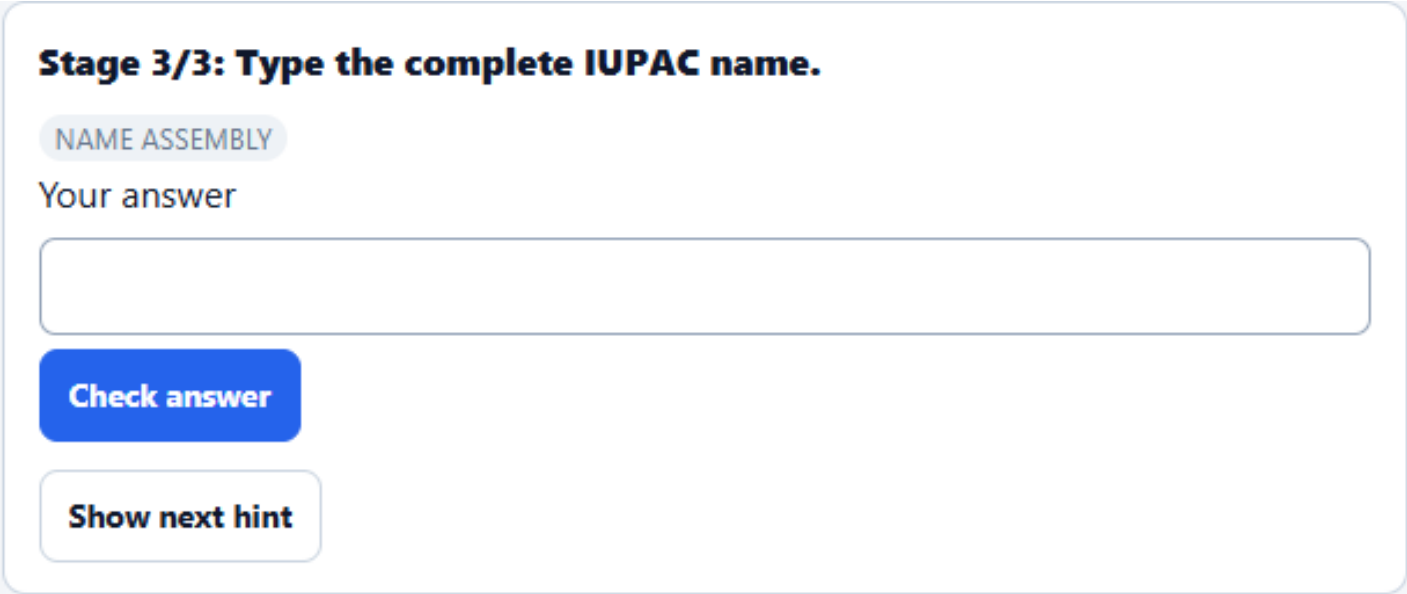
gp5b:stage-3 | Stage 3/3: Type the complete IUPAC name.

Answer type: text

Screenshot: gp5b-stage-3-molecule



Screenshot: gp5b-stage-3-question



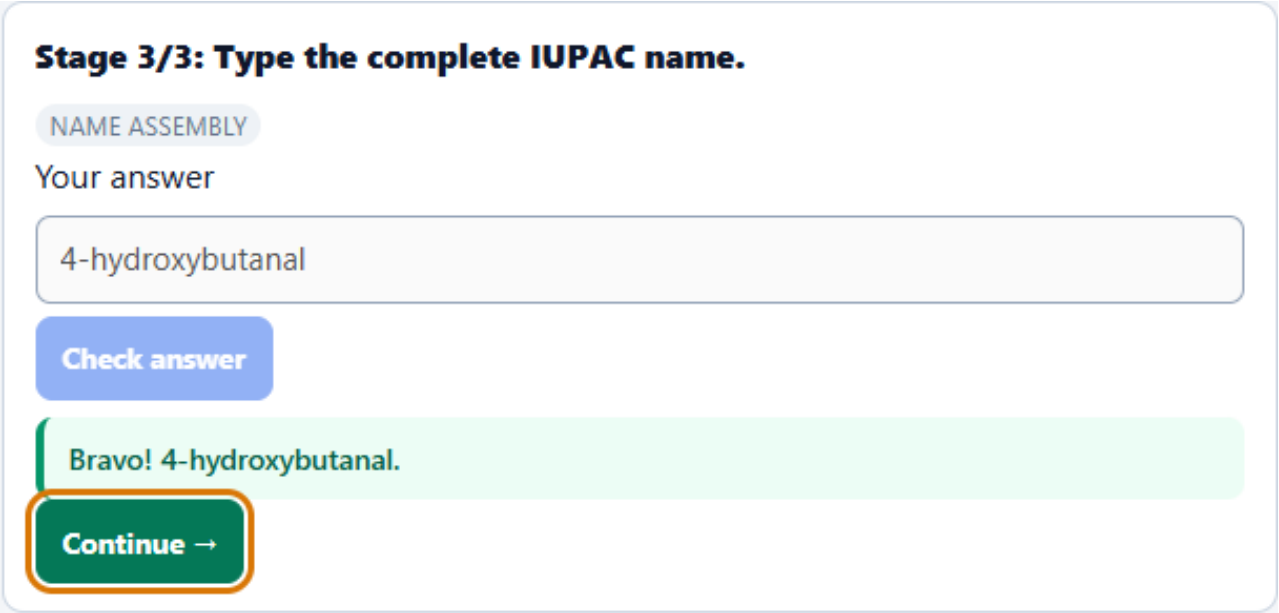
{"authorAnswer": ["4-hydroxybutanal"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Hint 1 (Rule): Terminal aldehydes are assumed to be C1 and take the suffix –al without a number.

Hint 2: Hint 2 (Structure): 4-hydroxy prefix + 4-carbon aldehyde parent (butanal).

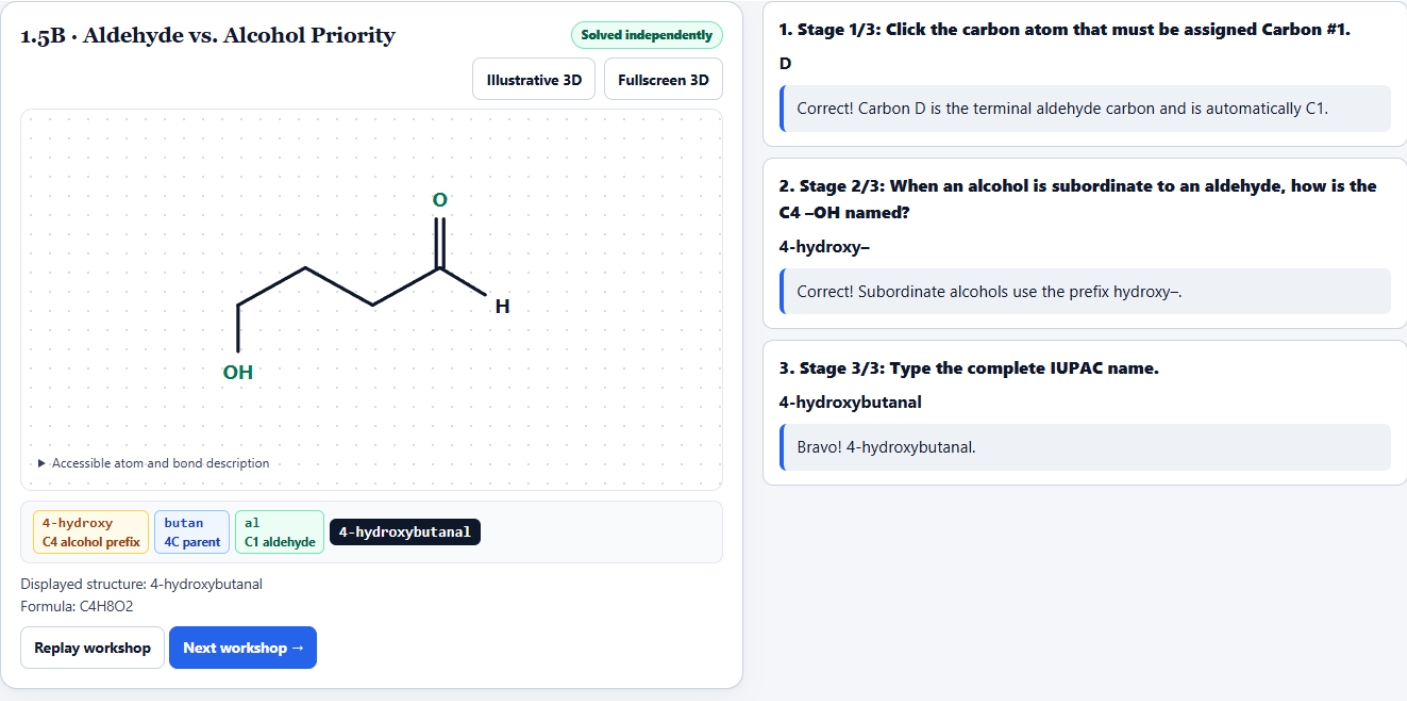
Explanation: Bravo! 4-hydroxybutanal.

Screenshot: gp5b-stage-3-feedback



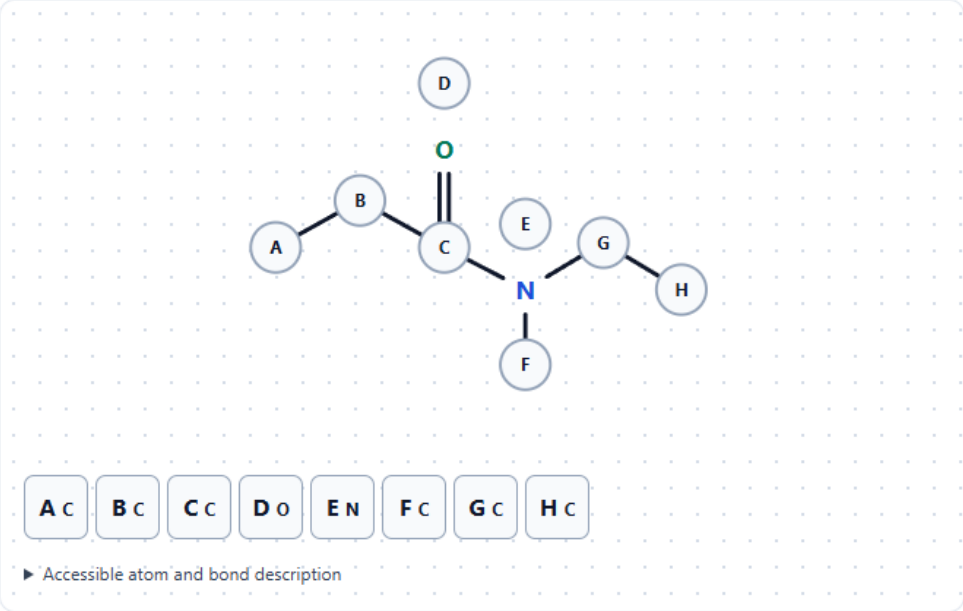
Completion summary: 4-hydroxybutanal

Screenshot: gp5b-complete



gp6a | 1.6A · N-Substituted Amide Construction

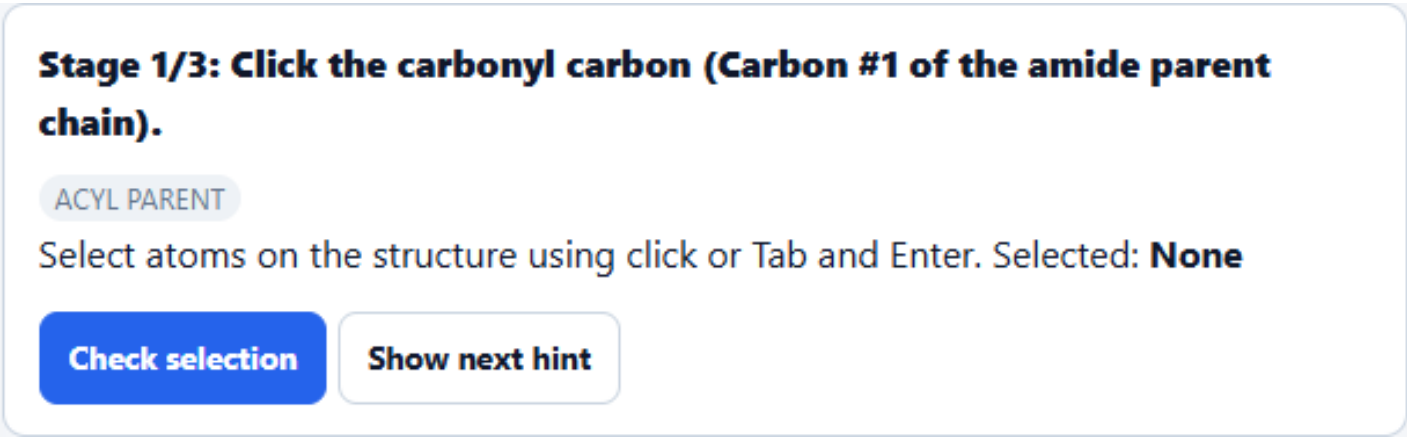
Screenshot: gp6a-stage-1-molecule



gp6a:stage-1 | Stage 1/3: Click the carbonyl carbon (Carbon #1 of the amide parent chain).

Answer type: atoms

Screenshot: gp6a-stage-1-question



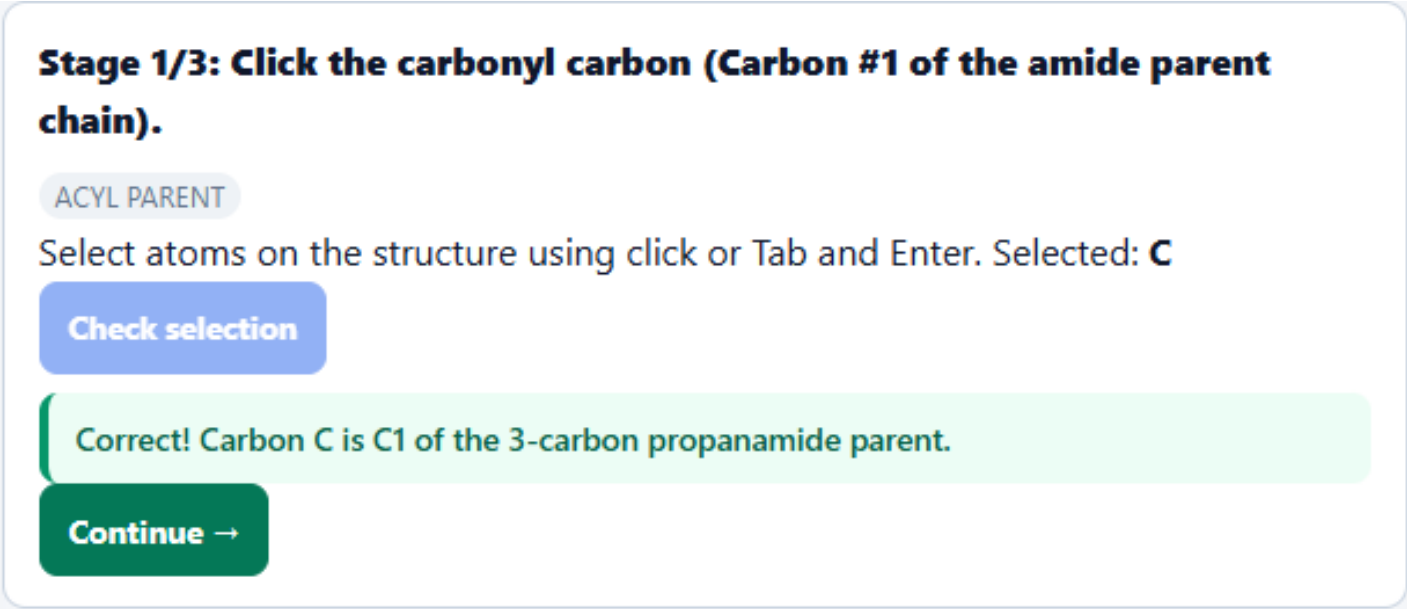
{"authorAnswer": [2], "tolerance": null, "highlightsAfterCorrectAnswer": {"path": [2, 1, 0], "num": [2, 1, 0], "prin": [3, 4]}}

Hint 1: The amide parent includes the carbonyl carbon; nitrogen-bound alkyl groups are named separately.

Hint 2: Choose the carbon double-bonded to oxygen and single-bonded to nitrogen.

Explanation: Correct! Carbon C is C1 of the 3-carbon propanamide parent.

Screenshot: gp6a-stage-1-feedback



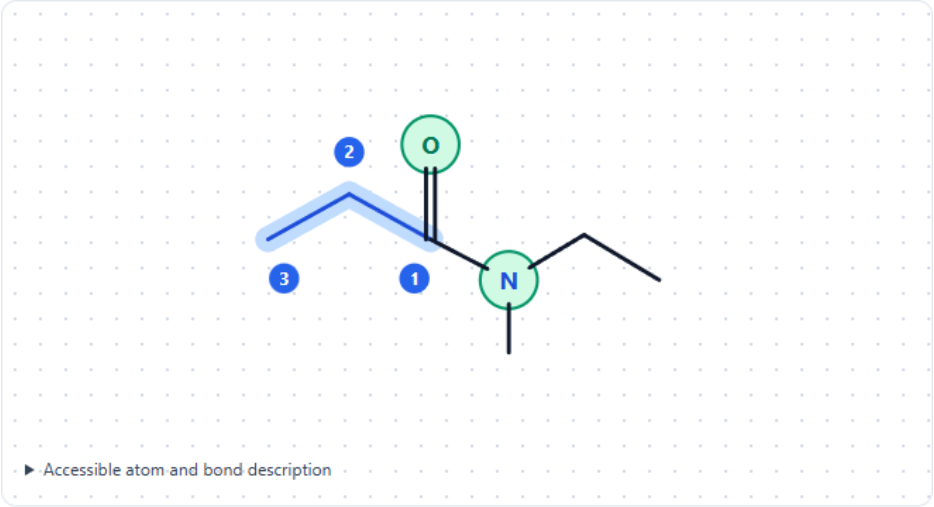
gp6a:stage-2 | Stage 2/3: How are the ethyl and methyl groups on nitrogen written?

Answer type: choice

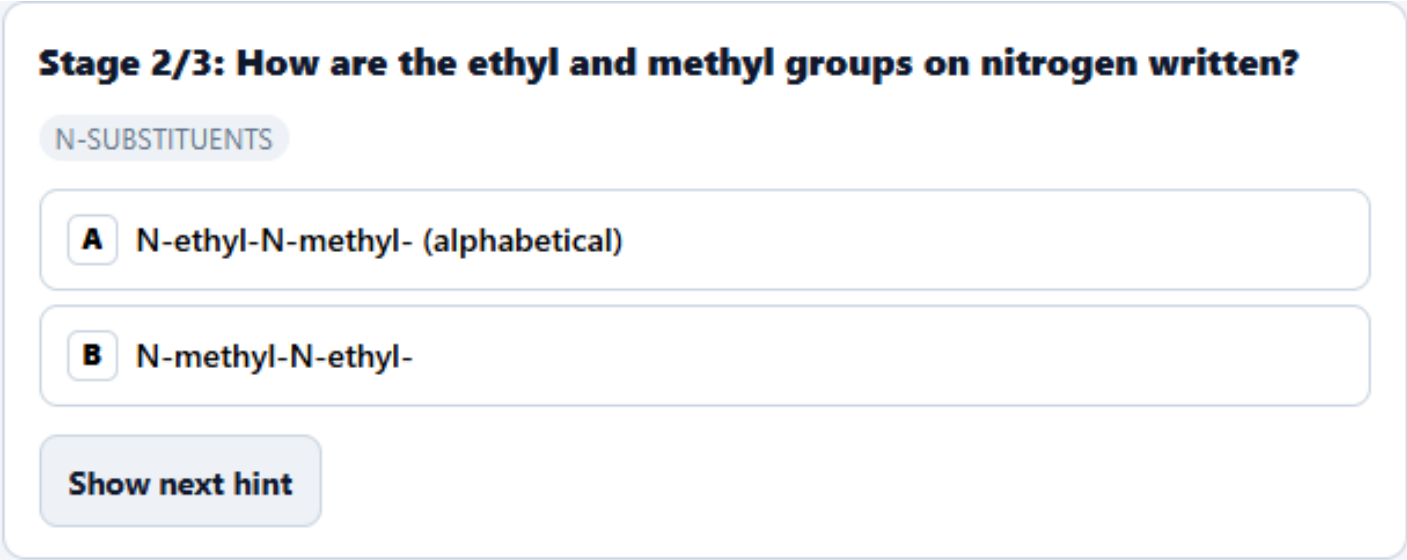
Option 0 (A): N-ethyl-N-methyl- (alphabetical)

Option 1 (B): N-methyl-N-ethyl-

Screenshot: gp6a-stage-2-molecule



Screenshot: gp6a-stage-2-question



{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {"path": [2, 1, 0], "num": [2, 1, 0], "sub": [5, 6, 7], "prin": [3, 4]}}

Hint 1: Substituents attached directly to nitrogen use the letter locant N, not a parent-chain carbon number.

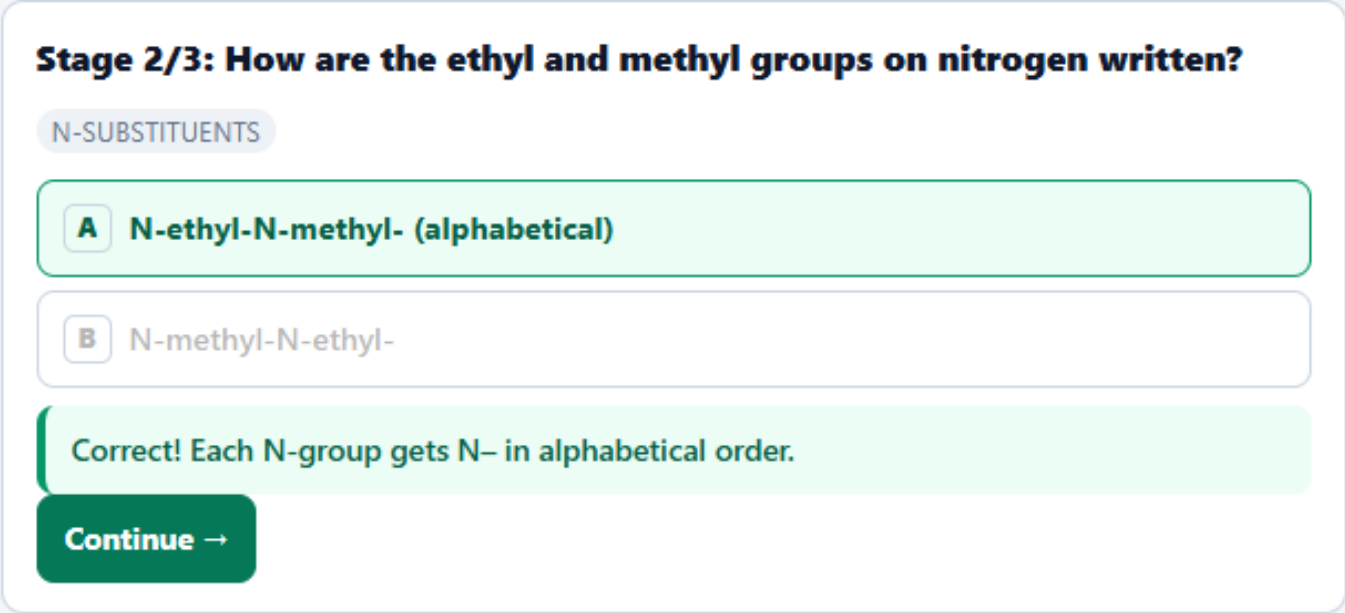
Hint 2: Give each nitrogen-bound group its own N locant and list ethyl before methyl alphabetically.

Explanation: Correct! Each N-group gets N– in alphabetical order.

Option 0 feedback: Correct! Each N-group gets N– in alphabetical order.

Option 1 feedback: Alphabetize ethyl before methyl.

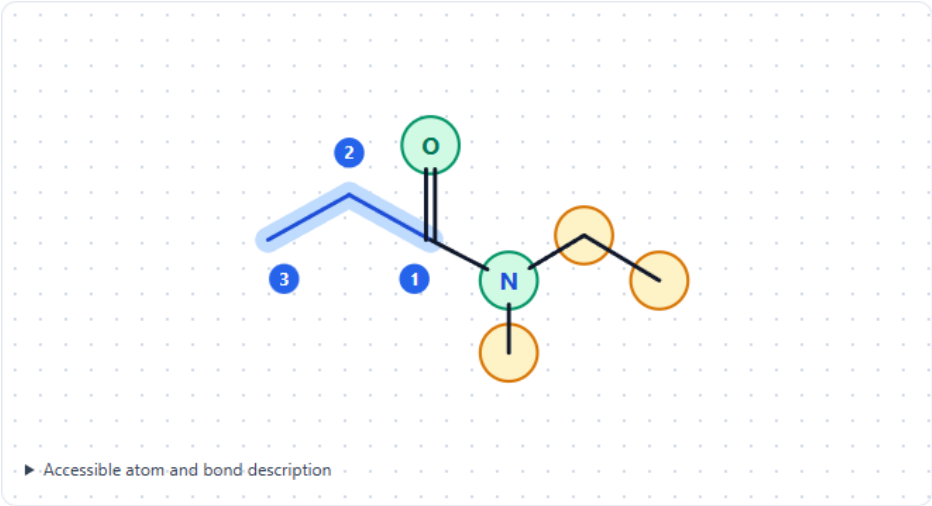
Screenshot: gp6a-stage-2-feedback



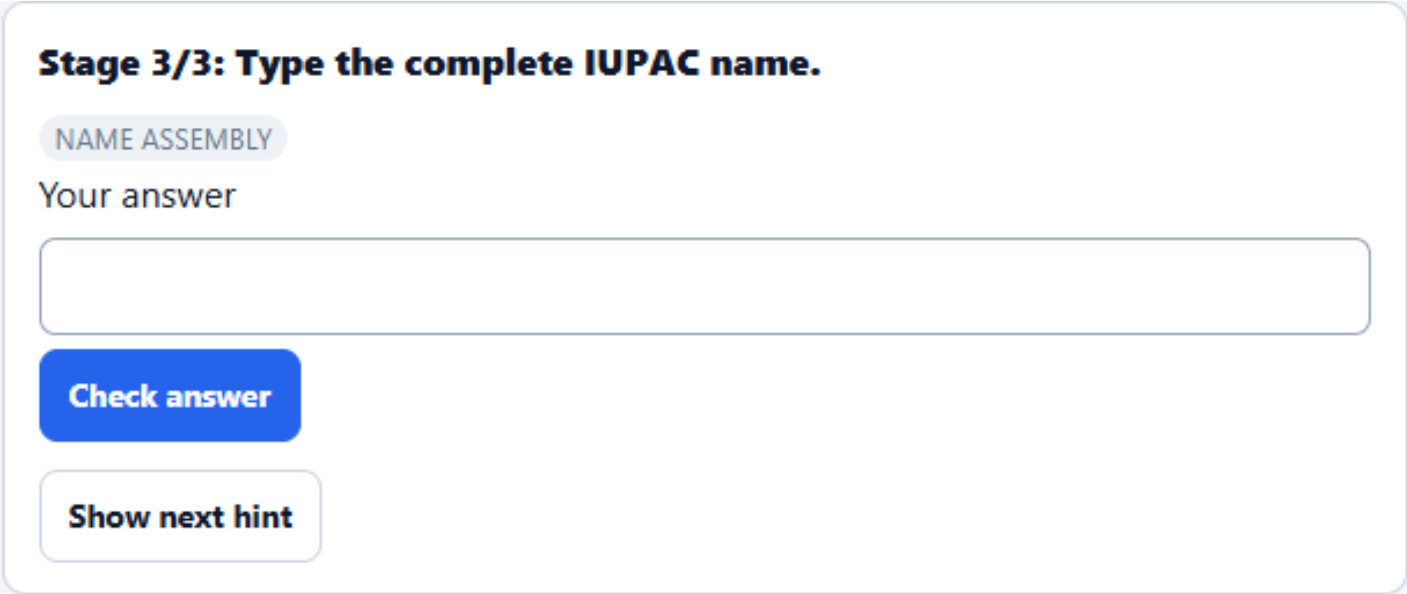
gp6a:stage-3 | Stage 3/3: Type the complete IUPAC name.

Answer type: text

Screenshot: gp6a-stage-3-molecule



Screenshot: gp6a-stage-3-question



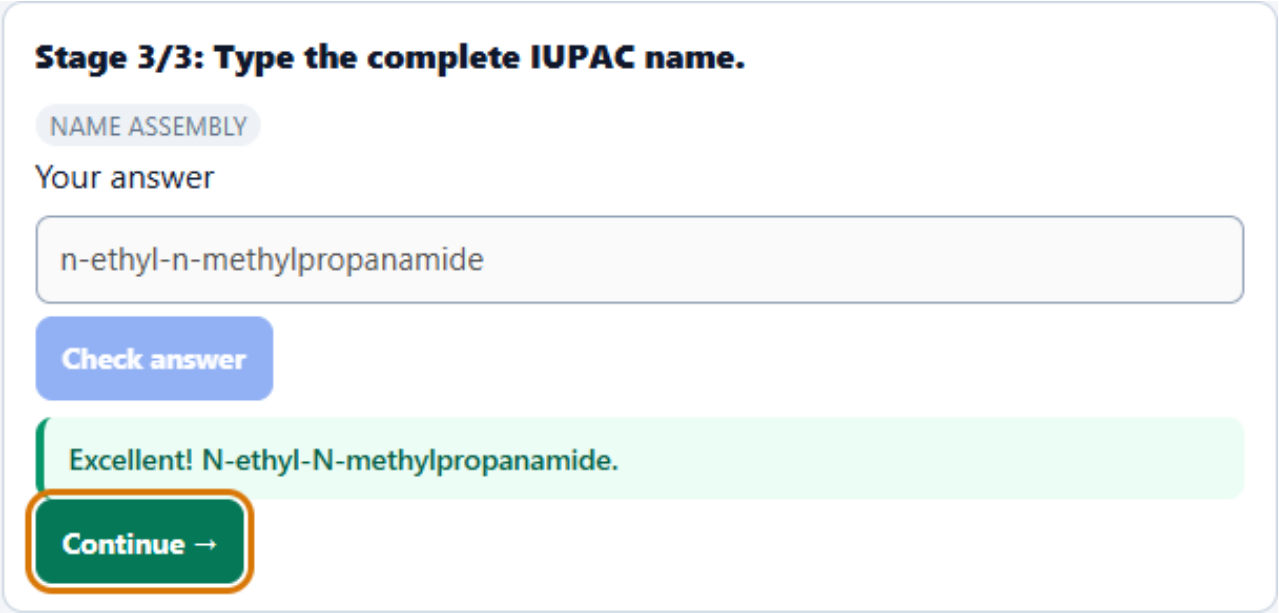
{"authorAnswer": ["n-ethyl-n-methylpropanamide"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Hint 1 (Rule): List the alphabetized N-alkyl prefixes directly before the acyl parent amide name.

Hint 2: Hint 2 (Structure): N-ethyl-N-methyl + 3-carbon amide parent (propanamide).

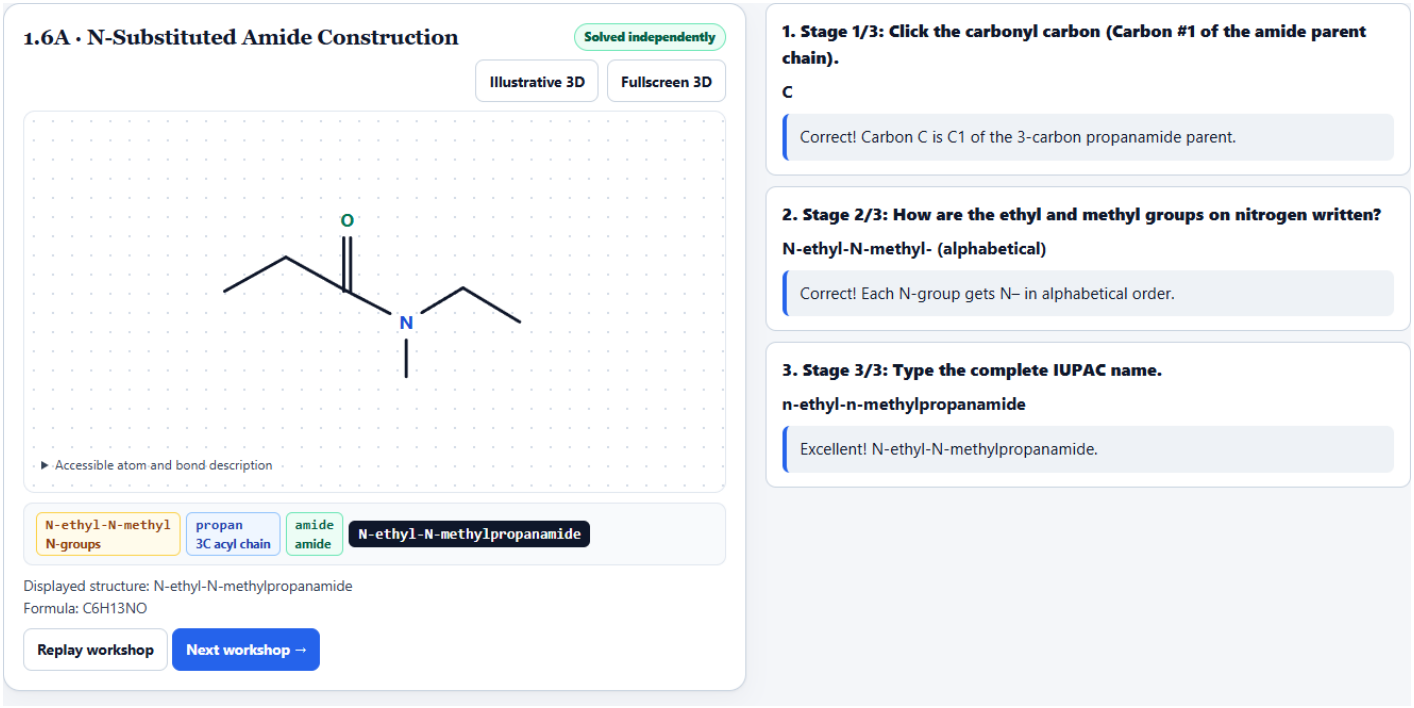
Explanation: Excellent! N-ethyl-N-methylpropanamide.

Screenshot: gp6a-stage-3-feedback



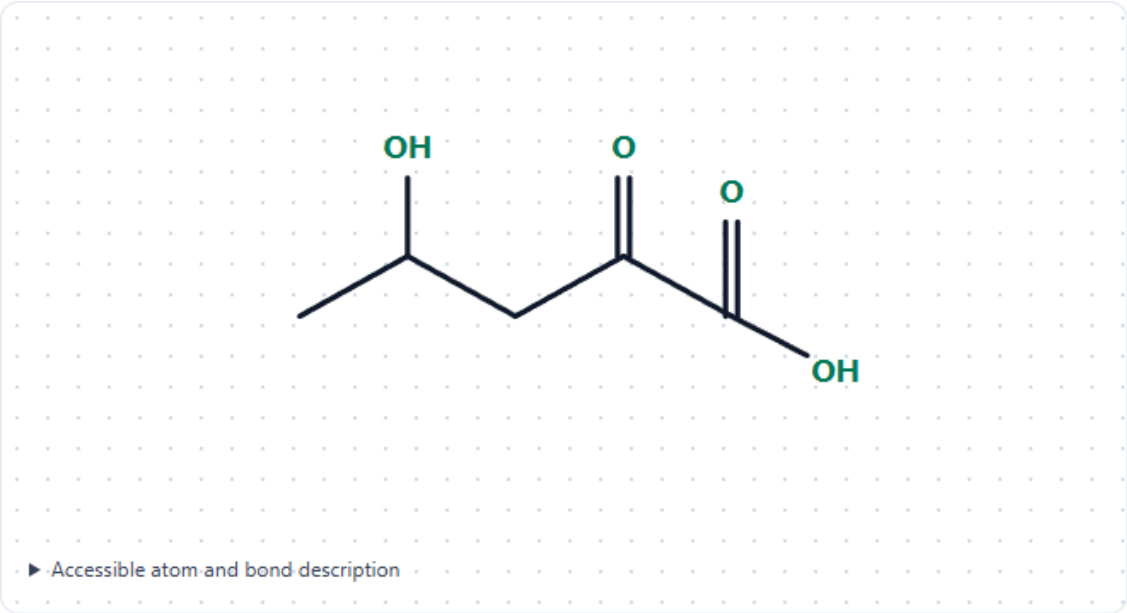
Completion summary: N-ethyl-N-methylpropanamide

Screenshot: gp6a-complete



gp6b | 1.6B · Multifunctional Priority Ladder

Screenshot: gp6b-stage-1-molecule



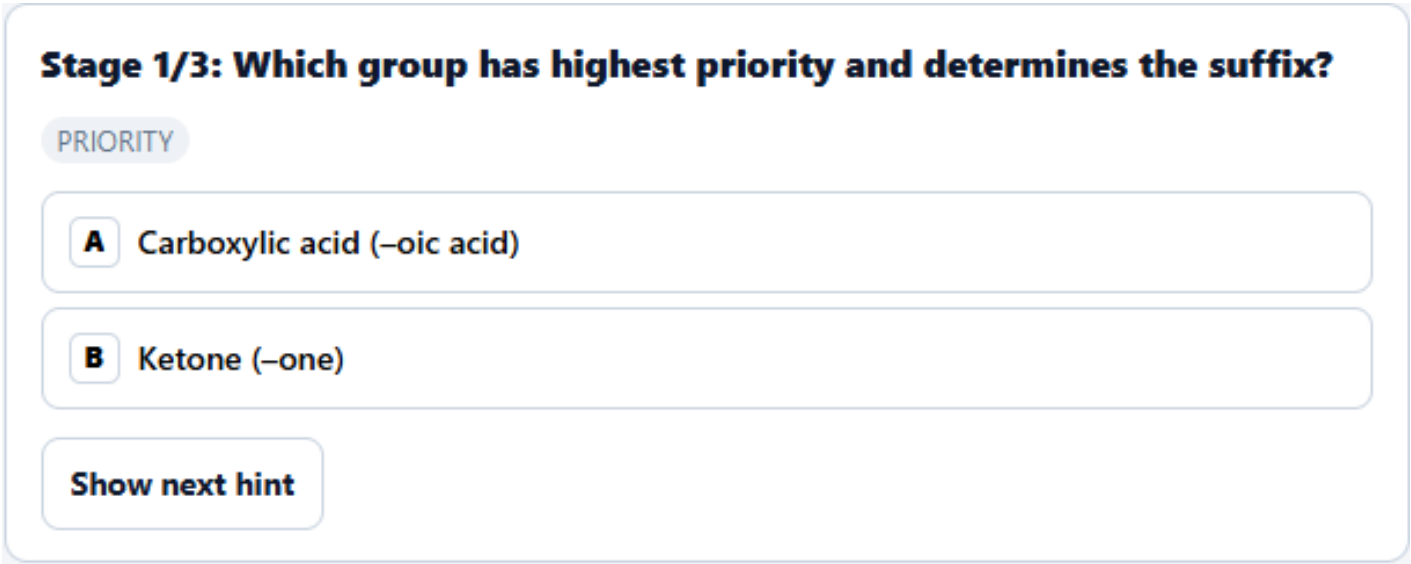
gp6b:stage-1 | Stage 1/3: Which group has highest priority and determines the suffix?

Answer type: choice

Option 0 (A): Carboxylic acid (–oic acid)

Option 1 (B): Ketone (–one)

Screenshot: gp6b-stage-1-question



{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {"path": [0, 1, 2, 3, 4], "prin": [7, 8]}}

Hint 1: Compare carboxylic acid, ketone, and alcohol naming seniority before choosing the suffix.

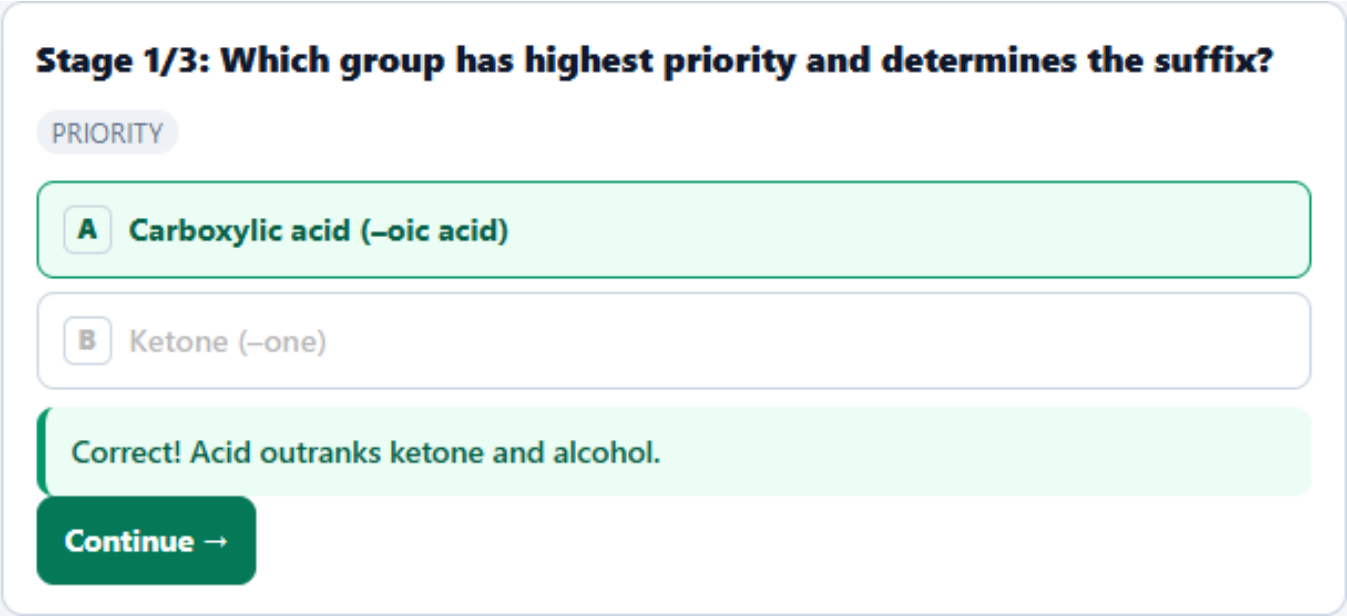
Hint 2: The carboxylic acid supplies -oic acid; the ketone and alcohol are named as prefixes.

Explanation: Correct! Acid outranks ketone and alcohol.

Option 0 feedback: Correct! Acid outranks ketone and alcohol.

Option 1 feedback: Carboxylic acid outranks ketone.

Screenshot: gp6b-stage-1-feedback



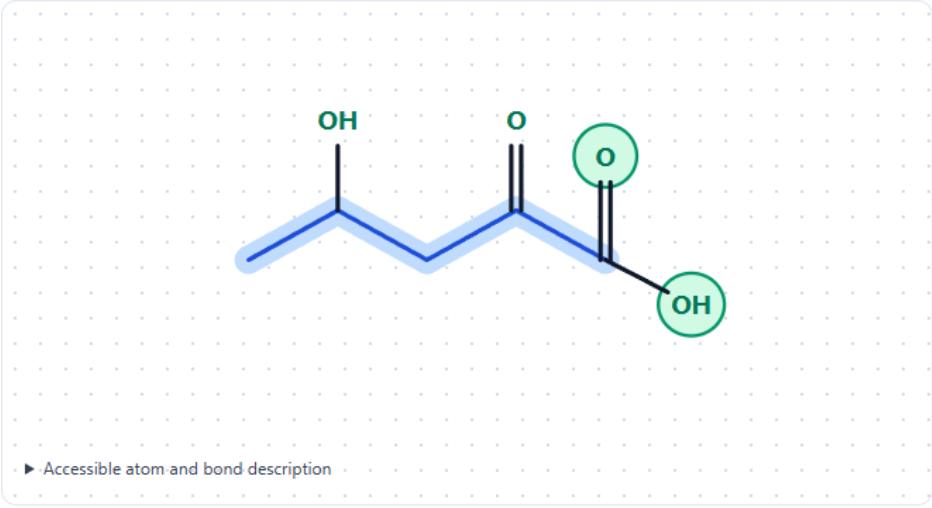
gp6b:stage-2 | Stage 2/3: With C1 at the acid, what are the prefixes for the C2 ketone and C4 alcohol?

Answer type: choice

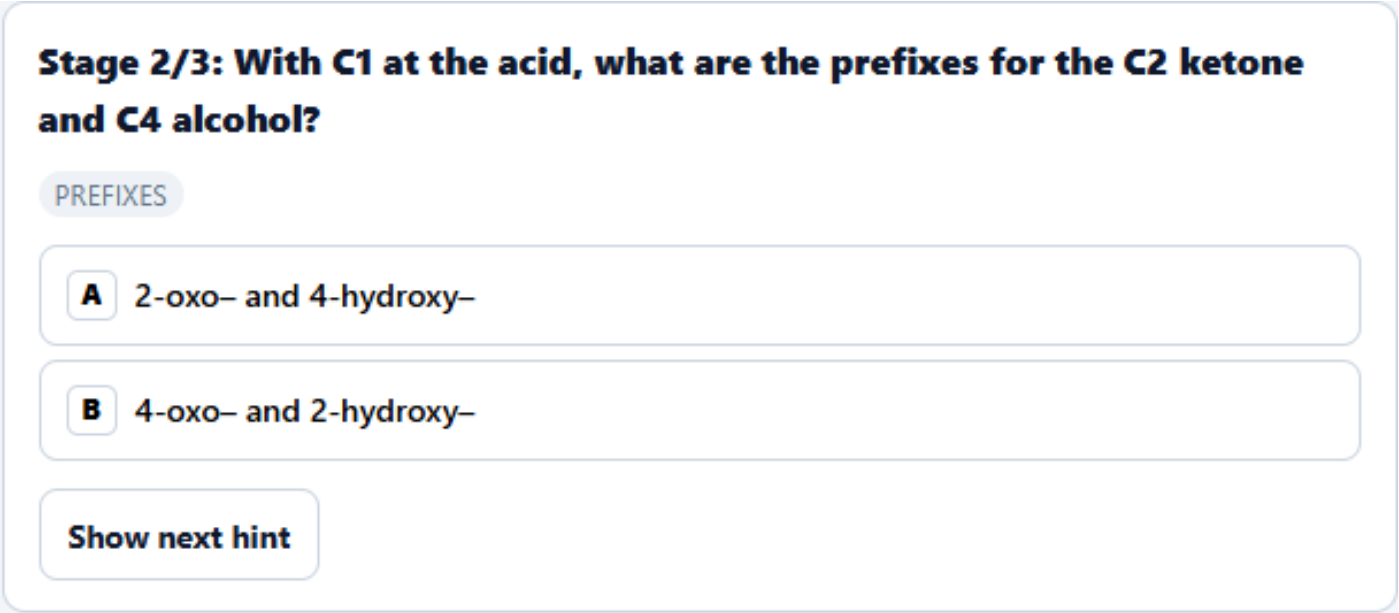
Option 0 (A): 2-oxo– and 4-hydroxy–

Option 1 (B): 4-oxo– and 2-hydroxy–

Screenshot: gp6b-stage-2-molecule



Screenshot: gp6b-stage-2-question



{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {"path": [0, 1, 2, 3, 4], "num": [4, 3, 2, 1, 0], "prin": [7, 8], "sub": [5, 6]}}

Hint 1: With the acid fixed as C1, a subordinate ketone uses oxo- and a subordinate alcohol uses hydroxy-.

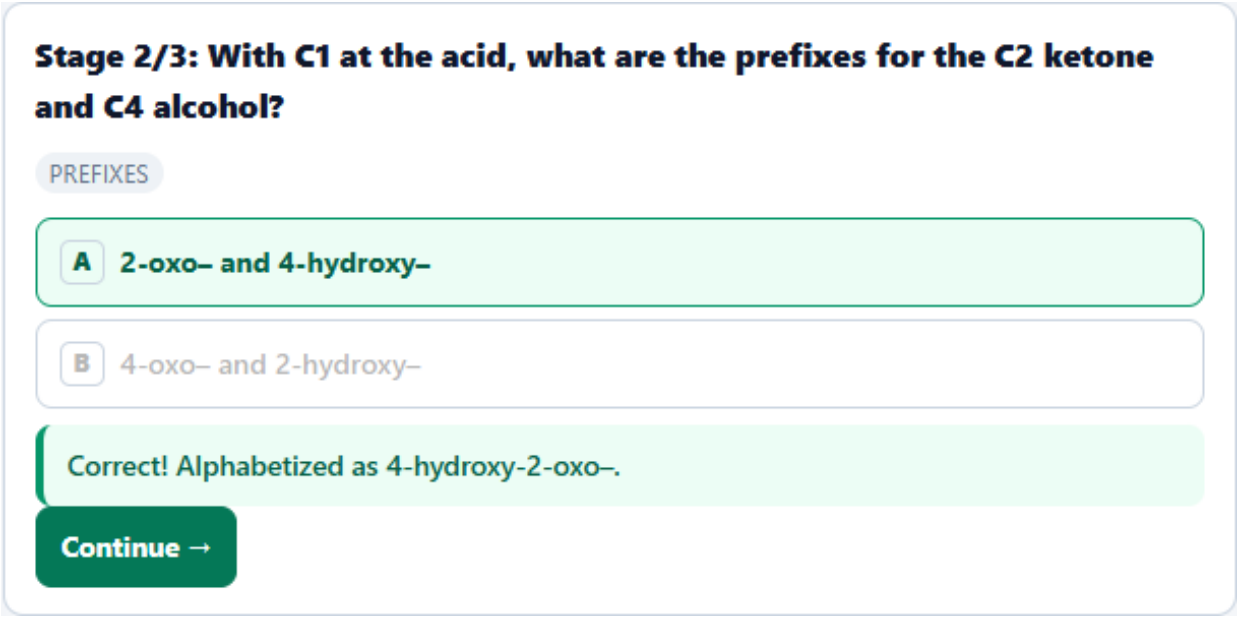
Hint 2: Attach 2 to oxo and 4 to hydroxy, then list the prefixes alphabetically.

Explanation: Correct! Alphabetized as 4-hydroxy-2-oxo–.

Option 0 feedback: Correct! Alphabetized as 4-hydroxy-2-oxo–.

Option 1 feedback: C1 is the acid on the right, so the ketone is C2 and the alcohol is C4!

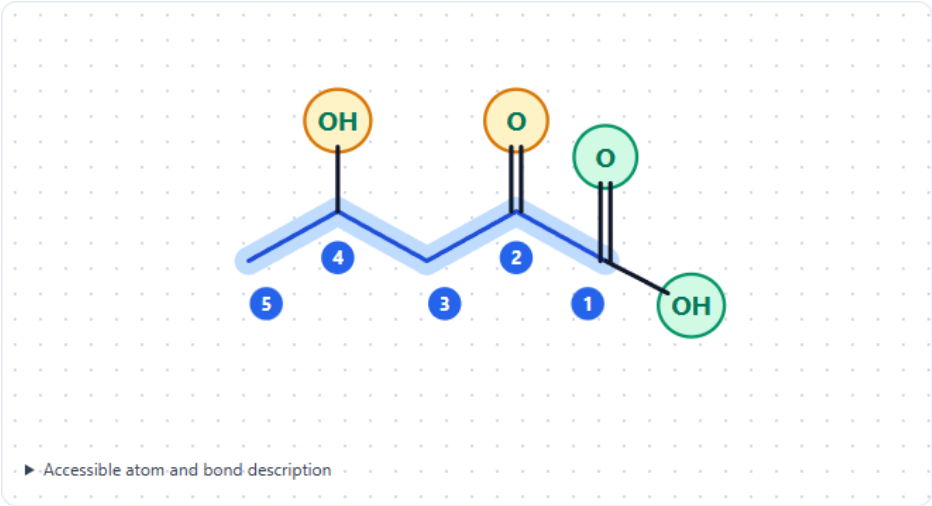
Screenshot: gp6b-stage-2-feedback



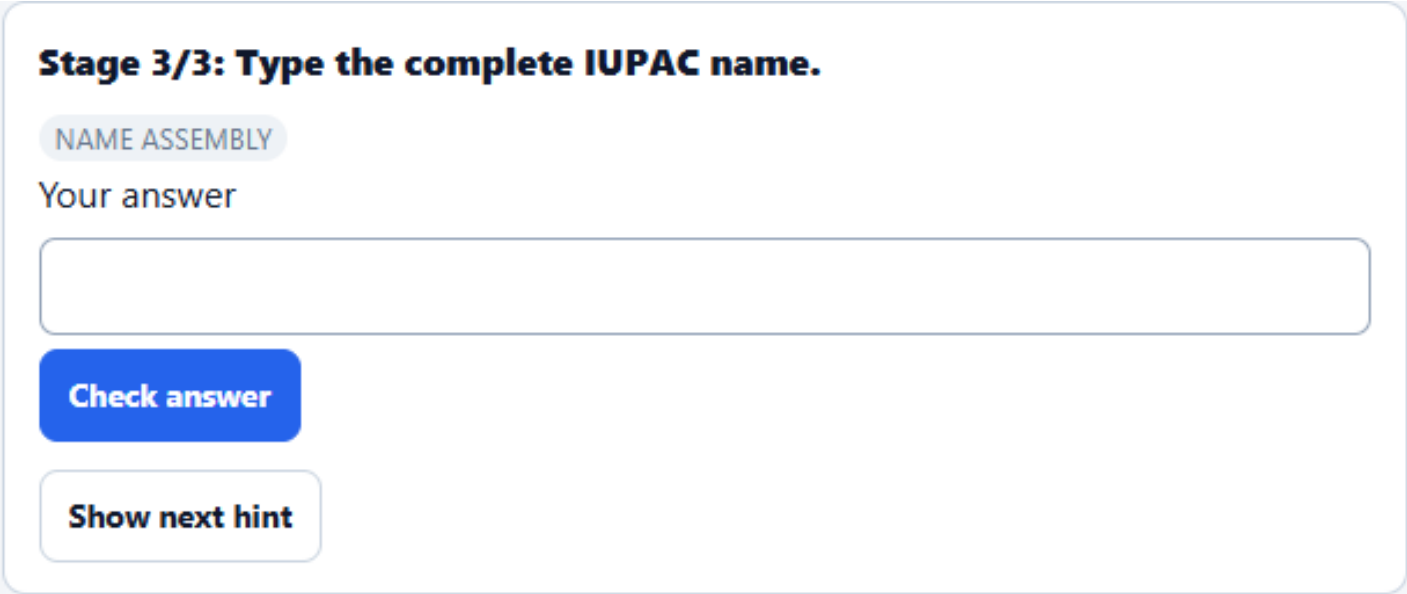
gp6b:stage-3 | Stage 3/3: Type the complete IUPAC name.

Answer type: text

Screenshot: gp6b-stage-3-molecule



Screenshot: gp6b-stage-3-question



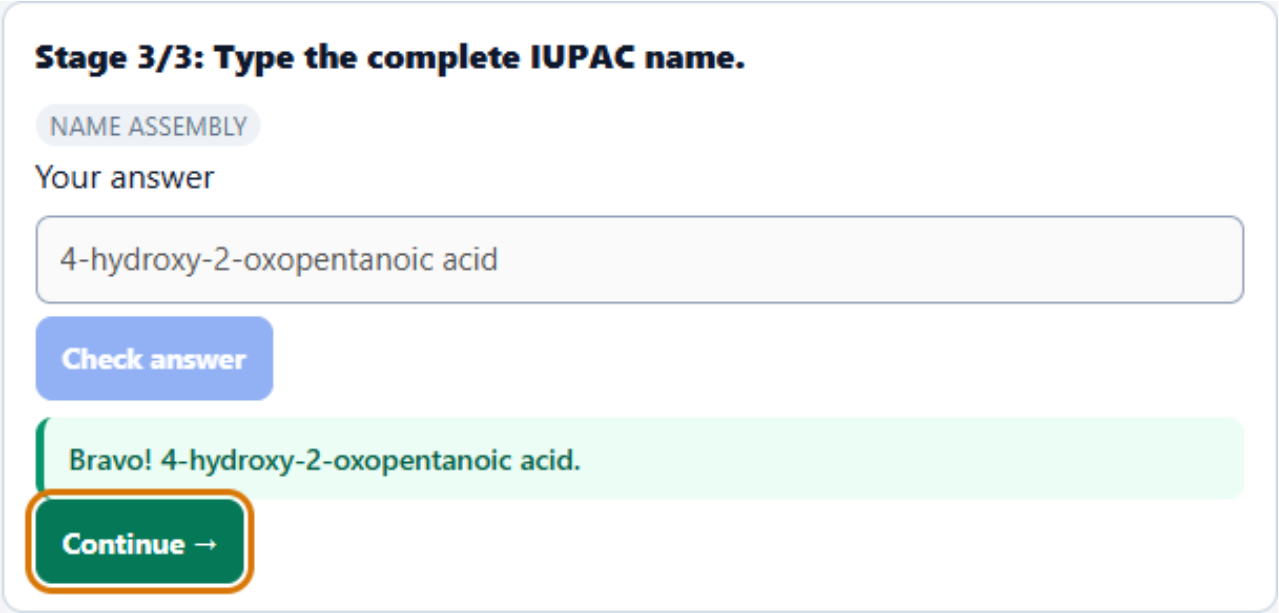
{"authorAnswer": ["4-hydroxy-2-oxopentanoic acid"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Hint 1 (Rule): Alphabetize the prefixes ("h" in hydroxy before "o" in oxo) before the parent acid name.

Hint 2: Hint 2 (Structure): 4-hydroxy-2-oxo + 5-carbon carboxylic acid (pentanoic acid).

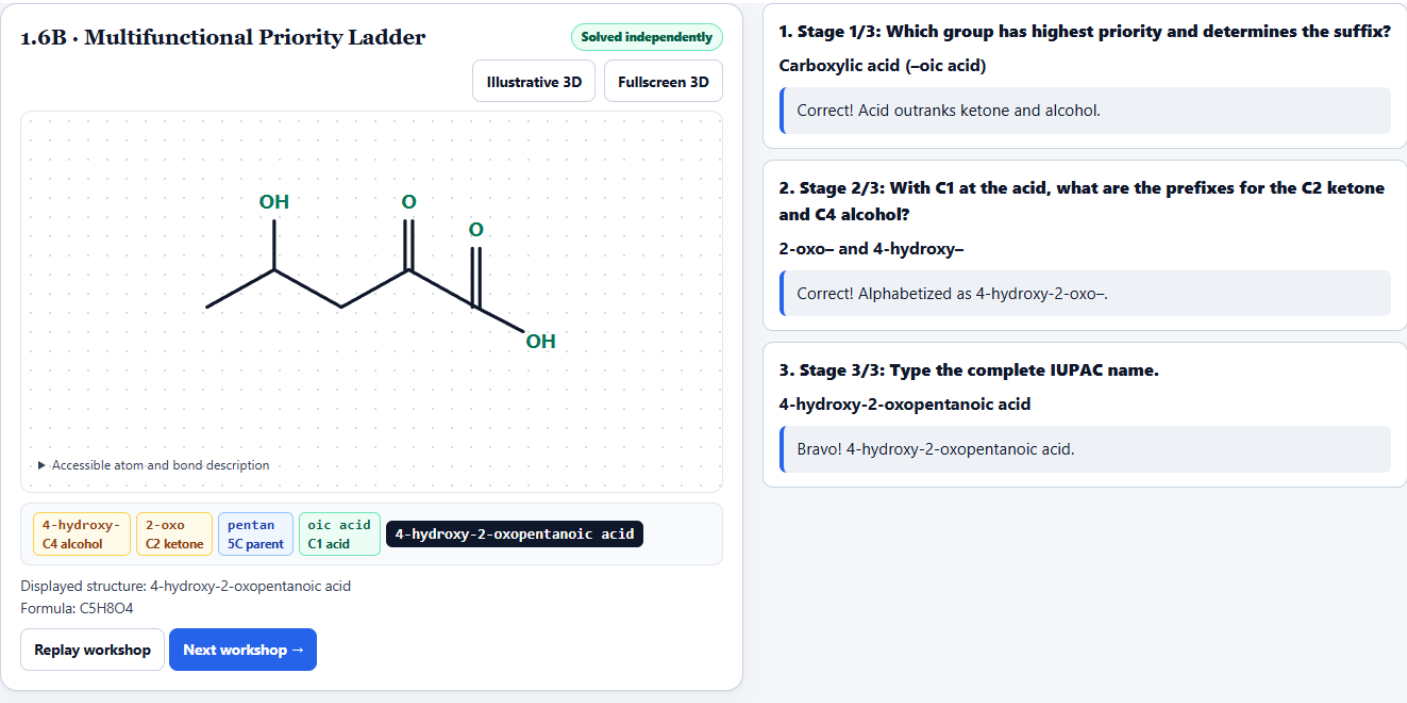
Explanation: Bravo! 4-hydroxy-2-oxopentanoic acid.

Screenshot: gp6b-stage-3-feedback



Completion summary: 4-hydroxy-2-oxopentanoic acid

Screenshot: gp6b-complete



Molecular repository: author claims to verify

fig15

Author claims: 4-ethyl-2,3-dimethylheptane | formula C11H24 | canonical composition C11H24 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=210.00, y=85.00; charge=0; explicit H override=none

4 (E): C; x=264.00, y=115.00; charge=0; explicit H override=none

5 (F): C; x=318.00, y=85.00; charge=0; explicit H override=none

6 (G): C; x=372.00, y=115.00; charge=0; explicit H override=none

7 (H): C; x=210.00, y=32.00; charge=0; explicit H override=none

8 (I): C; x=256.00, y=12.00; charge=0; explicit H override=none

9 (J): C; x=264.00, y=146.00; charge=0; explicit H override=none

10 (K): C; x=318.00, y=32.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [4, 5, 1], [5, 6, 1], [3, 7, 1], [7, 8, 1], [4, 9, 1], [5, 10, 1]]

enol

Author claims: but-3-en-2-ol | formula C4H8O | canonical composition C4H8O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=210.00, y=85.00; charge=0; explicit H override=none

4 (E): O; x=156.00, y=166.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 1], [2, 4, 1]]

cyhex

Author claims: 4-ethyl-1-methylcyclohex-1-ene | formula C9H16 | canonical composition C9H16 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=155.00, y=42.00; charge=0; explicit H override=none

1 (B): C; x=208.00, y=74.00; charge=0; explicit H override=none

2 (C): C; x=208.00, y=136.00; charge=0; explicit H override=none

3 (D): C; x=155.00, y=168.00; charge=0; explicit H override=none

4 (E): C; x=102.00, y=136.00; charge=0; explicit H override=none

5 (F): C; x=102.00, y=74.00; charge=0; explicit H override=none

6 (G): C; x=258.00, y=48.00; charge=0; explicit H override=none

7 (H): C; x=302.00, y=74.00; charge=0; explicit H override=none

8 (I): C; x=50.00, y=168.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [4, 5, 2], [5, 0, 1], [1, 6, 1], [6, 7, 1], [4, 8, 1]]

vicinal

Author claims: ethane-1,2-diol | formula C2H6O2 | canonical composition C2H6O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=48.00, y=52.00; charge=0; explicit H override=none

3 (D): O; x=102.00, y=30.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [0, 2, 1], [1, 3, 1]]

geminal

Author claims: propane-2,2-diol | formula C3H8O2 | canonical composition C3H8O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=75.00, y=32.00; charge=0; explicit H override=none

4 (E): O; x=129.00, y=32.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1], [1, 4, 1]]

ketone

Author claims: pentan-2-one | formula C5H10O | canonical composition C5H10O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=210.00, y=85.00; charge=0; explicit H override=none

4 (E): C; x=264.00, y=115.00; charge=0; explicit H override=none

5 (F): O; x=102.00, y=30.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [1, 5, 2]]

hyald

Author claims: 4-hydroxybutanal | formula C4H8O2 | canonical composition C4H8O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=210.00, y=85.00; charge=0; explicit H override=none

4 (E): O; x=210.00, y=30.00; charge=0; explicit H override=none

5 (F): H; x=260.00, y=115.00; charge=0; explicit H override=none

6 (G): O; x=48.00, y=168.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 2], [3, 5, 1], [0, 6, 1]]

hyoxo

Author claims: 4-hydroxy-2-oxopentanoic acid | formula C5H8O4 | canonical composition C5H8O4 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=210.00, y=85.00; charge=0; explicit H override=none

4 (E): C; x=264.00, y=115.00; charge=0; explicit H override=none

5 (F): O; x=102.00, y=30.00; charge=0; explicit H override=none

6 (G): O; x=210.00, y=30.00; charge=0; explicit H override=none

7 (H): O; x=264.00, y=52.00; charge=0; explicit H override=none

8 (I): O; x=316.00, y=142.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [1, 5, 1], [3, 6, 2], [4, 7, 2], [4, 8, 1]]

ester

Author claims: ethyl propanoate | formula C5H10O2 | canonical composition C5H10O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=156.00, y=52.00; charge=0; explicit H override=none

4 (E): O; x=208.00, y=142.00; charge=0; explicit H override=none

5 (F): C; x=258.00, y=112.00; charge=0; explicit H override=none

6 (G): C; x=308.00, y=142.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1], [4, 5, 1], [5, 6, 1]]

namide

Author claims: N-ethyl-N-methylpropanamide | formula C6H13NO | canonical composition C6H13NO | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=156.00, y=52.00; charge=0; explicit H override=none

4 (E): N; x=208.00, y=142.00; charge=0; explicit H override=none

5 (F): C; x=208.00, y=190.00; charge=0; explicit H override=none

6 (G): C; x=258.00, y=112.00; charge=0; explicit H override=none

7 (H): C; x=308.00, y=142.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1], [4, 5, 1], [4, 6, 1], [6, 7, 1]]

anhyd

Author claims: ethanoic anhydride | formula C4H6O3 | canonical composition C4H6O3 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=45.00, y=130.00; charge=0; explicit H override=none

1 (B): C; x=100.00, y=96.00; charge=0; explicit H override=none

2 (C): O; x=155.00, y=130.00; charge=0; explicit H override=none

3 (D): C; x=210.00, y=96.00; charge=0; explicit H override=none

4 (E): C; x=265.00, y=130.00; charge=0; explicit H override=none

5 (F): O; x=100.00, y=40.00; charge=0; explicit H override=none

6 (G): O; x=210.00, y=40.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [1, 5, 2], [3, 6, 2]]

branch

Author claims: 3-methylhexane | formula C7H16 | canonical composition C7H16 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=210.00, y=85.00; charge=0; explicit H override=none

4 (E): C; x=264.00, y=115.00; charge=0; explicit H override=none

5 (F): C; x=318.00, y=85.00; charge=0; explicit H override=none

6 (G): C; x=156.00, y=36.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [4, 5, 1], [2, 6, 1]]

fake_pent

Author claims: 2,4-dimethylhexane | formula C8H18 | canonical composition C8H18 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=210.00, y=85.00; charge=0; explicit H override=none

4 (E): C; x=264.00, y=115.00; charge=0; explicit H override=none

5 (F): C; x=318.00, y=85.00; charge=0; explicit H override=none

6 (G): C; x=102.00, y=32.00; charge=0; explicit H override=none

7 (H): C; x=210.00, y=32.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [4, 5, 1], [1, 6, 1], [3, 7, 1]]

ynol5

Author claims: pent-4-yn-2-ol | formula C5H8O | canonical composition C5H8O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=210.00, y=85.00; charge=0; explicit H override=none

4 (E): C; x=264.00, y=115.00; charge=0; explicit H override=none

5 (F): O; x=210.00, y=32.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 3], [1, 2, 1], [2, 3, 1], [3, 4, 1], [3, 5, 1]]

cyol

Author claims: 3-methylcyclohexan-1-ol | formula C7H14O | canonical composition C7H14O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=155.00, y=42.00; charge=0; explicit H override=none

1 (B): C; x=208.00, y=74.00; charge=0; explicit H override=none

2 (C): C; x=208.00, y=136.00; charge=0; explicit H override=none

3 (D): C; x=155.00, y=168.00; charge=0; explicit H override=none

4 (E): C; x=102.00, y=136.00; charge=0; explicit H override=none

5 (F): C; x=102.00, y=74.00; charge=0; explicit H override=none

6 (G): O; x=155.00, y=10.00; charge=0; explicit H override=none

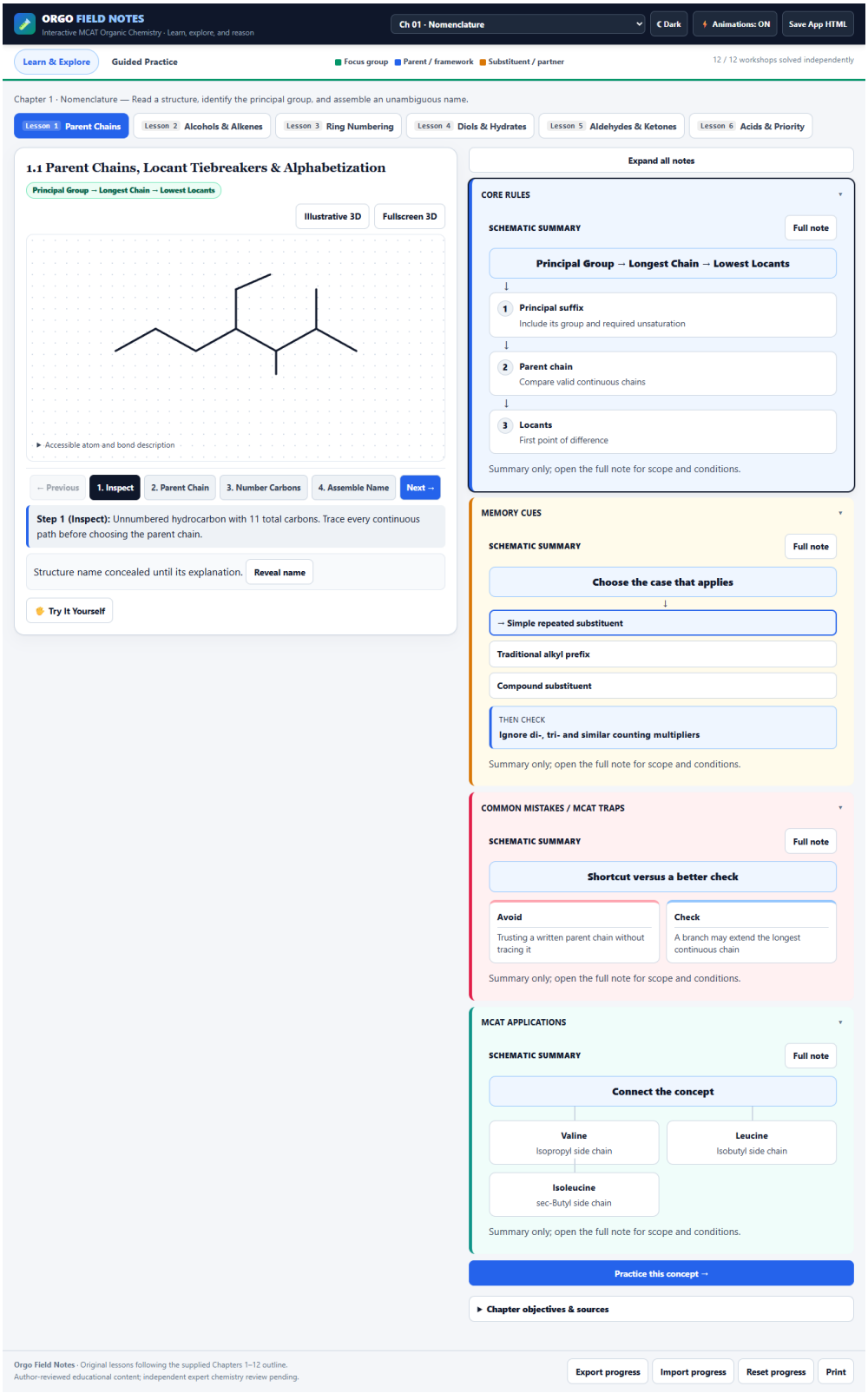
7 (H): C; x=260.00, y=136.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [4, 5, 1], [5, 0, 1], [0, 6, 1], [2, 7, 1]]

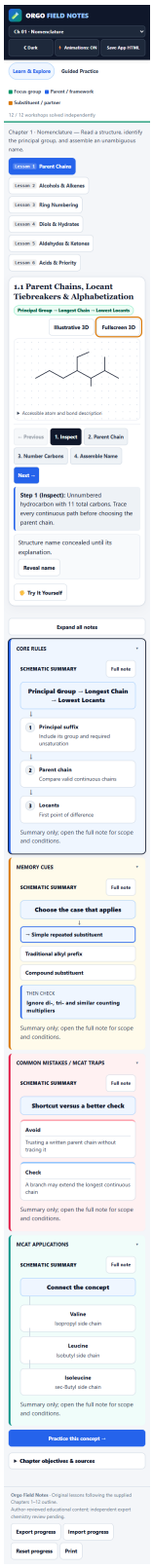
Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

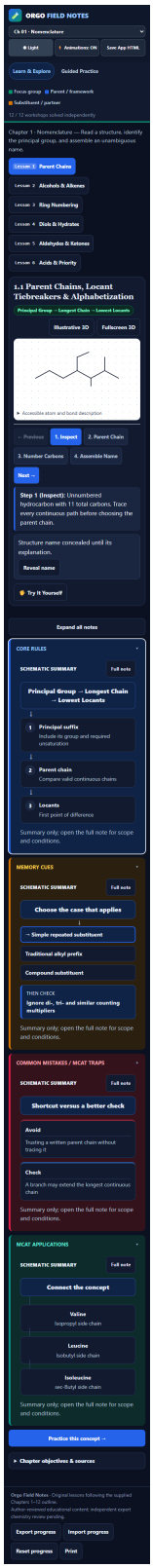
Screenshot: layout-desktop-light



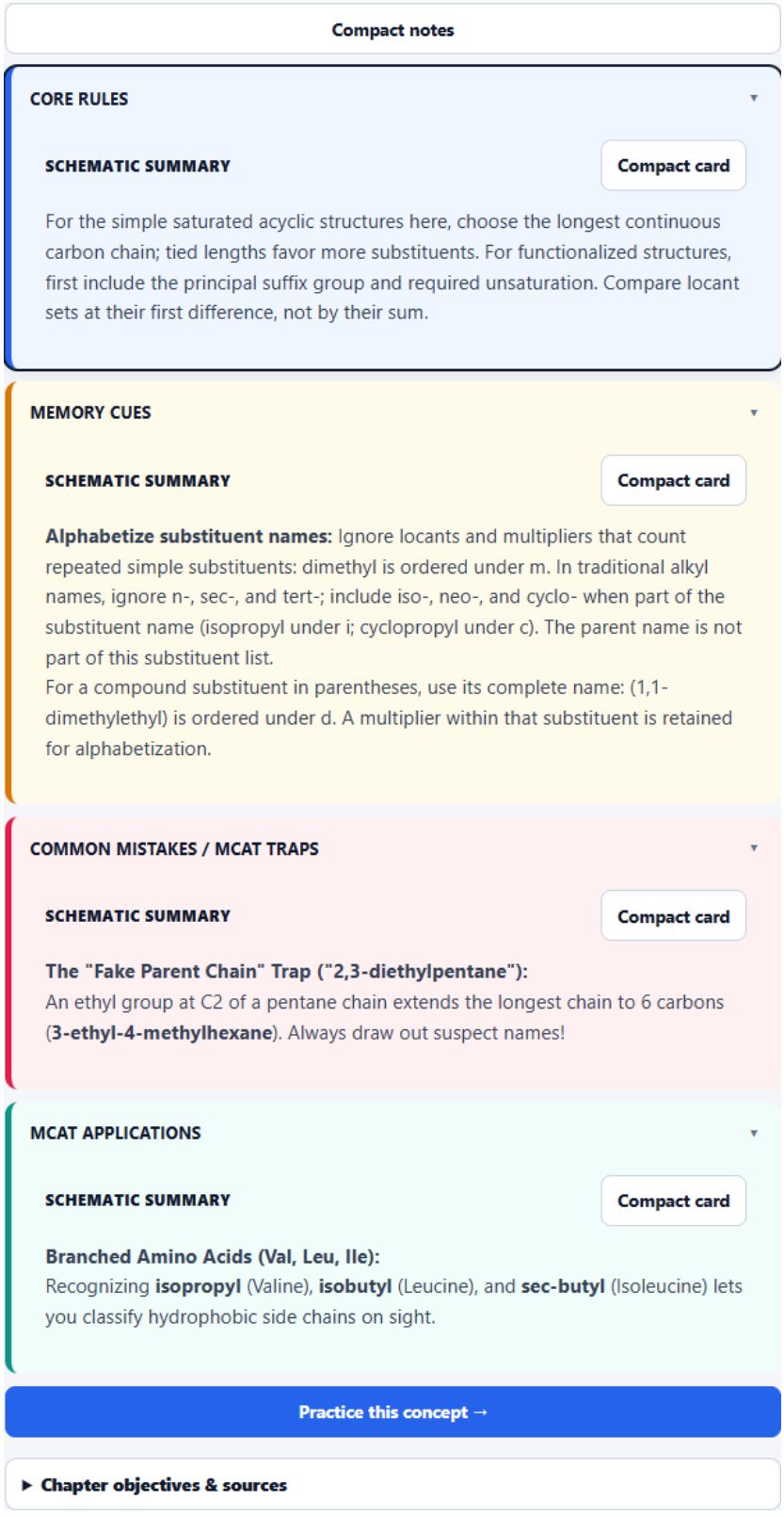
Screenshot: layout-mobile-light



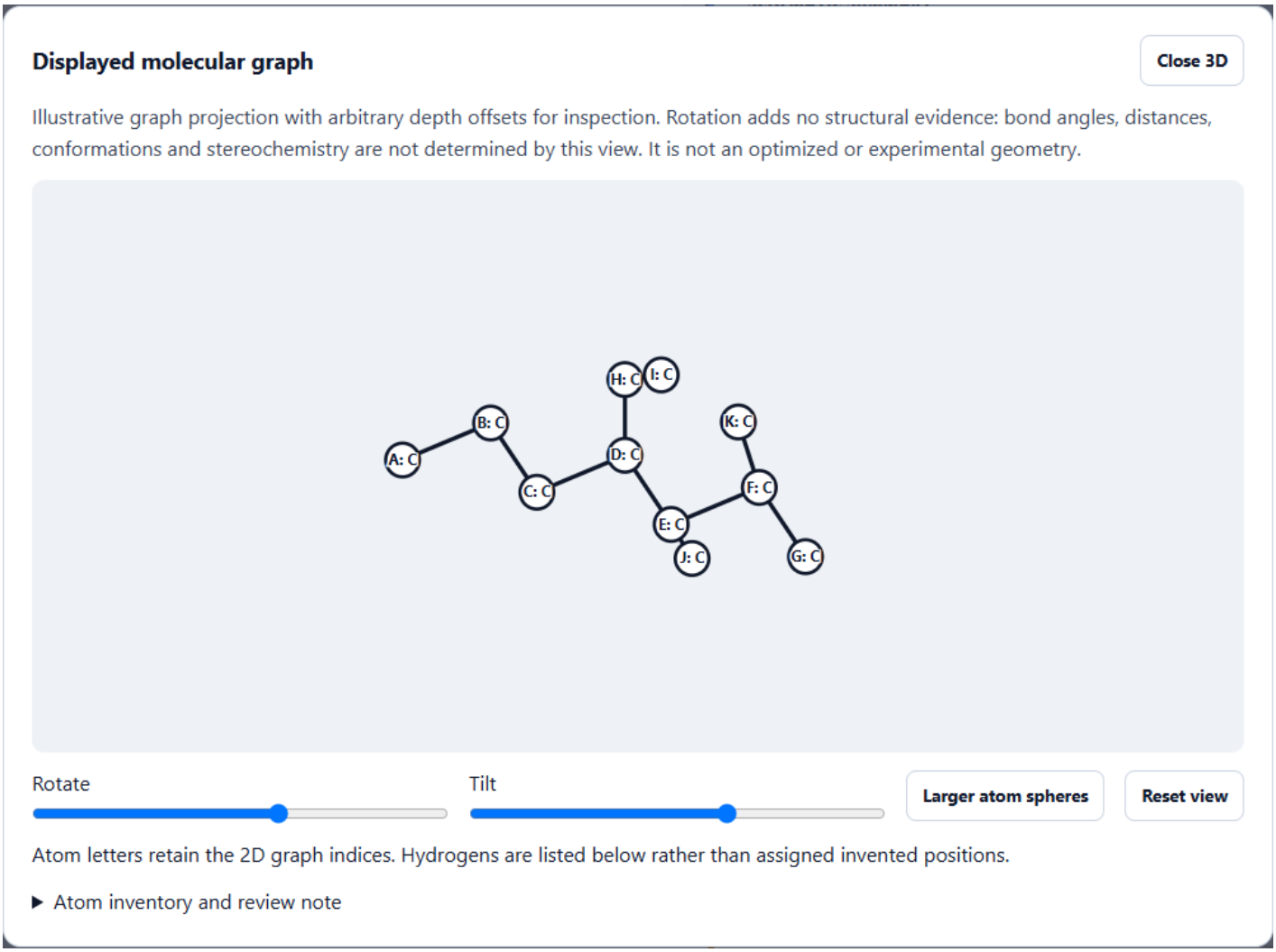
Screenshot: layout-mobile-dark



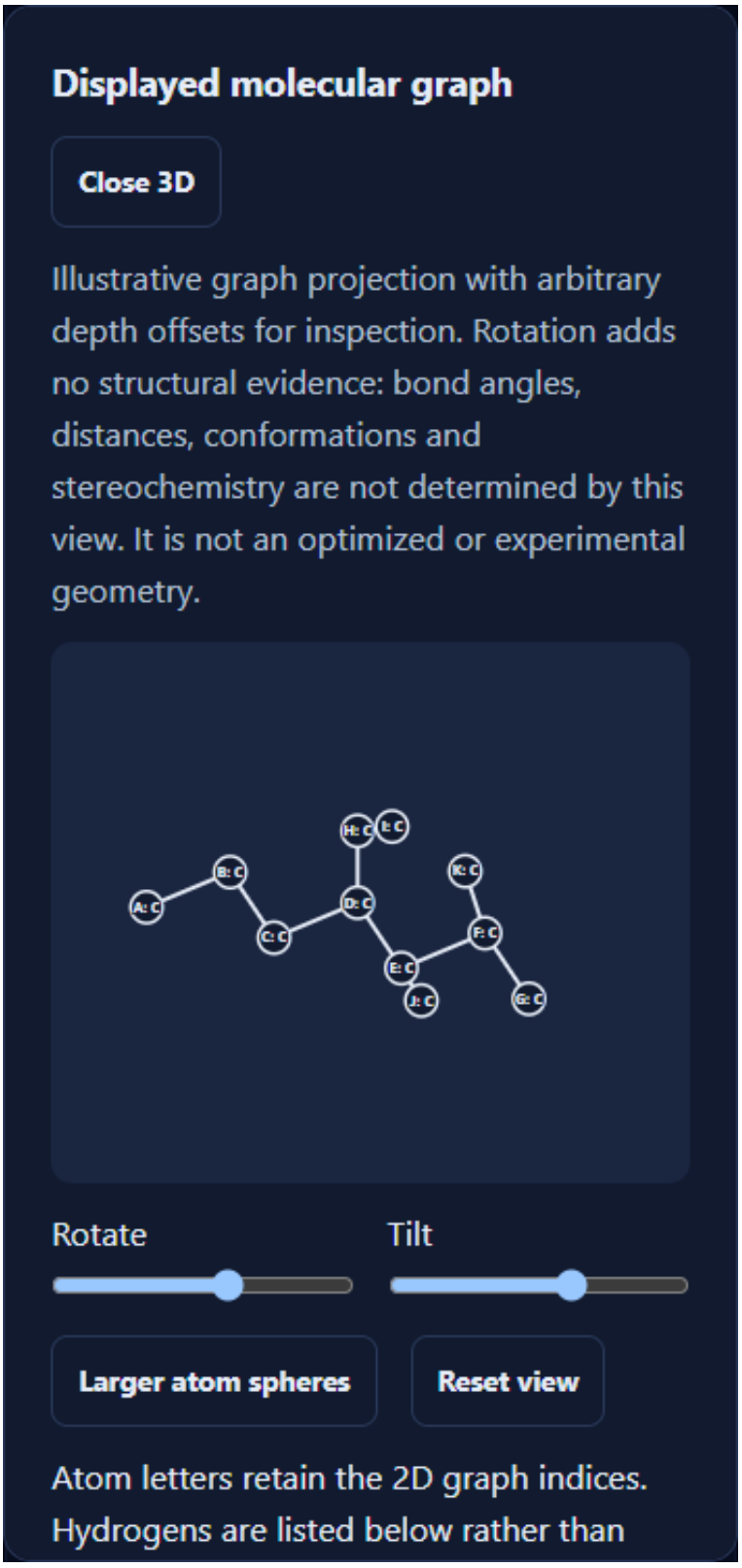
Screenshot: visual-notes-expanded



Screenshot: projection-desktop-light



Screenshot: projection-mobile-dark



Execution evidence

168 screenshots and 55 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.